Group I, Row II in coordinate space, the three-ρ cancellation, and the evolution of target (JIMWLK) and projectile (BFKL)

All terms proportional to $\log(\vee/\Lambda)$ (and $\log(\vee/k^+)$) in one note

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1 Question and answer

Question.

1. Fourier transform the three-ρ part of Group I, Row II (your expression, with the full bracket of logarithms) to coordinate space.
2. Take all three-ρ terms proportional to $\log(\vee/\Lambda)$ and show that they cancel. You suspected that the earlier input did not contain all of them.
3. Take all two-ρ terms proportional to $\log(\vee/\Lambda)$ and show that they reproduce the JIMWLK evolution of the target and the BFKL evolution of the projectile.

Answer.

1. **The Fourier transform** (Part I, every step written out).
   - The bracket in your message lacks the term $-\mathbf{k}^i \log(\vee/k^+)$, which `Group_I.tex` and `main.tex` have. The ξ-integral form of `main.tex` confirms the term (Table 2). It contains no Λ.
   - With the full bracket, $\mathfrak{B}^i = A^i \log(k^+/\Lambda) + R^i$, with $A^i = -E^i - 2\mathbf{k}^i - D^i$ (Eq. (3.12)). So the coefficients of $\log(\vee/\Lambda)$ and $\log(\vee/k^+)$ are A^i and $-A^i$. R^i has no large logarithm.

- Only the part of the kernel that is odd under $x \leftrightarrow y$ contributes. The transform is

$$\int d^2\mathbf{k}\, d^2\mathbf{p}\, e^{i\mathbf{k}\cdot(w-y)} e^{i\mathbf{p}\cdot(y-x)} \frac{A^i}{\mathbf{k}^2\mathbf{p}^2}\bigg|_{\text{odd}} = 2\pi i \left[\mathfrak{F}^i_\beta + \mathfrak{F}^i_\alpha\right],$$

$$\mathfrak{F}^i_\beta = \mathcal{K}^i(x-w)\int_z K(w,y,z) - \mathcal{K}^i(y-w)\int_z K(w,x,z), \qquad \mathfrak{F}^i_\alpha = -\left[\mathcal{K}^i(x-w) - \mathcal{K}^i(y-w)\right]\int_z K(x,y,z).$$

- The row in coordinate space is Eq. (4.13). Its prefactor is $-\frac{i}{2}\mathcal{N}$ for $\log(\vee/\Lambda)$ and $+\frac{i}{2}\mathcal{N}_P$ for $\log(\vee/k^+)$.

2. **The earlier input was complete for** $\log(\vee/\Lambda)$**.** `Evolution_three_rho.tex` has exactly $A^i \log(\vee/\Lambda)$ for this row, and the transform above is the one used before. The new pieces of the full bracket multiply $\log(\vee/k^+)$ or no large logarithm. So the three-ρ result does not change.

3. **The three-ρ terms proportional to** $\log(\vee/\Lambda)$ (Part II).

   - *As written they do not all cancel.* Of 48 colour networks, 28 cancel and 20 do not. Every left-over comes from Group I:
     - the $\mathfrak{F}_\alpha$ part of Row II in the $U(w)\mathbb{A}^{(3)}$ term;
     - part 2 of Row III (the edge kernel Φ), together with a missing term $\mathcal{D}$.
   - *With fixes F3 and F4 all 40 remaining networks cancel exactly*, and a pointwise test gives 10^{-16}.
   - *F4 is now understood* (Sec. 5). The $-2\mathbf{k}^i$ term of the bracket is exactly the commutator $[\mathbb{N}^{(2)}, \mathbb{A}^{(1)}]$ of the normalization with the LO amplitude. `main.tex` sets this commutator to zero, but it is not zero. Ordering the virtual factor in p^+ (soft part to the left of $\mathbb{A}^{(1)}$, hard part to the right) is exactly F4. With it the gluon exchanged between the two charges, $\mathfrak{F}_\alpha$, comes with $\log(\vee/k^+)$ (projectile) instead of $\log(k^+/\Lambda)$ (target).

4. **The two-ρ terms proportional to** $\log(\vee/\Lambda)$ **give JIMWLK⊗LO exactly** (Part III). This holds after fixes F1–F6, in all 13 kernel structures, pointwise in z and with exact coefficients. This is the evolution of the target: the coefficient of $\log(\vee/\Lambda)$ is $-\partial/\partial \log\Lambda$, which comes from gluons softer than the measured one.

5. **BFKL⊗LO is not the coefficient of** $\log(\vee/\Lambda)$**.** It is a different function (Table 10). The projectile evolves with $\log\vee$, i.e. with the coefficient of $\log(\vee/\Lambda)$ *plus* that of $\log(\vee/k^+)$. With fixes P1, F3–F6 (Part IV):

   - the symmetrized three-ρ terms of $\log\vee$ cancel (4 kernel structures, 44 networks);
   - the two-ρ terms of $\log\vee$ equal BFKL⊗LO exactly (9 kernel structures, exact coefficients).

6. **Together** (Part V):

$$\left.\frac{d^3N}{d^3k}\right|_{\text{LL}} = \underbrace{\log\frac{k^+}{\Lambda}}_{\delta Y_T}\,\text{JIMWLK}\otimes\text{LO} + \underbrace{\log\frac{\vee}{k^+}}_{\delta Y_P}\,\text{BFKL}\otimes\text{LO}, \qquad \delta Y_T + \delta Y_P = \log\frac{\vee}{\Lambda} = \delta Y.$$

   The target evolves over the rapidity interval below the measured gluon and the projectile over the interval above it. In terms of your two logarithms, $\log(\vee/\Lambda)$ multiplies JIMWLK⊗LO, and $\log(\vee/k^+)$ multiplies BFKL⊗LO−JIMWLK⊗LO.

fix	what	$\log(\vee/\Lambda)$	$\log\vee$
F1	`4rho.tex`, first part: drop the $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$ term	2ρ	2ρ
F2	`4rho.tex`, $dN^{(1)}$ term 2: repeated index; correct term $3f^{abc}f^{deg}\rho^{b'}(x')\rho^c(y)U^{ag}(y)U^{db'}(w')U^{eb}(z)$	2ρ	2ρ
F3	Group I Row III: part 2 (Φ terms) $\to \mathcal{D}$	3ρ, 2ρ	–
F4	Group I Row II, $U(w)\mathbb{A}^{(3)}$: $\mathfrak{F}_\alpha$ goes with $-\log(\vee/k^+)$, not $\log(k^+/\Lambda)$ (p^+ ordering, Sec. 5)	3ρ, 2ρ	3ρ, 2ρ
F5	Group III Row III part 2 (genuine two-ρ): overall sign	2ρ	2ρ
F6	Group I Row II part 1 (genuine two-ρ): bracket $\to \frac{1}{\pi}\int_z[K(x,w,z) - K(w,w,z)]$ (pole and constant only)	2ρ	–
P1	Group I Row III, $p^+ > k^+ + \Lambda$: large-p^+ limit $-2K_h \to +2K_h$	–	3ρ, 2ρ

Table 1: The fixes, and which coefficients they change. Groups II and III are correct as written, except F5. They agree row by row with an independent leading-log computation, in both regions $p^+ \ll k^+$ and $p^+ \gg k^+$.

Everything uses your convention $U^{\dagger bc} = U^{bc} = U^{cb}$.

How this note is organized. Part I is new. Parts II–IV and the appendices collect the earlier notes `ThreeRho_cancellation`, `TwoRho_evolution`, `LogVeeKplus_BFKL` and `ThreeRho_to_TwoRho`, with cross-references updated. Part IV now compares the coefficient of $\log\vee$ directly with BFKL⊗LO; these tables are new. Equation and table labels of the form `C:`, `T:`, `B:`, `R:` in the source refer to those notes.

2 Conventions

Units.

$$\mathcal{N} \equiv \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\frac{\vee}{\Lambda}, \qquad \mathcal{N}_P \equiv \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\frac{\vee}{k^+}. \tag{2.1}$$

So $\frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = i\mathcal{N}$, $\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = 2i\mathcal{N}$, $\frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = \frac{i}{2}\mathcal{N}$, and the same with $\mathcal{N}_P$ for $\log(\vee/k^+)$. A coefficient written as a number (e.g. $2i$) is in units of $\mathcal{N}$ or $\mathcal{N}_P$.

Logarithms. Your rules are $\log(\Lambda/k^+) = -\log(\vee/\Lambda) + \log(\vee/k^+)$ and $\log((\vee - k^+)/\Lambda) = \log(\vee/\Lambda) + \log(1 - k^+/\vee)$. We write $L_\Lambda = \log(\vee/\Lambda)$, $L_k = \log(\vee/k^+)$, and

$$\delta Y_T \equiv \log\frac{k^+}{\Lambda} = L_\Lambda - L_k, \qquad \delta Y_P \equiv \log\frac{\vee}{k^+} = L_k, \qquad \delta Y_T + \delta Y_P = \log\frac{\vee}{\Lambda}. \tag{2.2}$$

Kernels. $\mathcal{K}(X) \equiv X/X^2$ (a two-vector), $\mathcal{K}(-X) = -\mathcal{K}(X)$, $\mathcal{K}(X)\cdot\mathcal{K}(X) = 1/X^2$, and

$$K(a,b,z) \equiv \mathcal{K}(a-z)\cdot\mathcal{K}(b-z) = \frac{(a-z)\cdot(b-z)}{(a-z)^2(b-z)^2}, \qquad K(a,a,z) = \frac{1}{(a-z)^2}. \tag{2.3}$$

All transverse positions except the measured-gluon positions w (amplitude) and w' (conjugate amplitude) are integrated, and the phase $e^{-ik\cdot(w'-w)}$ is always present. The unobserved (soft or hard) gluon is at z.

Charges. The valence charges obey

$$\big[\rho^a(x), \rho^b(y)\big] = if^{abc}\rho^c(x)\,\delta^{(2)}(x-y). \tag{2.4}$$

Every ordered product of three charges is written as its fully symmetrized part $S(ABC) = \frac{1}{6}(ABC + ACB + BAC + BCA + CAB + CBA)$ plus two-ρ terms (App. A, Eqs. (A.4)–(A.7)):

$$ABC \to S(ABC), \qquad A\{B,C\} \to 2S(ABC), \qquad \{B,C\}A \to 2S(ABC) \qquad \text{(three-}\rho\text{ part)}. \tag{2.5}$$

$S(ABC)$ behaves as a product of commuting charges. After reordering, the two charges of a two-ρ term form a symmetric product and also behave as commuting variables.

Complex conjugate. A three-ρ term has an imaginary coefficient, and $S(\rho\rho\rho)$ is Hermitian. Its c.c. is therefore

$$\text{c.c. of a three-}\rho\text{ term} = -\big(\text{same term with } P \leftrightarrow P'\big), \tag{2.6}$$

where P, P' are the measured-gluon positions. A two-ρ term has a real coefficient, so its c.c. is the same term with $P \leftrightarrow P'$ (no minus sign).

Colour. $(T^a)_{bc} = -if^{abc}$, (F) $f^{acd}f^{bcd} = N_c\delta^{ab}$, and for adjoint Wilson lines

$$\text{(U1)} \quad U^{ab}(x)\,U^{ac}(x) = \delta^{bc}, \tag{2.7}$$

$$\text{(U2)} \quad f^{abc}\,U^{ad}(x)\,U^{be}(x) = f^{deh}\,U^{ch}(x). \tag{2.8}$$

Your notes use

$$U^{\dagger\,bc} = U^{bc} = U^{cb}, \tag{2.9}$$

and we follow this. Steps that need it are marked (U1*), (U2*). They are exact only for a symmetric U.

Part I
The Fourier transform of Group I, Row II

3 The term and its three pieces

The term. The three-ρ part of Group I, Row II, as in your message:

$$\begin{aligned}\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4f^{abc}}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2\mathbf{k}\,d^2\mathbf{p}\int_{x,y} e^{i\mathbf{k}\cdot(w-y)}\,e^{i\mathbf{p}\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,\mathfrak{B}^i(\mathbf{k},\mathbf{p})\\ &\times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{ab}(w)\{\rho^b(x),\rho^c(y)\} - U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big] + \text{c.c.}\end{aligned} \tag{3.1}$$

- *Notation.* Inside the transverse integral $\mathbf{k}$ and $\mathbf{p}$ (bold) are integration momenta. The plain k in $e^{-ik\cdot(w'-w)}$ and k^+ belong to the measured gluon. We also write $\mathbf{q} \equiv \mathbf{k} - \mathbf{p}$, and

$$L_\Lambda \equiv \log\frac{\vee}{\Lambda}, \qquad L_k \equiv \log\frac{\vee}{k^+}, \qquad L_T \equiv \log\frac{k^+}{\Lambda} = L_\Lambda - L_k. \tag{3.2}$$

- *c.c.* Group I always comes with its complex conjugate (as in `Evolution_three_rho.tex`); we write it explicitly.

- *Colour.* As written, the indices a and b each appear three times in the first term of the second bracket. That term is $U(w)\mathbb{A}^{(3)}$ with $\mathbb{A}^{(3)h} \propto f^{hbc}\{\rho^b(x), \rho^c(y)\}$, so we read it as $U^{ah}(w) f^{hbc}\{\rho^b(x), \rho^c(y)\}$. The second term, $f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x), \rho^e(y)\}$, is $\bar{\mathbb{A}}^{(3)a}$ (rotated charges) and is unchanged.

- $(x'-w')^i/(x'-w')^2 = \mathcal{K}^i(x'-w')$ is the LO emission kernel of the conjugate amplitude.

The bracket. With your rules $\log(\Lambda/k^+) = -L_\Lambda + L_k$ and $\log((\vee - k^+)/\Lambda) = L_\Lambda + \log(1 - k^+/\vee)$, the bracket of your message is

$$\mathfrak{B}^i_{\rm msg} = \big[-L_\Lambda + L_k\big]E^i - 2\mathbf{k}^i L_\Lambda - 2\mathbf{k}^i\,\ell_1 + C^i\,\ell_2 - D^i\big[L_\Lambda + \ell_3\big], \tag{3.3}$$

with the three momentum functions

$$E^i \equiv \frac{\mathbf{k}^2\mathbf{p}^i - (\mathbf{k}\cdot\mathbf{p})\,\mathbf{k}^i}{\mathbf{q}^2}, \qquad C^i \equiv \frac{(2\mathbf{p}^2 - 2\mathbf{k}\cdot\mathbf{p} + \mathbf{k}^2)\,\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{\mathbf{q}^2}, \qquad D^i \equiv \frac{\mathbf{p}^2\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{\mathbf{q}^2}, \tag{3.4}$$

and the three logarithms

$$\ell_1 \equiv \log\frac{k^+\mathbf{p}^2}{\vee\mathbf{k}^2 + k^+\mathbf{p}^2}, \qquad \ell_2 \equiv \log\Big[1 + \Big(\frac{\vee}{k^+} - 1\Big)\frac{\mathbf{k}^2}{\mathbf{q}^2}\Big], \qquad \ell_3 \equiv \log\Big(1 - \frac{k^+}{\vee}\Big). \tag{3.5}$$

A useful identity. Since $2\mathbf{p}^2 - 2\mathbf{k}\cdot\mathbf{p} + \mathbf{k}^2 = (\mathbf{k}-\mathbf{p})^2 + \mathbf{p}^2 = \mathbf{q}^2 + \mathbf{p}^2$,

$$C^i = \frac{\mathbf{q}^2\mathbf{k}^i + \mathbf{p}^2\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{\mathbf{q}^2} = \mathbf{k}^i + D^i. \tag{3.6}$$

3.1 A term that is missing in the message

In `Group_I.tex` and in the coordinate-space $\mathbb{A}^{(3)}_2$ of `main.tex`, the same bracket ends with one more term, $-\mathbf{k}^i\log(\vee/k^+)$:

$$\boxed{\mathfrak{B}^i = \mathfrak{B}^i_{\rm msg} - \mathbf{k}^i\,L_k} \tag{3.7}$$

`main.tex` also gives this bracket in momentum space *before* the longitudinal integral is done ($\xi = p^+/k^+$):

$$\begin{aligned}\mathfrak{B}^i_\xi = {}& \log\frac{\Lambda}{k^+}\,E^i - 2\mathbf{k}^i L_\Lambda - 2\mathbf{k}^i\ell_1 \\ & - \int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\,\frac{\xi}{\big(\xi\mathbf{q}^2 + (1-\xi)\mathbf{k}^2\big)(1-\xi)}\Big[\Big(\frac{2}{\xi}(\mathbf{p}^2 - \mathbf{k}\cdot\mathbf{p}) - \mathbf{p}^2 + 2\mathbf{k}\cdot\mathbf{p}\Big)\mathbf{k}^i + \frac{1}{\xi}\mathbf{k}^2\mathbf{p}^i\Big].\end{aligned} \tag{3.8}$$

Table 2 compares the three forms at random momenta (component $i = 1$, $k^+ = 1$, $\Lambda = 10^{-4}$). The ξ form agrees with $\mathfrak{B}$ up to terms of order $k^+/\vee$, and differs from $\mathfrak{B}_{\rm msg}$ by $\mathbf{k}^i L_k$, which grows with $\vee$. We therefore use $\mathfrak{B}$, Eq. (3.7).

$\vee$	$\mathfrak{B}^1_\xi$ (`main.tex`, ξ integral)	$\mathfrak{B}^1$ (with $-\mathbf{k}^1 L_k$)	$\mathfrak{B}^1_{\rm msg}$ (your message)
10^2	0.306467	0.304224	0.309890
10^2	11.741657	11.718828	9.624992
10^4	8.028061	8.028089	3.494700
10^4	−2.180515	−2.180523	−1.209622

Table 2: The bracket of Group I, Row II at random transverse momenta.

The extra term contains no Λ. It therefore does not change the terms proportional to $\log(\vee/\Lambda)$ (Parts II and III). It does change the terms proportional to $\log(\vee/k^+)$ (Part IV).

3.2 Splitting the bracket into $\log(\vee/\Lambda)$, $\log(\vee/k^+)$ and the rest

For $\vee \gg k^+$ (at fixed k^+ and transverse momenta) the three logarithms are, step by step:

$$\begin{aligned}
\ell_1 &= -\log\Big(\frac{\vee \mathbf{k}^2}{k^+\mathbf{p}^2}+1\Big) = -\log\frac{\vee}{k^+} - \log\frac{\mathbf{k}^2}{\mathbf{p}^2} - \log\Big(1+\frac{k^+\mathbf{p}^2}{\vee\mathbf{k}^2}\Big) = -L_k + \log\frac{\mathbf{p}^2}{\mathbf{k}^2} + O\Big(\frac{k^+}{\vee}\Big),\\
\ell_2 &= \log\Big[\frac{\vee}{k^+}\frac{\mathbf{k}^2}{\mathbf{q}^2}\Big(1+\frac{k^+}{\vee}\Big(\frac{\mathbf{q}^2}{\mathbf{k}^2}-1\Big)\Big)\Big] = L_k + \log\frac{\mathbf{k}^2}{\mathbf{q}^2} + O\Big(\frac{k^+}{\vee}\Big),\\
\ell_3 &= O\Big(\frac{k^+}{\vee}\Big).
\end{aligned} \tag{3.9}$$

Insert these and $C^i = \mathbf{k}^i + D^i$ into Eq. (3.7):

$$\mathfrak{B}^i = \big[-L_\Lambda + L_k\big]E^i - 2\mathbf{k}^i L_\Lambda - 2\mathbf{k}^i\Big[-L_k + \log\frac{\mathbf{p}^2}{\mathbf{k}^2}\Big] + \big(\mathbf{k}^i + D^i\big)\Big[L_k + \log\frac{\mathbf{k}^2}{\mathbf{q}^2}\Big] - D^i L_\Lambda - \mathbf{k}^i L_k + O\Big(\frac{k^+}{\vee}\Big). \tag{3.10}$$

Collecting the coefficients of L_Λ and L_k:

$$\begin{aligned}
L_\Lambda: &\quad -E^i - 2\mathbf{k}^i - D^i \;\equiv\; A^i,\\
L_k: &\quad +E^i + 2\mathbf{k}^i + \mathbf{k}^i + D^i - \mathbf{k}^i = E^i + 2\mathbf{k}^i + D^i \;\equiv\; B^i = -A^i,\\
\text{rest}: &\quad -2\mathbf{k}^i\log\frac{\mathbf{p}^2}{\mathbf{k}^2} + \big(\mathbf{k}^i + D^i\big)\log\frac{\mathbf{k}^2}{\mathbf{q}^2} \;\equiv\; R^i.
\end{aligned} \tag{3.11}$$

Hence

$$\boxed{\mathfrak{B}^i = A^i\,L_\Lambda - A^i\,L_k + R^i + O\Big(\frac{k^+}{\vee}\Big) = A^i\log\frac{k^+}{\Lambda} + R^i + O\Big(\frac{k^+}{\vee}\Big), \qquad A^i = -E^i - 2\mathbf{k}^i - D^i.} \tag{3.12}$$

Table 3 checks Eq. (3.12) numerically: the difference falls like $1/\vee$. With the bracket of the message, the coefficient of L_k would be $B^i_{\rm msg} = B^i + \mathbf{k}^i$, i.e. $A^i + B^i_{\rm msg} = \mathbf{k}^i \neq 0$.

point	$\vee/k^+$	$\max_i\lvert\mathfrak{B}^i - (A^iL_\Lambda + B^iL_k + R^i)\rvert$	$\max_i\lvert\mathfrak{B}^i\rvert$
1	10^2	8.0×10^{-4}	25.74
1	10^4	8.2×10^{-6}	25.74
1	10^6	8.2×10^{-8}	25.74
2	10^2	2.7×10^{-1}	18.65
2	10^4	2.8×10^{-3}	18.91
2	10^6	2.8×10^{-5}	18.92
3	10^2	6.1×10^{-3}	17.11
3	10^4	6.2×10^{-5}	17.12
3	10^6	6.2×10^{-7}	17.12

Table 3: The split (3.12) at random momenta ($\Lambda/k^+ = 10^{-3}$).

What the split means. A^i is the coefficient of $\log(\vee/\Lambda)$, B^i that of $\log(\vee/k^+)$. Their sum $A^i + B^i$ is the coefficient of $\log\vee$, which comes from virtual gluons much harder than the measured one ($p^+ \gg k^+$). As written, $A^i + B^i = 0$: in your notes the large logarithm of this row is $\log(k^+/\Lambda)$, i.e. the virtual gluon is always softer than the measured one. Sec. 5 shows that this is not quite right for the part $-2\mathbf{k}^i$ of A^i (fix F4).

The earlier input was complete. `Evolution_three_rho.tex`, which the three-ρ analysis used, has for this row exactly $-L_\Lambda\big[E^i + 2\mathbf{k}^i + D^i\big] = A^iL_\Lambda$. The full bracket adds only terms with L_k and terms without large logarithms, and none of them contains Λ. So no $\log(\vee/\Lambda)$ term was missing. The Fourier transform below is the one used before (App. A of `ThreeRho_to_TwoRho`), now with every step written out.

4 The Fourier transform, step by step

For a momentum function $X^i(\mathbf{k},\mathbf{p})$ define its transform

$$F[X]^i(w;x,y) \equiv \int d^2\mathbf{k}\,d^2\mathbf{p}\;e^{i\mathbf{k}\cdot(w-y)}\,e^{i\mathbf{p}\cdot(y-x)}\,\frac{X^i(\mathbf{k},\mathbf{p})}{\mathbf{k}^2\mathbf{p}^2}. \tag{4.1}$$

By Eq. (3.12) we need $F[A]$; then $F[B] = -F[A]$. The rest R is discussed in Sec. 4.6.

4.1 Only the part odd under $x \leftrightarrow y$ contributes

Call the two colour structures $O_1(x,y) = U^{ah}(w) f^{hbc}\{\rho^b(x), \rho^c(y)\}$ and $O_2(x,y) = f^{abc} U^{bd}(x) U^{ce}(y) \{\rho^d(x), \rho^e(y)\}$.

- $O_1(y,x) = U^{ah}(w) f^{hbc}\{\rho^b(y), \rho^c(x)\}$. Rename the dummy indices $b \leftrightarrow c$: $U^{ah}(w) f^{hcb}\{\rho^c(y), \rho^b(x)\}$. The anti-commutator is symmetric and $f^{hcb} = -f^{hbc}$, so $O_1(y,x) = -O_1(x,y)$.
- $O_2(y,x) = f^{abc} U^{bd}(y) U^{ce}(x) \{\rho^d(y), \rho^e(x)\}$. Rename $b \leftrightarrow c$, $d \leftrightarrow e$: $f^{acb} U^{ce}(y) U^{bd}(x) \{\rho^e(y), \rho^d(x)\} = -O_2(x,y)$.

Write any kernel as $F = F_{\rm odd} + F_{\rm even}$ with $F_{\rm odd/even}(x,y) = \frac{1}{2}[F(x,y) \mp F(y,x)]$. Renaming the integration variables $x \leftrightarrow y$ gives $\int_{x,y} F_{\rm even} O = \int_{x,y} F_{\rm even}(y,x) O(y,x) = -\int_{x,y} F_{\rm even} O$, so this integral vanishes. Hence only $F_{\rm odd}$ contributes:

$$\int_{x,y} F(x,y)\, O_{1,2}(x,y) = \int_{x,y} F_{\rm odd}(x,y)\, O_{1,2}(x,y). \tag{4.2}$$

(In momentum space $x \leftrightarrow y$ is $\mathbf{p} \leftrightarrow \mathbf{q}$: the phase $e^{i\mathbf{k}\cdot w} e^{-i\mathbf{p}\cdot x} e^{-i\mathbf{q}\cdot y}$ goes into $e^{i\mathbf{k}\cdot w} e^{-i\mathbf{q}\cdot x} e^{-i\mathbf{p}\cdot y}$.)

4.2 Three basic transforms

(T2) The propagator. $\int d^2\mathbf{k}\, e^{i\mathbf{k}\cdot X}/\mathbf{k}^2$ diverges at small $\mathbf{k}$. With a small mass m as regulator, do the angular integral first, $\int_0^{2\pi} d\theta\, e^{i|\mathbf{k}||X|\cos\theta} = 2\pi J_0(|\mathbf{k}||X|)$, and then use $\int_0^\infty dk\, k\, J_0(k|X|)/(k^2+m^2) = K_0(m|X|)$:

$$\int d^2\mathbf{k}\, \frac{e^{i\mathbf{k}\cdot X}}{\mathbf{k}^2+m^2} = 2\pi K_0(m|X|) = -\pi \log X^2 + C_0 + O(m^2X^2), \qquad C_0 = -2\pi\Big[\log\frac{m}{2} + \gamma_E\Big]. \tag{T2}$$

Here we used $K_0(t) = -\log(t/2) - \gamma_E + O(t^2)$. The constant C_0 does not depend on X; it cancels in all results below.

(T1) The emission kernel. Since $\mathbf{k}^i e^{i\mathbf{k}\cdot X} = -i\,\partial_X^i e^{i\mathbf{k}\cdot X}$, and $\partial^i \log X^2 = 2X^i/X^2$,

$$\int d^2\mathbf{k}\, e^{i\mathbf{k}\cdot X}\, \frac{\mathbf{k}^i}{\mathbf{k}^2} = -i\,\partial_X^i\big(-\pi\log X^2 + C_0\big) = 2\pi i\, \frac{X^i}{X^2} = 2\pi i\, \mathcal{K}^i(X). \tag{T1}$$

Inverting the Fourier transform, $\mathcal{K}^i(u) = \frac{1}{2\pi i}\int d^2\mathbf{k}\, e^{i\mathbf{k}\cdot u}\mathbf{k}^i/\mathbf{k}^2$ gives

$$\frac{\mathbf{k}^i}{\mathbf{k}^2} = \frac{2\pi i}{(2\pi)^2}\int d^2u\, e^{-i\mathbf{k}\cdot u}\, \mathcal{K}^i(u) = \frac{i}{2\pi}\int d^2u\, e^{-i\mathbf{k}\cdot u}\, \mathcal{K}^i(u). \tag{T1$'$}$$

(T3) The soft-gluon kernel. With $K(a,b,z) \equiv \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, write both $\mathcal{K}$'s with (T1), and use $\int d^2z\, e^{-i(\mathbf{k}+\mathbf{k}')\cdot z} = (2\pi)^2\delta^{(2)}(\mathbf{k}+\mathbf{k}')$:

$$\begin{aligned}\int_z K(a,b,z) &= \frac{1}{(2\pi i)^2}\int d^2\mathbf{k}\, d^2\mathbf{k}'\, \frac{\mathbf{k}\cdot\mathbf{k}'}{\mathbf{k}^2\mathbf{k}'^2}\, e^{i\mathbf{k}\cdot a + i\mathbf{k}'\cdot b}\int_z e^{-i(\mathbf{k}+\mathbf{k}')\cdot z} = \frac{(2\pi)^2}{(2\pi i)^2}\int d^2\mathbf{k}\, \frac{\mathbf{k}\cdot(-\mathbf{k})}{\mathbf{k}^2\,\mathbf{k}^2}\, e^{i\mathbf{k}\cdot(a-b)} \\ &= \int d^2\mathbf{k}\, \frac{e^{i\mathbf{k}\cdot(a-b)}}{\mathbf{k}^2} = -\pi\log(a-b)^2 + C_0. \end{aligned} \tag{T3}$$

So $\int d^2\mathbf{k}\, e^{i\mathbf{k}\cdot X}/\mathbf{k}^2$ is the soft-gluon kernel integrated over the gluon position, with the same infrared constant.

4.3 Partial fractions

$A^i = -(E^i + D^i) - 2\mathbf{k}^i$, and

$$E^i + D^i = \frac{\mathbf{k}^2\mathbf{p}^i - (\mathbf{k}\cdot\mathbf{p})\mathbf{k}^i + \mathbf{p}^2\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{\mathbf{q}^2} = \frac{2\mathbf{k}^2\mathbf{p}^i + (\mathbf{p}^2 - \mathbf{k}\cdot\mathbf{p})\,\mathbf{k}^i}{\mathbf{q}^2} = \frac{2\mathbf{k}^2\mathbf{p}^i - (\mathbf{p}\cdot\mathbf{q})\,\mathbf{k}^i}{\mathbf{q}^2}, \tag{4.3}$$

because $\mathbf{p}^2 - \mathbf{k}\cdot\mathbf{p} = -\mathbf{p}\cdot(\mathbf{k}-\mathbf{p}) = -\mathbf{p}\cdot\mathbf{q}$. Dividing by $\mathbf{k}^2\mathbf{p}^2$:

$$\frac{A^i}{\mathbf{k}^2\mathbf{p}^2} = \underbrace{-\frac{2\mathbf{p}^i}{\mathbf{q}^2\,\mathbf{p}^2}}_{(a)} + \underbrace{\frac{(\mathbf{p}\cdot\mathbf{q})\,\mathbf{k}^i}{\mathbf{q}^2\,\mathbf{k}^2\,\mathbf{p}^2}}_{(b)} \underbrace{-\frac{2\mathbf{k}^i}{\mathbf{k}^2\,\mathbf{p}^2}}_{(c)}. \tag{4.4}$$

Terms (a) and (b) come from $E^i + D^i$ (the pole at $\mathbf{q} = 0$); term (c) comes from $-2\mathbf{k}^i$. (Checked with `sympy`.)

4.4 The three transforms

Term (c). It factorizes. By (T1) and (T3),

$$F[(c)]^i = -2\int d^2\mathbf{k}\, e^{i\mathbf{k}\cdot(w-y)}\frac{\mathbf{k}^i}{\mathbf{k}^2}\int d^2\mathbf{p}\, \frac{e^{i\mathbf{p}\cdot(y-x)}}{\mathbf{p}^2} = -2\cdot 2\pi i\, \mathcal{K}^i(w-y)\int_z K(x,y,z) = 4\pi i\, \mathcal{K}^i(y-w)\int_z K(x,y,z). \tag{4.5}$$

Term (a). Change variables from $(\mathbf{k},\mathbf{p})$ to $(\mathbf{q},\mathbf{p})$, $\mathbf{k}=\mathbf{q}+\mathbf{p}$ (Jacobian 1). The phase becomes

$$\mathbf{k}\cdot(w-y)+\mathbf{p}\cdot(y-x)=\mathbf{q}\cdot(w-y)+\mathbf{p}\cdot(w-y)+\mathbf{p}\cdot(y-x)=\mathbf{q}\cdot(w-y)+\mathbf{p}\cdot(w-x), \tag{4.6}$$

so

$$F[(a)]^i=-2\int d^2\mathbf{p}\,e^{i\mathbf{p}\cdot(w-x)}\frac{\mathbf{p}^i}{\mathbf{p}^2}\int d^2\mathbf{q}\,\frac{e^{i\mathbf{q}\cdot(w-y)}}{\mathbf{q}^2}=-2\cdot 2\pi i\,\mathcal{K}^i(w-x)\int_z K(w,y,z)=4\pi i\,\mathcal{K}^i(x-w)\int_z K(w,y,z). \tag{4.7}$$

Term (b). Write $\mathbf{k}^i/\mathbf{k}^2$ with (T1′) and change variables to $(\mathbf{q},\mathbf{p})$. The phase becomes $\mathbf{q}\cdot(w-y-u)+\mathbf{p}\cdot(w-x-u)$, and the $\mathbf{p}$ and $\mathbf{q}$ integrals factorize:

$$\begin{aligned}F[(b)]^i&=\frac{i}{2\pi}\int d^2u\,\mathcal{K}^i(u)\int d^2\mathbf{p}\,e^{i\mathbf{p}\cdot(w-x-u)}\frac{\mathbf{p}^j}{\mathbf{p}^2}\int d^2\mathbf{q}\,e^{i\mathbf{q}\cdot(w-y-u)}\frac{\mathbf{q}^j}{\mathbf{q}^2}\\&=\frac{i}{2\pi}(2\pi i)^2\int d^2u\,\mathcal{K}^i(u)\,\mathcal{K}(w-u-x)\cdot\mathcal{K}(w-u-y)\\&=-2\pi i\int_z\mathcal{K}^i(w-z)\,K(x,y,z)\qquad(z\equiv w-u).\end{aligned} \tag{4.8}$$

4.5 The result

Now take the part odd under $x\leftrightarrow y$:

- (b) is even, because $K(x,y,z)=K(y,x,z)$. It drops out.
- (a): $\frac{1}{2}\big[F[(a)](x,y)-F[(a)](y,x)\big]=2\pi i\big[\mathcal{K}^i(x-w)\int_z K(w,y,z)-\mathcal{K}^i(y-w)\int_z K(w,x,z)\big]$.
- (c): $\frac{1}{2}\big[F[(c)](x,y)-F[(c)](y,x)\big]=-2\pi i\big[\mathcal{K}^i(x-w)-\mathcal{K}^i(y-w)\big]\int_z K(x,y,z)$.

Therefore

$$\boxed{F[A]^i_{\rm odd}=2\pi i\,\mathfrak{F}^i(w;x,y),\qquad \mathfrak{F}^i=\mathfrak{F}^i_\beta+\mathfrak{F}^i_\alpha} \tag{4.9}$$

with

$$\begin{aligned}\mathfrak{F}^i_\beta(w;x,y)&=\mathcal{K}^i(x-w)\int_z K(w,y,z)-\mathcal{K}^i(y-w)\int_z K(w,x,z),\\\mathfrak{F}^i_\alpha(w;x,y)&=-\big[\mathcal{K}^i(x-w)-\mathcal{K}^i(y-w)\big]\int_z K(x,y,z).\end{aligned} \tag{4.10}$$

$\mathfrak{F}_\beta$ comes from E^i+D^i (the pole at $\mathbf{q}=0$): the soft gluon at z connects the measured gluon at w with a charge. $\mathfrak{F}_\alpha$ comes from $-2\mathbf{k}^i$: the soft gluon connects the two charges x and y.

Infrared finiteness. By (T3) each $\int_z K$ is $-\pi\log(\cdot)^2+C_0$. In the sum $\mathfrak{F}_\beta+\mathfrak{F}_\alpha$ the constant multiplies $\mathcal{K}^i(x-w)(C_0-C_0)-\mathcal{K}^i(y-w)(C_0-C_0)=0$, so

$$\mathfrak{F}^i(w;x,y)=-\pi\Big[\mathcal{K}^i(x-w)\log\frac{(w-y)^2}{(x-y)^2}-\mathcal{K}^i(y-w)\log\frac{(w-x)^2}{(x-y)^2}\Big], \tag{4.11}$$

which is finite. In the rest of this note all kernels are kept as integrands in the soft-gluon position z; the large-z behaviour of the individual pieces cancels in the sums (JIMWLK⊗LO and BFKL⊗LO).

Prefactor. With Eq. (4.9) the prefactor of Eq. (3.1) becomes

$$-\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\cdot 2\pi i=-\frac{i}{2}\cdot\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}, \tag{4.12}$$

i.e. $-\frac{i}{2}\mathcal{N}$ for the L_Λ term and, because $B^i=-A^i$, $+\frac{i}{2}\mathcal{N}_P$ for the L_k term (units of Sec. 2).

Group I, Row II in coordinate space. Collecting,

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm I.II,\,3\rho}=\Big(-\frac{i}{2}\mathcal{N}+\frac{i}{2}\mathcal{N}_P\Big)\int e^{-ik\cdot(w'-w)}\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,y)\Big\{\\+\,U^{ab'}(x')\,f^{hbc}\,U^{ah}(w)\,\rho^{b'}(x')\,\{\rho^b(x),\rho^c(y)\}-U^{ab'}(x')\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\{\rho^d(x),\rho^e(y)\}\\-\,U^{ab'}(w')\,f^{hbc}\,U^{ah}(w)\,\rho^{b'}(x')\,\{\rho^b(x),\rho^c(y)\}+U^{ab'}(w')\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\{\rho^d(x),\rho^e(y)\}\Big\}\\+\,(\text{finite, Sec. 4.6})+\text{c.c.}\end{aligned} \tag{4.13}$$

Since $-\frac{i}{2}(\mathcal{N}-\mathcal{N}_P)\propto\log(\vee/\Lambda)-\log(\vee/k^+)=\log(k^+/\Lambda)$, the whole row is proportional to $\log(k^+/\Lambda)$, as found in Eq. (3.12). The sub-terms are numbered 1–4 in the order shown: 1 and 3 contain $U(w)\mathbb{A}^{(3)}$, 2 and 4 the rotated $\bar{\mathbb{A}}^{(3)}$. The $\log(\vee/\Lambda)$ part is exactly the input of the three-ρ analysis (Part II and App. A).

With the bracket of your message, the L_k term would carry $\mathfrak{F}+\frac{1}{2}\mathfrak{F}_\alpha$ instead of $\mathfrak{F}$ (the extra $+\mathbf{k}^iL_k$ transforms like term (c) with weight $-\frac{1}{2}$).

4.6 The rest R^i

$$R^i = -2\mathbf{k}^i \log\frac{\mathbf{p}^2}{\mathbf{k}^2} + \big(\mathbf{k}^i + D^i\big)\log\frac{\mathbf{k}^2}{\mathbf{q}^2}, \qquad \frac{R^i}{\mathbf{k}^2\mathbf{p}^2} = -\frac{2\mathbf{k}^i}{\mathbf{k}^2\mathbf{p}^2}\log\frac{\mathbf{p}^2}{\mathbf{k}^2} + \Big[\frac{\mathbf{k}^i}{\mathbf{k}^2\mathbf{p}^2} + \frac{\mathbf{k}^i}{\mathbf{q}^2\mathbf{k}^2} + \frac{\mathbf{p}^i}{\mathbf{q}^2\mathbf{p}^2}\Big]\log\frac{\mathbf{k}^2}{\mathbf{q}^2}. \tag{4.14}$$

(The second form uses $D^i/(\mathbf{k}^2\mathbf{p}^2) = \mathbf{k}^i/(\mathbf{q}^2\mathbf{k}^2) + \mathbf{p}^i/(\mathbf{q}^2\mathbf{p}^2)$.) R^i contains neither $\vee$, nor Λ, nor k^+. It is part of the finite NLO correction and does not enter the evolution. Only its part odd under $\mathbf{p} \leftrightarrow \mathbf{q}$ contributes. Its transform involves $\log^2$ of distances and three-point integrals that do not factorize; as `main.tex` does for its finite terms, we keep it in momentum space:

$$\left.\frac{d^3N}{d^3k}\right|_{\text{I.II, finite}} = -\frac{1}{(2\pi)^3}\frac{g^4 f^{abc}}{64\pi^6}\frac{1}{k^+}\int e^{-ik\cdot(w'-w)}\int d^2\mathbf{k}\,d^2\mathbf{p}\int_{x,y} e^{i\mathbf{k}\cdot(w-y)}e^{i\mathbf{p}\cdot(y-x)}\,\frac{\mathcal{K}^i(x'-w')\,R^i}{\mathbf{k}^2\mathbf{p}^2}\,[\dots][\dots] + \text{c.c.}, \tag{4.15}$$

with the colour brackets of Eq. (3.1).

5 The α part is an operator-ordering commutator

The three-ρ terms proportional to $\log(\vee/\Lambda)$ cancel only if the part $\mathfrak{F}_\alpha$ is removed from the $U(w)\mathbb{A}^{(3)}$ term of Row II (fix F4, Part II). Here we show what this part is.

Ingredients (`main.tex`). With $\rho^a(\mathbf{p}) \equiv \int_x e^{-i\mathbf{p}\cdot x}\rho^a(x)$, the charge algebra reads

$$[\rho^a(\mathbf{k}), \rho^b(\mathbf{p})] = \int_{x,y} e^{-i\mathbf{k}\cdot x - i\mathbf{p}\cdot y}\, if^{abc}\rho^c(x)\,\delta^{(2)}(x-y) = if^{abc}\rho^c(\mathbf{k}+\mathbf{p}). \tag{5.1}$$

The LO amplitude and the normalization are (Eqs. (A1coeffM), (Ncoeffpos) of `main.tex`; for $\mathbb{N}$ use (T3))

$$\mathbb{A}_{2i}^{(1)a}(k^+,\mathbf{k}) = -\frac{\sqrt{2}\,g\,\mathbf{k}^i\rho^a(-\mathbf{k})}{\sqrt{k^+}\,\mathbf{k}^2}, \tag{5.2}$$

$$\mathbb{N}^{(2)} = -\frac{g^2}{(2\pi)^3}L_\Lambda\int_{x,y,z} K(x,y,z)\,\rho^b(x)\rho^b(y) = -\frac{g^2}{8\pi^3}L_\Lambda\int d^2\mathbf{p}\,\frac{\rho^b(\mathbf{p})\rho^b(-\mathbf{p})}{\mathbf{p}^2} \equiv L_\Lambda\, n. \tag{5.3}$$

$\mathbb{A}_{2i}^{(3)a}$ of `main.tex` contains a three-charge term and the $f\{\rho,\rho\}$ term whose bracket is $\mathfrak{B}^i$:

$$T_3 = \int d^2\mathbf{p}\,\frac{g^3\,\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})\,\mathbf{k}^i}{4\sqrt{2}\pi^3\,\mathbf{k}^2\mathbf{p}^2\sqrt{k^+}}\,L_\Lambda, \tag{5.4}$$

$$T_f = \int d^2\mathbf{p}\,\frac{ig^3 f^{abc}\{\rho^b(-\mathbf{p}),\rho^c(\mathbf{p}-\mathbf{k})\}}{8\sqrt{2}\pi^3\,\mathbf{p}^2\mathbf{k}^2\sqrt{k^+}}\,\mathfrak{B}^i. \tag{5.5}$$

Step 1: the three-charge term is $\mathbb{N}^{(2)}\mathbb{A}^{(1)}$. Multiplying,

$$\mathbb{N}^{(2)}\mathbb{A}_{2i}^{(1)a} = \Big(-\frac{g^2L_\Lambda}{8\pi^3}\Big)\Big(-\frac{\sqrt{2}\,g\,\mathbf{k}^i}{\sqrt{k^+}\mathbf{k}^2}\Big)\int d^2\mathbf{p}\,\frac{\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{\mathbf{p}^2} = \frac{\sqrt{2}}{8}\,\frac{g^3\mathbf{k}^iL_\Lambda}{\pi^3\sqrt{k^+}\mathbf{k}^2}\int d^2\mathbf{p}\,\frac{\rho^b(\mathbf{p})\rho^b(-\mathbf{p})\rho^a(-\mathbf{k})}{\mathbf{p}^2} = T_3, \tag{5.6}$$

since $\sqrt{2}/8 = 1/(4\sqrt{2})$.

Step 2: a commutator. By Eq. (5.1),

$$\begin{aligned}\big[\rho^b(\mathbf{p})\rho^b(-\mathbf{p}),\rho^a(-\mathbf{k})\big] &= \rho^b(\mathbf{p})\big[\rho^b(-\mathbf{p}),\rho^a(-\mathbf{k})\big] + \big[\rho^b(\mathbf{p}),\rho^a(-\mathbf{k})\big]\rho^b(-\mathbf{p})\\ &= if^{bac}\rho^b(\mathbf{p})\rho^c(-\mathbf{p}-\mathbf{k}) + if^{bac}\rho^c(\mathbf{p}-\mathbf{k})\rho^b(-\mathbf{p}).\end{aligned} \tag{5.7}$$

Under $\int d^2\mathbf{p}/\mathbf{p}^2$ we may replace $\mathbf{p}\to-\mathbf{p}$ in the first term. Then both terms have the same arguments:

$$\int\frac{d^2\mathbf{p}}{\mathbf{p}^2}\big[\rho^b(\mathbf{p})\rho^b(-\mathbf{p}),\rho^a(-\mathbf{k})\big] = \int\frac{d^2\mathbf{p}}{\mathbf{p}^2}\,if^{bac}\big\{\rho^b(-\mathbf{p}),\rho^c(\mathbf{p}-\mathbf{k})\big\} = -i\int\frac{d^2\mathbf{p}}{\mathbf{p}^2}\,f^{abc}\big\{\rho^b(-\mathbf{p}),\rho^c(\mathbf{p}-\mathbf{k})\big\}. \tag{5.8}$$

Step 3: the $-2\mathbf{k}^iL_\Lambda$ term of $\mathfrak{B}^i$ is $[\mathbb{N}^{(2)},\mathbb{A}^{(1)}]$. With Step 2,

$$\big[\mathbb{N}^{(2)},\mathbb{A}_{2i}^{(1)a}\big] = \frac{\sqrt{2}\,g^3\mathbf{k}^iL_\Lambda}{8\pi^3\sqrt{k^+}\mathbf{k}^2}\int\frac{d^2\mathbf{p}}{\mathbf{p}^2}\big[\rho^b(\mathbf{p})\rho^b(-\mathbf{p}),\rho^a(-\mathbf{k})\big] = -\frac{ig^3\mathbf{k}^iL_\Lambda}{4\sqrt{2}\pi^3\sqrt{k^+}\mathbf{k}^2}\int\frac{d^2\mathbf{p}}{\mathbf{p}^2}f^{abc}\{\rho^b(-\mathbf{p}),\rho^c(\mathbf{p}-\mathbf{k})\}. \tag{5.9}$$

This is T_f with $\mathfrak{B}^i\to-2\mathbf{k}^iL_\Lambda$, because $\frac{i}{8\sqrt{2}}\cdot(-2) = -\frac{i}{4\sqrt{2}}$. More generally, the part $-2\mathbf{k}^i\,X$ of $\mathfrak{B}^i$ equals $X\,[n,\mathbb{A}_2^{(1)}]$ for any longitudinal factor X. In $\mathfrak{B}^i$ the terms of this type are $-2\mathbf{k}^iL_\Lambda$ and $-2\mathbf{k}^i(-L_k)$ from ℓ_1 (the $\mathbf{k}^i$ in C^i and the extra $-\mathbf{k}^iL_k$ cancel). So

$$\boxed{\mathbb{A}_2^{(3)} \supset L_\Lambda\, n\,\mathbb{A}_2^{(1)} + L_T\,\big[n,\mathbb{A}_2^{(1)}\big], \qquad L_T = \log\frac{k^+}{\Lambda}} \tag{5.10}$$

and in coordinate space the commutator term is the $\mathfrak{F}_\alpha$ part of Row II: the soft gluon is exchanged between the two charges, $\int_z K(x,y,z)$, as in n. This identity was also checked with explicit charge matrices (SU(2) charges on three sites, random kernels): $[\sum K_{ij}\rho_i^b\rho_j^b, \sum c_l\rho_l^a] = -\frac{i}{2}\sum K_{ij}(c_j-c_i)f^{abc}\{\rho_i^b,\rho_j^c\}$ holds to 3.9×10^{-16}.

Step 4: what fix F4 does. Split $L_\Lambda = L_T + L_P$ with $L_T = \log(k^+/\Lambda)$ (virtual gluons softer than the measured one) and $L_P = \log(\vee/k^+) = L_k$ (harder ones). Order the virtual factor in p^+: the soft part to the left of $\mathbb{A}^{(1)}$, the hard part to the right,

$$L_T\, n\, \mathbb{A}_2^{(1)} + L_P\, \mathbb{A}_2^{(1)}\, n = L_\Lambda\, n\, \mathbb{A}_2^{(1)} - L_P\, \big[n, \mathbb{A}_2^{(1)}\big]. \tag{5.11}$$

Eq. (5.10) minus Eq. (5.11) is $(L_T + L_P)[n, \mathbb{A}_2^{(1)}] = L_\Lambda[n, \mathbb{A}_2^{(1)}]$. So the p^+-ordered form is the form of the notes with the commutator removed from the coefficient of $\log(\vee/\Lambda)$, while the coefficient of $\log(\vee/k^+)$ is unchanged. In the cross section (sub-terms 1, 3 of Eq. (4.13)) this is exactly fix F4:

$$\text{written: } \big(\mathfrak{F}_\alpha + \mathfrak{F}_\beta\big)\log\frac{k^+}{\Lambda} \qquad\longrightarrow\qquad \text{F4: } \mathfrak{F}_\beta \log\frac{k^+}{\Lambda} - \mathfrak{F}_\alpha \log\frac{\vee}{k^+}. \tag{5.12}$$

The soft gluon between the two valence charges ($\mathfrak{F}_\alpha$) is a virtual correction of the projectile; with F4 it comes with the projectile interval $\log(\vee/k^+)$ and belongs to BFKL (Part IV). The soft gluon between the measured gluon and a charge ($\mathfrak{F}_\beta$) comes with the target interval $\log(k^+/\Lambda)$ and belongs to JIMWLK (Part III).

Where `main.tex` loses the order. After the unitarity relations, `main.tex` sets $\mathbb{N}^{(2)}\mathbb{A}^{(1)} = \mathbb{A}^{(1)}\mathbb{N}^{(2)}$, arguing that $\mathbb{A}^{(1)}$ is linear in ρ. As operators they do not commute: by Step 3 their commutator is the α term itself. The equations of `main.tex` therefore do not fix where the virtual factor stands relative to $\mathbb{A}^{(1)}$. Three independent checks show that the ordered form (5.11) is the right one:

1. the symmetrized three-ρ terms proportional to $\log(\vee/\Lambda)$ cancel with it and do not cancel without it (Part II);
2. the two-ρ terms proportional to $\log(\vee/\Lambda)$ then equal JIMWLK⊗LO (Part III);
3. the coefficient of $\log\vee$ then equals BFKL⊗LO; in particular Row II contributes $-\mathfrak{F}_\alpha$ to it, as the independent region-$p^+ \gg k^+$ computation requires (Part IV).

The rotated term $\bar{\mathbb{A}}^{(3)}$ (sub-terms 2, 4) is correct as written; its $\mathfrak{F}_\alpha$ part is needed for the cancellation (networks α_3, α_4 of Part II).

Part II
The three-ρ terms proportional to $\log(\vee/\Lambda)$: the cancellation

This part adds up the symmetrized parts $S(\rho\rho\rho)$ of all three-ρ terms proportional to $\log(\vee/\Lambda)$. Group I, Row II enters with the coordinate-space kernel of Part I: its four monomials a–d are the z integrands of $\mathfrak{F}_\beta$ (a, c) and $\mathfrak{F}_\alpha$ (b, d). The step-by-step reordering of every term, which also gives the two-ρ terms used in Part III, is in Apps. A–F.

6 Overview

Question. Apps. A–F reorder every three-ρ term proportional to $\log(\vee/\Lambda)$ into its fully symmetrized part $S(\rho\rho\rho)$ plus two-ρ terms. Here we add up all the symmetrized parts:

1. the three-ρ remainders of the four-ρ terms, with two Wilson lines (`A1`, `A2`) and with three Wilson lines (`B1`, `B2`); no c.c.;
2. Group I, Rows II–V, with c.c.;
3. Group II, Row I (both terms), no c.c.;
4. Group III, Rows I, II, III, V, VI, with c.c.

Does the sum vanish?

Answer.

1. **As written, no.** In common integration variables the three-ρ terms split into 160 entries (one source, one subterm, one kernel monomial, amplitude or c.c.). The entries fall into six kernel classes and 48 colour networks. 28 networks cancel exactly. 20 networks do not; each of them is left with $\pm 2i\mathcal{N}$ times one colour structure (Table 4, Sec. 9).

2. **Every left-over traces back to Group I.** Rows II and III of Group I contain terms with no partner, and one Group I term is missing:

   - Row II: the part $\mathfrak{F}_\alpha$ of the two-charge kernel (soft gluon between the two charges x, y) in the term $U(w)\mathbb{A}^{(3)}$ (networks $\alpha_1, \alpha_2, \alpha'_1, \alpha'_2$);
   - Row III, part 2: the edge kernel Φ appears nowhere else (networks $\gamma_{1\ldots4}$, $\gamma'_{1\ldots4}$);
   - a missing term $\mathcal{D}$ (soft gluon between the measured gluon and a charge), Eq. (11.4) (networks $\beta_2, \beta_4, \beta_{11}, \beta_{12}$ and $\beta'_2, \beta'_4, \beta'_{11}, \beta'_{12}$).

   The four-ρ remainders, Group II and Group III are correct as written: they agree row by row with an independent leading-log computation (Sec. 10.2).

3. **Two changes in Group I make everything cancel** (Sec. 11):

   (1) in Row II, the term $U^{ah}(w)f^{hbc}\{\rho^b(x), \rho^c(y)\}$ keeps only the part $\mathfrak{F}_\beta$ of the kernel;

   (2) in Row III, part 2 (the Φ terms) is replaced by $\mathcal{D}$.

   After them all 40 remaining networks cancel with exact coefficients (Sec. 12), and the corrected Group I agrees with the independent computation piece by piece.

4. **Independent numerical test.** We evaluate the total of all symmetrized three-ρ terms at random points without any bookkeeping. As written, it is 5–40% of the sum of the absolute values of the terms; after the two changes, 10^{-16} (Sec. 13).

5. **Convention.** All of this uses $U^{bc} = U^{cb}$, as the notes do. Sec. 14 explains what changes for a non-symmetric U.

class	kernel	entries	networks	cancel	do not	networks after the changes
α	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	32	8	6	2	8
α'	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	32	8	6	2	8
β	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$	44	12	8	4	12
β'	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$	44	12	8	4	12
γ	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi_w(x)$	4	4	0	4	0
γ'	$\mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\Phi_{w'}(x')$	4	4	0	4	0

Table 4: The six kernel classes (Sec. 7.1). "Cancel" means the coefficients of the network add up to zero exactly. After the two changes the γ, γ' networks are empty and all others cancel.

7 Method

7.1 Common variables and the six kernel classes

After symmetrization the charges are commuting variables. The three charge positions and the soft-gluon position are integrated, so they may be renamed freely. In every kernel monomial

- one charge is tied to w by its LO emission kernel $\mathcal{K}(\,\cdot - w)$: we call its position x;
- one charge is tied to w' by $\mathcal{K}(\,\cdot - w')$: we call it x';
- the third charge is called y, and the soft gluon z.

Before this, the measured-gluon positions are given their standard names: in Row V of Group I (first term $z \to w$, $x \to z$; second term $x \to w$) and in Rows III, V, VI of Group III ($y' \to w'$). In a c.c. entry we first swap $w \leftrightarrow w'$, Eq. (2.6). After the renaming every kernel monomial is one of

$$
\begin{aligned}
K_\alpha &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) && \text{soft gluon between the charges } x,\ y,\\
K_{\alpha'} &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) && \text{soft gluon between the charges } x',\ y,\\
K_\beta &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z) && \text{soft gluon between the measured gluon at } w \text{ and } y,\\
K_{\beta'} &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z) && \text{soft gluon between the measured gluon at } w' \text{ and } y,\\
K_\gamma &= \mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi_w(x) && \text{Row III of Group I, part 2,}
\end{aligned}
$$

$$K_{\gamma'} = \mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\;\Phi_{w'}(x') \qquad \text{its c.c.} \tag{7.1}$$

Bringing a monomial to this form may require $\mathcal{K}(a-b) = -\mathcal{K}(b-a)$; that sign is called ε and is included in the coefficients. $\Phi_w(x)$ is the function $\Phi(x)$ of Row III (it depends on x, z and w); $\Phi_{w'}$ is the same with $w \to w'$. With $A = x-w$, $B = x-z$, so that $z-w = A-B$, one has $2\big[(z-w)\cdot(x-z)A^2-(z-w)\cdot(x-w)B^2\big] = 2A\cdot B(A^2+B^2)-4A^2B^2 = (A^2-B^2)^2-(z-w)^2(A^2+B^2)$, and therefore

$$\Phi_w(x) = \frac{1}{(z-w)^2}\left[\frac{2(z-w)\cdot(x-z)\,(x-w)^2 - 2(z-w)\cdot(x-w)\,(x-z)^2}{\big[(x-w)^2-(x-z)^2\big]^2} - 1\right] = -\frac{(x-w)^2+(x-z)^2}{\big[(x-w)^2-(x-z)^2\big]^2}. \tag{7.2}$$

7.2 Colour networks

After the renaming, the colour structure of each entry is simplified with (U1) where possible and its dummy colour indices are renamed $a, b, c, \ldots$. Inside one class all entries carry the same kernel. They can therefore only cancel if their colour structures are equal up to a sign. We call each distinct colour structure a *network*. Two structures belong to the same network if they agree after (i) renaming dummy colour indices, (ii) using $f^{abc} = -f^{bac}$ etc., and (iii) using $U^{bc} = U^{cb}$.

Example (network α_1). Entry `GII.I1.1b` has the structure $M = f^{abc}U^{de}(w')U^{da}(w)\rho^e(x')\rho^b(y)\rho^c(x)$. Rename $d \leftrightarrow e$ and use $U^{ed}(w') = U^{de}(w')$, $U^{ea}(w) = U^{ae}(w)$: $M = f^{abc}U^{de}(w')U^{ae}(w)\rho^d(x')\rho^b(y)\rho^c(x)$. Rename $a \leftrightarrow b$ and use $f^{bac} = -f^{abc}$ and $U^{be} = U^{eb}$: $M = -f^{abc}U^{de}(w')U^{eb}(w)\rho^d(x')\rho^a(y)\rho^c(x)$. This is -1 times the structure of entry `A1.1`, the representative of α_1. Hence `GII.I1.1b` enters α_1 with sign -1.

How the computer does it. The sign is fixed by evaluating each structure for random symmetric adjoint U's and random charges. Two independent random sets are used, and the ratio to the representative must be exactly $+1$ or -1 in both. Structures that are not equal give ratios that are not ±1.

7.3 Criterion

The sum of all symmetrized three-ρ terms is

$$\mathcal{N}\int e^{-ik\cdot(w'-w)} \sum_{\text{classes } K} K \sum_{\text{networks } C\in K} \Big(\sum_{\text{entries } e\in C} c_e\, s_e\Big)\, C, \tag{7.3}$$

where c_e is the coefficient of entry e (in units of $\mathcal{N}$) and $s_e = \pm1$ its sign relative to the network. It vanishes if every network sum $\sum_e c_e s_e$ vanishes. Conversely, a non-zero network sum is a genuine left-over. The six kernels are different functions of x, x', y, z. In the γ networks with $U(z)$ the z-integral cannot even be done, and the kernel Φ is not integrable at all (Sec. 10), so the γ entries cannot be traded for other classes by integrating. The networks are independent functions of the Wilson lines and charges. Sec. 13 confirms the conclusions pointwise, without using classes or networks.

8 Every term in common variables

For every source we list the input (prefactor, factor 1 or 2 from the symmetrization, c.c., phase, kernel monomials, sub-terms), the renaming of each kernel monomial, and the resulting entries. Entry labels are `source.n[m][*]`: sub-term n, kernel monomial m (when there is more than one), and `*` for the c.c. The coefficient of an entry, in units of $\mathcal{N}$, is

$$c_e = (\text{prefactor})^{(*)} \times (1 \text{ or } 2) \times (\text{sign of sub-term } n) \times (\text{coefficient of monomial } m) \times \varepsilon, \tag{8.1}$$

where the prefactor is complex conjugated for a c.c. entry. The colour structure is the sub-term's colour factor and charges, written in the common variables.

Three-ρ remainders of the four-ρ terms (no c.c.)

8.0.1 `A1`: Two Wilson lines, first term

Prefactor $-i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; no c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel: $K = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. The symmetrized term (commuting charges):

$$\begin{aligned} -i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\Big\{ & \\ (1)\quad & + f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^d(x')\,\rho^a(x)\,\rho^c(y) \\ (2)\quad & - f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,\rho^d(x')\,\rho^a(x)\,\rho^c(y)\Big\} \end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$x \to y,\ y \to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z)$	$+$	α

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
`A1.1`	α	$-i$	$f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^{d}(x')\,\rho^{a}(y)\,\rho^{c}(x)$
`A1.2`	α	$+i$	$f^{abc}\,U^{db}(w)\,U^{de}(x')\,\rho^{e}(x')\,\rho^{a}(y)\,\rho^{c}(x)$

8.0.2 `A2`: Two Wilson lines, second term

Prefactor $-i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; no c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel: $K = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
&-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{\\
&\quad(1)\quad +\,f^{abc}\,U^{eb}(w')\,U^{ed}(y)\,\rho^{c}(x')\,\rho^{a}(x)\,\rho^{d}(y)\\
&\quad(2)\quad -\,f^{abc}\,U^{be}(w')\,U^{ed}(w)\,\rho^{c}(x')\,\rho^{a}(x)\,\rho^{d}(y)\Big\}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$x \to y,\ y \to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
`A2.1`	α'	$-i$	$f^{abc}\,U^{db}(w')\,U^{de}(x)\,\rho^{c}(x')\,\rho^{a}(y)\,\rho^{e}(x)$
`A2.2`	α'	$+i$	$f^{abc}\,U^{bd}(w')\,U^{de}(w)\,\rho^{c}(x')\,\rho^{a}(y)\,\rho^{e}(x)$

8.0.3 `B1`: Three Wilson lines, kernel K_1

Prefactor $i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; no c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel: $K = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{\\
&\quad(1)\quad -\,f^{abc}\,U^{db}(w')\,U^{ec}(y)\,U^{af}(x)\,\rho^{d}(x')\,\rho^{f}(x)\,\rho^{e}(y)\\
&\quad(2)\quad +\,f^{abc}\,U^{ec}(y)\,U^{bd}(x')\,U^{af}(x)\,\rho^{d}(x')\,\rho^{f}(x)\,\rho^{e}(y)\Big\}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$x \to y,\ y \to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z)$	$+$	α

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
`B1.1`	α	$-i$	$f^{abc}\,U^{db}(w')\,U^{ec}(x)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$
`B1.2`	α	$+i$	$f^{abc}\,U^{dc}(x)\,U^{be}(x')\,U^{ah}(y)\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)$

8.0.4 `B2`: Three Wilson lines, kernel K_3

Prefactor $i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; no c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel: $K = \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
&i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{\\
&\quad(1)\quad +\,f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,\rho^{f}(y)\\
&\quad(2)\quad -\,f^{abc}\,U^{ec}(x')\,U^{bd}(x)\,U^{af}(y)\,\rho^{e}(x')\,\rho^{d}(x)\,\rho^{f}(y)\Big\}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	none	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
B2.1	α'	$+i$	$f^{abc}\,U^{dc}(x')\,U^{eb}(w)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
B2.2	α'	$-i$	$f^{abc}\,U^{dc}(x')\,U^{be}(x)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$

Group I (with c.c.)

8.0.5 GI.II: Group I, Row II (two-charge part of $\mathbb{A}^{(3)}$)

Prefactor $-\frac{i}{2}\mathcal{N}$; $\rho\{\rho,\rho\}\to 2S(\rho\rho\rho)$, factor 2; with c.c.; phase $e^{-ik\cdot(w'-w)}$. The kernel $\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,y)$ is written with the z-integral representation of $\mathfrak{F}^i$ (Part I, Eq. (4.10)): four monomials, a–d. Kernel $K=\sum$ of the monomials

$$\begin{aligned}
&\text{a:}\quad \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\\
&\text{b:}\quad -\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\
&\text{c:}\quad -\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(x-z)\\
&\text{d:}\quad \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)
\end{aligned}$$

The symmetrized term (commuting charges):

$$\begin{aligned}
-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{\\
(1)\quad &+ f^{hbc}\,U^{ab'}(x')\,U^{ah}(w)\,\rho^{b'}(x')\,\rho^{b}(x)\,\rho^{c}(y)\\
(2)\quad &- f^{abc}\,U^{ab'}(x')\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\rho^{d}(x)\,\rho^{e}(y)\\
(3)\quad &- f^{hbc}\,U^{ab'}(w')\,U^{ah}(w)\,\rho^{b'}(x')\,\rho^{b}(x)\,\rho^{c}(y)\\
(4)\quad &+ f^{abc}\,U^{ab'}(w')\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\rho^{d}(x)\,\rho^{e}(y)\Big\}+\text{c.c.}
\end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X)=-\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	none	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$	$+$	β
b	amp.	none	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	$+$	α
c	amp.	$x\to y,\ y\to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$	$+$	β
d	amp.	$x\to y,\ y\to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z)$	$+$	α
a	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$	$+$	β'
b	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	$+$	α'
c	c.c.	$x'\to x,\ x\to y,\ y\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$	$+$	β'
d	c.c.	$x'\to x,\ x\to y,\ y\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
GI.II.1a	β	$-i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(x)\,\rho^{c}(y)$
GI.II.1b	α	$+i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(x)\,\rho^{c}(y)$
GI.II.1c	β	$+i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(y)\,\rho^{c}(x)$
GI.II.1d	α	$-i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(y)\,\rho^{c}(x)$
GI.II.2a	β	$+i$	$f^{abc}\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.II.2b	α	$-i$	$f^{abc}\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.II.2c	β	$-i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GI.II.2d	α	$+i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GI.II.3a	β	$+i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(x)\,\rho^{c}(y)$
GI.II.3b	α	$-i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(x)\,\rho^{c}(y)$
GI.II.3c	β	$-i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(y)\,\rho^{c}(x)$
GI.II.3d	α	$+i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(y)\,\rho^{c}(x)$
GI.II.4a	β	$-i$	$f^{abc}\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.II.4b	α	$+i$	$f^{abc}\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.II.4c	β	$+i$	$f^{abc}\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GI.II.4d	α	$-i$	$f^{abc}\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GI.II.1a*	β'	$+i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$
GI.II.1b*	α'	$-i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$
GI.II.1c*	β'	$-i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(y)\,\rho^{c}(x')$
GI.II.1d*	α'	$+i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(y)\,\rho^{c}(x')$

entry	class	coeff.	colour structure
GI.II.2a*	β'	$-i$	$f^{abc}\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.II.2b*	α'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.II.2c*	β'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
GI.II.2d*	α'	$-i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
GI.II.3a*	β'	$-i$	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$
GI.II.3b*	α'	$+i$	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$
GI.II.3c*	β'	$+i$	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(y)\,\rho^{c}(x')$
GI.II.3d*	α'	$-i$	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(y)\,\rho^{c}(x')$
GI.II.4a*	β'	$+i$	$f^{abc}\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.II.4b*	α'	$-i$	$f^{abc}\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.II.4c*	β'	$-i$	$f^{abc}\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
GI.II.4d*	α'	$+i$	$f^{abc}\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$

8.0.6 GI.III: Group I, Row III ($\bar{\mathbb{B}}_3^\dagger\mathbb{A}$)

Prefactor $i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(w'-w)}$. $\Phi(x)$ is the edge kernel of the notes (it depends on x, z and w); monomial b is part 2 of the row. Kernel $K = \sum$ of the monomials

$$\begin{aligned} &\text{a:}\quad \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\\ &\text{b:}\quad 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x)\end{aligned}$$

The symmetrized term (commuting charges):

$$\begin{aligned} i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{ &\\ (1)\quad &+\ f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\ (2)\quad &-\ f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\ (3)\quad &-\ f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\ (4)\quad &+\ f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\Big\} + \text{c.c.}\end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	none	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	$+$	α
b	amp.	none	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi_w(x)$	$+$	γ
a	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	$+$	α'
b	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\,\Phi_{w'}(x')$	$+$	γ'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
GI.III.1a	α	$+i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.1b	γ	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.2a	α	$-i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.2b	γ	$-2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.3a	α	$-i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.3b	γ	$-2i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.4a	α	$+i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.4b	γ	$+2i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.III.1a*	α'	$-i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.1b*	γ'	$-2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.2a*	α'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.2b*	γ'	$+2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.3a*	α'	$+i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.3b*	γ'	$+2i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.4a*	α'	$-i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.III.4b*	γ'	$-2i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$

8.0.7 GI.IV: Group I, Row IV ($\bar{\mathbb{A}}^\dagger\mathbb{B}_2$)

Prefactor $2i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel $K = \sum$ of the monomials

$$\begin{aligned} &\text{a:}\quad \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)\\ &\text{b:}\quad \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\end{aligned}$$

The symmetrized term (commuting charges):

$$
\begin{aligned}
2i\mathcal{N} \int e^{-ik\cdot(w'-w)}\, K \Big\{ & \\
(1) \quad & + f^{cdc'}\, U^{ab'}(x')\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, \rho^{b'}(x')\, \rho^{e}(y)\, \rho^{c'}(x) \\
(2) \quad & - f^{abc}\, U^{ab'}(x')\, U^{be}(y)\, U^{cd}(x)\, \rho^{b'}(x')\, \rho^{e}(y)\, \rho^{d}(x) \\
(3) \quad & - f^{cdc'}\, U^{ab'}(w')\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, \rho^{b'}(x')\, \rho^{e}(y)\, \rho^{c'}(x) \\
(4) \quad & + f^{abc}\, U^{ab'}(w')\, U^{be}(y)\, U^{cd}(x)\, \rho^{b'}(x')\, \rho^{e}(y)\, \rho^{d}(x) \Big\} + \text{c.c.}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)$	$-$	β
b	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	$+$	α
a	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)$	$-$	β'
b	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
`GI.IV.1a`	β	$-2i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$
`GI.IV.1b`	α	$+i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$
`GI.IV.2a`	β	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`GI.IV.2b`	α	$-i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`GI.IV.3a`	β	$+2i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$
`GI.IV.3b`	α	$-i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$
`GI.IV.4a`	β	$-2i$	$f^{abc}\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`GI.IV.4b`	α	$+i$	$f^{abc}\,U^{ad}(w')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`GI.IV.1a*`	β'	$+2i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$
`GI.IV.1b*`	α'	$-i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$
`GI.IV.2a*`	β'	$-2i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
`GI.IV.2b*`	α'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
`GI.IV.3a*`	β'	$-2i$	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$
`GI.IV.3b*`	α'	$+i$	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$
`GI.IV.4a*`	β'	$+2i$	$f^{abc}\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
`GI.IV.4b*`	α'	$-i$	$f^{abc}\,U^{ad}(w)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$

8.0.8 `GI.V1`: Group I, Row V, first term (measured gluon at z)

Prefactor $-\frac{i}{2}\mathcal{N}$; $\rho\{\rho,\rho\} \to 2S(\rho\rho\rho)$, factor 2; with c.c.; phase $e^{-ik\cdot(w'-z)}$. The measured gluon of the amplitude sits at z and the soft gluon at x; we first rename $z \to w$, $x \to z$. Kernel: $K = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
-i\mathcal{N} \int e^{-ik\cdot(w'-z)}\, K \Big\{ & \\
(1) \quad & + f^{adc}\, U^{ab'}(x')\, U^{db}(x)\, U^{ce}(z)\, \rho^{b'}(x')\, \rho^{e}(y)\, \rho^{b}(y') \\
(2) \quad & - f^{abc}\, U^{ab'}(x')\, U^{cd}(y)\, U^{be}(y')\, \rho^{b'}(x')\, \rho^{d}(y)\, \rho^{e}(y') \\
(3) \quad & - f^{adc}\, U^{ab'}(w')\, U^{db}(x)\, U^{ce}(z)\, \rho^{b'}(x')\, \rho^{e}(y)\, \rho^{b}(y') \\
(4) \quad & + f^{abc}\, U^{ab'}(w')\, U^{cd}(y)\, U^{be}(y')\, \rho^{b'}(x')\, \rho^{d}(y)\, \rho^{e}(y') \Big\} + \text{c.c.}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$y\to x,\ y'\to y,\ x\to z,\ z\to w$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$	$+$	β
	c.c.	$x'\to x,\ y\to x',\ y'\to y,\ w'\to w,\ x\to z,\ z\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$	$+$	β'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
`GI.V1.1`	β	$-i$	$f^{abc}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)$
`GI.V1.2`	β	$+i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
`GI.V1.3`	β	$+i$	$f^{abc}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)$

entry	class	coeff.	colour structure
GI.V1.4	β	$-i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$
GI.V1.1*	β'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')\,\rho^{e}(y)$
GI.V1.2*	β'	$-i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$
GI.V1.3*	β'	$-i$	$f^{abc}\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')\,\rho^{e}(y)$
GI.V1.4*	β'	$+i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$

8.0.9 GI.V2: Group I, Row V, second term (measured gluon at x)

Prefactor $-\frac{i}{2}\mathcal{N}$; $\rho\{\rho,\rho\}\to 2S(\rho\rho\rho)$, factor 2; with c.c.; phase $e^{-ik\cdot(w'-x)}$. The measured gluon of the amplitude sits at x; we first rename $x\to w$. Kernel: $K=\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
-i\mathcal{N}\int e^{-ik\cdot(w'-x)}\,K\,\Big\{\\
(1)\quad &+\ f^{adc}\,U^{ab'}(x')\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{b}(y')\\
(2)\quad &-\ f^{abc}\,U^{ab'}(x')\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\rho^{d}(y)\,\rho^{e}(y')\\
(3)\quad &-\ f^{adc}\,U^{ab'}(w')\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{b}(y')\\
(4)\quad &+\ f^{abc}\,U^{ab'}(w')\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\rho^{d}(y)\,\rho^{e}(y')\Big\}+\text{c.c.}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X)=-\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$y'\to x,\ x\to w$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)$	$-$	β
	c.c.	$x'\to x,\ y'\to x',\ w'\to w,\ x\to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)$	$-$	β'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
GI.V2.1	β	$+i$	$f^{abc}\,U^{ad}(x')\,U^{be}(w)\,U^{ch}(z)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$
GI.V2.2	β	$-i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GI.V2.3	β	$-i$	$f^{abc}\,U^{ad}(w')\,U^{be}(w)\,U^{ch}(z)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$
GI.V2.4	β	$+i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GI.V2.1*	β'	$-i$	$f^{abc}\,U^{ad}(x)\,U^{be}(w')\,U^{ch}(z)\,\rho^{d}(x)\,\rho^{h}(y)\,\rho^{e}(x')$
GI.V2.2*	β'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
GI.V2.3*	β'	$+i$	$f^{abc}\,U^{ad}(w)\,U^{be}(w')\,U^{ch}(z)\,\rho^{d}(x)\,\rho^{h}(y)\,\rho^{e}(x')$
GI.V2.4*	β'	$-i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$

Group II (no c.c.)

8.0.10 GII.I1: Group II, Row I, first term

Prefactor $-i\mathcal{N}$; $\{\rho,\rho\}\rho\to 2S(\rho\rho\rho)$, factor 2; no c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel $K=\sum$ of the monomials

$$
\begin{aligned}
&\text{a:}\quad \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)\\
&\text{b:}\quad \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)
\end{aligned}
$$

The symmetrized term (commuting charges):

$$
\begin{aligned}
-2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{\\
(1)\quad &+\ f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{b}(y)\,\rho^{d}(x)\\
(2)\quad &-\ f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{b}(y)\,\rho^{e}(x)\\
(3)\quad &-\ f^{cbd}\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,U^{ac}(w)\,\rho^{d'}(x')\,\rho^{e'}(y)\,\rho^{d}(x)\\
(4)\quad &+\ f^{acd}\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\,\rho^{e'}(y)\,\rho^{e}(x)\Big\}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X)=-\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)$	$-$	β
b	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	$+$	α

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure	
GII.I1.1a	β	$+2i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(y)\,\rho^{c}(x)$	
GII.I1.1b	α	$-i$	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(y)\,\rho^{c}(x)$	
GII.I1.2a	β	$-2i$	$f^{abc}\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	
GII.I1.2b	α	$+i$	$f^{abc}\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	
GII.I1.3a	β	$-2i$	$f^{abc}\,U^{bd}(z)\,U^{eh}(x')\,U^{dl}(y)\,U^{ea}(w)\,\rho^{h}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	
GII.I1.3b	α	$+i$	$f^{abc}\,U^{bd}(z)\,U^{eh}(x')\,U^{dl}(y)\,U^{ea}(w)\,\rho^{h}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	
GII.I1.4a	β	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	(U1)
GII.I1.4b	α	$-i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	(U1)

8.0.11 GII.I2: Group II, Row I, second term

Prefactor $i\mathcal{N}$; $\rho\{\rho,\rho\} \to 2S(\rho\rho\rho)$, factor 2; no c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel $K = \sum$ of the monomials

$$\begin{array}{ll} \text{a:} & \mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(z-w')\cdot\mathcal{K}(y-z) \\ \text{b:} & \tfrac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) \end{array}$$

The symmetrized term (commuting charges):

$$\begin{aligned} 2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{ & \\ (1)\quad & + f^{c'bd'}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x)\,\rho^{c}(x')\,\rho^{b}(y) \\ (2)\quad & - f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{d'}(x)\,\rho^{d}(x')\,\rho^{e}(y) \\ (3)\quad & - f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,\rho^{e'}(x)\,\rho^{c}(x')\,\rho^{b}(y) \\ (4)\quad & + f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{e'}(x)\,\rho^{d}(x')\,\rho^{e}(y)\Big\} \end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	$x \to x',\ x' \to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)$	$-$	β'
b	amp.	$x \to x',\ x' \to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure	
GII.I2.1a	β'	$-2i$	$f^{abc}\,U^{da}(w')\,U^{de}(w)\,\rho^{c}(x')\,\rho^{e}(x)\,\rho^{b}(y)$	
GII.I2.1b	α'	$+i$	$f^{abc}\,U^{da}(w')\,U^{de}(w)\,\rho^{c}(x')\,\rho^{e}(x)\,\rho^{b}(y)$	
GII.I2.2a	β'	$+2i$	$f^{abc}\,U^{da}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\,\rho^{c}(x')\,\rho^{h}(x)\,\rho^{l}(y)$	
GII.I2.2b	α'	$-i$	$f^{abc}\,U^{da}(w')\,U^{be}(z)\,U^{dh}(x)\,U^{el}(y)\,\rho^{c}(x')\,\rho^{h}(x)\,\rho^{l}(y)$	
GII.I2.3a	β'	$+2i$	$f^{abc}\,U^{db}(z)\,U^{ce}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{h}(x)\,\rho^{d}(y)$	
GII.I2.3b	α'	$-i$	$f^{abc}\,U^{db}(z)\,U^{ce}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{h}(x)\,\rho^{d}(y)$	
GII.I2.4a	β'	$-2i$	$f^{abc}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	(U1)
GII.I2.4b	α'	$+i$	$f^{abc}\,U^{cd}(x')\,U^{ae}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	(U1)

Group III (with c.c.)

8.0.12 GIII.I: Group III, Row I

Prefactor $-2i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel $K = \sum$ of the monomials

$$\begin{array}{ll} \text{a:} & \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) \\ \text{b:} & \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) \end{array}$$

The symmetrized term (commuting charges):

$$\begin{aligned} -2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{ & \\ (1)\quad & + f^{cbd}\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{d}(x) \\ (2)\quad & - f^{acd}\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{e}(x) \\ (3)\quad & - f^{cbd}\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{d}(x) \\ (4)\quad & + f^{acd}\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{e}(x)\Big\} + \text{c.c.} \end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)$	$-$	β
b	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	$+$	α
a	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)$	$-$	β'
b	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure	
`GIII.I.1a`	β	$+2i$	$f^{abc}\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	
`GIII.I.1b`	α	$-i$	$f^{abc}\,U^{de}(x')\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	
`GIII.I.2a`	β	$-2i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	(U1*)
`GIII.I.2b`	α	$+i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	(U1*)
`GIII.I.3a`	β	$-2i$	$f^{abc}\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ha}(w)\,\rho^{l}(x')\,\rho^{e}(y)\,\rho^{c}(x)$	
`GIII.I.3b`	α	$+i$	$f^{abc}\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w')\,U^{ha}(w)\,\rho^{l}(x')\,\rho^{e}(y)\,\rho^{c}(x)$	
`GIII.I.4a`	β	$+2i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(w')\,U^{ch}(x)\,\rho^{e}(x')\,\rho^{d}(y)\,\rho^{h}(x)$	(U1)
`GIII.I.4b`	α	$-i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(w')\,U^{ch}(x)\,\rho^{e}(x')\,\rho^{d}(y)\,\rho^{h}(x)$	(U1)
`GIII.I.1a*`	β'	$-2i$	$f^{abc}\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$	
`GIII.I.1b*`	α'	$+i$	$f^{abc}\,U^{de}(x)\,U^{hl}(y)\,U^{hb}(z)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{l}(y)\,\rho^{c}(x')$	
`GIII.I.2a*`	β'	$+2i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$	(U1*)
`GIII.I.2b*`	α'	$-i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$	(U1*)
`GIII.I.3a*`	β'	$+2i$	$f^{abc}\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\,\rho^{l}(x)\,\rho^{e}(y)\,\rho^{c}(x')$	
`GIII.I.3b*`	α'	$-i$	$f^{abc}\,U^{bd}(z)\,U^{de}(y)\,U^{hl}(w)\,U^{ha}(w')\,\rho^{l}(x)\,\rho^{e}(y)\,\rho^{c}(x')$	
`GIII.I.4a*`	β'	$-2i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\,\rho^{e}(x)\,\rho^{d}(y)\,\rho^{h}(x')$	(U1)
`GIII.I.4b*`	α'	$+i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(w)\,U^{ch}(x')\,\rho^{e}(x)\,\rho^{d}(y)\,\rho^{h}(x')$	(U1)

8.0.13 `GIII.II`: Group III, Row II

Prefactor $-2i\mathcal{N}$; ordered product $\rho\rho\rho\to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel $K=\sum$ of the monomials

$$\begin{array}{ll}\text{a:} & \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)\\ \text{b:} & \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\end{array}$$

The symmetrized term (commuting charges):

$$\begin{aligned}-2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{&\\ (1)\quad &+\ f^{cbd}\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,U^{ac}(w)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{d}(x)\\ (2)\quad &-\ f^{acd}\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\\ (3)\quad &-\ f^{cbd}\,U^{ab'}(x')\,U^{ac}(w)\,\rho^{b}(y)\,\rho^{b'}(x')\,\rho^{d}(x)\\ (4)\quad &+\ f^{acd}\,U^{ab'}(x')\,U^{bc}(z)\,U^{de}(x)\,\rho^{b}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\Big\}+\text{c.c.}\end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X)=-\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
a	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)$	$-$	β
b	amp.	none	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$	$+$	α
a	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)$	$-$	β'
b	c.c.	$x'\to x,\ x\to x',\ w'\to w,\ w\to w'$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$	$+$	α'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure	
`GIII.II.1a`	β	$+2i$	$f^{abc}\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)\,\rho^{e}(y)\,\rho^{l}(x')\,\rho^{c}(x)$	
`GIII.II.1b`	α	$-i$	$f^{abc}\,U^{de}(y)\,U^{hl}(x')\,U^{bd}(z)\,U^{ha}(w)\,\rho^{e}(y)\,\rho^{l}(x')\,\rho^{c}(x)$	
`GIII.II.2a`	β	$-2i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\,\rho^{d}(y)\,\rho^{e}(x')\,\rho^{h}(x)$	(U1)
`GIII.II.2b`	α	$+i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(x')\,U^{ch}(x)\,\rho^{d}(y)\,\rho^{e}(x')\,\rho^{h}(x)$	(U1)
`GIII.II.3a`	β	$-2i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{b}(y)\,\rho^{e}(x')\,\rho^{c}(x)$	
`GIII.II.3b`	α	$+i$	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{b}(y)\,\rho^{e}(x')\,\rho^{c}(x)$	
`GIII.II.4a`	β	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\,\rho^{e}(y)\,\rho^{d}(x')\,\rho^{h}(x)$	
`GIII.II.4b`	α	$-i$	$f^{abc}\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\,\rho^{e}(y)\,\rho^{d}(x')\,\rho^{h}(x)$	
`GIII.II.1a*`	β'	$-2i$	$f^{abc}\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')\,\rho^{e}(y)\,\rho^{l}(x)\,\rho^{c}(x')$	
`GIII.II.1b*`	α'	$+i$	$f^{abc}\,U^{de}(y)\,U^{hl}(x)\,U^{bd}(z)\,U^{ha}(w')\,\rho^{e}(y)\,\rho^{l}(x)\,\rho^{c}(x')$	
`GIII.II.2a*`	β'	$+2i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\,\rho^{d}(y)\,\rho^{e}(x)\,\rho^{h}(x')$	(U1)
`GIII.II.2b*`	α'	$-i$	$f^{abc}\,U^{bd}(y)\,U^{ae}(x)\,U^{ch}(x')\,\rho^{d}(y)\,\rho^{e}(x)\,\rho^{h}(x')$	(U1)

entry	class	coeff.	colour structure
GIII.II.3a*	β'	$+2i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{b}(y)\,\rho^{e}(x)\,\rho^{c}(x')$
GIII.II.3b*	α'	$-i$	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{b}(y)\,\rho^{e}(x)\,\rho^{c}(x')$
GIII.II.4a*	β'	$-2i$	$f^{abc}\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,\rho^{h}(x')$
GIII.II.4b*	α'	$+i$	$f^{abc}\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\,\rho^{e}(y)\,\rho^{d}(x)\,\rho^{h}(x')$

8.0.14 GIII.III: Group III, Row III, first part

Prefactor $-i\mathcal{N}$; $\rho\{\rho,\rho\} \to 2S(\rho\rho\rho)$, factor 2; with c.c.; phase $e^{-ik\cdot(y'-w)}$. The measured gluon of the conjugate amplitude sits at y'; we first rename $y' \to w'$. Kernel: $K = \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
&-2i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K\Big\{\\
&\quad(1)\quad+\ f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ac}(w)\,\rho^{e'}(x')\,\rho^{c}(y)\,\rho^{b}(x)\\
&\quad(2)\quad-\ f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x)\\
&\quad(3)\quad-\ f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,\rho^{e'}(x')\,\rho^{c}(y)\,\rho^{b}(x)\\
&\quad(4)\quad+\ f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x)\Big\}+\text{c.c.}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$y\to x,\ x\to y,\ y'\to w'$	$\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)$	$+$	β'
	c.c.	$x'\to x,\ y\to x',\ x\to y,\ y'\to w,\ w\to w'$	$\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	$+$	β

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure	
GIII.III.1	β'	$-2i$	$f^{abc}\,U^{da}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)$	
GIII.III.2	β'	$+2i$	$f^{abc}\,U^{da}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	(U1*)
GIII.III.3	β'	$+2i$	$f^{abc}\,U^{db}(z)\,U^{ae}(x')\,U^{ch}(w)\,\rho^{e}(x')\,\rho^{h}(x)\,\rho^{d}(y)$	
GIII.III.4	β'	$-2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	(U1)
GIII.III.1*	β	$+2i$	$f^{abc}\,U^{da}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')\,\rho^{e}(y)$	
GIII.III.2*	β	$-2i$	$f^{abc}\,U^{da}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$	(U1*)
GIII.III.3*	β	$-2i$	$f^{abc}\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(w')\,\rho^{e}(x)\,\rho^{h}(x')\,\rho^{d}(y)$	
GIII.III.4*	β	$+2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$	(U1)

8.0.15 GIII.V: Group III, Row V

Prefactor $-2i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(y'-w)}$. Phase $e^{-ik\cdot(y'-w)}$; we first rename $y' \to w'$. Kernel: $K = \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
&-2i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K\Big\{\\
&\quad(1)\quad+\ f^{c'ca}\,U^{e'c'}(y')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\\
&\quad(2)\quad-\ f^{c'ca}\,U^{e'c'}(y')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\\
&\quad(3)\quad-\ f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\\
&\quad(4)\quad+\ f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\Big\}+\text{c.c.}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$x\to y,\ y\to x,\ y'\to w'$	$\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)$	$+$	β'
	c.c.	$x'\to x,\ x\to y,\ y\to x',\ y'\to w,\ w\to w'$	$\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	$+$	β

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
GIII.V.1	β'	$-2i$	$f^{abc}\,U^{da}(w')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
GIII.V.2	β'	$+2i$	$f^{abc}\,U^{da}(w')\,U^{ce}(w)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$
GIII.V.3	β'	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{be}(y)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$

entry	class	coeff.	colour structure
GIII.V.4	β'	$-2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(w)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$
GIII.V.1*	β	$+2i$	$f^{abc}\,U^{da}(w)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
GIII.V.2*	β	$-2i$	$f^{abc}\,U^{da}(w)\,U^{ce}(w')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{h}(y)\,\rho^{e}(x')$
GIII.V.3*	β	$-2i$	$f^{abc}\,U^{ad}(x)\,U^{be}(y)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
GIII.V.4*	β	$+2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(w')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{h}(y)\,\rho^{e}(x')$

8.0.16 GIII.VI: Group III, Row VI

Prefactor $-2i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(y'-w)}$. Phase $e^{-ik\cdot(y'-w)}$; we first rename $y' \to w'$. Kernel: $K = \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$. The symmetrized term (commuting charges):

$$
\begin{aligned}
-2i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K\,\Big\{ &\\
(1)\quad &+ f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x)\\
(2)\quad &- f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x)\\
(3)\quad &- f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x)\\
(4)\quad &+ f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x)\Big\} + \text{c.c.}
\end{aligned}
$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$y\to x,\ x\to y,\ y'\to w'$	$\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)$	$+$	β'
	c.c.	$x'\to x,\ y\to x',\ x\to y,\ y'\to w,\ w\to w'$	$\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')$	$+$	β

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure	
GIII.VI.1	β'	$-2i$	$f^{abc}\,U^{da}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	(U1*)
GIII.VI.2	β'	$+2i$	$f^{abc}\,U^{da}(w')\,U^{be}(z)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{h}(x)\,\rho^{e}(y)$	
GIII.VI.3	β'	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	(U1)
GIII.VI.4	β'	$-2i$	$f^{abc}\,U^{db}(z)\,U^{ae}(x')\,U^{ch}(x)\,\rho^{e}(x')\,\rho^{h}(x)\,\rho^{d}(y)$	
GIII.VI.1*	β	$+2i$	$f^{abc}\,U^{da}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$	(U1*)
GIII.VI.2*	β	$-2i$	$f^{abc}\,U^{da}(w)\,U^{be}(z)\,U^{ch}(x')\,\rho^{d}(x)\,\rho^{h}(x')\,\rho^{e}(y)$	
GIII.VI.3*	β	$-2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')\,\rho^{h}(y)$	(U1)
GIII.VI.4*	β	$+2i$	$f^{abc}\,U^{db}(z)\,U^{ae}(x)\,U^{ch}(x')\,\rho^{e}(x)\,\rho^{h}(x')\,\rho^{d}(y)$	

9 The colour networks, as written

For each class we list every network: its colour structure C (from the first entry in it), the entries in it with their coefficient c_e and sign s_e, and the sum $\sum_e c_e s_e$ in units of $\mathcal{N}$. A sum written in red does not vanish. Entries in red are the ones removed by the corrections of Sec. 11.

9.1 Class α: soft gluon between the charges x and y

Kernel $K_\alpha = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. 32 entries, 8 colour networks; 6 cancel, 2 do not.

	colour structure C	entries: coefficient × sign	sum
α_1	$f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^{d}(x')\,\rho^{a}(y)\,\rho^{c}(x)$	A1.1 $(-i)$; GI.II.3b $(-i)$; GI.II.3d $(+i)(-1)$; GII.I1.1b $(-i)(-1)$	$-2i$ ✗
α_2	$f^{abc}\,U^{db}(w)\,U^{de}(x')\,\rho^{e}(x')\,\rho^{a}(y)\,\rho^{c}(x)$	A1.2 $(+i)$; GI.II.1b $(+i)$; GI.II.1d $(-i)(-1)$; GIII.II.3b $(+i)(-1)$	$+2i$ ✗
α_3	$f^{abc}\,U^{db}(w')\,U^{ec}(x)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$	B1.1 $(-i)$; GI.II.4b $(+i)$; GI.II.4d $(-i)(-1)$; GI.III.4a $(+i)(-1)$; GI.IV.4b $(+i)(-1)$; GIII.I.4b $(-i)(-1)$	0 ✓
α_4	$f^{abc}\,U^{dc}(x)\,U^{be}(x')\,U^{ah}(y)\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)$	B1.2 $(+i)$; GI.II.2b $(-i)$; GI.II.2d $(+i)(-1)$; GI.III.2a $(-i)(-1)$; GI.IV.2b $(-i)(-1)$; GII.I1.4b $(-i)(-1)$; GIII.I.2b $(+i)(-1)$; GIII.II.2b $(+i)(-1)$	0 ✓
α_5	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	GI.III.1a $(+i)$; GIII.II.4b $(-i)$	0 ✓

	colour structure C	entries: coefficient × sign	sum
α_6	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.III.3a $(-i)$; GII.I1.2b $(+i)$	0 ✓
α_7	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')$ $\rho^{l}(y)\,\rho^{c}(x)$	GI.IV.1b $(+i)$; GII.I1.3b $(+i)$; GIII.I.1b $(-i)$; GIII.II.1b $(-i)$	0 ✓
α_8	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')$ $\rho^{l}(y)\,\rho^{c}(x)$	GI.IV.3b $(-i)$; GIII.I.3b $(+i)$	0 ✓

9.2 Class α': soft gluon between the charges x' and y

Kernel $K_{\alpha'} = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$. 32 entries, 8 colour networks; 6 cancel, 2 do not.

	colour structure C	entries: coefficient × sign	sum
α'_1	$f^{abc}\,U^{db}(w')\,U^{de}(x)\,\rho^{c}(x')\,\rho^{a}(y)\,\rho^{e}(x)$	A2.1 $(-i)$; GI.II.1b* $(-i)$; GI.II.1d* $(+i)(-1)$; GIII.II.3b* $(-i)(-1)$	$-2i$ ✗
α'_2	$f^{abc}\,U^{bd}(w')\,U^{de}(w)\,\rho^{c}(x')\,\rho^{a}(y)\,\rho^{e}(x)$	A2.2 $(+i)$; GI.II.3b* $(+i)$; GI.II.3d* $(-i)(-1)$; GII.I2.1b $(+i)(-1)$	$+2i$ ✗
α'_3	$f^{abc}\,U^{dc}(x')\,U^{eb}(w)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	B2.1 $(+i)$; GI.II.4b* $(-i)$; GI.II.4d* $(+i)(-1)$; GI.III.4a* $(-i)(-1)$; GI.IV.4b* $(-i)(-1)$; GIII.I.4b* $(+i)(-1)$	0 ✓
α'_4	$f^{abc}\,U^{dc}(x')\,U^{be}(x)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	B2.2 $(-i)$; GI.II.2b* $(+i)$; GI.II.2d* $(-i)(-1)$; GI.III.2a* $(+i)(-1)$; GI.IV.2b* $(+i)(-1)$; GII.I2.4b $(+i)(-1)$; GIII.I.2b* $(-i)(-1)$; GIII.II.2b* $(-i)(-1)$	0 ✓
α'_5	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GI.III.1a* $(-i)$; GIII.II.4b* $(+i)$	0 ✓
α'_6	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GI.III.3a* $(+i)$; GII.I2.3b $(-i)$	0 ✓
α'_7	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)$ $\rho^{l}(y)\,\rho^{c}(x')$	GI.IV.1b* $(-i)$; GII.I2.2b $(-i)$; GIII.I.1b* $(+i)$; GIII.II.1b* $(+i)$	0 ✓
α'_8	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)$ $\rho^{l}(y)\,\rho^{c}(x')$	GI.IV.3b* $(+i)$; GIII.I.3b* $(-i)$	0 ✓

9.3 Class β: soft gluon between the measured gluon at w and the charge y

Kernel $K_\beta = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(y-z)$. 44 entries, 12 colour networks; 8 cancel, 4 do not.

	colour structure C	entries: coefficient × sign	sum
β_1	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(x)\,\rho^{c}(y)$	GI.II.1a $(-i)$; GI.II.1c $(+i)(-1)$; GIII.II.3a $(-2i)(-1)$	0 ✓
β_2	$f^{abc}\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.II.2a $(+i)$; GI.II.2c $(-i)(-1)$; GI.IV.2a $(+2i)(-1)$; GI.V1.2 $(+i)(-1)$; GI.V2.2 $(-i)$; GII.I1.4a $(+2i)(-1)$; GIII.I.2a $(-2i)(-1)$; GIII.II.2a $(-2i)(-1)$; GIII.III.4* $(+2i)$; GIII.V.3* $(-2i)$; GIII.VI.3* $(-2i)$	$-2i$ ✗
β_3	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,\rho^{e}(x')\,\rho^{b}(x)\,\rho^{c}(y)$	GI.II.3a $(+i)$; GI.II.3c $(-i)(-1)$; GII.I1.1a $(+2i)(-1)$	0 ✓
β_4	$f^{abc}\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.II.4a $(-i)$; GI.II.4c $(+i)(-1)$; GI.IV.4a $(-2i)(-1)$; GI.V1.4 $(-i)(-1)$; GI.V2.4 $(+i)$; GIII.I.4a $(+2i)(-1)$; GIII.V.4* $(+2i)$	$+2i$ ✗
β_5	$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')$ $\rho^{l}(y)\,\rho^{c}(x)$	GI.IV.1a $(-2i)$; GII.I1.3a $(-2i)$; GIII.I.1a $(+2i)$; GIII.II.1a $(+2i)$	0 ✓
β_6	$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')$ $\rho^{l}(y)\,\rho^{c}(x)$	GI.IV.3a $(+2i)$; GIII.I.3a $(-2i)$	0 ✓
β_7	$f^{abc}\,U^{ad}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^{d}(x')\,\rho^{h}(x)$ $\rho^{e}(y)$	GI.V1.1 $(-i)$; GI.V2.1 $(+i)(-1)$; GIII.VI.2* $(-2i)(-1)$	0 ✓
β_8	$f^{abc}\,U^{ad}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^{d}(x')\,\rho^{h}(x)$ $\rho^{e}(y)$	GI.V1.3 $(+i)$; GI.V2.3 $(-i)(-1)$; GIII.III.1* $(+2i)(-1)$	0 ✓
β_9	$f^{abc}\,U^{ad}(w')\,U^{eb}(z)\,U^{ch}(x)\,\rho^{d}(x')\,\rho^{e}(y)$ $\rho^{h}(x)$	GII.I1.2a $(-2i)$; GIII.III.3* $(-2i)(-1)$	0 ✓
β_{10}	$f^{abc}\,U^{ad}(x')\,U^{eb}(z)\,U^{ch}(x)\,\rho^{e}(y)\,\rho^{d}(x')$ $\rho^{h}(x)$	GIII.II.4a $(+2i)$; GIII.VI.4* $(+2i)(-1)$	0 ✓

	colour structure C	entries: coefficient × sign	sum
β_{11}	$f^{abc}\,U^{da}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GIII.III.2* $(-2i)$; GIII.V.1* $(+2i)$; GIII.VI.1* $(+2i)$	$+2i$ ✗
β_{12}	$f^{abc}\,U^{da}(w)\,U^{ce}(w')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{h}(y)$ $\rho^{e}(x')$	GIII.V.2* $(-2i)$	$-2i$ ✗

9.4 Class β': soft gluon between the measured gluon at w' and the charge y

Kernel $K_{\beta'} = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-z)$. 44 entries, 12 colour networks; 8 cancel, 4 do not.

	colour structure C	entries: coefficient × sign	sum
β'_1	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$	GI.II.1a* $(+i)$; GI.II.1c* $(-i)(-1)$; GIII.II.3a* $(+2i)(-1)$	0 ✓
β'_2	$f^{abc}\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GI.II.2a* $(-i)$; GI.II.2c* $(+i)(-1)$; GI.IV.2a* $(-2i)(-1)$; GI.V1.2* $(-i)(-1)$; GI.V2.2* $(+i)$; GII.I2.4a $(-2i)(-1)$; GIII.I.2a* $(+2i)(-1)$; GIII.II.2a* $(+2i)(-1)$; GIII.III.4 $(-2i)$; GIII.V.3 $(+2i)$; GIII.VI.3 $(+2i)$	$+2i$ ✗
β'_3	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,\rho^{e}(x)\,\rho^{b}(x')\,\rho^{c}(y)$	GI.II.3a* $(-i)$; GI.II.3c* $(+i)(-1)$; GII.I2.1a $(-2i)(-1)$	0 ✓
β'_4	$f^{abc}\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GI.II.4a* $(+i)$; GI.II.4c* $(-i)(-1)$; GI.IV.4a* $(+2i)(-1)$; GI.V1.4* $(+i)(-1)$; GI.V2.4* $(-i)$; GIII.I.4a* $(-2i)(-1)$; GIII.V.4 $(-2i)$	$-2i$ ✗
β'_5	$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)$ $\rho^{l}(y)\,\rho^{c}(x')$	GI.IV.1a* $(+2i)$; GII.I2.2a $(+2i)$; GIII.I.1a* $(-2i)$; GIII.II.1a* $(-2i)$	0 ✓
β'_6	$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x)$ $\rho^{l}(y)\,\rho^{c}(x')$	GI.IV.3a* $(-2i)$; GIII.I.3a* $(+2i)$	0 ✓
β'_7	$f^{abc}\,U^{ad}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')$ $\rho^{e}(y)$	GI.V1.1* $(+i)$; GI.V2.1* $(-i)(-1)$; GIII.VI.2 $(+2i)(-1)$	0 ✓
β'_8	$f^{abc}\,U^{ad}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x)\,\rho^{h}(x')$ $\rho^{e}(y)$	GI.V1.3* $(-i)$; GI.V2.3* $(+i)(-1)$; GIII.III.1 $(-2i)(-1)$	0 ✓
β'_9	$f^{abc}\,U^{db}(z)\,U^{ce}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{h}(x)$ $\rho^{d}(y)$	GII.I2.3a $(+2i)$; GIII.III.3 $(+2i)(-1)$	0 ✓
β'_{10}	$f^{abc}\,U^{ad}(x)\,U^{eb}(z)\,U^{ch}(x')\,\rho^{e}(y)\,\rho^{d}(x)$ $\rho^{h}(x')$	GIII.II.4a* $(-2i)$; GIII.VI.4 $(-2i)(-1)$	0 ✓
β'_{11}	$f^{abc}\,U^{da}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GIII.III.2 $(+2i)$; GIII.V.1 $(-2i)$; GIII.VI.1 $(-2i)$	$-2i$ ✗
β'_{12}	$f^{abc}\,U^{da}(w')\,U^{ce}(w)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{h}(y)$ $\rho^{e}(x)$	GIII.V.2 $(+2i)$	$+2i$ ✗

9.5 Class γ: the edge kernel of Row III, part 2

Kernel $K_{\gamma} = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\;\Phi_{w}(x)$. 4 entries, 4 colour networks; 0 cancel, 4 do not.

	colour structure C	entries: coefficient × sign	sum
γ_1	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.III.1b $(+2i)$	$+2i$ ✗
γ_2	$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.III.2b $(-2i)$	$-2i$ ✗
γ_3	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.III.3b $(-2i)$	$-2i$ ✗
γ_4	$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^{d}(x')\,\rho^{e}(x)$ $\rho^{h}(y)$	GI.III.4b $(+2i)$	$+2i$ ✗

9.6 Class γ': the edge kernel of Row III, part 2 (c.c.)

Kernel $K_{\gamma'} = \mathcal{K}(x-w)\cdot\mathcal{K}(y-z)\;\Phi_{w'}(x')$. 4 entries, 4 colour networks; 0 cancel, 4 do not.

	colour structure C	entries: coefficient × sign	sum
γ'_1	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GI.III.1b* $(-2i)$	$-2i$ ✗
γ'_2	$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(y)\,\rho^{d}(x)\,\rho^{e}(x')$ $\rho^{h}(y)$	GI.III.2b* $(+2i)$	$+2i$ ✗

	colour structure C	entries: coefficient × sign	sum
γ'_3	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^d(x)\,\rho^e(x')$ $\rho^h(y)$	`GI.III.3b*` $(+2i)$	$+2i$ ✗
γ'_4	$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^d(x)\,\rho^e(x')$ $\rho^h(y)$	`GI.III.4b*` $(-2i)$	$-2i$ ✗

10 Where the left-overs come from

10.1 The non-cancelling networks one by one

Table 5 splits the sum of each non-cancelling network into the entries that stay, the entries removed by correction 1 (Row II) and correction 2 (Row III, part 2), and the entries of the new term $\mathcal{D}$.

network	sum as written	entries that stay	removed by (1)	removed by (2)	entries of $\mathcal{D}$	sum after
α_1	$-2i$	0	$-2i$	–	–	0
α_2	$2i$	0	$2i$	–	–	0
α'_1	$-2i$	0	$-2i$	–	–	0
α'_2	$2i$	0	$2i$	–	–	0
β_2	$-2i$	$-2i$	–	–	$2i$	0
β_4	$2i$	$2i$	–	–	$-2i$	0
β_{11}	$2i$	$2i$	–	–	$-2i$	0
β_{12}	$-2i$	$-2i$	–	–	$2i$	0
β'_2	$2i$	$2i$	–	–	$-2i$	0
β'_4	$-2i$	$-2i$	–	–	$2i$	0
β'_{11}	$-2i$	$-2i$	–	–	$2i$	0
β'_{12}	$2i$	$2i$	–	–	$-2i$	0
γ_1	$2i$	–	–	$2i$	–	empty
γ_2	$-2i$	–	–	$-2i$	–	empty
γ_3	$-2i$	–	–	$-2i$	–	empty
γ_4	$2i$	–	–	$2i$	–	empty
γ'_1	$-2i$	–	–	$-2i$	–	empty
γ'_2	$2i$	–	–	$2i$	–	empty
γ'_3	$2i$	–	–	$2i$	–	empty
γ'_4	$-2i$	–	–	$-2i$	–	empty

Table 5: The 20 networks that do not cancel as written (units of $\mathcal{N}$).

- **α_1, α_2 and α'_1, α'_2.** Without the Row II entries these networks cancel, e.g. in α_1: `A1.1` $(-i)$+`GII.I1.1b` $(-i)(-1) = 0$. The two Row II entries (monomials b, d of sub-term 3) add $-2i$. They come from the part

$$\mathfrak{F}^i_\alpha(w;x,y) = -\big[\mathcal{K}^i(x-w) - \mathcal{K}^i(y-w)\big] \int_z \mathcal{K}(x-z)\cdot\mathcal{K}(y-z) \tag{10.1}$$

  of the kernel (soft gluon between the two charges) in the term $U^{ah}(w) f^{hbc}\{\rho^b(x), \rho^c(y)\}$, i.e. $U(w)\mathbb{A}^{(3)}$. The same part in the term $\bar{\mathbb{A}}^{(3)}$ (sub-terms 2, 4) is needed: its entries sit in networks α_3, α_4, which cancel.

- **$\gamma_{1\ldots4}$, $\gamma'_{1\ldots4}$.** Each contains a single entry, from part 2 of Row III. No other term has the kernel Φ, so these entries cannot cancel. In γ_1, γ_3 the colour depends on z through $U(z)$, so the z-integral cannot turn them into another class either. Moreover, by Eq. (7.2), $\Phi_w(x) = -[(x-w)^2 + (x-z)^2]/[(x-w)^2 - (x-z)^2]^2$ has a double pole wherever $|x-w| = |x-z|$: in x on the whole perpendicular bisector of w and z, in z on the whole circle of radius $|x-w|$ around x. Neither the x- nor the z-integral of these terms exists, so they cannot cancel anything after integration either.

- **$\beta_2, \beta_4, \beta_{11}, \beta_{12}$ and $\beta'_2, \beta'_4, \beta'_{11}, \beta'_{12}$.** Each sums to $\pm 2i$. Networks β_{11}, β_{12} contain only Group III entries, which are correct (Sec. 10.2). So a term is missing, with kernel class β: a soft gluon between the measured gluon at w and a charge. The term $\mathcal{D}$ of Eq. (11.4) supplies exactly $\mp 2i$ in each of these networks and nothing elsewhere.

10.2 Comparison with an independent leading-log computation

In earlier work on the same cross section we built the $\log(\vee/\Lambda)$ terms row by row directly from the leading-log light-cone wave function, keeping the operator order, and checked those rows numerically against an exact evaluation of the operator products. That construction does not use the intermediate steps of the notes; we call it the *reference*. Its symmetrized three-ρ terms cancel (residual $\sim 10^{-14}$), also for non-symmetric U. Here we use it only to locate errors: for every row we compare the symmetrized three-ρ content of the notes with that of the reference, kernel monomial by kernel monomial, for random symmetric Wilson lines and charges. A four-ρ term of a row contributes through its three-ρ remainder.

piece	max |notes − reference|	size	verdict
four-ρ terms of the notes, symmetrized: max |sum| (must vanish)	8.0×10^{-15}	116	cancel
`A1+A2+B1+B2` vs. three-ρ part of the four-ρ terms	1.1×10^{-14}	2.2	agree
Group II, Row I (both terms + its four-ρ part)	2.4×10^{-15}	7.0	agree
Group II, Row II (four-ρ part only)	2.7×10^{-15}	7.8	agree
Group II, Row III (four-ρ part only)	1.3×10^{-15}	0.0	both vanish
Group III, Row I	4.4×10^{-15}	9.1	agree
Group III, Row II	3.1×10^{-15}	8.9	agree
Group III, Row III	2.7×10^{-15}	12.6	agree
Group III, Row IV (four-ρ part only)	1.8×10^{-15}	7.3	agree
Group III, Row V	9.8×10^{-15}	4.4	agree
Group III, Row VI	1.8×10^{-15}	7.4	agree
Group I, Row I (normalization; four-ρ part)	1.3×10^{-15}	6.2	agree
Group I, Row IV, sub-terms 1, 3	1.8×10^{-15}	7.5	agree
Group I, Row V, sub-terms 1, 3	5.3×10^{-15}	9.6	agree
Group I, Row II, sub-terms 1, 3 ($U(w)\mathbb{A}^{(3)}$)	3.2×10^{0}	5.8	differ
Group I, Row III, sub-terms 1, 3	6.4×10^{0}	4.3	differ
Group I, sub-terms 2, 4 of Rows II–V (together)	2.9×10^{0}	2.8	differ
Group I, all rows, as written	7.4×10^{0}	20.5	differ

Table 6: Notes vs. reference, with the notes as written (maximum over kernel monomials and three random configurations; "size" is the largest value of the reference piece). Sub-terms 1, 3 of a Group I row are those with the direct amplitude ($U(w)\mathbb{A}^{(3)}$, $\bar{\mathbb{B}}_3^\dagger U\mathbb{A}$, ...); sub-terms 2, 4 contain the rotated charges and, by unitarity, combine into $\bar{\mathbb{A}}^{(3)\dagger}$.

Everything outside Group I agrees with the reference to machine precision. Inside Group I, Row I and the direct parts of Rows IV and V agree as well. The direct parts of Rows II and III and the unitarity combination differ, which matches the three items of Table 5.

11 The two corrections

11.1 Correction 1: Row II of Group I

Split the two-charge kernel of Row II as $\mathfrak{F}^i = \mathfrak{F}^i_\beta + \mathfrak{F}^i_\alpha$, with $\mathfrak{F}^i_\alpha$ of Eq. (10.1) and

$$\mathfrak{F}^i_\beta(w;x,y) = \mathcal{K}^i(x-w)\int_z \mathcal{K}(w-z)\cdot\mathcal{K}(y-z) - \mathcal{K}^i(y-w)\int_z \mathcal{K}(w-z)\cdot\mathcal{K}(x-z). \tag{11.1}$$

Correction 1: the $U(w)\mathbb{A}^{(3)}$ term carries $\mathfrak{F}_\beta$ only,

$$\begin{aligned}\text{Row II} = &-\frac{i}{2}\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\mathcal{K}^i(x'-w')\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x') \\ &\times\Big[\mathfrak{F}^i_{\mathcal{B}}(w;x,y)\,U^{ah}(w) f^{hbc}\{\rho^b(x),\rho^c(y)\} - \mathfrak{F}^i(w;x,y)\,f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big] + \text{c.c.}\end{aligned} \tag{11.2}$$

In the tables this deletes the 8 entries `GI.II.1b` $(+i)$, `GI.II.1d` $(-i)$, `GI.II.3b` $(-i)$, `GI.II.3d` $(+i)$, `GI.II.1b*` $(-i)$, `GI.II.1d*` $(+i)$, `GI.II.3b*` $(+i)$, `GI.II.3d*` $(-i)$.

An equivalent form. One may instead remove $\mathfrak{F}_\alpha$ from *both* terms of Row II and add the commutator term

$$\begin{aligned}&-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(s-w)\,\mathcal{K}(u-z)\cdot\mathcal{K}(s-z) \\ &\quad\times f^{eag}\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\,\big\{U^{eh}(u)\rho^h(u),\,U^{gl}(s)\rho^l(s)\big\} + \text{c.c.},\end{aligned} \tag{11.3}$$

which is $-[\bar{\mathbb{N}}_2, \bar{\mathbb{A}}^{(1)}]$ of the unitarity relation times the conjugate LO amplitude. Its symmetrized three-ρ part equals that of $\mathfrak{F}_\alpha$ in the $\bar{\mathbb{A}}^{(3)}$ term. The two forms therefore have the same three-ρ terms (we checked: the corrected Group I agrees with the reference to 1.4×10^{-14} either way), but different two-ρ terms. The three-ρ check cannot decide between them. In the reference this structure arises in the commutator form (11.3), from the operator order of $\bar{\mathbb{N}}_2$ and $\bar{\mathbb{A}}^{(1)}$. Part III found that both forms give the same two-ρ terms (Sec. 21). Sec. 5 shows what correction 1 means: the $\mathfrak{F}_\alpha$ part is the commutator $[\mathbb{N}^{(2)}, \mathbb{A}^{(1)}]$, and correction 1 is the p^+ ordering of the virtual factor.

11.2 Correction 2: Row III of Group I

Delete part 2 of Row III (the 8 entries `GI.III.1b` $(+2i)$, `GI.III.2b` $(-2i)$, `GI.III.3b` $(-2i)$, `GI.III.4b` $(+2i)$, `GI.III.1b*` $(-2i)$, `GI.III.2b*` $(+2i)$, `GI.III.3b*` $(+2i)$, `GI.III.4b*` $(-2i)$) and add

$$\mathcal{D} = -2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$$

$$\times\, f^{abd}\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\, U^{de}(x)\rho^{e}(x)\, \big[U^{bc}(w) - U^{bc}(y)\big]\rho^{c}(y) \;+\; \text{c.c.} \tag{11.4}$$

$\mathcal{D}$ has the colour structure of Row III with $U^{bc}(z) \to U^{bc}(w)$. The charge y emits the measured gluon at w at leading order ($\mathcal{K}(y-w)$, $[U(w)-U(y)]\rho(y)$), and a soft gluon at z is exchanged between the measured gluon and the rotated charge $U(x)\rho(x)$ ($\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$). This is Row III in the region where the gluon emitted by $\mathbb{A}^{(1)}(p^+,z)$ carries almost all of k^+ and becomes the measured gluon, so $z \to w$ in its Wilson line. We believe part 2 of Row III is the notes' version of this region. Its edge kernel Φ should be replaced by the kernel of Eq. (11.4). In the reference, sub-terms 1, 3 of $\mathcal{D}$ are a separate virtual row (soft gluon between the measured gluon and $U\rho$), and sub-terms 2, 4 belong to the unitarity combination (Table 7). For the two-ρ terms of $\mathcal{D}$, use the charge order of Row III, $\rho(x')\rho(x)\rho(y)$.

Consequence for the two-ρ terms. Apps. B–F derive the two-ρ terms from the terms as written. Three groups of them change; all other two-ρ terms stay as they are.

- The two-ρ terms of Row III, part 2, go away.
- The two-ρ terms coming from $\mathfrak{F}_\alpha$ in the $U(w)\mathbb{A}^{(3)}$ term of Row II go away. With the form (11.3), the $\mathfrak{F}_\alpha$ terms go away from both terms of Row II, and those of the commutator term are added instead.
- $\mathcal{D}$ adds its own two-ρ terms.

The entries of $\mathcal{D}$:

`D`: Replacement $\mathcal{D}$ of Row III, part 2

Prefactor $-2i\mathcal{N}$; ordered product $\rho\rho\rho \to S(\rho\rho\rho)$, factor 1; with c.c.; phase $e^{-ik\cdot(w'-w)}$. Kernel: $K = \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$. The symmetrized term (commuting charges):

$$\begin{aligned}
&-2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K\,\Big\{\\
&\quad(1)\quad +\, f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(w)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\
&\quad(2)\quad -\, f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\
&\quad(3)\quad -\, f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(w)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\
&\quad(4)\quad +\, f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\Big\} + \text{c.c.}
\end{aligned}$$

Renaming to common variables (all replacements simultaneous; ε is the sign from $\mathcal{K}(-X) = -\mathcal{K}(X)$):

mon.	term	renaming	kernel in common variables	ε	class
	amp.	$x \to y,\ y \to x$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(w-z)$	$+$	β
	c.c.	$x' \to x,\ x \to y,\ y \to x',\ w' \to w,\ w \to w'$	$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(y-z)\cdot\mathcal{K}(w'-z)$	$+$	β'

Entries (coefficient in units of $\mathcal{N}$; colour structure in common variables):

entry	class	coeff.	colour structure
`D.1`	β	$-2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(w)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`D.2`	β	$+2i$	$f^{abc}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`D.3`	β	$+2i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(w)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`D.4`	β	$-2i$	$f^{abc}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(x)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$
`D.1*`	β'	$+2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(w')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
`D.2*`	β'	$-2i$	$f^{abc}\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
`D.3*`	β'	$-2i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(w')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$
`D.4*`	β'	$+2i$	$f^{abc}\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(x')\,\rho^{d}(x)\,\rho^{e}(y)\,\rho^{h}(x')$

12 The networks after the corrections

The corrections change only the networks of Table 5. Here they are again, with the entries of $\mathcal{D}$ in blue and the deleted entries left out. In total 152 entries remain ($160 - 8 - 8 + 8$).

Class α. 2 of the 8 networks change:

	colour structure C	entries: coefficient × sign	sum
α_1	$f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^{d}(x')\,\rho^{a}(y)\,\rho^{c}(x)$	`A1.1` $(-i)$; `GII.I1.1b` $(-i)(-1)$	0 ✓
α_2	$f^{abc}\,U^{db}(w)\,U^{de}(x')\,\rho^{e}(x')\,\rho^{a}(y)\,\rho^{c}(x)$	`A1.2` $(+i)$; `GIII.II.3b` $(+i)(-1)$	0 ✓

Class α'. 2 of the 8 networks change:

	colour structure C	entries: coefficient × sign	sum
α'_1	$f^{abc}\,U^{db}(w')\,U^{de}(x)\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)$	`A2.1` $(-i)$; `GIII.II.3b*` $(-i)(-1)$	0 ✓
α'_2	$f^{abc}\,U^{bd}(w')\,U^{de}(w)\,\rho^c(x')\,\rho^a(y)\,\rho^e(x)$	`A2.2` $(+i)$; `GII.I2.1b` $(+i)(-1)$	0 ✓

Class β. 4 of the 12 networks change:

	colour structure C	entries: coefficient × sign	sum
β_2	$f^{abc}\,U^{ad}(x')\,U^{be}(x)\,U^{ch}(y)\,\rho^d(x')\,\rho^e(x)$ $\rho^h(y)$	`GI.II.2a` $(+i)$; `GI.II.2c` $(-i)(-1)$; `GI.IV.2a` $(+2i)(-1)$; `GI.V1.2` $(+i)(-1)$; `GI.V2.2` $(-i)$; `GII.I1.4a` $(+2i)(-1)$; `GIII.I.2a` $(-2i)(-1)$; `GIII.II.2a` $(-2i)(-1)$; `GIII.III.4*` $(+2i)$; `GIII.V.3*` $(-2i)$; `GIII.VI.3*` $(-2i)$; `D.2` $(+2i)$	0 ✓
β_4	$f^{abc}\,U^{ad}(w')\,U^{be}(x)\,U^{ch}(y)\,\rho^d(x')\,\rho^e(x)$ $\rho^h(y)$	`GI.II.4a` $(-i)$; `GI.II.4c` $(+i)(-1)$; `GI.IV.4a` $(-2i)(-1)$; `GI.V1.4` $(-i)(-1)$; `GI.V2.4` $(+i)$; `GIII.I.4a` $(+2i)(-1)$; `GIII.V.4*` $(+2i)$; `D.4` $(-2i)$	0 ✓
β_{11}	$f^{abc}\,U^{da}(w)\,U^{ce}(x')\,U^{bh}(y)\,\rho^d(x)\,\rho^e(x')$ $\rho^h(y)$	`GIII.III.2*` $(-2i)$; `GIII.V.1*` $(+2i)$; `GIII.VI.1*` $(+2i)$; `D.1` $(-2i)$	0 ✓
β_{12}	$f^{abc}\,U^{da}(w)\,U^{ce}(w')\,U^{bh}(y)\,\rho^d(x)\,\rho^h(y)$ $\rho^e(x')$	`GIII.V.2*` $(-2i)$; `D.3` $(+2i)$	0 ✓

Class β'. 4 of the 12 networks change:

	colour structure C	entries: coefficient × sign	sum
β'_2	$f^{abc}\,U^{ad}(x)\,U^{be}(x')\,U^{ch}(y)\,\rho^d(x)\,\rho^e(x')$ $\rho^h(y)$	`GI.II.2a*` $(-i)$; `GI.II.2c*` $(+i)(-1)$; `GI.IV.2a*` $(-2i)(-1)$; `GI.V1.2*` $(-i)(-1)$; `GI.V2.2*` $(+i)$; `GII.I2.4a` $(-2i)(-1)$; `GIII.I.2a*` $(+2i)(-1)$; `GIII.II.2a*` $(+2i)(-1)$; `GIII.III.4` $(-2i)$; `GIII.V.3` $(+2i)$; `GIII.VI.3` $(+2i)$; `D.2*` $(-2i)$	0 ✓
β'_4	$f^{abc}\,U^{ad}(w)\,U^{be}(x')\,U^{ch}(y)\,\rho^d(x)\,\rho^e(x')$ $\rho^h(y)$	`GI.II.4a*` $(+i)$; `GI.II.4c*` $(-i)(-1)$; `GI.IV.4a*` $(+2i)(-1)$; `GI.V1.4*` $(+i)(-1)$; `GI.V2.4*` $(-i)$; `GIII.I.4a*` $(-2i)(-1)$; `GIII.V.4` $(-2i)$; `D.4*` $(+2i)$	0 ✓
β'_{11}	$f^{abc}\,U^{da}(w')\,U^{ce}(x)\,U^{bh}(y)\,\rho^d(x')\,\rho^e(x)$ $\rho^h(y)$	`GIII.III.2` $(+2i)$; `GIII.V.1` $(-2i)$; `GIII.VI.1` $(-2i)$; `D.1*` $(+2i)$	0 ✓
β'_{12}	$f^{abc}\,U^{da}(w')\,U^{ce}(w)\,U^{bh}(y)\,\rho^d(x')\,\rho^h(y)$ $\rho^e(x)$	`GIII.V.2` $(+2i)$; `D.3*` $(-2i)$	0 ✓

Class γ. All 4 entries are deleted; the class is empty.

Class γ'. All 4 entries are deleted; the class is empty.

So **after the two corrections every symmetrized three-ρ term cancels**: all 40 networks add up to zero with exact coefficients. The corrected Group I also agrees with the reference piece by piece:

corrected notes	reference row	max |difference|	size
Row II, sub-terms 1, 3, with correction 1	$U(w)\mathbb{A}^{(3)}$	8.9×10^{-16}	5.8
Row III, sub-terms 1, 3, part 1 only	$\bar{\mathbb{B}}_3^{\dagger}U\mathbb{A}$	2.3×10^{-15}	4.3
$\mathcal{D}$, sub-terms 1, 3	row I′ (soft gluon between w and $U\rho$)	4.4×10^{-15}	6.4
sub-terms 2, 4 of Rows II–V and of $\mathcal{D}$	$\bar{\mathbb{A}}^{(3)\dagger}$ (unitarity)	8.9×10^{-16}	2.8
Group I, all rows, corrected	all	1.4×10^{-14}	20.5

Table 7: Group I after the corrections vs. the reference.

13 Pointwise numerical check

This check does not use the classes or the networks. We pick random points w, w', z and three random charge positions p_1, p_2, p_3. We pick random $SU(3)$ adjoint Wilson lines at every point and random classical charges. For every term of the notes, written exactly as in the sources (Φ as in the notes; for Row II the integrand of the z-representation of $\mathfrak{F}$,

taken at the same z), we sum over the 3! ways of placing its charges on p_1, p_2, p_3 and add everything up. The table gives $|\text{sum}|/\sum|\text{terms}|$.

	symmetric U ($U^{bc} = U^{cb}$)			general adjoint U	
trial	as written	corrected	corrected (form (11.3))	as written	corrected
1	1.1×10^{-1}	3.4×10^{-16}	3.0×10^{-16}	4.0×10^{-2}	3.0×10^{-3}
2	1.9×10^{-1}	6.1×10^{-17}	2.7×10^{-17}	2.0×10^{-2}	4.7×10^{-2}
3	3.9×10^{-1}	1.3×10^{-16}	1.8×10^{-16}	3.4×10^{-1}	3.4×10^{-3}
4	5.2×10^{-2}	8.5×10^{-17}	8.1×10^{-17}	9.9×10^{-2}	1.7×10^{-2}

With the notes' convention the corrected sum vanishes to machine precision, and the sum as written does not.

14 Remark on $U^{bc} = U^{cb}$

The notes identify U^{bc} with U^{cb} throughout, and everything above uses this. A true adjoint Wilson line is real and orthogonal but not symmetric: $U^\dagger = U^T \neq U$. For a general U even the corrected sum does not vanish (last column above). Already the remainders `A1`–`B2` then differ from the three-ρ part of the four-ρ terms they come from, and the four-ρ terms of Groups II and III differ from the reference. In the reference, which keeps U and U^T apart, the three-ρ terms cancel for general U too. So the cancellation itself does not need $U = U^T$. To use non-symmetric Wilson lines, the index order of every U in the notes has to be restored first. That is a separate task from the one done here.

Part III
The two-ρ terms proportional to $\log(\vee/\Lambda)$: JIMWLK evolution of the target

This part collects every two-ρ term proportional to $\log(\vee/\Lambda)$ and compares the sum with the JIMWLK and BFKL evolution of the LO cross section. The two-ρ terms of Group I, Row II are those of App. C, with the coordinate-space kernel of Part I and fix F4.

15 Overview

Question. Collect all two-ρ terms proportional to $\log(\vee/\Lambda)$:

1. those from reordering the three-ρ terms (Apps. B–F, including the remainders A1–B2 of the four-ρ terms);
2. those from reordering the four-ρ terms (`4rho.tex`, "Two ρ contribution");
3. the genuine two-ρ terms of `Evolution_two_rho.tex`. The $\log(\vee/\Lambda)$ part of Row V of Group I (part 2b) still has to be extracted (Sec. 17).

Do they reproduce the BFKL and JIMWLK evolution of the leading-order cross section (Appendices E and F), with $\delta Y = \log(\vee/\Lambda)$? If not, why not, and what must be fixed?

Answer.

1. **The $\log(\vee/\Lambda)$ terms give JIMWLK⊗LO, not BFKL⊗LO.** With your rule $\log(\Lambda/k^+) = -\log(\vee/\Lambda) + \log(\vee/k^+)$, the coefficient of $\log(\vee/\Lambda)$ comes only from the soft gluon with $p^+ \ll k^+$. That gluon evolves the target, i.e. it gives JIMWLK. BFKL comes from gluons with $p^+ \gg k^+$ and multiplies $\log(\vee/k^+)$ (Sec. 16). The terms themselves show this directly: in every $\log(\vee/\Lambda)$ two-ρ term the measured gluon is emitted by a charge, whereas BFKL⊗LO (Eq. (LOBFKL)) contains terms in which it is emitted by the new gluon at z, e.g. $\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)$.
2. **As written, the sum is not JIMWLK⊗LO.** 32 of the 38 kernel structures differ (Table 8).
3. **Group II and Group III are correct row by row**, except the genuine two-ρ term of Group III, Row III (part 2), whose overall sign must be reversed. Every other row of Groups II and III agrees exactly with the independent leading-log computation (Table 9).
4. **After six fixes the sum equals JIMWLK⊗LO exactly**, pointwise in the soft-gluon position z, in all 13 kernel structures and with exact coefficients (Sec. 20). No two-ρ term is left over. The fixes:

   F1 `4rho.tex`, first part: drop the term $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)\ldots$

F2 `4rho.tex`, second part, $dN^{(1)}$ term 2: the index b' appears four times. The correct term is $3\,f^{abc}f^{deg}\rho^{b'}(x')\rho^{c}(y)U^{ag}(y)U^{db'}(w')U^{eb}(z)$.

F3 Group I, Row III: part 2 is replaced by $\mathcal{D}$ (Part II). Its two-ρ terms follow from the charge order $\rho(x')\rho(x)\rho(y)$ of Row III; every other order fails.

F4 Group I, Row II: $\mathfrak{F}\to\mathfrak{F}_\beta$ in the $U(w)\mathbb{A}^{(3)}$ term (Part II). The commutator form gives the *same* two-ρ terms, so either form can be used. F4 is the p^+ ordering of the virtual factor (Sec. 5).

F5 Group III, Row III, part 2 (genuine two-ρ): overall sign reversed.

F6 Group I, Row II (genuine two-ρ): the bracket $[-\frac{2}{\epsilon}-2\gamma-\log\frac{(x-w)^2\mu^2}{4}]$ must be $\frac{1}{\pi}\int_z[K(x,w,z)-K(w,w,z)]$. In dimensional regularization this is $\frac{1}{\epsilon}-\log(4\pi^2\mu^2(x-w)^2)$, the same form as in Group II, Row IV. The dependence on $(x-w)^2$ is correct; only the pole and constant differ.

5. **Bookkeeping.** The lists of `4rho.tex` contain the two-ρ terms that come *directly* from reordering the four-ρ terms. Together with the two-ρ terms of A1–B2 they give the complete two-ρ content of Eq. (4rho) (checked against the exact decomposition, Sec. 18.3). Both pieces are needed, and neither may be counted twice.

6. **Appendix F** as pasted still contains the two errors reported earlier: M at the wrong points in $dN^{(1)}$–$dN^{(4)}$, and the sign of $dN^{(5)}$. The corrected form equals $-\delta Y\,H_{\text{JIMWLK}}$ acting on LO exactly (Sec. 19). The sum of the two-ρ terms matches the corrected form.

Everything uses your convention $U^{bc}=U^{cb}$.

16 Which evolution multiplies $\log(\vee/\Lambda)$?

The coefficient of $\log(\vee/\Lambda)$ is the soft-gluon limit. Every $\log(\vee/\Lambda)$ comes from a p^+ integral of the type

$$\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}\,f\Big(\frac{p^+}{k^+}\Big)=f_T\log\frac{k^+}{\Lambda}+f_P\log\frac{\vee}{k^+}+(\text{no large logarithm}),\qquad f_T=f(0),\quad f_P=f(\infty).\tag{16.1}$$

With your rule $\log(\Lambda/k^+)=-\log(\vee/\Lambda)+\log(\vee/k^+)$ this is

$$f_T\,\log\frac{\vee}{\Lambda}+(f_P-f_T)\log\frac{\vee}{k^+}+\dots\tag{16.2}$$

So the coefficient of $\log(\vee/\Lambda)$ is always f_T, the limit $p^+\ll k^+$. Your second rule, $\log((\vee-k^+)/\Lambda)=\log(\vee/\Lambda)+\log(1-k^+/\vee)$, agrees with this when f is constant ($f_T=f_P$).

Soft gluons evolve the target. A gluon softer than the measured one ($\Lambda<p^+\ll k^+$) is emitted by all harder charges (the valence charges and the measured gluon) and scatters on the target. Integrating it out is one step of JIMWLK evolution of the target Wilson lines. The projectile's $\langle\rho\rho\rangle$ changes only through gluons harder than the measured one ($k^+\ll p^+<\vee$), which come with $\log(\vee/k^+)$. Hence, at leading logarithm,

$$\begin{aligned}\frac{d^3N}{d^3k}\bigg|_{\text{LL}}&=\log\frac{k^+}{\Lambda}\,\big[\text{JIMWLK}\otimes\text{LO}\big]+\log\frac{\vee}{k^+}\,\big[\text{BFKL}\otimes\text{LO}\big]\\&=\log\frac{\vee}{\Lambda}\,\big[\text{JIMWLK}\otimes\text{LO}\big]+\log\frac{\vee}{k^+}\,\big[\text{BFKL}\otimes\text{LO}-\text{JIMWLK}\otimes\text{LO}\big].\end{aligned}\tag{16.3}$$

This was derived from the wave function in `BFKL_JIMWLK.pdf`. The coefficient of $\log(\vee/\Lambda)$ must therefore be JIMWLK⊗LO alone. BFKL⊗LO is the coefficient of $\log\vee$, i.e. of $\log(\vee/k^+)$ plus $\log(\vee/\Lambda)$.

Direct check on the terms. After renaming, every $\log(\vee/\Lambda)$ two-ρ term has one of the kernels

$$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,K(a,b,z)\qquad\text{or}\qquad\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,K(a,b,z),\tag{16.4}$$

so the measured gluon is always emitted by a charge (Sec. 20). Eq. (LOBFKL) contains $-2\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\,\mathrm{Tr}\{\dots\}$ and $\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')[2K(x,y,z)-K(x,x,z)]\mathrm{Tr}\{\dots\}$, in which the measured gluon is emitted by the new gluon at z. No $\log(\vee/\Lambda)$ term has such a kernel. Table 10 compares the two at test points: BFKL⊗LO is a different (and here much larger) function than JIMWLK⊗LO, which the notes' terms reproduce exactly (Sec. 20).

17 Group I, Row V: the $\log(\vee/\Lambda)$ part of part 2b

Part 2b of Row V (`Evolution_two_rho.tex`) has the p^+ integral still to be done:

$$\frac{d^3N_{2b}}{d^3k}=\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^4}\frac{1}{k^+}\int e^{-ik\cdot(w'-z)}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\int_{x,y,z,x'}\frac{1}{p^+(y-x)^2+(k^+-p^+)(y-z)^2}$$

$$\times \frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2}\Big[\cdots - \frac{p^+}{k^+-p^+}X^{im}_{\rm up} + \cdots - \frac{k^+-p^+}{p^+}X^{im}_{\rm low} + \ldots\Big]$$
$$\times \big[U^{ab'}(x') - U^{ab'}(w')\big]N_c\big[U^{af}(y) - U^{af}(x)\big]\rho^{b'}(x')\rho^f(y), \tag{17.1}$$

with

$$X^{im}_{\rm low} = \frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}, \qquad X^{im}_{\rm up} = \frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}. \tag{17.2}$$

The dots are the other four terms of the bracket. They have the factors $\frac{p^+(k^+-p^+)}{k^{+2}}$, $\frac{k^+-p^+}{k^+}$, $\frac{p^+}{k^+}$ and 1, are finite at both ends, and give no logarithm.

Lower end, $p^+ \to \Lambda$. The denominator becomes $k^+(y-z)^2$ and the phase becomes 1:

$$\int_\Lambda \frac{dp^+}{2\pi}\,\frac{-\frac{k^+}{p^+}X_{\rm low}}{k^+(y-z)^2} = -\frac{1}{2\pi}\frac{X_{\rm low}}{(y-z)^2}\log\frac{k^+}{\Lambda} + \ldots \tag{17.3}$$

Upper end, $p^+ \to k^+ - \Lambda$. The denominator becomes $k^+(y-x)^2$ and the phase becomes $e^{-ik\cdot(z-x)}$. With $e^{-ik\cdot(w'-z)}$ this gives $e^{-ik\cdot(w'-x)}$:

$$\int^{k^+-\Lambda} \frac{dp^+}{2\pi}\,\frac{-\frac{k^+}{k^+-p^+}X_{\rm up}}{k^+(y-x)^2} = -\frac{1}{2\pi}\frac{X_{\rm up}}{(y-x)^2}\log\frac{k^+}{\Lambda} + \ldots \tag{17.4}$$

Both ends are soft-gluon limits: either the gluon at x ($p^+ \to 0$) or the one at z ($k^+ - p^+ \to 0$) is soft. With $\frac{g^4}{8\pi^4}\cdot\frac{1}{2\pi} = \frac{g^4}{16\pi^5}$ and $\log(k^+/\Lambda) = \log(\vee/\Lambda) - \log(\vee/k^+)$, the $\log(\vee/\Lambda)$ part of 2b is

$$\left.\frac{d^3N_{2b}}{d^3k}\right|_{\log(\vee/\Lambda)} = -\mathcal{N}\int\Big[e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i(z-x)^mX^{im}_{\rm low}}{(x'-w')^2(z-x)^2(y-z)^2} + e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i(z-x)^mX^{im}_{\rm up}}{(x'-w')^2(z-x)^2(y-x)^2}\Big]$$
$$\times \big[U^{ab'}(x') - U^{ab'}(w')\big]N_c\big[U^{af}(y) - U^{af}(x)\big]\rho^{b'}(x')\rho^f(y). \tag{17.5}$$

These are exactly the two kernels of part 2a. Adding part 2a, whose colour factor is $-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(x)$, the $N_cU^{af}(x)$ terms cancel:

$$\left.\frac{d^3N}{d^3k}\right|_{\rm I.V,\,2\rho} = -\mathcal{N}\int\Big\{e^{-ik\cdot(w'-z)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\Big[\mathcal{K}(z-x)\cdot\mathcal{K}(x-z) + \tfrac{1}{2}\mathcal{K}(z-x)\cdot\mathcal{K}(y-x)\Big]$$
$$+ e^{-ik\cdot(w'-x)}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-x)\Big[\mathcal{K}(z-x)\cdot\mathcal{K}(x-z) - \tfrac{1}{2}\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\Big]\Big\}$$
$$\times \big[U^{ab'}(x') - U^{ab'}(w')\big]\Big[-f^{adc}f^{fbe}U^{db}(x)U^{ce}(z) + N_cU^{af}(y)\Big]\rho^{b'}(x')\rho^f(y) \;+\; \text{c.c.} \tag{17.6}$$

In the first line the measured gluon is at z and the soft gluon at x; in the second it is the other way round. We rename accordingly ($z \to w$, $x \to z$ and $x \to w$) and write $\mathcal{K}(z-x)\cdot\mathcal{K}(x-z) = -1/(x-z)^2$.

18 All two-ρ terms

18.1 Genuine two-ρ terms (`Evolution_two_rho.tex`)

In units of $\mathcal{N}$ (the factors are $\frac{g^4N_c}{16\pi^4} = \pi N_c\cdot\frac{g^4}{16\pi^5}$ and $\frac{g^4}{8\pi^5} = 2\cdot\frac{g^4}{16\pi^5}$, $\frac{g^4}{8\pi^4} = 2\pi\cdot\frac{g^4}{16\pi^5}$):

Group I, Row II (with c.c.):

$$-\pi N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,B(x,w)\,\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\big[U^{ab}(w) - U^{ab}(x)\big]\rho^b(x) + \text{c.c.},$$
$$B(x,w) = -\tfrac{2}{\epsilon} - 2\gamma - \log\tfrac{(x-w)^2\mu^2}{4} \quad \text{(as written)}. \tag{18.1}$$

Group I, Row V (with c.c.): Eq. (17.6).

Group II, Row I (no c.c.). The kernel of the notes simplifies: $\big[(x-z)^2 - \frac{(x-w)^2(x'-z)^2}{(x'-w')^2}\big]/\big[(x'-w')^2(x-z)^2(x-w)^2 - (x-w)^4(x'-z)^2\big] = \frac{1}{(x-w)^2(x'-w')^2}$. Then

$$2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\Big[K(w,w',z) - \tfrac{1}{2}K(x',w,z) - \tfrac{1}{2}K(x,w',z) + \tfrac{1}{4}K(x,x',z)\Big]$$
$$\times\Big[f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^d(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{bc}(z)U^{de}(x)\rho^{d'}(x')\rho^e(x)$$
$$- f^{ac'd'}f^{cbd}U^{ac}(w)U^{bc'}(z)U^{e'd'}(x')\rho^{e'}(x')\rho^d(x) + N_cU^{e'd}(x')U^{de}(x)\rho^{e'}(x')\rho^e(x)\Big]. \tag{18.2}$$

Group II, Row IV (no c.c.; $y \to w$, $y' \to w'$):

$$2\pi N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,B'(w,w')\,\big[U^{e'c}(w')-U^{e'c}(x')\big]\big[U^{ec}(w)-U^{ec}(x)\big]\rho^{e'}(x')\rho^{e}(x),$$
$$B'(w,w')=\tfrac{1}{\bar\epsilon}-\log\big(4\pi^2\mu^2(w-w')^2\big). \tag{18.3}$$

Group III, Row III, part 2 (with c.c.; $y' \to w'$):

$$\begin{aligned}&2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[-K(w,w',z)+\tfrac12 K(x,w',z)\Big]\\&\times\Big[f^{c'd'a}U^{e'c'}(w')U^{d'b}(z)-f^{c'd'a}U^{bd'}(z)U^{c'e'}(x')\Big]\rho^{e'}(x')\\&\times\Big[f^{cbd}U^{ac}(w)\rho^{d}(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^{e}(x)\Big]+\text{c.c.}\end{aligned} \tag{18.4}$$

The two integrated brackets. B and B' are already integrated over the soft-gluon position in dimensional regularization ($d=2-2\epsilon$, $\frac{1}{\bar\epsilon}=\frac{1}{\epsilon}-\gamma+\log 4\pi$). Using $2X\cdot Y=X^2+Y^2-(X-Y)^2$ with $X=a-z$, $Y=b-z$, and dropping the scaleless integrals,

$$\begin{aligned}\int d^{2-2\epsilon}z\,\mu^{2\epsilon}\,K(a,b,z)&=-\frac{(a-b)^2}{2}\int\frac{d^{2-2\epsilon}z\,\mu^{2\epsilon}}{(a-z)^2(b-z)^2}=\pi\Big[\frac{1}{\epsilon}-\gamma-\log\big(\pi\mu^2(a-b)^2\big)\Big]\\&=\pi\Big[\frac{1}{\bar\epsilon}-\log\big(4\pi^2\mu^2(a-b)^2\big)\Big],\qquad\int d^{2-2\epsilon}z\,\mu^{2\epsilon}\,K(a,a,z)=0.\end{aligned} \tag{18.5}$$

So $B'(w,w')=\frac{1}{\pi}\int_z K(w,w',z)$ exactly, as written. The same formula gives the required $B(x,w)=\frac{1}{\pi}\int_z K(x,w,z)=\frac{1}{\bar\epsilon}-\log(4\pi^2\mu^2(x-w)^2)$ (fix F6). The notes' B has the same $(x-w)^2$ dependence but a different pole and constant. For the pointwise comparison of Sec. 20 we use the equivalent integrands

$$\begin{aligned}B(x,w)&\to\tfrac{1}{\pi}\big[K(x,w,z)-K(w,w,z)\big],\\B'(w,w')&\to\tfrac{1}{\pi}\big[K(w,w',z)-\tfrac12K(w,w,z)-\tfrac12K(w',w',z)\big]=-\tfrac{1}{2\pi}\big[K(w,w,z)-2K(w,w',z)+K(w',w',z)\big],\end{aligned} \tag{18.6}$$

which have the same integrals by Eq. (18.5). (The second is the dipole kernel; both are free of infrared divergences.)

18.2 From reordering the three-ρ terms

These are listed term by term in Apps. B–F. They were checked here once more, independently, by an exact (Weyl) decomposition of every ordered three-ρ product: both agree in all kernel structures. Part II changed two Group I rows, and their two-ρ terms change accordingly:

- Row III: in all two-ρ terms of Row III, $K_{\text{I.III}}\to K^{(1)}$ (part 2 removed), and the two-ρ terms of $\mathcal{D}$ (below) are added.
- Row II: in the two-ρ terms of sub-terms 1 and 3 (the $U(w)\mathbb{A}^{(3)}$ term), $\mathfrak{F}^i\to\mathfrak{F}^i_\beta$. Using the commutator form instead gives identical two-ρ terms (Sec. 21).

The two-ρ terms of $\mathcal{D}$

The three-ρ term.

$$\begin{aligned}\frac{d^3N}{d^3k}\Big|_{\text{D}}=&-2i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathcal{D}}(x',x,y)\,\Big\{\\&+f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(w)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\&-f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\&-f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(w)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\&+f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\Big\}+\text{c.c.}\end{aligned} \tag{18.7}$$

with $K_{\mathcal{D}}(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(w)\, \rho^{b'}(x')\, \rho^{e}(x)\, \rho^{c}(y).$

$$
\begin{aligned}
[A,B]:\ & [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
& \tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x',y)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x')\, U^{bc}(w)\, f^{b'eh}\, \{\rho^{h}(x'),\rho^{c}(y)\} \\
& \qquad = -\tfrac{1}{4}\, iN_c\; K_{\mathcal{D}}(x',x',y)\, U^{ab}(w)\, U^{ac}(x')\, \{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U2), (F)} \qquad (18.8) \\
[A,C]:\ & [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
& \tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(w)\, f^{b'ch}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
& \qquad = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{cl}(x)\, U^{be}(w)\, \{\rho^{h}(x'),\rho^{l}(x)\} \qquad (18.9) \\
[B,C]:\ & [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
& \tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(w)\, f^{ech}\, \{\rho^{h}(x),\rho^{b'}(x')\} \\
& \qquad = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{cd}(x)\, U^{be}(w)\, \{\rho^{h}(x),\rho^{l}(x')\} \qquad (18.10)
\end{aligned}
$$

Sub-term 2: $- f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(y)\, \rho^{b'}(x')\, \rho^{e}(x)\, \rho^{c}(y).$

$$
\begin{aligned}
[A,B]:\ & [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
& \tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x',y)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x')\, U^{bc}(y)\, f^{b'eh}\, \{\rho^{h}(x'),\rho^{c}(y)\} \\
& \qquad = \tfrac{1}{4}\, iN_c\; K_{\mathcal{D}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U2), (F)} \qquad (18.11) \\
[A,C]:\ & [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
& \tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(x')\, f^{b'ch}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
& \qquad = -\tfrac{1}{4}\, iN_c\; K_{\mathcal{D}}(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \{\rho^{c}(x'),\rho^{b}(x)\} \quad \text{(U2), (F)} \qquad (18.12) \\
[B,C]:\ & [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
& \tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(x)\, f^{ech}\, \{\rho^{h}(x),\rho^{b'}(x')\} \\
& \qquad = \tfrac{1}{4}\, iN_c\; K_{\mathcal{D}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \{\rho^{c}(x),\rho^{b}(x')\} \quad \text{(U2), (F)} \qquad (18.13)
\end{aligned}
$$

Sub-term 3: $- f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(w)\, \rho^{b'}(x')\, \rho^{e}(x)\, \rho^{c}(y).$

$$
\begin{aligned}
[A,B]:\ & [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
& \tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x',y)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x')\, U^{bc}(w)\, f^{b'eh}\, \{\rho^{h}(x'),\rho^{c}(y)\} \\
& \qquad = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(w)\, \{\rho^{h}(x'),\rho^{l}(y)\} \qquad (18.14) \\
[A,C]:\ & [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
& \tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(w)\, f^{b'ch}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
& \qquad = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(w)\, \{\rho^{h}(x'),\rho^{l}(x)\} \qquad (18.15) \\
[B,C]:\ & [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
& \tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(w)\, f^{ech}\, \{\rho^{h}(x),\rho^{b'}(x')\} \\
& \qquad = -\tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{cd}(x)\, U^{be}(w)\, \{\rho^{h}(x),\rho^{l}(x')\} \qquad (18.16)
\end{aligned}
$$

Sub-term 4: $+ f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(y)\, \rho^{b'}(x')\, \rho^{e}(x)\, \rho^{c}(y).$

$$
\begin{aligned}
[A,B]:\ & [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
& \tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x',y)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x')\, U^{bc}(y)\, f^{b'eh}\, \{\rho^{h}(x'),\rho^{c}(y)\} \\
& \qquad = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(y)\, \{\rho^{h}(x'),\rho^{l}(y)\} \qquad (18.17) \\
[A,C]:\ & [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
& \tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(x')\, f^{b'ch}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
& \qquad = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(x')\, \{\rho^{h}(x'),\rho^{l}(x)\} \qquad (18.18) \\
[B,C]:\ & [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
& \tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K_{\mathcal{D}}(x',x,x)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(x)\, f^{ech}\, \{\rho^{h}(x),\rho^{b'}(x')\} \\
& \qquad = -\tfrac{1}{4}\, iN_c\; K_{\mathcal{D}}(x',x,x)\, U^{ab}(w')\, U^{ac}(x)\, \{\rho^{c}(x),\rho^{b}(x')\} \quad \text{(U2), (F)} \qquad (18.19)
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\mathrm{D}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{ & \\ & -\tfrac{1}{2}\,N_c\,K_{\mathcal{D}}(x',x',y)\,U^{ab}(w)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\} \\ & +\tfrac{1}{2}\,K_{\mathcal{D}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w)\,\{\rho^h(x'),\rho^l(x)\} \\ & +\tfrac{1}{2}\,K_{\mathcal{D}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(x)\,U^{be}(w)\,\{\rho^h(x),\rho^l(x')\} \\ & +\tfrac{1}{2}\,N_c\,K_{\mathcal{D}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\} \\ & -\tfrac{1}{2}\,N_c\,K_{\mathcal{D}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(x)\} \\ & +\tfrac{1}{2}\,N_c\,K_{\mathcal{D}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^c(x),\rho^b(x')\} \\ & -\tfrac{1}{2}\,K_{\mathcal{D}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(w)\,\{\rho^h(x'),\rho^l(y)\} \\ & -\tfrac{1}{2}\,K_{\mathcal{D}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(x)\,U^{be}(w)\,\{\rho^h(x'),\rho^l(x)\} \\ & -\tfrac{1}{2}\,K_{\mathcal{D}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(x)\,U^{be}(w)\,\{\rho^h(x),\rho^l(x')\} \\ & +\tfrac{1}{2}\,K_{\mathcal{D}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y)\,\{\rho^h(x'),\rho^l(y)\} \\ & +\tfrac{1}{2}\,K_{\mathcal{D}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(x)\,U^{be}(x')\,\{\rho^h(x'),\rho^l(x)\} \\ & -\tfrac{1}{2}\,N_c\,K_{\mathcal{D}}(x',x,x)\,U^{ab}(w')\,U^{ac}(x)\,\{\rho^c(x),\rho^b(x')\}\Big\} + (w'\leftrightarrow w)\end{aligned} \tag{18.20}$$

The kernel at the coincident points:

$$\begin{aligned}K_{\mathcal{D}}(x',x',y) &= \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\\ K_{\mathcal{D}}(x',x,x') &= \mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\\ K_{\mathcal{D}}(x',x,x) &= \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\end{aligned}$$

18.3 From reordering the four-ρ terms

The two-ρ terms of `4rho.tex` in the present notation (kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)$ is the LO kernel of Eq. (4rho) with $u\to y$; $K(a,b,z)$ as above; units of $\mathcal{N}$):

term	coeff.	kernel	colour
first part, 1	$+2\,N_c$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',x',z)$	$U^{ca}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^b(y)$
first part, 2	$-2\,N_c$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',x',z)$	$U^{ca}(x')\,U^{cb}(y)\,\rho^a(x')\,\rho^b(y)$
first part, 3	$+2\,N_c$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$U^{ca}(y)\,U^{cb}(w')\,\rho^b(x')\,\rho^a(y)$
first part, 4	$-2\,N_c$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$U^{ca}(y)\,U^{cb}(x')\,\rho^b(x')\,\rho^a(y)$
first part, 5	$+\frac{3}{2}\,N_c$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$U^{bc}(x')\,U^{ac}(y)\,\rho^b(x')\,\rho^a(y)$
first part, 6 (drop)	$-8\,N_c$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$	$U^{bc}(x')\,U^{ca}(y)\,\rho^b(x')\,\rho^a(y)$
$dN^{(1)}$, 1	-1	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$f^{abc}\,f^{def}\,U^{eb}(w)\,U^{b'f}(x')\,U^{da}(y)\,\rho^{b'}(x')\,\rho^c(y)$
$dN^{(1)}$, 2 (as printed)	$+3$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{deb'}\,U^{ab'}(y)\,U^{db'}(w')\,U^{eb}(z)\,\rho^{b'}(x')\,\rho^c(y)$
$dN^{(1)}$, 2 (corrected)	$+3$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{deg}\,U^{ag}(y)\,U^{db'}(w')\,U^{eb}(z)\,\rho^{b'}(x')\,\rho^c(y)$
$dN^{(1)}$, 3	$+\frac{1}{2}$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$f^{abc}\,f^{ade}\,U^{db'}(w')\,U^{b'b}(w)\,\rho^e(x')\,\rho^c(y)$
$dN^{(1)}$, 4	-1	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$f^{abc}\,f^{ade}\,U^{b'b}(w)\,U^{b'd}(x')\,\rho^e(x')\,\rho^c(y)$
$dN^{(1)}$, 5	$+2$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$f^{abc}\,f^{ade}\,U^{b'b}(y)\,U^{b'd}(x')\,\rho^e(x')\,\rho^c(y)$
$dN^{(1)}$, 6	$+1$	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(x',y,z)$	$f^{abc}\,f^{dae}\,U^{b'b}(w')\,U^{b'd}(x')\,\rho^e(x')\,\rho^c(y)$
$dN^{(1)}$, 7	-3	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{af}(z)\,U^{bd'}(w)\,U^{df}(y)\,U^{b'd'}(w')\,\rho^{b'}(x')\,\rho^e(y)$
$dN^{(1)}$, 8	$+2$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{af}(z)\,U^{bd'}(y)\,U^{df}(y)\,U^{b'd'}(w')\,\rho^{b'}(x')\,\rho^e(y)$
$dN^{(1)}$, 9	-3	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{af}(w)\,U^{bd'}(y)\,U^{dd'}(z)\,U^{b'f}(w')\,\rho^{b'}(x')\,\rho^e(y)$
$dN^{(1)}$, 10	$+3$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{ad'}(z)\,U^{df}(w)\,U^{fb'}(w')\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$
$dN^{(1)}$, 11	$+1$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fd}(w)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^e(y)$
$dN^{(1)}$, 12	$+\frac{1}{2}$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$f^{abc}\,f^{d'fl}\,U^{ad'}(z)\,U^{fb}(w)\,U^{b'l}(x')\,\rho^{b'}(x')\,\rho^c(y)$
$dN^{(1)}$, 13	$+\frac{1}{2}$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x',y,z)$	$f^{abc}\,f^{d'fl}\,U^{ad'}(z)\,U^{bf}(w')\,U^{b'l}(y)\,\rho^c(x')\,\rho^{b'}(y)$

term	coeff.	kernel	colour
$dN^{(1)}$, 14	-1	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(x',y,z)$	$f^{abc}\,f^{d'fl}\,U^{fb}(y)\,U^{b'l}(x')\,U^{d'a}(z)\,\rho^{b'}(x')\,\rho^{c}(y)$
$dN^{(1)}$, 15	-1	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fb}(y)\,U^{fb'}(w)\,U^{d'a}(z)\,U^{d'd}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(1)}$, 16	$+1$	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fb}(y)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'd}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(1)}$, 17	$+2$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{ad'}(z)\,U^{bf}(y)\,U^{dd'}(y)\,U^{b'f}(x')\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(1)}$, 18	$+1$	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(w)\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(1)}$, 19	-2	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(w')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(1)}$, 20	-2	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(1)}$, 21	-1	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{fd}(y)\,U^{fb'}(x')\,U^{d'a}(z)\,U^{d'b}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(2)}$, 1	$+1$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{d'a}(w)\,U^{d'b'}(x')\,U^{fb}(z)\,U^{fd}(y)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(2)}$, 2	$+1$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{cde}\,U^{d'b}(w)\,U^{d'b'}(x')\,U^{fa}(y)\,U^{fd}(z)\,\rho^{b'}(x')\,\rho^{e}(y)$
$dN^{(2)}$, 3	-1	$\mathcal{K}(y-w')\cdot\mathcal{K}(x'-w)\,K(x',y,z)$	$f^{fd'l}\,f^{deb'}\,U^{df}(z)\,U^{cl}(y)\,U^{d'e}(x')\,\rho^{b'}(x')\,\rho^{c}(y)$
$dN^{(2)}$, 4	$+1$	$\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(y,y,z)$	$f^{abc}\,f^{deb'}\,U^{ad}(z)\,U^{bb'}(y)\,U^{ef}(x')\,\rho^{f}(x')\,\rho^{c}(y)$

Check. The complete two-ρ content of Eq. (4rho) (with its c.c.) was computed by exact Weyl decomposition of every four-ρ product. It equals [the lists above] + [the two-ρ terms of A1, A2, B1, B2 from App. B] exactly, in all kernel structures, once the two marked changes are made. As printed, the $-8N_c$ term has nothing to cancel, and term 2 of $dN^{(1)}$ cannot be evaluated (the index b' occurs four times). Only the assignment shown in blue reproduces the exact result.

19 The target: JIMWLK⊗LO

Operator form. With $\delta Y=\log(\vee/\Lambda)$ and $\mathcal{N}=\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\cdot\frac{\alpha_s}{2\pi^2}\delta Y$ (LO prefactor times $\frac{\alpha_s}{2\pi^2}\delta Y$),

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{JIMWLK}\otimes\text{LO}} &= -\mathcal{N}\int e^{-ik\cdot(w'-w)}\int_{x,x'}\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\\ &\quad\times\tilde{H}\Big\{\rho^{b'}(x')\big[U(x')-U(w')\big]^{ab'}\big[U(x)-U(w)\big]^{ab}\rho^{b}(x)\Big\},\\ \tilde{H}&=\int_{u,v,z}K(u,v,z)\Big\{J_R^h(u)J_R^h(v)+J_L^m(u)J_L^m(v)-U^{mh}(z)\big[J_R^h(u)J_L^m(v)+J_L^m(u)J_R^h(v)\big]\Big\},\end{aligned}\tag{19.1}$$

with $J_R^h(u)\,U(p)=\delta^{(2)}(u-p)\,U(p)T^h$, $J_L^m(u)=U^{mg}(u)J_R^g(u)$, $(T^h)_{bc}=-if^{hbc}$. This is your H_{JIMWLK} of Eq. (HLO), and $-\delta Y\,H_{\text{JIMWLK}}$ is "$d/dY=-H$".

Corrected Appendix F. With $P\equiv\frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$, $M(u,v)\equiv(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))$ and $\sigma(x)=\sigma(y)=+1$, $\sigma(w)=\sigma(w')=-1$:

$$\frac{dN^{(1)}}{d^2wd^2w'dy}=P\sum_{v\in\{x,w\}}\sum_{u\in\{y,w'\}}\sigma(u)\sigma(v)\,K(u,v,t)\,\mathrm{Tr}\big\{T^eU^\dagger(v)U(u)T^e\,M(u,v)\big\},$$

$$\frac{dN^{(2)}}{d^2wd^2w'dy}=P\sum_{u\in\{x,w\}}\sum_{v\in\{y,w'\}}\sigma(u)\sigma(v)\,K(u,v,t)\,\mathrm{Tr}\big\{T^eU^\dagger(u)U(v)T^e\,M(u,v)^T\big\},$$

$$\frac{dN^{(3)}}{d^2wd^2w'dy}=-P\sum_{v\in\{y,w'\}}\sigma(v)\,K(v,v,t)\,\mathrm{Tr}\big\{T^e(U^\dagger(x)-U^\dagger(w))U(v)T^e\,M(v,v)\big\},$$

$$\frac{dN^{(4)}}{d^2wd^2w'dy}=-P\sum_{v\in\{x,w\}}\sigma(v)\,K(v,v,t)\,\mathrm{Tr}\big\{T^eU^\dagger(v)(U(y)-U(w'))T^e\,M(v,v)\big\},$$

$$\begin{aligned}\frac{dN^{(5)}}{d^2wd^2w'dy}=+P\Big[&\sum_{u\in\{y,w'\}}\sigma(u)K(u,u,t)\,\mathrm{Tr}\{(U^\dagger(u)-U^\dagger(t))U(u)T^d\}\,\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(u)T^d\}\\ &-\sum_{u\in\{x,w\}}\sigma(u)K(u,u,t)\,\mathrm{Tr}\{(U^\dagger(u)-U^\dagger(t))U(u)T^d\}\,\mathrm{Tr}\{U^\dagger(u)(U(y)-U(w'))T^d\}\Big].\end{aligned}\tag{19.2}$$

The two corrections (red) are the ones reported in `BFKL_JIMWLK.pdf`, and the pasted Appendix F still has both. (i) In $dN^{(1)}$–$dN^{(4)}$ the matrix M must be taken at the points u, v where the derivatives act; the notes always write $(U^\dagger(x) - U^\dagger(t))(U(y) - U(t))$. (ii) The overall sign of $dN^{(5)}$ is $+$. Table 10 checks this numerically. At test points, with $\langle \rho^a \rho^b \rangle \propto \delta^{ab}$, the corrected sum equals the operator form above to all printed digits. The sum as written does not.

Kernel by kernel. Carrying out the derivatives and simplifying (with $U^{bc} = U^{cb}$) gives JIMWLK⊗LO as a sum over kernel structures, each times a combination of independent colour structures:

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{JIMWLK}\otimes\text{LO}} = \mathcal{N} \int e^{-ik\cdot(w'-w)} \int_{x,x',z} \Big\{ \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x,x,z) \Big[\\
&\quad + 2N_c\, U^{ab}(w')\, U^{bc}(x)\, \rho^a(x')\, \rho^c(x) \\
&\quad - 2N_c\, U^{ab}(x')\, U^{ac}(x)\, \rho^b(x')\, \rho^c(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{ce}(x)\, U^{bd}(z)\, \rho^l(x')\, \rho^h(x) \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{ce}(x)\, U^{bd}(z)\, \rho^l(x')\, \rho^h(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x,x',z) \Big[\\
&\quad + 2N_c\, U^{ab}(x)\, U^{bc}(x')\, \rho^c(x')\, \rho^a(x) \\
&\quad + 2\, f^{abc}\, f^{ade}\, U^{hb}(x)\, U^{hd}(x')\, \rho^e(x')\, \rho^c(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{ae}(x')\, U^{cl}(x)\, U^{bd}(z)\, \rho^l(x)\, \rho^h(x') \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{ad}(x)\, U^{cl}(x')\, U^{be}(z)\, \rho^l(x')\, \rho^h(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x',x',z) \Big[\\
&\quad + 2N_c\, U^{ab}(w)\, U^{bc}(x')\, \rho^a(x)\, \rho^c(x') \\
&\quad - 2N_c\, U^{ab}(x)\, U^{ac}(x')\, \rho^b(x)\, \rho^c(x') \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{al}(x)\, U^{cd}(x')\, U^{be}(z)\, \rho^h(x')\, \rho^l(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{al}(w)\, U^{cd}(x')\, U^{be}(z)\, \rho^h(x')\, \rho^l(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x',w,z) \Big[\\
&\quad - 2N_c\, U^{ab}(w)\, U^{bc}(x')\, \rho^c(x')\, \rho^a(x) \\
&\quad - 2\, f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x') \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{ea}(w)\, U^{db}(z)\, U^{hl}(x')\, \rho^l(x')\, \rho^c(x) \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{ae}(x')\, U^{bd}(z)\, U^{cl}(w)\, \rho^l(x)\, \rho^h(x') \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x,w',z) \Big[\\
&\quad - 2N_c\, U^{ab}(w')\, U^{bc}(x)\, \rho^a(x')\, \rho^c(x) \\
&\quad + 2\, f^{abc}\, f^{dce}\, U^{hd}(x)\, U^{ha}(w')\, \rho^b(x')\, \rho^e(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{da}(w')\, U^{eb}(z)\, U^{hl}(x)\, \rho^c(x')\, \rho^l(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{ad}(x)\, U^{be}(z)\, U^{cl}(w')\, \rho^l(x')\, \rho^h(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(w,w,z) \Big[\\
&\quad + 2N_c\, U^{ab}(x')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\quad - 2N_c\, U^{ab}(w')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{be}(z)\, U^{ch}(w)\, \rho^l(x')\, \rho^d(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{be}(z)\, U^{ch}(w)\, \rho^l(x')\, \rho^d(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(w',w',z) \Big[\\
&\quad + 2N_c\, U^{ab}(x)\, U^{ac}(w')\, \rho^c(x')\, \rho^b(x) \\
&\quad - 2N_c\, U^{ab}(w)\, U^{ac}(w')\, \rho^c(x')\, \rho^b(x) \\
&\quad - 2\, f^{abc}\, f^{deh}\, U^{al}(x)\, U^{be}(z)\, U^{ch}(w')\, \rho^d(x')\, \rho^l(x)
\end{aligned}
$$

$$
\begin{aligned}
&\quad + 2\, f^{abc}\, f^{deh}\, U^{al}(w)\, U^{be}(z)\, U^{ch}(w')\, \rho^{d}(x')\, \rho^{l}(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(w,w',z)\, \Big[\\
&\quad + 2N_c\, U^{ab}(w')\, U^{cb}(w)\, \rho^{a}(x')\, \rho^{c}(x) \\
&\quad + 2\, f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x')\, \rho^{e}(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{la}(w')\, U^{be}(z)\, U^{cd}(w)\, \rho^{l}(x')\, \rho^{h}(x) \\
&\quad + 2\, f^{abc}\, f^{deh}\, U^{la}(w)\, U^{be}(z)\, U^{cd}(w')\, \rho^{h}(x')\, \rho^{l}(x) \Big] \Big\}
\end{aligned}
\tag{19.3}
$$

Only 8 of the 13 kernel structures of Sec. 20 appear in JIMWLK⊗LO. The notes' terms in the other structures must cancel among themselves.

20 The comparison

Method. Every two-ρ term is renamed as in Sec. 2. After renaming its kernel is one of 13 structures $\mathcal{K}(\cdot - w)\cdot\mathcal{K}(\cdot - w')\, K(a,b,z)$. Within one kernel structure, the colour factors are simplified (Wilson lines at equal points, $U^{bc} = U^{cb}$, f contractions). Several of the resulting colour structures are linearly dependent through Jacobi-type identities. We therefore express all of them in a basis of independent colour structures. The coefficients are fixed exactly, as polynomials in N_c, from evaluations with random symmetric SU(3) and SU(4) Wilson lines and random charges; the identities used are listed in App. H. The comparison is pointwise in z: no integration over the soft-gluon position is needed.

As written, and fix by fix. Table 8 shows how many of the 38 kernel structures (including the 6 edge-kernel structures of Row III, part 2) differ from JIMWLK⊗LO, as the fixes are applied one after the other.

state	structures that differ	structures
as written	32	38
+ `4rho.tex`: drop the $-8N_c$ term, fix the index of term 2	30	38
+ Row III part 2 $\to \mathcal{D}$	10	38
+ Row II: $\mathfrak{F}_\alpha$ removed from $U(w)\mathbb{A}^{(3)}$	6	38
+ Group III Row III part 2: sign	0	38

Table 8: Two-ρ total minus JIMWLK⊗LO, pointwise. Fix F6 (the constant in B) does not show up here: it concerns only the pole and constant after the z integration (Sec. 18.1).

Row by row. The rows can also be compared one by one with the leading-log computation of the earlier notes. There, each row was built directly from the wave function in the region $p^+ \ll k^+$; that computation is independent of the steps of the notes. Its sum is JIMWLK⊗LO. So everything outside Group I is correct except the sign of Group III,

row	max |notes − reference|, as written	after the fixes	size	
Group II, Row I	3.6×10^{-14}	3.6×10^{-14}	13.7	agree
Group II, Row II	2.0×10^{-15}	2.0×10^{-15}	5.7	agree
Group II, Row III	4.4×10^{-16}	4.4×10^{-16}	3.9	agree
Group II, Row IV	9.6	9.6	19.2	agree after $\int_z$; the sum agrees pointwise
Group III, Row I	1.7×10^{-13}	1.7×10^{-13}	11.5	agree
Group III, Row II	9.9×10^{-14}	9.9×10^{-14}	11.5	agree
Group III, Row III	4.5×10^{1}	5.3×10^{-15}	22.3	agree
Group III, Row IV	3.9×10^{-14}	3.9×10^{-14}	11.8	agree
Group III, Row V	3.6×10^{-15}	3.6×10^{-15}	12.9	agree
Group III, Row VI	1.8×10^{-15}	1.8×10^{-15}	11.1	agree
Group I, all rows	1.3×10^{1}	9.6	19.5	agree after $\int_z$; the sum agrees pointwise

Table 9: Two-ρ content of each row (genuine terms plus those from reordering), compared pointwise with the reference row at a random configuration. Group II, Row IV and Group I differ pointwise only by terms $\propto K(w,w,z)$, $K(w',w',z)$ without $U(z)$. Such terms move between the two rows; their integral vanishes, and the sum of Group I and Group II Row IV agrees pointwise.

Row III, part 2 (F5). The other fixes are in Group I and in the four-ρ lists.

The result, structure by structure. After all fixes: for every kernel structure and every independent colour structure B, the coefficients contributed by the rows of the notes add up to the coefficient in JIMWLK⊗LO (last column). Coefficients are in units of $\mathcal{N}$, multiplying [kernel]$\times B$. Row labels: I.II = Group I, Row II etc.; 4ρ = the lists of `4rho.tex`; A,B = the two-ρ terms of the remainders A1–B2; $\mathcal{D}$ = the new term of Row III. In total there are 154 independent colour structures, and all of them match.

Structure 1: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$	I.III: $\frac{N_c}{2}$; I.IV: $-\frac{N_c}{2}$; III.I: $\frac{N_c}{2}$; 4ρ: $2N_c$; A,B: $-\frac{N_c}{2}$	$2N_c$	$2N_c$ ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.III: $-\frac{N_c}{2}$; I.IV: $\frac{N_c}{2}$; II.I: $\frac{N_c}{2}$; III.I: $-\frac{N_c}{2}$; III.II: $-\frac{N_c}{2}$; 4ρ: $-2N_c$; A,B: $\frac{N_c}{2}$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: $\frac{1}{2}$; III.II: $\frac{1}{2}$; 4ρ: 1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$; 4ρ: -1	-2	-2 ✓
$U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$	II.I: $\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$	0	0 ✓
$U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	III.II: $-\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$	I.IV: $\frac{1}{2}$; II.I: $\frac{1}{2}$; III.I: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$	I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓

Structure 2: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(x)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)$	I.II: $-2N_c$; I.III: N_c; I.IV: N_c; II.I: $\frac{N_c}{2}$; III.I: $-N_c$; III.II: N_c; 4ρ: $\frac{3N_c}{2}$	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)$	4ρ: 2	2	2 ✓
$f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	I.III: $\frac{1}{2}$; III.II: $\frac{1}{2}$; 4ρ: 1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$; 4ρ: -1	-2	-2 ✓
$f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$	II.I: $\frac{1}{2}$; 4ρ: $\frac{1}{2}$; A,B: -1	0	0 ✓
$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)$	III.II: $\frac{1}{2}$; 4ρ: -1; A,B: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	III.II: $-\frac{1}{2}$; 4ρ: 1; A,B: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$	I.II: 1; I.III: $-\frac{1}{2}$; I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; A,B: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$	I.II: 1; I.III: $-\frac{1}{2}$; I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; 4ρ: -1; A,B: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	I.III: $-\frac{1}{2}$; I.IV: $\frac{1}{2}$; II.I: $\frac{1}{2}$; III.I: $-\frac{1}{2}$; III.II: $\frac{1}{2}$; 4ρ: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: $\frac{1}{2}$; I.IV: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; III.II: $-\frac{1}{2}$; 4ρ: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$	I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$	I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓

Structure 3: $\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\;K(x,y,z)$

JIMWLK⊗LO has no term with this kernel: the terms of the notes must cancel among themselves.

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(y)\,U^{ac}(x)\,\rho^b(y)\,\rho^c(x)$	I.II: $2N_c$; I.III: $-N_c$; I.IV: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: N_c; A,B: $-N_c$	0	0 ✓
$U^{ab}(z)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(y)$	I.III: N_c; III.II: $-N_c$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(y)\,\rho^e(x)$	II.I: $\frac{1}{2}$; A,B: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hd}(x)\,\rho^a(y)\,\rho^e(x)$	III.II: $-\frac{1}{2}$; A,B: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{hd}(x)\,\rho^a(y)\,\rho^e(x)$	III.II: $\frac{1}{2}$; A,B: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{hd}(w)\,\rho^a(y)\,\rho^e(x)$	II.I: $-\frac{1}{2}$; A,B: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x)\,U^{al}(y)\,\rho^l(y)\,\rho^h(x)$	I.II: 1; I.III: $-\frac{1}{2}$; I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; A,B: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ec}(x)\,U^{db}(w)\,U^{al}(y)\,\rho^h(x)\,\rho^l(y)$	I.II: -1; I.III: $\frac{1}{2}$; I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$; A,B: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x)\,U^{bl}(z)\,\rho^h(x)\,\rho^l(y)$	I.III: $-\frac{1}{2}$; II.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(y)\,\rho^h(x)$	I.III: $\frac{1}{2}$; II.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(y)\,\rho^m(y)\,\rho^e(x)$	I.IV: $\frac{1}{2}$; II.I: $\frac{1}{2}$; III.I: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(y)$ $\rho^m(y)\,\rho^e(x)$	I.IV: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(y)$ $\rho^m(y)\,\rho^e(x)$	I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(y)$ $\rho^m(y)\,\rho^e(x)$	I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓

Structure 4: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$	I.III: $\frac{N_c}{2}$; I.IV: $-\frac{N_c}{2}$; III.I: $\frac{N_c}{2}$; 4ρ: $2N_c$; A,B: $-\frac{N_c}{2}$	$2N_c$	$2N_c$ ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$	I.III: $-\frac{N_c}{2}$; I.IV: $\frac{N_c}{2}$; II.I: $\frac{N_c}{2}$; III.I: $-\frac{N_c}{2}$; III.II: $-\frac{N_c}{2}$; 4ρ: $-2N_c$; A,B: $\frac{N_c}{2}$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	I.III: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$; 4ρ: -1	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')$ $\rho^l(x)$	I.III: $\frac{1}{2}$; II.I: $\frac{1}{2}$; 4ρ: 1	2	2 ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	III.II: $-\frac{N_c}{2}$; A,B: $\frac{N_c}{2}$	0	0 ✓
$U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$	II.I: $\frac{N_c}{2}$; A,B: $-\frac{N_c}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')$ $\rho^e(x')\,\rho^l(x)$	I.IV: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')$ $\rho^e(x')\,\rho^l(x)$	I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓

Structure 5: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$

JIMWLK⊗LO has no term with this kernel: the terms of the notes must cancel among themselves.

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	I.II: $-N_c$; III.II: N_c	0	0 ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.II: N_c; I.IV: $-N_c$; I.V: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: N_c; III.III: N_c; III.V: $-N_c$; III.VI: $-N_c$; $\mathcal{D}$: N_c	0	0 ✓
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	I.II: N_c; II.I: $-N_c$	0	0 ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.II: $-N_c$; I.IV: N_c; I.V: N_c; III.I: $-N_c$; III.V: N_c; $\mathcal{D}$: $-N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')$ $\rho^h(x)$	I.V: 1; III.VI: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(w)\,\rho^l(x')$ $\rho^h(x)$	I.V: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')$ $\rho^h(x)$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^l(x')\,\rho^h(x)$	III.II: 1; III.VI: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')$ $\rho^h(x)$	III.III: -1; III.V: 1; III.VI: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^l(x')$ $\rho^h(x)$	III.V: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$	I.IV: -1; II.I: -1; III.I: 1; III.II: 1	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$	I.IV: 1; III.I: -1	0	0 ✓

Structure 6: $\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\;K(y,w,z)$

JIMWLK⊗LO has no term with this kernel: the terms of the notes must cancel among themselves.

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(y)\,U^{ac}(x)\,\rho^c(x)\,\rho^b(y)$	I.II: $-N_c$; I.IV: N_c; I.V: N_c; II.I: N_c; III.I: $-N_c$; III.II: $-N_c$; III.III: $-N_c$; III.V: N_c; III.VI: N_c; $\mathcal{D}$: $-N_c$	0	0 ✓
$U^{ab}(z)\,U^{bc}(x)\,\rho^a(y)\,\rho^c(x)$	III.II: N_c; III.VI: $-N_c$	0	0 ✓
$f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w)\,\rho^e(x)\,\rho^c(y)$	I.II: -1; III.II: 1	0	0 ✓

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$f^{abc}\, f^{bde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^e(x)\, \rho^c(y)$	I.II: 1; I.I: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bd}(x)\, U^{cl}(y)\, \rho^h(x)\, \rho^l(y)$	I.II: -1; I.IV: 1; I.V: 1; III.I: -1; III.V: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(x)\, U^{bl}(z)\, U^{cd}(w)\, \rho^h(x)\, \rho^l(y)$	I.V: -1; III.VI: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bl}(z)\, U^{cd}(w)\, \rho^h(x)\, \rho^l(y)$	I.V: 1; III.III: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{lb}(z)\, U^{cd}(x)\, \rho^l(y)\, \rho^h(x)$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{da}(w)\, U^{ce}(x)\, U^{bl}(y)\, \rho^l(y)\, \rho^h(x)$	III.III: -1; III.V: 1; III.VI: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{da}(w)\, U^{ce}(w')\, U^{bl}(y)\, \rho^l(y)\, \rho^h(x)$	III.V: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\, f^{cde}\, U^{hd}(x)\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)$ $\rho^m(y)\, \rho^e(x)$	I.IV: -1; II.I: -1; III.I: 1; III.II: 1	0	0 ✓
$f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)$ $\rho^m(y)\, \rho^e(x)$	I.IV: 1; III.I: -1	0	0 ✓

Structure 7: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\; K(x',w,z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(w)\, U^{bc}(x')\, \rho^c(x')\, \rho^a(x)$	III.III: $-N_c$; III.V: N_c; III.VI: $-N_c$; $\mathcal{D}$: $-N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x')$	I.II: -1; III.II: -1	-2	-2 ✓
$f^{abc}\, f^{deh}\, U^{ea}(w)\, U^{db}(z)\, U^{hl}(x')\, \rho^l(x')$ $\rho^c(x)$	I.IV: -1; II.I: -1; III.I: 1; III.II: -1	-2	-2 ✓
$f^{abc}\, f^{deh}\, U^{ae}(x')\, U^{bd}(z)\, U^{cl}(w)\, \rho^l(x)$ $\rho^h(x')$	I.V: -1; III.VI: -1	-2	-2 ✓
$U^{ab}(x)\, U^{ac}(x')\, \rho^b(x)\, \rho^c(x')$	I.II: N_c; I.IV: $-N_c$; I.V: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: $-N_c$; III.III: N_c; III.V: $-N_c$; III.VI: N_c; $\mathcal{D}$: N_c	0	0 ✓
$f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x')$	I.II: 1; II.I: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bl}(x)\, U^{cd}(x')\, \rho^l(x)$ $\rho^h(x')$	I.II: -1; I.IV: 1; I.V: 1; III.I: -1; III.V: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{bd}(z)\, U^{cl}(w)\, \rho^l(x)$ $\rho^h(x')$	I.V: 1; III.III: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{db}(z)\, U^{cl}(x)\, \rho^h(x')$ $\rho^l(x)$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{eb}(z)\, U^{cl}(x)\, \rho^h(x')\, \rho^l(x)$	III.II: 1; III.VI: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{la}(w)\, U^{ce}(w')\, U^{bd}(x')\, \rho^h(x')$ $\rho^l(x)$	III.V: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{le}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{md}(x')$ $\rho^h(x')\, \rho^c(x)$	I.IV: 1; III.I: -1	0	0 ✓

Structure 8: $\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\; K(y,w',z)$

JIMWLK⊗LO has no term with this kernel: the terms of the notes must cancel among themselves.

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(y)\, U^{ac}(x)\, \rho^b(y)\, \rho^c(x)$	I.II: $-N_c$; I.IV: N_c; I.V: N_c; II.I: N_c; III.I: $-N_c$; III.II: $-N_c$; III.III: $-N_c$; III.V: N_c; III.VI: N_c; $\mathcal{D}$: $-N_c$	0	0 ✓
$U^{ab}(z)\, U^{bc}(x)\, \rho^c(x)\, \rho^a(y)$	III.II: N_c; III.VI: $-N_c$	0	0 ✓
$f^{abc}\, f^{dbe}\, U^{hd}(x)\, U^{ha}(w')\, \rho^c(y)\, \rho^e(x)$	I.II: 1; III.II: -1	0	0 ✓
$f^{abc}\, f^{dbe}\, U^{hd}(w)\, U^{ha}(w')\, \rho^c(y)\, \rho^e(x)$	I.II: -1; II.I: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{be}(x)\, U^{cl}(y)\, \rho^l(y)\, \rho^h(x)$	I.II: 1; I.IV: -1; I.V: -1; III.I: 1; III.V: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ad}(x)\, U^{bl}(z)\, U^{ce}(w')\, \rho^l(y)\, \rho^h(x)$	I.V: 1; III.VI: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{bl}(z)\, U^{ce}(w')\, \rho^l(y)\, \rho^h(x)$	I.V: -1; III.III: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{lb}(z)\, U^{ce}(x)\, U^{ad}(w)\, \rho^h(x)\, \rho^l(y)$	II.I: 1; III.III: -1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ea}(w')\, U^{cd}(x)\, U^{bl}(y)\, \rho^h(x)\, \rho^l(y)$	III.III: 1; III.V: -1; III.VI: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\, f^{deh}\, U^{ea}(w')\, U^{cd}(w)\, U^{bl}(y)\, \rho^l(y)\, \rho^h(x)$	III.V: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\, f^{dce}\, U^{hd}(x)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(y)$ $\rho^m(y)\, \rho^e(x)$	I.IV: 1; II.I: 1; III.I: -1; III.II: -1	0	0 ✓
$f^{abc}\, f^{dce}\, U^{hd}(w)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(y)$ $\rho^m(y)\, \rho^e(x)$	I.IV: -1; III.I: 1	0	0 ✓

Structure 9: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$

JIMWLK⊗LO has no term with this kernel: the terms of the notes must cancel among themselves.

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$	I.II: $-N_c$; III.II: N_c	0	0 ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.II: N_c; I.IV: $-N_c$; I.V: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: N_c; III.III: N_c; III.V: $-N_c$; III.VI: $-N_c$; $\mathcal{D}$: N_c	0	0 ✓
$U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$	I.II: N_c; II.I: $-N_c$	0	0 ✓
$U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.II: $-N_c$; I.IV: N_c; I.V: N_c; III.I: $-N_c$; III.V: N_c; $\mathcal{D}$: $-N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$	I.V: -1; III.VI: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)$	I.V: 1; III.III: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^l(x)\,\rho^h(x')$	II.I: 1; III.III: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^l(x)\,\rho^h(x')$	III.II: -1; III.VI: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x')$	III.III: 1; III.V: -1; III.VI: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^h(x')\,\rho^l(x)$	III.V: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$	I.IV: 1; II.I: 1; III.I: -1; III.II: -1	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$	I.IV: -1; III.I: 1	0	0 ✓

Structure 10: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$	III.III: $-N_c$; III.V: N_c; III.VI: $-N_c$; $\mathcal{D}$: $-N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$	I.II: 1; III.II: 1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	I.IV: 1; II.I: 1; III.I: -1; III.II: 1	2	2 ✓
$f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$	I.V: 1; III.VI: 1	2	2 ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.II: N_c; I.IV: $-N_c$; I.V: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: $-N_c$; III.III: N_c; III.V: $-N_c$; III.VI: N_c; $\mathcal{D}$: N_c	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)$	I.II: -1; II.I: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)$	I.II: 1; I.IV: -1; I.V: -1; III.I: 1; III.V: -1; $\mathcal{D}$: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)$	I.V: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{db}(z)\,U^{cl}(x')\,U^{ae}(w)\,\rho^l(x')\,\rho^h(x)$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^l(x')\,\rho^h(x)$	III.II: -1; III.VI: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{la}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^l(x')\,\rho^h(x)$	III.V: 1; $\mathcal{D}$: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$	I.IV: -1; III.I: 1	0	0 ✓

Structure 11: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	I.II: N_c; II.IV: N_c	$2N_c$	$2N_c$ ✓
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	I.II: $-N_c$; II.IV: $-N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)$	I.V: -2	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)$	I.V: 2	2	2 ✓

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.II: $-N_c$; I.V: $2N_c$; II.IV: $-N_c$	0	0 ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.II: N_c; I.V: $-2N_c$; II.IV: N_c	0	0 ✓

Structure 12: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$	I.II: N_c; II.IV: N_c	$2N_c$	$2N_c$ ✓
$U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)$	I.II: $-N_c$; II.IV: $-N_c$	$-2N_c$	$-2N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')$ $\rho^l(x)$	I.V: -2	-2	-2 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')$ $\rho^l(x)$	I.V: 2	2	2 ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.II: $-N_c$; I.V: $2N_c$; II.IV: $-N_c$	0	0 ✓
$U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.II: N_c; I.V: $-2N_c$; II.IV: N_c	0	0 ✓

Structure 13: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$

colour structure B	contributions of the notes (row: coefficient)	sum	JIMWLK
$U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$	II.IV: $2N_c$	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	II.I: 2	2	2 ✓
$f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')$ $\rho^h(x)$	III.III: 2	2	2 ✓
$f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')$ $\rho^l(x)$	III.III: 2	2	2 ✓
$U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$	II.I: $2N_c$; II.IV: $2N_c$; III.III: $-4N_c$	0	0 ✓
$U^{ab}(w')\,U^{cb}(x)\,\rho^a(x')\,\rho^c(x)$	II.IV: $-2N_c$; III.III: $2N_c$	0	0 ✓
$U^{ab}(x')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)$	II.IV: $-2N_c$; III.III: $2N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')$ $\rho^l(x)$	II.I: -2; III.III: 2	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')$ $\rho^h(x)$	II.I: -2; III.III: 2	0	0 ✓

21 Numerical checks

Variants. The full sum after all fixes, minus JIMWLK⊗LO, pointwise at random points:

variant	total minus JIMWLK⊗LO
correction of Row II as $\mathfrak{F}\to\mathfrak{F}_\beta$ in $U(w)\mathbb{A}^{(3)}$	0 of 40 structures differ, largest difference 5.1×10^{-14}
correction of Row II in commutator form (Eq. (11.3))	0 of 40 structures differ, largest difference 7.2×10^{-14}
$\mathcal{D}$ with charge order $\rho(x')\rho(x)\rho(y)$	0 of 40 structures differ, largest difference 5.1×10^{-14}
$\mathcal{D}$ with charge order $\rho(x')\rho(y)\rho(x)$	4 of 40 structures differ, largest difference 1.9×10^{1}
$\mathcal{D}$ with charge order $\rho(x)\rho(x')\rho(y)$	4 of 40 structures differ, largest difference 1.9×10^{1}
$\mathcal{D}$ with charge order $\rho(x)\rho(y)\rho(x')$	8 of 40 structures differ, largest difference 1.9×10^{1}
$\mathcal{D}$ with charge order $\rho(y)\rho(x')\rho(x)$	8 of 40 structures differ, largest difference 1.9×10^{1}
$\mathcal{D}$ with charge order $\rho(y)\rho(x)\rho(x')$	12 of 40 structures differ, largest difference 1.9×10^{1}
other random configuration (seed 21)	0 of 40 structures differ, largest difference 8.4×10^{-14}
other random configuration (seed 33)	0 of 40 structures differ, largest difference 6.6×10^{-14}
general (non-symmetric) adjoint U	28 of 40 structures differ, largest difference 1.1×10^{1}

The commutator form of the Row II correction gives the same two-ρ terms as $\mathfrak{F}\to\mathfrak{F}_\beta$ in $U(w)\mathbb{A}^{(3)}$. This supersedes the remark at the end of correction 1 in Sec. 11, which expected the two forms to have different two-ρ terms. $\mathcal{D}$ must have the charge order of Row III. For a general (non-symmetric) adjoint U the notes do not apply, as in the three-ρ check: they use $U^{bc}=U^{cb}$ throughout.

Appendix F and BFKL at test points. In the colour-singlet projection $\langle\rho^a(p)\rho^b(q)\rangle=\delta^{ab}\mu(p,q)/(N_c^2-1)$, at random points (x,y,w,w',t), in units of $P\mu$:

U	seed	JIMWLK⊗LO (operator form)	App. F corrected	App. F as written	BFKL⊗LO (Eq. LOBFKL)
general	1	4.223482	4.223482	5.866983	95.2475
general	2	88.007301	88.007301	52.780006	91.6463
general	3	0.014910	0.014910	0.007840	0.1332
symmetric	1	−47.715089	−47.715089	−28.081101	79.3073
symmetric	2	−230.981876	−230.981876	−179.490327	−107.5655
symmetric	3	0.016879	0.016879	0.012758	0.0900

Table 10: The corrected Appendix F equals the operator form. As written it does not, and BFKL⊗LO is a different function (same units, $-N_c\times$ the bracket of Eq. (LOBFKL)).

Part IV
The coefficient of $\log\vee$: BFKL evolution of the projectile

22 Overview

What is compared. $\vee$ appears only as the upper limit of the p^+ integrals. The coefficient of $\log\vee$, i.e. the coefficient of $\log(\vee/\Lambda)$ plus that of $\log(\vee/k^+)$, therefore comes from unobserved gluons much harder than the measured one ($p^+ \gg k^+$). Such a gluon is part of the projectile wave function, so the coefficient of $\log\vee$ must be BFKL⊗LO. For each row we call it $P_{\rm row} = A_{\rm row} + B_{\rm row}$. Here A is the coefficient of $\log(\vee/\Lambda)$ (Parts II, III) and B that of $\log(\vee/k^+)$.

Result.

- As written, the coefficient of $\log\vee$ is not BFKL⊗LO: 15 of the 19 two-ρ structures differ, and 18 of the 18 symmetrized three-ρ structures are left over (pointwise count).
- With the fixes of Table 1 (P1, F4, F5, and F1, F2 for the four-ρ lists), the symmetrized three-ρ terms cancel exactly (Sec. 26). The two-ρ terms equal BFKL⊗LO exactly, pointwise in z, in all 9 kernel structures, with exact coefficients (Sec. 27).
- Group I, Row II contributes $P_{\rm I.II} = -\mathfrak{F}_\alpha$ (in the $U(w)\mathbb{A}^{(3)}$ sub-terms). This is the p^+-ordered virtual factor of Sec. 5: the gluon exchanged between the two valence charges is a virtual correction of the projectile.
- In the colour-singlet projection, BFKL⊗LO equals Eq. (LOBFKL) of your Appendix E (Sec. 28).

The rows are listed with A, P and B (Sec. 24), because $B = P - A$ is what your notes write as $\log(\vee/k^+)$ terms. The step-by-step reordering of the region-P three-ρ terms is in App. G.

23 Which terms multiply $\log(\vee/k^+)$

The rule. Every longitudinal function L in the rows is a function of Λ, k^+ and $\vee$ (and of transverse distances). With your rules it is written uniquely as

$$L = A_L \log\frac{\vee}{\Lambda} + B_L \log\frac{\vee}{k^+} + (\text{no large logarithm}), \tag{23.1}$$

where "no large logarithm" means a function that stays finite for $\Lambda \to 0$, $\vee \to \infty$ (e.g. $\log(1-k^+/\vee)$, $\log(1-\Lambda/k^+)$, $\log\frac{(x-z)^2}{(x-w)^2}$). Table 11 lists every function that appears in the rows.

Table 11: The longitudinal functions of the rows. $a = (x-w)^2$, $b = (x-z)^2$ (primed for x', w'). A: coefficient of $\log(\vee/\Lambda)$. B: coefficient of $\log(\vee/k^+)$. $P = A + B$: coefficient of $\log\vee$.

function L	A	B	remark
$\log\frac{\vee}{\Lambda}$	1	0	
$\log\frac{\vee-k^+}{\Lambda} = \log\frac{\vee}{\Lambda} + \log(1-\frac{k^+}{\vee})$	1	0	your second rule
$\log\frac{\Lambda}{k^+}$	−1	1	your first rule
$\log\frac{\vee}{k^+}$, $\log(\frac{\vee}{k^+}-1)$	0	1	
$\log\frac{\vee}{\Lambda+k^+}$, $\log\frac{\vee}{k^+-\Lambda}$	0	1	
$\log\frac{k^+-\Lambda}{\vee}$, $\log\frac{k^+}{\vee}$	0	−1	

function L	A	B	remark
$\log\frac{\vee}{\Lambda}+\log\frac{k^+-\Lambda}{\vee}=\log\frac{k^+-\Lambda}{\Lambda}$	1	-1	Group I Rows III (3a), V; Group II Row IV
$L_P(a,b)\equiv\log\frac{(\vee-k^+)b+k^+a}{\Lambda b+k^+a}$	0	1	$\to\log\frac{\vee}{k^+}+\log\frac{b}{a}$
$L_o(a,b)\equiv\log\left\lvert\frac{\vee b-k^+a}{(k^++\Lambda)b-k^+a}\right\rvert$	0	1	$\to\log\frac{\vee}{k^+}+\log\left\lvert\frac{b}{b-a}\right\rvert$
$L_{o'}(a,b)\equiv\log\left\lvert\frac{k^+a-\vee b}{k^+a-\Lambda b}\right\rvert$	0	1	$\to\log\frac{\vee}{k^+}+\log\frac{b}{a}$
$\log\left\lvert\frac{(k^+-\Lambda)b-k^+a}{\Lambda b-k^+a}\right\rvert$, $\mathcal{A}$ (arctan),	0	0	finite
$\log\frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2}$			
$\log\frac{k^+p^2}{\vee k^2+k^+p^2}$ (momentum space)	0	-1	Group I Row II
$\log\left[1+(\frac{\vee}{k^+}-1)\frac{k^2}{(k-p)^2}\right]$ (momentum space)	0	1	Group I Row II
$\int_0^{\vee-k^+}dp^+\,[\frac{1}{p^+}]_+\,e^{-i\beta p^+}\,G(p^+)$	0	$-G(0)$	hidden; Eq. (23.5)
$\int^{\vee-k^+}dp^+\,\frac{1}{p^+}\,G$ in the variable w'	0	$G(\infty)$	hidden; Sec. 24.3.2

$B=P-A$. In every row, $\vee$ appears only as the upper limit of the p^+ integral, $\int^{\vee-k^+}dp^+$ or $\int^{\vee}dp^+$. Hence the coefficient of $\log\vee$ is the large-p^+ limit of the integrand:

$$P_{\text{row}}\equiv A_{\text{row}}+B_{\text{row}}=\frac{\partial}{\partial\log\vee}\frac{d^3N_{\text{row}}}{d^3k}=\lim_{p^+\to\infty}p^+\times(\text{integrand}). \tag{23.2}$$

So the $\log(\vee/k^+)$ terms of a row are

$$\boxed{B_{\text{row}}=P_{\text{row}}-A_{\text{row}}} \tag{23.3}$$

For most rows B is read off directly from the notes with Table 11. For the rows left as plus-prescription integrals (Group II Row IV, Group III Rows III, V, VI) P is obtained from Eq. (23.2) before the p^+ integration.

Which evolution is $\sum P$. For $p^+\gg k^+$ the unobserved gluon is harder than the measured one. It is part of the projectile wave function: the measured gluon is emitted by the valence charges or by the gluon at z, and both scatter on the target. Integrating it out is one step of the evolution of the projectile, which is BFKL:

$$\sum_{\text{rows}}P_{\text{row}}=\text{BFKL}\otimes\text{LO},\qquad\sum_{\text{rows}}A_{\text{row}}=\text{JIMWLK}\otimes\text{LO},\qquad\sum_{\text{rows}}B_{\text{row}}=\text{BFKL}\otimes\text{LO}-\text{JIMWLK}\otimes\text{LO}. \tag{23.4}$$

The second equation is the result of Part III. The first is checked here row by row against an independent leading-log computation of the region $p^+\gg k^+$ (Sec. 29). Both hold after the fixes.

Remark on A. A_{row} is not always the $p^+\to0$ limit of the integrand. A pole at $p^+=k^+$, cut off at $k^+\pm\Lambda$, also gives $\log(1/\Lambda)$. These "edge" logarithms appear in Group I Row III (the Φ terms). Eq. (23.3) holds regardless, because P only involves the upper limit.

The plus prescription hides a $\log(\vee/k^+)$. For the rows written with $[1/p^+]_+$ and a phase,

$$\int_0^P dp^+\,\frac{e^{-i\beta p^+}-1}{p^+}=-\log(|\beta|P)-\gamma_E-\tfrac{i\pi}{2}\operatorname{sgn}\beta+O\Big(\frac{1}{\beta P}\Big), \tag{23.5}$$

with $P=\vee-k^+$ and $\beta\propto1/k^+$. This contains $-\log(\vee/k^+)$, so $\int_0^{\vee-k^+}dp^+\,[\frac{1}{p^+}]_+(\dots)$ contains $-\log(\vee/k^+)\times$(the residue at $p^+=0$), i.e. $-A$.

24 The $\log(\vee/k^+)$ terms, row by row

For each row: the $\log(\vee/\Lambda)$ terms A (as in your `Evolution` files), the coefficient of $\log\vee$, P, and the $\log(\vee/k^+)$ terms $B=P-A$. Notation used in several rows (Group I Row IV, Group II Row I, Group III Rows I, II), with $X=x'-w'$ and $Y=y-z$:

$$K_{\mathfrak{B}}=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac12K(x,y,z)\Big]\qquad\text{(your }\log(\vee/\Lambda)\text{ bracket)}, \tag{24.1}$$

$$K^P_{\mathfrak{B}}=\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,K(x,y,z)-\tfrac12\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,K(x,y,z)\qquad\text{(coefficient of }\log\vee), \tag{24.2}$$

$$\begin{aligned}K^L_{\mathfrak{B}}&=K^P_{\mathfrak{B}}-K_{\mathfrak{B}}\qquad\text{(coefficient of }\log(\vee/k^+))\\&=\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,K(x,y,z)-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(z-w)\end{aligned}$$

$$- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,K(x,y,z). \tag{24.3}$$

$K^L_{\mathfrak{B}} = \mathcal{K}^i(x'-w')\mathcal{K}^j(y-z)\big[J_1^{ij} - \frac{\delta^{ij}}{2(z-w)^2}\big]$, where J_1^{ij} is the bracket multiplying $\log\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{\Lambda(x-z)^2+k^+(x-w)^2}$ in your notes. In $K^P_{\mathfrak{B}}$ the measured gluon is emitted by the harder gluon at z, which is emitted by x (first term). In $K_{\mathfrak{B}}$ the softer gluon at z is emitted by the measured gluon (first term). $\bar{K}_{\mathfrak{B}}$, $\bar{K}^P_{\mathfrak{B}}$, $\bar{K}^L_{\mathfrak{B}}$ are the same with $w\leftrightarrow w'$ in the second factor and $(x'-w')\to(x'-w)$ (Group II Row I, second term).

24.1 Group I (with c.c.)

24.1.1 Row I

Only four-ρ terms, all $\propto\log(\vee/\Lambda)$ with $A=P$: $B=0$.

24.1.2 Row II

Part 1 (two-ρ). Your bracket is $\big[\frac{11}{3}+\log\frac{\vee}{k^+}-2\log\frac{\vee}{\Lambda}+\log(\frac{\vee}{k^+}-1)\big]$, and $\log\frac{\vee}{k^+}+\log(\frac{\vee}{k^+}-1)=2\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})$. The prefactor $\frac{g^4N_c}{32\pi^4k^+}$ is $\frac{\pi N_c}{2}$, so

$$A_{\mathrm{I.II},1} = -\pi N_c\mathcal{N}\int \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,B(x,w)\,\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\big[U^{ab}(w)-U^{ab}(x)\big]\rho^b(x)+\text{c.c.},$$
$$B_{\mathrm{I.II},1} = +\pi N_c\mathcal{N}_P\int \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,B(x,w)\,\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\big[U^{ab}(w)-U^{ab}(x)\big]\rho^b(x)+\text{c.c.}, \tag{24.4}$$

and $P_{\mathrm{I.II},1}=0$. As in Part III (fix F6), the bracket must be $B(x,w)=\frac{1}{\pi}\int_z[K(x,w,z)-K(w,w,z)]=\frac{1}{\bar\epsilon}-\log(4\pi^2\mu^2(x-w)^2)$ instead of $-\frac{2}{\epsilon}-2\gamma-\log\frac{(x-w)^2\mu^2}{4}$.

Part 2 (three-ρ, momentum space). In your bracket (with $\mathbf{k}$ the Fourier variable inside the bracket), Table 11 gives, term by term:

$\log\frac{\Lambda}{k^+}\,\frac{k^2p^i-k\cdot p\,k^i}{(k-p)^2}$	$\to\ A:-\frac{k^2p^i-k\cdot p\,k^i}{(k-p)^2},\quad B:+\frac{k^2p^i-k\cdot p\,k^i}{(k-p)^2},$
$-2k^i\log\frac{\vee}{\Lambda}$	$\to\ A:-2k^i,$
$-2k^i\log\frac{k^+p^2}{\vee k^2+k^+p^2}$	$\to\ B:+2k^i,$
$\frac{(2p^2-2k\cdot p+k^2)k^i+k^2p^i}{(k-p)^2}\log\big[1+(\frac{\vee}{k^+}-1)\frac{k^2}{(k-p)^2}\big]$	$\to\ B:+\frac{(2p^2-2k\cdot p+k^2)k^i+k^2p^i}{(k-p)^2},$
$-\frac{p^2k^i+k^2p^i}{(k-p)^2}\log\frac{\vee-k^+}{\Lambda}$	$\to\ A:-\frac{p^2k^i+k^2p^i}{(k-p)^2},$
$-k^i\log\frac{\vee}{k^+}$	$\to\ B:-k^i.$

So the $\log(\vee/k^+)$ bracket is

$$B\text{-bracket}=\frac{2k^2p^i+(3p^2-5k\cdot p+2k^2)k^i}{(k-p)^2},\qquad A\text{-bracket}=-\frac{2k^2p^i+(p^2-k\cdot p)k^i}{(k-p)^2}-2k^i, \tag{24.5}$$

and their sum (the coefficient of $\log\vee$) is $\frac{2(k-p)^2k^i}{(k-p)^2}-2k^i=0$. Hence $P_{\mathrm{I.II},2}=0$ and

$$B_{\mathrm{I.II},2}=-A_{\mathrm{I.II},2}=+\frac{i}{2}\mathcal{N}_P\int\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,y)\,\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')$$
$$\times\Big[U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\}-f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big]+\text{c.c.}, \tag{24.6}$$

with $\mathfrak{F}^i=\mathfrak{F}^i_\alpha+\mathfrak{F}^i_\beta$ the coordinate-space kernel of Part I, Eq. (4.10).

Fix F4 does not change B. F4 removes $\mathfrak{F}_\alpha$ from the $\log(\vee/\Lambda)$ terms of the $U(w)\mathbb{A}^{(3)}$ sub-terms ($A\to A-\alpha$) and leaves the $\log(\vee/k^+)$ terms as they are, so $P=0\to-\alpha$: with the p^+-ordered virtual factor, $\mathfrak{F}_\alpha$ multiplies $-\log(\vee/k^+)$ (Sec. 5, Eq. (5.12)). The region-P comparison (Sec. 29) confirms $P_{\mathrm{I.II}}=-\alpha$ independently. The $\log(\vee/k^+)$ terms of Row II are correct as written, provided the bracket contains $-\mathbf{k}^i\log(\vee/k^+)$ (Eq. (3.7)).

24.1.3 Row III

Prefactor $\frac{ig^4f^{abd}}{16\pi^5k^+}\to i\,f^{abd}$ and colour operator

$$\mathcal{O}_{\mathrm{III}}=\big[U^{ab'}(x')\rho^{b'}(x')-U^{ab'}(w')\rho^{b'}(x')\big]\big[U^{de}(x)U^{bc}(z)\rho^e(x)\rho^c(y)-U^{de}(x)U^{bc}(y)\rho^e(x)\rho^c(y)\big].$$

With $a=(x-w)^2$, $b=(x-z)^2$, $X=x'-w'$, $Y=y-z$, $Z_w=z-w$, the kernels of your parts are

$$K_a=\frac{X\cdot Y\,(a+b)}{X^2Y^2(b-a)^2}=-\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x),\qquad K_b=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,K(x,y,z),$$

$$K_c=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(z-w),\qquad K_h=\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,K(x,y,z),$$
$$K_d=\frac{X\cdot Y\,Z_w\cdot(x-z)\,a}{X^2Y^2Z_w^2(a-b)^2},\quad K_e=\frac{X\cdot Y\,Z_w\cdot(x-w)\,b}{X^2Y^2Z_w^2(a-b)^2},\quad K_f=\frac{X\cdot Y}{X^2Y^2Z_w^2},$$

and $g_1,\dots,g_4$ the four terms with $\frac{1}{a-b}$ in your part 2 (multiplying $L_o(a,b)$, in order: $+\frac{X\cdot Z_w\,Y\cdot(x-w)}{(a-b)}$, $-\frac{X\cdot(x-w)\,Y\cdot Z_w\,b}{(a-b)a}$, $-\frac{X\cdot Z_w\,Y\cdot(x-z)\,a}{(a-b)b}$, $+\frac{X\cdot(x-z)\,Y\cdot Z_w}{(a-b)}$, all over $X^2Y^2Z_w^2$). Your longitudinal functions give (units if^{abd}, Table 11):

part	your term	A	B
1	$-[\log\frac{\vee}{\Lambda}+\log(1-\frac{k^+}{\vee})]\,K_a$	$-K_a$	0
1	$+L_o(a,b)\,K_a$	0	$+K_a$
1	$-[\log\frac{\vee}{k^+-\Lambda}-\log\frac{\vee}{\Lambda}]\,K_a$	$+K_a$	$-K_a$
1	$+\log\frac{\vee}{\Lambda}\,K_b$	$+K_b$	0
1	$-2\,L_{o'}(a,b)\,K_b$	0	$-2K_b$
2	$-2\log\frac{\vee}{\Lambda+k^+}\,K_c$	0	$-2K_c$
2	$+2[\log\frac{\vee}{\Lambda}+\log(1-\frac{k^+}{\vee})]\,(K_d-K_e)$	$2(K_d-K_e)$	0
2	$+2L_o(a,b)\,[g_1+g_2+g_3+g_4+K_e-K_d]$	0	$2(g_1+g_2+g_3+g_4)+2(K_e-K_d)$
3a	$[\log\frac{\vee}{\Lambda}+\log\frac{k^+-\Lambda}{\vee}]\,(2K_d-2K_e-2K_f)$	$2(K_d-K_e-K_f)$	$-2(K_d-K_e-K_f)$
3b	$\mathcal{A}$, $\log\frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2}$ terms	0	0

Two identities simplify the sums: $4(K_d-K_e)-2K_f=-2K_a$ (use $Z_w=(x-w)-(x-z)$ and $Z_w^2=a+b-2(x-w)\cdot(x-z)$), and $-K_c+g_1+g_2+g_3+g_4=-K_h$. Then

$$A_{\rm I.III}=i\,(K_b-2K_a)\,f^{abd}\mathcal{O}_{\rm III}+\text{c.c.}\qquad\text{(your Evolution_three_rho, with } -2K_a=2\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\Phi(x)),$$
$$B_{\rm I.III}=i\,(-2K_b-2K_h+2K_a)\,f^{abd}\mathcal{O}_{\rm III}+\text{c.c.},$$
$$P_{\rm I.III}=A+B=i\,(-K_b-2K_h)\,f^{abd}\mathcal{O}_{\rm III}+\text{c.c.}\tag{24.7}$$

Direct check of P. Only two of your p^+ integrals reach $\vee$. The one with $\frac{[k^+(x-w)^2+p^+(x-z)^2]}{[p^+(x-z)^2-k^+(x-w)^2]}\frac{1}{p^+k^+}$ tends to $\frac{1}{p^+k^+}$ and gives $-K_b$. The one with $\frac{[k^+(x-w)^m-p^+(x-z)^m]}{[k^+(x-w)^2-p^+(x-z)^2]}\frac{1}{k^+-p^+}[\delta_{ik}\delta_{jm}-\dots]$ tends to $-\frac{1}{p^+}\frac{(x-z)^m}{(x-z)^2}\delta_{ik}\delta_{jm}$ and gives $-2K_h$. This is the same P, so your p^+ integration is correct (also checked by numerical integration).

Fix P1. The region-P comparison (Sec. 29) requires $P_{\rm I.III}=i(-K_b+2K_h)$. The term $-K_b$ is right, but the $\bar{\mathbb{B}}_3^\dagger$ term over $k^+ + \Lambda < p^+ < \vee$ has the wrong sign at large p^+. In the comparison, the structures with K_h match the independent computation exactly once the sign is reversed, and they are off by exactly a factor -1 otherwise.

Fix F3 carried over. In Part II the Φ terms of A ($-2iK_a$) were replaced by $\mathcal{D}$,

$$\mathcal{D}=-2i\int\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,K(x,w,z)\,f^{abd}\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\big[U^{de}(x)U^{bc}(w)-U^{de}(x)U^{bc}(y)\big]\rho^e(x)\rho^c(y)+\text{c.c.}$$

(the soft gluon exchanged between the measured gluon and the rotated charge x). Hence, corrected,

$$A_{\rm I.III}=iK_b\,f^{abd}\mathcal{O}_{\rm III}+\mathcal{D},\qquad P_{\rm I.III}=i(-K_b+2K_h)\,f^{abd}\mathcal{O}_{\rm III}\quad(+\text{c.c.}),$$
$$\boxed{B_{\rm I.III}=i(-2K_b+2K_h)\,f^{abd}\mathcal{O}_{\rm III}-\mathcal{D}\quad(+\text{c.c.})}\tag{24.8}$$

The terms $\pm K_a$ (the edge kernel Φ) are absent. In your notes they come from the poles at $p^+=k^+$ and multiply $\log\frac{k^+}{\Lambda}=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}$.

24.1.4 Row IV

Prefactor $\frac{ig^4}{8\pi^5k^+}\to 2i$. Colour operator

$$\mathcal{O}_{\rm IV}=\big[U^{ab'}(x')\rho^{b'}(x')-U^{ab'}(w')\rho^{b'}(x')\big]\big[f^{cdc'}U^{ac}(w)U^{bd}(z)U^{be}(y)\rho^e(y)\rho^{c'}(x)-f^{abc}U^{be}(y)U^{cd}(x)\rho^e(y)\rho^d(x)\big].$$

Your result has $L_P(a,b)\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}J_1^{ij}$ $(B=1)$, $-\frac{1}{2}\log\frac{\vee}{\Lambda+k^+}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ $(B=1)$ and the blue $[\log\frac{\vee}{\Lambda}+\log(1-\frac{k^+}{\vee})]K_{\mathfrak{B}}$ $(A=1)$. Therefore

$$A_{\rm I.IV}=2i\,K_{\mathfrak{B}}\,\mathcal{O}_{\rm IV}+\text{c.c.},\qquad P_{\rm I.IV}=2i\,K_{\mathfrak{B}}^P\,\mathcal{O}_{\rm IV}+\text{c.c.},$$
$$\boxed{B_{\rm I.IV}=2i\,K_{\mathfrak{B}}^L\,\mathcal{O}_{\rm IV}+\text{c.c.}}\tag{24.9}$$

P agrees with the large-p^+ limit of your integrand, $\frac{1}{k^+}\big[\frac{(x-z)^j(z-w)^i}{(z-w)^2}-\frac{(x-z)^j(x-w)^i}{2(x-w)^2}\big]\frac{1}{p^+(x-z)^2}$. Row IV is correct.

24.1.5 Row V

Here $0 < p^+ < k^+$, so $P_{\rm I.V} = 0$. All your logarithms are $\log\frac{\vee}{\Lambda} + \log\frac{k^+-\Lambda}{\vee} = \log\frac{k^+-\Lambda}{\Lambda}$ ($A = 1$, $B = -1$). Hence, for parts 1, 2a and 2b,

$$\boxed{B_{\rm I.V} = -A_{\rm I.V}} \tag{24.10}$$

with $A_{\rm I.V}$ the $\log(\vee/\Lambda)$ terms of your Evolution files:

$$\begin{aligned} B_{\rm I.V,1} = &+\frac{i}{2}\mathcal{N}_P\int\Big[e^{-ik\cdot(w'-z)}\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x) \\ &\quad + e^{-ik\cdot(w'-x)}\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\Big] \\ &\times\big[U^{ab'}(x') - U^{ab'}(w')\big]\rho^{b'}(x')\Big[f^{adc}U^{db}(x)U^{ce}(z)\{\rho^e(y),\rho^b(y')\} - f^{abc}U^{cd}(y)U^{be}(y')\{\rho^d(y),\rho^e(y')\}\Big] + \text{c.c.}, \end{aligned} \tag{24.11}$$

and $B_{\rm I.V,2}$ is minus Eq. (17.6) (parts 2a and 2b together, with $\mathcal{N} \to \mathcal{N}_P$).

24.2 Group II (no c.c.)

24.2.1 Row I

Part 1 (three-ρ). The two terms have the structure of Group I Row IV, with prefactors $-\frac{ig^4}{16\pi^5 k^+} \to -i$ and $+i$:

$$\begin{aligned} B_{\rm II.I,1} = &-i\mathcal{N}_P\int K^L_{\mathfrak{B}}\,\mathcal{O}^{(1)} + i\mathcal{N}_P\int \bar{K}^L_{\mathfrak{B}}\,\mathcal{O}^{(2)}, \\ \mathcal{O}^{(1)} = &\Big[U^{ac'}(w')\{\rho^{c'}(x'),\rho^b(y)\} - U^{bc'}(z)U^{ad'}(x')U^{c'e'}(y)\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big] \\ &\times\Big[f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\Big], \\ \mathcal{O}^{(2)} = &\Big[f^{c'bd'}U^{ac'}(w')\rho^{d'}(x) - f^{ac'd'}U^{bc'}(z)U^{d'e'}(x)\rho^{e'}(x)\Big] \\ &\times\Big[U^{ac}(w)\{\rho^c(x'),\rho^b(y)\} - U^{bc}(z)U^{ad}(x')U^{ce}(y)\{\rho^d(x'),\rho^e(y)\}\Big]. \end{aligned} \tag{24.12}$$

A and P are the same with $K^L \to K$ and $K^L \to K^P$.

Part 2 (two-ρ). Colour operator (your reduced form)

$$\begin{aligned} \mathcal{O}_{\rm II.I} = &f^{c'bd'}f^{cbd}U^{ac'}(w')U^{ac}(w)\rho^{d'}(x')\rho^d(x) - f^{c'bd'}f^{acd}U^{ac'}(w')U^{bc}(z)U^{de}(x)\rho^{d'}(x')\rho^e(x) \\ &- f^{ac'd'}f^{cbd}U^{ac}(w)U^{bc'}(z)U^{e'd'}(x')\rho^{e'}(x')\rho^d(x) + N_cU^{e'd}(x')U^{de}(x)\rho^{e'}(x')\rho^e(x). \end{aligned}$$

With a, b, a', b' as in Table 11 and $D = a'b - ab'$, the $\log(\vee/k^+)$ content of your six parts 2a–2f is:

- 2a: $L_P(a,b) - L_P(a',b')$, so $B = 1 - 1 = 0$;
- 2b: blue $\log\frac{\vee - k^+}{\Lambda}$ ($B = 0$; it gives A), and $-\frac{b}{aD}L_P(a,b) + \frac{b'}{a'D}L_P(a',b')$, so $B = \frac{-a'b+ab'}{aa'D} = -\frac{1}{aa'}$ times K_B;
- 2c: $\frac{a'}{b'D}L_P(a',b') - \frac{a}{bD}L_P(a,b)$, so $B = \frac{a'b-ab'}{bb'D} = \frac{1}{bb'}$ times K_C;
- 2d, 2e, 2f: the coefficients of $\log\frac{\vee}{\Lambda+k^+}$ and of the two L_P cancel, $B = 0$.

Here K_B and K_C are your brackets multiplying $\frac{(p^++k^+)^2}{(p^+)^2}$ and $\frac{(p^++k^+)^2}{(k^+)^2}$. They factorize:

$$\begin{aligned} \frac{K_B}{aa'} &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(w-z) - \tfrac{1}{2}\mathcal{K}(x-z)\Big]\cdot\Big[\mathcal{K}(w'-z) - \tfrac{1}{2}\mathcal{K}(x'-z)\Big], \\ \frac{K_C}{bb'} &= K(x,x',z)\Big[\mathcal{K}(z-w) - \tfrac{1}{2}\mathcal{K}(x-w)\Big]\cdot\Big[\mathcal{K}(z-w') - \tfrac{1}{2}\mathcal{K}(x'-w')\Big]. \end{aligned} \tag{24.13}$$

With the prefactor $\frac{g^4}{8\pi^5 k^+} \to 2$:

$$A_{\rm II.I,2} = 2\int\frac{K_B}{aa'}\,\mathcal{O}_{\rm II.I}, \qquad P_{\rm II.I,2} = 2\int\frac{K_C}{bb'}\,\mathcal{O}_{\rm II.I}, \qquad \boxed{B_{\rm II.I,2} = 2\int\Big[\frac{K_C}{bb'} - \frac{K_B}{aa'}\Big]\mathcal{O}_{\rm II.I}} \tag{24.14}$$

In A the softer gluon z is emitted by the measured gluon (or by x, with weight $-\frac{1}{2}$). In P the measured gluon is emitted by the harder gluon z (or by x, with weight $-\frac{1}{2}$). The two are related by exchanging the roles of z and w, w'.

24.2.2 Rows II and III

Only four-ρ terms $\propto \log(\vee/\Lambda)$: $B = 0$.

24.2.3 Row IV

Colour operator $\mathcal{O}_{\rm II.IV}(p',p) = \big[U^{e'c}(p') - U^{e'c}(x')\big]\big[U^{ec}(p) - U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)$. Your blue factor is $\log\frac{\vee}{\Lambda} + \log\frac{k^+}{\vee}$ ($A = 1$, $B = -1$). The ξ integral $\int_1^{k^+/\vee}\frac{d\xi}{\xi^2}$ hides a $+\log(\vee/k^+)$ at its lower end $\xi \to k^+/\vee$, i.e. $p^+ \to \vee$. It is easiest to read off before the w, w' and z integrations. For $p^+ \gg k^+$ the product of your two brackets is $(\frac{p^+ + k^+}{k^+})^2\delta_{mm'}\delta_{kk'}$, times $\frac{p^+k^+}{(p^+ + k^+)^4}$. This gives $\frac{1}{k^+p^+}$ with $y, y' \to z$. Hence ($\frac{g^4N_c}{8\pi^4k^+} \to 2\pi N_c$ and $\frac{g^4N_c}{8\pi^5k^+} \to 2N_c$)

$$
\begin{aligned}
A_{\rm II.IV} &= 2N_c\int_{x,x'}\Big[\int_z K(w,w',z)\Big]\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{O}_{\rm II.IV}(w',w),\\
P_{\rm II.IV} &= 2N_c\int_{x,x',z} K(w,w',z)\,K(x,x',z)\,\mathcal{O}_{\rm II.IV}(z,z),\\
B_{\rm II.IV} &= P_{\rm II.IV} - A_{\rm II.IV}.
\end{aligned}
\tag{24.15}
$$

Here $\int_z K(w,w',z) = \pi[\frac{1}{\bar\epsilon} - \log(4\pi^2\mu^2(w-w')^2)]$ is your bracket. In P both measured gluons are emitted by the harder gluon at z, which comes from x and x'.

24.3 Group III (with c.c.)

The overall sign of Rows III, V and VI is that of `Evolution_three_rho.tex` ($-\frac{4}{(2\pi)^3}$, $-\frac{2}{(2\pi)^3}$), which is opposite to `Group_III.tex`. This applies to the $\log(\vee/k^+)$ terms as well, and to part 2 of Row III (fix F5).

24.3.1 Rows I and II

Same structure as Group I Row IV, with prefactor $-\frac{ig^4}{8\pi^5k^+} \to -2i$:

$$
B_{\rm III.I(II)} = -2i\,\mathcal{N}_P\int K^L_{\mathfrak{B}}\,\mathcal{O}_{\rm III.I(II)} + \text{c.c.},
\tag{24.16}
$$

with

$$
\begin{aligned}
\mathcal{O}_{\rm III.I} &= \big[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y) - U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)\big]\\
&\quad\times\big[f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\big],\\
\mathcal{O}_{\rm III.II} &= \big[U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\big]\\
&\quad\times\big[f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\big].
\end{aligned}
$$

24.3.2 Row III

Part 1 (three-ρ). Your result is the plus-prescription integral $\mathcal{I}_1$ plus the blue term $[\log\frac{\vee}{\Lambda} + \log(1 - \frac{k^+}{\vee})]$. The blue term has $A = 1$, $B = 0$, and gives

$$
A_{\rm III.III,1} = -i\,\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K(y',x,z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\;f^{c'd'a}\mathcal{O}_4\mathcal{O}_5 + \text{c.c.}
$$

with

$$
\begin{aligned}
\mathcal{O}_4 &= U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x'),\\
\mathcal{O}_5 &= U^{ac}(w)\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)U^{ad}(y)U^{ce}(x)\{\rho^d(y),\rho^e(x)\}.
\end{aligned}
$$

The integral $\mathcal{I}_1 = \int_0^{\vee-k^+}\frac{dp^+}{2\pi}(\dots)$ contains two hidden $\log(\vee/k^+)$:

1. the $[\frac{1}{p^+}]_+$ term with its phase gives $-\log(\vee/k^+)\times A_{\rm III.III,1}$ (Eq. (23.5));
2. the term $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ behaves as $\frac{1}{k^+p^+}$ in the variable w', where $y' = z + \xi(w'-z)$, $\xi = \frac{k^+}{k^+ + p^+}$, $d^2y' = \xi^2 d^2w'$, and the phase becomes $e^{-ik\cdot(w'-w)}$. This gives $+\log(\vee/k^+)\times P_{\rm III.III,1}$ with

$$
P_{\rm III.III,1} = -i\,\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K(x',x,z)\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\;f^{c'd'a}\,\mathcal{O}_4\big|_{y'\to z}\,\mathcal{O}_5 + \text{c.c.}
$$

So $B_{\rm III.III,1} = P_{\rm III.III,1} - A_{\rm III.III,1}$. In P the measured gluon w' is emitted by the harder gluon z, which is exchanged between x' and x.

Part 2 (two-ρ). In the same way, from the blue term and the plus-prescription integral (with the sign of fix F5; prefactor $\frac{g^4}{8\pi^5 k^+} \to 2$):

$$A_{\mathrm{III.III,2}} = -2\mathcal{N}\int e^{-ik\cdot(y'-w)}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-y')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y'-z)+\tfrac{1}{2}K(x,y',z)\Big]\mathcal{O}_6\,\mathcal{O}_7 + \text{c.c.},$$
$$P_{\mathrm{III.III,2}} = -2\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K(x',x,z)\,\mathcal{K}(w'-z)\cdot\Big[\mathcal{K}(z-w)-\tfrac{1}{2}\mathcal{K}(x-w)\Big]\mathcal{O}_6\big|_{y'\to z}\,\mathcal{O}_7 + \text{c.c.}, \tag{24.17}$$

with $\mathcal{O}_6 = f^{c'd'a}[U^{e'c'}(y')U^{d'b}(z) - U^{bd'}(z)U^{c'e'}(x')]\rho^{e'}(x')$ and $\mathcal{O}_7 = f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)$. P comes from the term $-\frac{p^+}{k^{+2}}[\frac{(x-z)^{k'}(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}(x-w)^m}{2(x-w)^2}]$ of your bracket, which in the variable w' behaves as $\frac{1}{k^+p^+(x-z)^2}[\dots]$. $B_{\mathrm{III.III,2}} = P - A$.

24.3.3 Row IV

Only four-ρ terms $\propto \log(\vee/\Lambda)$: $B = 0$.

24.3.4 Rows V and VI

Both have the longitudinal structure of Row III part 1, with twice its prefactor:

$$A_{\mathrm{III.V(VI)}} = -2i\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K(y',x,z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\,\mathcal{C}_{\mathrm{V(VI)}} + \text{c.c.},$$
$$P_{\mathrm{III.V(VI)}} = -2i\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K(x',x,z)\,\mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,\mathcal{C}_{\mathrm{V(VI)}}\big|_{y'\to z} + \text{c.c.}, \tag{24.18}$$

with $\mathcal{C}_{\mathrm{V}} = f^{c'ca}[U^{e'c'}(y')\rho^{e'}(x') - U^{c'e'}(x')\rho^{e'}(x')][U^{ce}(x)U^{ad}(y)\rho^e(x)\rho^d(y) - U^{ad}(w)U^{ce}(x)\rho^e(x)\rho^d(y)]$ and $\mathcal{C}_{\mathrm{VI}} = f^{c'd'a}[U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')][U^{bc}(z)U^{ad}(y)U^{ce}(x)\rho^d(y)\rho^e(x) - U^{ad}(y)\rho^d(y)\rho^b(x)]$. $B = P - A$.

24.4 Summary of the rows

row	$\log(\vee/k^+)$ terms B	as written	status
I.I, II.II, II.III, III.IV, Eq. (4rho)	0	–	correct
I.II part 1	$-A$	explicit	F6 (bracket)
I.II part 2	$-A$	explicit	correct (F4 affects A only)
I.III	$P - A$	explicit	P1 ($+2K_h$), F3 ($\mathcal{D}$)
I.IV	$2iK^L_{\mathfrak{B}}$	explicit	correct
I.V	$-A$	explicit	correct
II.I parts 1, 2	$P - A$	explicit	correct
II.IV	$P - A$	$-A$ explicit, P hidden	correct
III.I, III.II	$-2iK^L_{\mathfrak{B}}$	explicit	correct
III.III part 1, III.V, III.VI	$P - A$	hidden	correct (sign of Evolution_three_rho)
III.III part 2	$P - A$	hidden	F5 (sign)

25 Four-ρ terms

All four-ρ terms of the notes (Eq. (4rho): Group I Row I, Group II Rows II and III, Group III Row IV and the four-ρ parts of the other rows) multiply $\log(\vee/\Lambda)$, with $A = P$. Their B is zero. The same holds for their three-ρ and two-ρ remainders (A1–B2 and the lists of `4rho.tex`). Physically, in these terms both gluons are emitted independently by the valence charges, and the p^+ integrand is $\propto 1/p^+$ over the whole range $\Lambda < p^+ < \vee$. As a check, the symmetrized four-ρ part of Eq. (4rho) equals the four-ρ part of the independent region-P computation, and both vanish in total (12 kernel structures).

26 The symmetrized three-ρ terms of $\log\vee$ cancel

All symmetrized three-ρ terms of the coefficient of $\log\vee$, after the fixes, sorted by kernel structure and colour network. Every entry is renamed so that the charge tied to w is x, the one tied to w' is x', and the third is y; if several namings are equivalent, the entry is averaged over them. Contributions are labelled by row: A1–B2 are the three-ρ remainders of the four-ρ terms (for them $P = A$), the other labels are the rows of Sec. 24 (their region-P sources, App. G; for I.II the term $-\mathfrak{F}_\alpha$ of fix F4; for I.III the corrected kernel of P1). Coefficients are in units of $\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5 k^+}\log\vee$ and include the c.c. where the row has one. In every table the sum of each row is zero: there are 4 kernel structures and 44 colour networks.

Structure 1: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,y,z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^{d}(x')\,\rho^{a}(y)\,\rho^{c}(x)$	A1–B2: $-i$; I.II: $2i$; II.I: $-i$	0 ✓
$f^{abc}\,U^{db}(w)\,U^{de}(x')\,\rho^{e}(x')\,\rho^{a}(y)\,\rho^{c}(x)$	A1–B2: i; I.II: $-2i$; III.II: i	0 ✓
$f^{abc}\,U^{db}(w')\,U^{ec}(x)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{h}(y)\,\rho^{e}(x)$	A1–B2: $-i$; I.III: i; I.IV: i; III.I: $-i$	0 ✓
$f^{abc}\,U^{dc}(x)\,U^{be}(x')\,U^{ah}(y)\,\rho^{e}(x')\,\rho^{h}(y)\,\rho^{d}(x)$	A1–B2: i; I.III: $-i$; I.IV: $-i$; II.I: $-i$; III.I: i; III.II: i	0 ✓
$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	I.III: $-i$; III.II: i	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	I.III: i; II.I: $-i$	0 ✓
$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	I.IV: $-i$; II.I: $-i$; III.I: i; III.II: i	0 ✓
$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	I.IV: i; III.I: $-i$	0 ✓

Structure 2: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',y,z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc}\,U^{db}(w')\,U^{de}(x)\,\rho^{c}(x')\,\rho^{a}(y)\,\rho^{e}(x)$	A1–B2: $-i$; I.II: $2i$; III.II: $-i$	0 ✓
$f^{abc}\,U^{bd}(w')\,U^{de}(w)\,\rho^{c}(x')\,\rho^{a}(y)\,\rho^{e}(x)$	A1–B2: i; I.II: $-2i$; II.I: i	0 ✓
$f^{abc}\,U^{dc}(x')\,U^{eb}(w)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	A1–B2: i; I.III: $-i$; I.IV: $-i$; III.I: i	0 ✓
$f^{abc}\,U^{dc}(x')\,U^{be}(x)\,U^{ah}(y)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	A1–B2: $-i$; I.III: i; I.IV: i; II.I: i; III.I: $-i$; III.II: $-i$	0 ✓
$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)$	I.III: i; III.II: $-i$	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)$	I.III: $-i$; II.I: i	0 ✓
$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)$	I.IV: i; II.I: i; III.I: $-i$; III.II: $-i$	0 ✓
$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)$	I.IV: $-i$; III.I: i	0 ✓

Structure 3: $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,y,z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc}\,U^{ad}(x')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	I.III: $-i$; III.II: i; III.III: $-i$; III.V: i	0 ✓
$f^{abc}\,U^{ad}(x')\,U^{ce}(y)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	I.III: $-i$; III.II: i; III.III: $-i$; III.V: i	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{ce}(x)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(x)\,\rho^{h}(y)$	I.III: i; II.I: $-i$; III.III: i; III.V: $-i$	0 ✓
$f^{abc}\,U^{ad}(w')\,U^{ce}(y)\,U^{bh}(z)\,\rho^{d}(x')\,\rho^{e}(y)\,\rho^{h}(x)$	I.III: i; II.I: $-i$; III.III: i; III.V: $-i$	0 ✓
$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(x)\,\rho^{e}(x')\,\rho^{l}(x)\,\rho^{c}(y)$	I.IV: $-i$; II.I: $-i$; III.I: i; III.II: i	0 ✓
$f^{abc}\,U^{de}(x')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	I.IV: $-i$; II.I: $-i$; III.I: i; III.II: i	0 ✓
$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(x)\,\rho^{e}(x')\,\rho^{l}(x)\,\rho^{c}(y)$	I.IV: i; III.I: $-i$	0 ✓
$f^{abc}\,U^{de}(w')\,U^{da}(w)\,U^{hb}(z)\,U^{hl}(y)\,\rho^{e}(x')\,\rho^{l}(y)\,\rho^{c}(x)$	I.IV: i; III.I: $-i$	0 ✓

Structure 4: $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',y,z)$

colour structure	contributions (row: coefficient)	sum
$f^{abc}\,U^{ad}(x)\,U^{ce}(y)\,U^{bh}(z)\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)$	I.III: i; III.II: $-i$; III.III: i; III.V: $-i$	0 ✓
$f^{abc}\,U^{ad}(x)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)$	I.III: i; III.II: $-i$; III.III: i; III.V: $-i$	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{ce}(y)\,U^{bh}(z)\,\rho^{h}(x')\,\rho^{e}(y)\,\rho^{d}(x)$	I.III: $-i$; II.I: i; III.III: $-i$; III.V: i	0 ✓
$f^{abc}\,U^{ad}(w)\,U^{ce}(x')\,U^{bh}(z)\,\rho^{h}(y)\,\rho^{e}(x')\,\rho^{d}(x)$	I.III: $-i$; II.I: i; III.III: $-i$; III.V: i	0 ✓
$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)$	I.IV: i; II.I: i; III.I: $-i$; III.II: $-i$	0 ✓
$f^{abc}\,U^{de}(x)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(x')\,\rho^{c}(y)\,\rho^{l}(x')\,\rho^{e}(x)$	I.IV: i; II.I: i; III.I: $-i$; III.II: $-i$	0 ✓
$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(y)\,\rho^{c}(x')\,\rho^{l}(y)\,\rho^{e}(x)$	I.IV: $-i$; III.I: i	0 ✓
$f^{abc}\,U^{de}(w)\,U^{da}(w')\,U^{hb}(z)\,U^{hl}(x')\,\rho^{c}(y)\,\rho^{l}(x')\,\rho^{e}(x)$	I.IV: $-i$; III.I: i	0 ✓

27 The two-ρ terms of $\log\vee$: comparison with BFKL⊗LO

All two-ρ terms of the coefficient of $\log\vee$, after the fixes:

- the two-ρ terms of the four-ρ terms: those of the remainders A1–B2 (App. B) and the lists of `4rho.tex` with F1, F2 (label `4rho`); for all of them $P = A$;
- the two-ρ terms of the region-P three-ρ sources (App. G), with P1 in Row III;
- Group I, Row II: the two-ρ terms of $-\mathfrak{F}_\alpha$ in the $U(w)\mathbb{A}^{(3)}$ sub-terms (fix F4, Sec. 5);
- the genuine two-ρ terms with $P \neq 0$: Group II Rows I and IV, and Group III Row III part 2 (with F5) (Sec. 24).

They are grouped by kernel structure and colour network. For each structure the colour networks are reduced to an independent basis, with coefficients that are exact polynomials in N_c (checked with SU(3) and SU(4); the relations are in App. H). The last column is BFKL⊗LO, Eq. (28.1), in the same basis, and a check mark means that the sum equals it. Units as in Sec. 26. The genuine Group II Row IV term is written as a z integrand, $K(w,w',z) - \frac{1}{2}K(w,w,z) - \frac{1}{2}K(w',w',z)$.

Structure 1: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$	A1–B2: $-\frac{N_c}{2}$; II.I: $-\frac{N_c}{2}$	$-N_c$	$-N_c$ ✓
$U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	A1–B2: $\frac{N_c}{2}$; III.II: $\frac{N_c}{2}$	N_c	N_c ✓
$U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$	A1–B2: $-\frac{N_c}{2}$; `4rho`: $2N_c$; I.III: $-\frac{N_c}{2}$; I.IV: $\frac{N_c}{2}$; III.I: $-\frac{N_c}{2}$	N_c	N_c ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	A1–B2: $\frac{N_c}{2}$; `4rho`: $-2N_c$; I.III: $\frac{N_c}{2}$; I.IV: $-\frac{N_c}{2}$; II.I: $-\frac{N_c}{2}$; III.I: $\frac{N_c}{2}$; III.II: $\frac{N_c}{2}$	$-N_c$	$-N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	`4rho`: 1; I.III: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	`4rho`: -1; I.III: $\frac{1}{2}$; II.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$	I.IV: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)$ $\rho^l(x')\,\rho^e(x)$	I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓

Structure 2: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)$	A1–B2: -1; `4rho`: $\frac{1}{2}$; I.II: 2; II.I: $\frac{1}{2}$	2	2 ✓
$f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)$	A1–B2: $\frac{1}{2}$; `4rho`: -1; I.II: -1; III.II: $-\frac{1}{2}$	-2	-2 ✓
$f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x)$	A1–B2: $-\frac{1}{2}$; `4rho`: 1; I.II: 1; III.II: $\frac{1}{2}$	2	2 ✓
$f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)$	`4rho`: 2	2	2 ✓
$U^{ab}(x)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)$	`4rho`: $\frac{3N_c}{2}$; I.III: $-N_c$; I.IV: $-N_c$; II.I: $\frac{N_c}{2}$; III.I: N_c; III.II: $-N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)$	A1–B2: $-\frac{1}{2}$; I.III: $\frac{1}{2}$; I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)$	A1–B2: $\frac{1}{2}$; `4rho`: -1; I.III: $\frac{1}{2}$; I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	`4rho`: 1; I.III: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	`4rho`: -1; I.III: $\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	`4rho`: $-\frac{1}{2}$; I.III: $\frac{1}{2}$; I.IV: $-\frac{1}{2}$; II.I: $\frac{1}{2}$; III.I: $\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	`4rho`: $\frac{1}{2}$; I.III: $-\frac{1}{2}$; I.IV: $\frac{1}{2}$; II.I: $-\frac{1}{2}$; III.I: $-\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')$ $\rho^h(x')\,\rho^c(x)$	I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)$ $\rho^c(x')\,\rho^h(x)$	I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓

Structure 3: $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$

BFKL⊗LO has no term with this kernel: the terms cancel among themselves.

colour structure	contributions (row: coefficient)	sum	BFKL
$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$	A1–B2: $-N_c$; I.III: N_c; I.IV: N_c; II.I: N_c; III.I: $-N_c$; III.II: $-N_c$	0	0 ✓
$U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.III: $-N_c$; III.II: N_c	0	0 ✓
$f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')$	A1–B2: $-\frac{1}{2}$; I.II: 1; II.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hd}(x')\,\rho^a(x)\,\rho^e(x')$	A1–B2: $\frac{1}{2}$; I.II: -1; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{hd}(x')\,\rho^a(x)\,\rho^e(x')$	A1–B2: $-\frac{1}{2}$; I.II: 1; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{hd}(w)\,\rho^a(x)\,\rho^e(x')$	A1–B2: $\frac{1}{2}$; I.II: -1; II.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')$	A1–B2: $-\frac{1}{2}$; I.III: $\frac{1}{2}$; I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)$	A1–B2: $\frac{1}{2}$; I.III: $-\frac{1}{2}$; I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$	I.III: $\frac{1}{2}$; II.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x')$	I.III: $-\frac{1}{2}$; II.I: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$	I.IV: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$; III.I: $\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$	I.IV: $\frac{1}{2}$; II.I: $\frac{1}{2}$; III.I: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$	I.IV: $\frac{1}{2}$; III.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$	I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓

Structure 4: $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	A1–B2: $\frac{N_c}{2}$; III.II: $\frac{N_c}{2}$	N_c	N_c ✓
$U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)$	A1–B2: $-\frac{N_c}{2}$; II.I: $-\frac{N_c}{2}$	$-N_c$	$-N_c$ ✓
$U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$	A1–B2: $-\frac{N_c}{2}$; `4rho`: $2N_c$; I.III: $-\frac{N_c}{2}$; I.IV: $\frac{N_c}{2}$; III.I: $-\frac{N_c}{2}$	N_c	N_c ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')$	A1–B2: $\frac{N_c}{2}$; `4rho`: $-2N_c$; I.III: $\frac{N_c}{2}$; I.IV: $-\frac{N_c}{2}$; II.I: $-\frac{N_c}{2}$; III.I: $\frac{N_c}{2}$; III.II: $\frac{N_c}{2}$	$-N_c$	$-N_c$ ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	`4rho`: -1; I.III: $\frac{1}{2}$; III.II: $\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	`4rho`: 1; I.III: $-\frac{1}{2}$; II.I: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$	I.IV: $\frac{1}{2}$; II.I: $\frac{1}{2}$; III.I: $-\frac{1}{2}$; III.II: $-\frac{1}{2}$	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$	I.IV: $-\frac{1}{2}$; III.I: $\frac{1}{2}$	0	0 ✓

Structure 5: $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	II.I: $-N_c$	$-N_c$	$-N_c$ ✓
$U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)$	III.II: N_c	N_c	N_c ✓
$U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)$	III.III: N_c	N_c	N_c ✓
$U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)$	III.VI: $-N_c$	$-N_c$	$-N_c$ ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.III: N_c; I.IV: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: N_c; III.III: N_c; III.V: $-N_c$; III.VI: $-N_c$	0	0 ✓
$U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.III: $-N_c$; I.IV: N_c; III.I: $-N_c$; III.V: N_c	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: -1; III.II: -1; III.III: -1; III.V: 1; III.VI: 2	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: 1; II.I: 1; III.III: -1; III.V: -1	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$	I.IV: -1; II.I: -1; III.I: 1; III.II: 1	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)$	I.IV: 1; III.I: -1	0	0 ✓

Structure 6: $\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)$	II.I: -2	-2	-2 ✓

colour structure	contributions (row: coefficient)	sum	BFKL
$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')$	III.II: -2	-2	-2 ✓
$f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')$	III.III: 2	2	2 ✓
$f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')$	III.VI: -2	-2	-2 ✓
$U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.III: $-N_c$; III.II: N_c; III.III: N_c; III.V: N_c; III.VI: $-2N_c$	0	0 ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	II.I: $2N_c$; III.II: $-2N_c$; III.III: $-2N_c$; III.VI: $2N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	I.III: -1; III.II: -1; III.III: -1; III.V: 1; III.VI: 2	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x)$	I.III: 1; II.I: -1; III.III: 1; III.V: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x')$	I.III: 1; II.I: 1; III.III: -1; III.V: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^l(x')\,\rho^c(x)$	I.IV: -1; II.I: 1; III.I: 1; III.II: -1	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$	I.IV: -1; II.I: -1; III.I: 1; III.II: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^h(x')\,\rho^c(x)$	I.IV: 1; III.I: -1	0	0 ✓
$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^m(x)\,\rho^e(x')$	I.IV: 1; III.I: -1	0	0 ✓

Structure 7: $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	II.I: -2	-2	-2 ✓
$f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)$	III.II: 2	2	2 ✓
$f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)$	III.III: -2	-2	-2 ✓
$f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)$	III.VI: 2	2	2 ✓
$U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	I.III: $-N_c$; III.II: N_c; III.III: N_c; III.V: N_c; III.VI: $-2N_c$	0	0 ✓
$U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)$	II.I: $2N_c$; III.II: $-2N_c$; III.III: $-2N_c$; III.VI: $2N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: 1; III.II: 1; III.III: 1; III.V: -1; III.VI: -2	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: -1; II.I: 1; III.III: -1; III.V: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)$	I.III: -1; II.I: -1; III.III: 1; III.V: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	I.IV: 1; II.I: -1; III.I: -1; III.II: 1	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)$	I.IV: 1; II.I: 1; III.I: -1; III.II: -1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^c(x')\,\rho^h(x)$	I.IV: -1; III.I: 1	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^m(x')\,\rho^e(x)$	I.IV: -1; III.I: 1	0	0 ✓

Structure 8: $\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')$	II.I: $-N_c$	$-N_c$	$-N_c$ ✓
$U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')$	III.II: N_c	N_c	N_c ✓
$U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')$	III.III: N_c	N_c	N_c ✓
$U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')$	III.VI: $-N_c$	$-N_c$	$-N_c$ ✓
$U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.III: N_c; I.IV: $-N_c$; II.I: $-N_c$; III.I: N_c; III.II: N_c; III.III: N_c; III.V: $-N_c$; III.VI: $-N_c$	0	0 ✓
$U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)$	I.III: $-N_c$; I.IV: N_c; III.I: $-N_c$; III.V: N_c	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	I.III: 1; III.II: 1; III.III: 1; III.V: -1; III.VI: -2	0	0 ✓
$f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)$	I.III: -1; II.I: -1; III.III: 1; III.V: 1	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$	I.IV: 1; II.I: 1; III.I: -1; III.II: -1	0	0 ✓
$f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^e(x')\,\rho^l(x)$	I.IV: -1; III.I: 1	0	0 ✓

Structure 9: $K(w,w',z)\;K(x,x',z)$

colour structure	contributions (row: coefficient)	sum	BFKL
$\rho^a(x)\,\rho^a(x')$	II.IV: $2N_c$	$2N_c$	$2N_c$ ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')$	II.I: 1	1	1 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)$	II.I: 1	1	1 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')$	III.III: 1	1	1 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)$	III.III: 1	1	1 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')$	III.III: 1	1	1 ✓
$f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)$	III.III: 1	1	1 ✓
$U^{ab}(x)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$	II.I: $2N_c$; II.IV: $2N_c$; III.III: $-4N_c$	0	0 ✓
$U^{ab}(z)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')$	II.IV: $-2N_c$; III.III: $2N_c$	0	0 ✓
$U^{ab}(z)\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)$	II.IV: $-2N_c$; III.III: $2N_c$	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^c(x)\,\rho^l(x')$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^l(x)\,\rho^h(x')$	II.I: -1; III.III: 1	0	0 ✓
$f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)$	II.I: -1; III.III: 1	0	0 ✓

28 The BFKL terms

Operator form. Let $O(x',x) = \rho^{b'}(x')\,M^{b'b}(x',x)\,\rho^b(x)$ with $M^{b'b}(p',p) = \big[U(w')-U(p')\big]^{ab'}\big[U(w)-U(p)\big]^{ab}$. The leading-order cross section is $\frac{d^3N_{\rm LO}}{d^3k} \propto \int \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\langle O(x',x)\rangle$. For $p^+ \gg k^+$ the gluon at z is added to the projectile: its charge $j^a(z)$ adds to ρ^a, and the valence charges acquire the virtual corrections. At order α_s the result is

$$
\begin{aligned}
\text{BFKL}\otimes\text{LO} = 2\mathcal{N}_P \int_{x,x',u,v,z} & e^{-ik\cdot(w'-w)}\,K(u,v,z)\,\Big\{ \\
& -\tfrac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\Big[\rho^h(u)\rho^h(v)\,O(x',x) - 2\rho^h(u)\,O(x',x)\,\rho^h(v) + O(x',x)\,\rho^h(u)\rho^h(v)\Big] \\
& +\mathcal{K}(z-w')\cdot\mathcal{K}(z-w)\;\rho^{e'}(u)\rho^e(v)\,(T^{b'}T^b)_{e'e}\,M^{b'b}(z,z) \\
& +\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\;\rho^{e'}(u)\rho^{b'}(x')\rho^e(v)\,(T^b)_{e'e}\,M^{b'b}(x',z) \\
& +\mathcal{K}(z-w')\cdot\mathcal{K}(x-w)\;\rho^{e'}(u)\rho^b(x)\rho^e(v)\,(T^{b'})_{e'e}\,M^{b'b}(z,x)\Big\},
\end{aligned}
\tag{28.1}
$$

with $(T^a)_{bc} = -if^{abc}$. The first line is the virtual correction (the double commutator $-\frac{1}{2}[\rho^h(u),[\rho^h(v),O]]$). The second line has both measured gluons emitted by the gluon at z. The last two lines have one measured gluon emitted by a valence charge and the other by the gluon at z.

Fully written out. After reordering the charges (Eq. (2.4)) and simplifying the colour algebra, Eq. (28.1) is a sum of two-ρ terms. Its symmetrized three-ρ and four-ρ parts vanish. In the basis of Sec. 27 (the same structures and colour basis as the comparison there):

$$
\begin{aligned}
\text{BFKL}\otimes\text{LO} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)} \int_{x,x',z} & \Big\{ \\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)\,\Big[\\
& \quad -N_c\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x) \\
& \quad +N_c\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
& \quad +N_c\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
& \quad -N_c\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\Big] \\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)\,\Big[\\
& \quad +2\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x) \\
& \quad -2\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
& \quad +2\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^c(x')\,\rho^e(x) \\
& \quad +2\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\Big] \\
& +\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)\,\Big[\\
& \quad +N_c\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&\quad - N_c\, U^{ab}(w')\, U^{bc}(w)\, \rho^a(x')\, \rho^c(x) \\
&\quad + N_c\, U^{ab}(w)\, U^{bc}(x')\, \rho^a(x)\, \rho^c(x') \\
&\quad - N_c\, U^{ab}(x)\, U^{ac}(x')\, \rho^b(x)\, \rho^c(x') \Big] \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\; K(x,x,z)\, \Big[\\
&\quad - N_c\, U^{ab}(w')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\quad + N_c\, U^{ab}(x')\, U^{ac}(w)\, \rho^b(x')\, \rho^c(x) \\
&\quad + N_c\, U^{ab}(w')\, U^{ca}(z)\, \rho^b(x')\, \rho^c(x) \\
&\quad - N_c\, U^{ab}(x')\, U^{ca}(z)\, \rho^b(x')\, \rho^c(x) \Big] \\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\; K(x,x',z)\, \Big[\\
&\quad - 2\, f^{abc}\, f^{bde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^e(x')\, \rho^c(x) \\
&\quad - 2\, f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, \rho^b(x)\, \rho^e(x') \\
&\quad + 2\, f^{abc}\, f^{ade}\, U^{hb}(w')\, U^{eh}(z)\, \rho^d(x)\, \rho^c(x') \\
&\quad - 2\, f^{abc}\, f^{ade}\, U^{hb}(x')\, U^{eh}(z)\, \rho^d(x)\, \rho^c(x') \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\; K(x,x',z)\, \Big[\\
&\quad - 2\, f^{abc}\, f^{bde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^c(x')\, \rho^e(x) \\
&\quad + 2\, f^{abc}\, f^{bde}\, U^{hd}(x)\, U^{ha}(w')\, \rho^c(x')\, \rho^e(x) \\
&\quad - 2\, f^{abc}\, f^{dae}\, U^{hb}(w)\, U^{eh}(z)\, \rho^d(x')\, \rho^c(x) \\
&\quad + 2\, f^{abc}\, f^{dae}\, U^{hb}(x)\, U^{eh}(z)\, \rho^d(x')\, \rho^c(x) \Big] \\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\; K(x',x',z)\, \Big[\\
&\quad - N_c\, U^{ab}(w')\, U^{ac}(w)\, \rho^c(x)\, \rho^b(x') \\
&\quad + N_c\, U^{ab}(x)\, U^{ac}(w')\, \rho^b(x)\, \rho^c(x') \\
&\quad + N_c\, U^{ab}(w)\, U^{ca}(z)\, \rho^b(x)\, \rho^c(x') \\
&\quad - N_c\, U^{ab}(x)\, U^{ca}(z)\, \rho^b(x)\, \rho^c(x') \Big] \\
&+ K(w,w',z)\; K(x,x',z)\, \Big[\\
&\quad + 2N_c\, \rho^a(x)\, \rho^a(x') \\
&\quad + f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(w)\, \rho^c(x)\, \rho^e(x') \\
&\quad + f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{hd}(w)\, \rho^c(x')\, \rho^e(x) \\
&\quad + f^{abc}\, f^{dbe}\, U^{ha}(w)\, U^{eh}(z)\, \rho^d(x)\, \rho^c(x') \\
&\quad + f^{abc}\, f^{dbe}\, U^{ha}(w)\, U^{eh}(z)\, \rho^d(x')\, \rho^c(x) \\
&\quad + f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{eh}(z)\, \rho^c(x)\, \rho^d(x') \\
&\quad + f^{abc}\, f^{dbe}\, U^{ha}(w')\, U^{eh}(z)\, \rho^c(x')\, \rho^d(x) \Big] \Big\}
\end{aligned}
\tag{28.2}
$$

Colour-singlet projection: Eq. (LOBFKL). For $\langle \rho^a(p)\rho^b(q)\rangle = \delta^{ab}\mu(p,q)/(N_c^2-1)$, Eq. (28.1) reduces to Eq. (LOBFKL) of your Appendix E:

$$
\begin{aligned}
-\frac{\alpha_s^2 N_c\,\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}\mu(x,y)\Big[&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,K_D(x,y,z)\,\mathrm{Tr}\big\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\big\} \\
&-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')\,K(x,y,z)\,\mathrm{Tr}\big\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\big\} \\
&+\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')\,(2K(x,y,z)-K(x,x,z))\,\mathrm{Tr}\big\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\big\} \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')\,(2K(x,y,z)-K(y,y,z))\,\mathrm{Tr}\big\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\big\}\Big],
\end{aligned}
\tag{28.3}
$$

with $K_D(x,y,z) = K(x,x,z) - 2K(x,y,z) + K(y,y,z)$. Its lines match those of Eq. (28.1): the first is the virtual (dipole) term, the second the term with both measured gluons from z, and the last two are the mixed terms. Table 12 compares the two numerically at random points, for symmetric and for general adjoint U. They agree to machine precision, so **Appendix E is correct as written** (unlike Appendix F; see Part III).

U	seed	Eq. (28.1), singlet projection	Eq. (LOBFKL)
symmetric	1	79.30732887	79.30732887
symmetric	2	−107.56552336	−107.56552336
symmetric	3	0.08995397	0.08995397
general	1	95.24753301	95.24753301
general	2	91.64627107	91.64627107
general	3	0.13321489	0.13321489

Table 12: BFKL⊗LO at random points (x, y, w, w', z), in units of $P\mu$ with $P = \alpha_s^2 \delta Y/(8\pi^6(N_c^2-1))$.

The coefficient of $\log\vee$**.** The coefficient of $\log(\vee/\Lambda)$ is JIMWLK⊗LO (Part III); adding that of $\log(\vee/k^+)$ gives BFKL⊗LO, Eq. (28.1) (Sec. 27): the projectile evolves with $\log\vee$, and the target with $\log(k^+/\Lambda)$.

29 Numerical checks

Row by row, region P**.** The coefficient of $\log\vee$ of each row of the notes, P_{row}, compared with an independent leading-log computation of the region $p^+ \gg k^+$. That computation is the one of `BFKL_JIMWLK_RowWise.pdf`, checked against an exact operator model of the wave function. Its rows are organized by intermediate state, as yours are. Group I is compared as a whole, because its rows are organized differently there. Pointwise in z at random points, for the symmetrized two-ρ and three-ρ parts:

row	charges	as written	corrected
Group I (all rows)	2	12 / 18	0 / 18 (max 2.3×10^{-13})
Group II, Row I	2	0 / 19	0 / 19 (max 2.9×10^{-14})
Group II, Row II	2	0 / 10	0 / 10 (max 2.0×10^{-15})
Group II, Row III	2	0 / 10	0 / 10 (max 4.4×10^{-16})
Group II, Row IV	2	0 / 1	0 / 1 (max 8.9×10^{-15})
Group III, Row I	2	0 / 18	0 / 18 (max 7.6×10^{-14})
Group III, Row II	2	0 / 18	0 / 18 (max 8.7×10^{-14})
Group III, Row III	2	0 / 9	0 / 9 (max 5.3×10^{-15})
Group III, Row IV	2	0 / 10	0 / 10 (max 3.9×10^{-14})
Group III, Row V	2	0 / 8	0 / 8 (max 5.3×10^{-15})
Group III, Row VI	2	0 / 8	0 / 8 (max 3.6×10^{-15})
Group I (all rows)	3	18 / 27	0 / 27 (max 3.5×10^{-14})
Group II, Row I	3	0 / 27	0 / 27 (max 1.0×10^{-14})
Group II, Row II	3	0 / 21	0 / 21 (max 1.6×10^{-15})
Group II, Row III	3	0 / 21	0 / 21 (max 3.3×10^{-16})
Group III, Row I	3	0 / 27	0 / 27 (max 1.1×10^{-14})
Group III, Row II	3	0 / 27	0 / 27 (max 1.2×10^{-14})
Group III, Row III	3	0 / 6	0 / 6 (max 9.4×10^{-16})
Group III, Row IV	3	0 / 21	0 / 21 (max 5.3×10^{-15})
Group III, Row V	3	0 / 6	0 / 6 (max 4.9×10^{-15})
Group III, Row VI	3	0 / 6	0 / 6 (max 1.6×10^{-15})

Table 13: Coefficient of $\log\vee$ of each row vs the independent region-P computation: number of differing kernel structures / total.

"corrected" means: Group I with P1 and with the α part of Row II removed from P (Sec. 24.1.2), i.e. fix F4 (Sec. 5). Each entry gives the number of kernel structures that differ, out of the total.

Cumulative fixes. Number of kernel structures (pointwise in z) where the $\log(\vee/k^+)$ terms differ from BFKL⊗LO−JIMWLK⊗LO:

state	two-ρ: differing / total	three-ρ (symmetrized): left over / total
as written	35 / 49	30 / 42
+ P1 (Group I Row III: sign of K_h)	31 / 49	24 / 42
+ F3 carried over (Row III part 2 → $\mathcal{D}$)	11 / 49	0 / 30
+ F5 carried over (Group III Row III part 2: sign)	0 / 37	0 / 30

Table 14: The $\log(\vee/k^+)$ terms as written and after each fix.

Other configurations. At two other random configurations (seeds 21 and 33) the corrected two-ρ total again agrees in all structures (largest difference 8.3×10^{-14}). For a general (non-symmetric) adjoint U the notes do not apply, because they use $U^{bc} = U^{cb}$ throughout: 35 of 37 structures differ.

Part V
Summary: the leading-logarithmic cross section

30 Target and projectile evolution

For every row the large-logarithm content is $A\,\log(\vee/\Lambda) + B\,\log(\vee/k^+)$, with $P = A + B$. Parts III and IV give, after the fixes,

$$\sum_{\text{rows}} A = \text{JIMWLK} \otimes \text{LO}, \qquad \sum_{\text{rows}} P = \text{BFKL} \otimes \text{LO}, \tag{30.1}$$

for the two-ρ terms, and zero for the symmetrized three- and four-ρ terms. Hence

$$\begin{aligned} \left.\frac{d^3N}{d^3k}\right|_{\text{LL}} &= \log\frac{\vee}{\Lambda}\sum A + \log\frac{\vee}{k^+}\sum B = \Big(\log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+}\Big)\sum A + \log\frac{\vee}{k^+}\sum P \\ &= \log\frac{k^+}{\Lambda}\ \text{JIMWLK} \otimes \text{LO} \ +\ \log\frac{\vee}{k^+}\ \text{BFKL} \otimes \text{LO}. \end{aligned} \tag{30.2}$$

- $\log(k^+/\Lambda) = \delta Y_T$ is the rapidity interval between the target and the measured gluon. Gluons in it are softer than the measured gluon. They are emitted by all harder partons (the valence charges and the measured gluon) and scatter on the target, so integrating them out evolves the target Wilson lines (JIMWLK).
- $\log(\vee/k^+) = \delta Y_P$ is the interval between the measured gluon and the projectile. Gluons in it are harder than the measured gluon, and integrating them out evolves the projectile charge correlator (BFKL). In BFKL⊗LO the measured gluon can be emitted by the new gluon at z (kernels $\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')$ and $\mathcal{K}(z-w)\cdot\mathcal{K}(\cdot-w')$); no $\log(\vee/\Lambda)$ term has such a kernel.
- Each side evolves over its own interval, and the total is $\delta Y = \log(\vee/\Lambda)$. There is no single δY for both: moving the target end, $-\partial/\partial\log\Lambda$, gives JIMWLK⊗LO; moving the projectile end, $\partial/\partial\log\vee$, gives BFKL⊗LO.

All totals, pointwise. Without any bookkeeping by kernel structure or colour network, the totals at random points (pointwise in z; random symmetric Wilson lines and charges) are:

coefficient	terms	as written: differing / total	corrected: differing / total
$\log(\vee/\Lambda)$	two-ρ vs JIMWLK⊗LO	32 / 38	0 / 26 (max 3.6×10^{-14})
$\log(\vee/\Lambda)$	symmetrized three-ρ (sum = 0)	36 / 36	0 / 24 (max 4.9×10^{-15})
$\log\vee$	two-ρ vs BFKL⊗LO	15 / 19	0 / 19 (max 3.2×10^{-14})
$\log\vee$	symmetrized three-ρ (sum = 0)	18 / 18	0 / 18 (max 2.7×10^{-15})

Table 15: Number of kernel structures in which the total differs from the target (JIMWLK⊗LO, BFKL⊗LO, or zero for the symmetrized three-ρ terms), and the largest remaining difference after the fixes.

Group I, Row II as an example. With F4, the $U(w)\mathbb{A}^{(3)}$ sub-terms of Row II are $\mathfrak{F}_\beta\,\delta Y_T - \mathfrak{F}_\alpha\,\delta Y_P$ (Eq. (5.12)). The soft gluon between the measured gluon and a charge ($\mathfrak{F}_\beta$) is part of JIMWLK; the gluon between the two charges ($\mathfrak{F}_\alpha$) is part of the virtual term of BFKL. As written, both had δY_T.

Appendices

A The reordering identity

Let A, B, C be three charges, e.g. $A = \rho^d(x')$, $B = \rho^a(x)$, $C = \rho^c(y)$. Their fully symmetrized product is

$$S(ABC) \equiv \frac{1}{6}\big(ABC + ACB + BAC + BCA + CAB + CBA\big). \tag{A.1}$$

Step 1: every ordering in terms of ABC. Moving operators one step at a time,

$$\begin{aligned}
ACB &= ABC - A[B,C],\\
BAC &= ABC - [A,B]C,\\
BCA &= BAC - B[A,C] = ABC - [A,B]C - B[A,C],\\
CAB &= ACB - [A,C]B = ABC - A[B,C] - [A,C]B,\\
CBA &= BCA - [B,C]A = ABC - [A,B]C - B[A,C] - [B,C]A.
\end{aligned} \tag{A.2}$$

Adding the six orderings,

$$6\,S(ABC) = 6\,ABC - 3[A,B]C - 2A[B,C] - 2B[A,C] - [A,C]B - [B,C]A. \tag{A.3}$$

Step 2: symmetrize the products of two. Each commutator is linear in ρ (Eq. (2.4)). For a commutator X and a charge Y, $XY = \frac{1}{2}\{X,Y\} + \frac{1}{2}[X,Y]$ and $YX = \frac{1}{2}\{X,Y\} - \frac{1}{2}[X,Y]$. Here $[X,Y]$ is a double commutator, i.e. a single ρ. Inserting this in Eq. (A.3), the anticommutators come with weights 3 for $\{[A,B],C\}$, $2+1=3$ for $\{[A,C],B\}$ and $2+1=3$ for $\{[B,C],A\}$. Hence

$$\boxed{ABC = S(ABC) + \tfrac{1}{4}\Big(\big\{[A,B],C\big\} + \big\{[A,C],B\big\} + \big\{[B,C],A\big\}\Big) + R_1} \tag{A.4}$$

with the one-ρ remainder

$$R_1 = \tfrac{1}{12}\Big(3\big[[A,B],C\big] + 2\big[A,[B,C]\big] + 2\big[B,[A,C]\big] + \big[[A,C],B\big] + \big[[B,C],A\big]\Big). \tag{A.5}$$

Step 3: drop the one-ρ terms. Each term of R_1 is a double commutator, $\propto ff\,\rho\,\delta\delta$. It is linear in ρ and multiplies c-numbers (kernels and Wilson lines). Its expectation value vanishes for a colour-neutral projectile, $\langle\rho^a(x)\rangle = 0$. We drop it.

Anticommutator forms. Several rows of the notes contain $\{\rho,\rho\}$. Applying Eq. (A.4) to ABC and ACB and adding, the terms $\{[B,C],A\}$ and $\{[C,B],A\}$ cancel, and $S(ACB) = S(ABC)$:

$$\boxed{A\{B,C\} = 2S(ABC) + \tfrac{1}{2}\big\{[A,B],C\big\} + \tfrac{1}{2}\big\{[A,C],B\big\} + (\text{one }\rho)} \tag{A.6}$$

When the single charge stands on the right, use $\{B,C\}A = A\{B,C\} - [A,\{B,C\}]$ and $[A,\{B,C\}] = \{[A,B],C\} + \{B,[A,C]\}$:

$$\boxed{\{B,C\}A = 2S(ABC) - \tfrac{1}{2}\big\{[A,B],C\big\} - \tfrac{1}{2}\big\{[A,C],B\big\} + (\text{one }\rho)} \tag{A.7}$$

One piece. For $X = \rho^x(p_X)$, $Y = \rho^y(p_Y)$, $Z = \rho^z(p_Z)$, Eq. (2.4) gives

$$\big\{[X,Y],Z\big\} = if^{xyh}\big\{\rho^h(p_X),\rho^z(p_Z)\big\}\,\delta^{(2)}(p_X - p_Y), \tag{A.8}$$

and $p_Y \to p_X$ everywhere. The two-ρ terms are written with the anticommutator $\{\rho,\rho\}$. A different ordering of two charges differs from it by one ρ, which we drop.

The symmetric three-ρ part. For every term, the part $S(\dots)$ is the same expression with $\rho\rho\rho$ replaced by $S(\rho\rho\rho)$ (and by $2S(\rho\rho\rho)$ for the anticommutator forms). These parts behave as products of commuting charges. We do not list them again; their cancellation is checked in Part II.

Complex conjugate. All prefactors below are $\pm i$, $\pm i/2$ or $\pm 2i$ times $\mathcal{N}$. The commutator supplies one more i, so every two-ρ term has a *real* coefficient. The kernels, f, U and δ are real, and $\{\rho,\rho\}$ is Hermitian. The complex conjugate of a two-ρ term therefore only turns $e^{-ik\cdot(P'-P)}$ into $e^{ik\cdot(P'-P)}$, where P' and P are the measured-gluon positions in the conjugate amplitude and the amplitude. Renaming $P \leftrightarrow P'$ gives back the original phase:

$$\text{c.c.} = \big(\text{same term with } P \leftrightarrow P'\big). \tag{A.9}$$

This holds for Groups I and III, with $(P',P) = (w',w)$, except (w',z) and (w',x) in Row V of Group I and (y',w) in Rows III, V and VI of Group III. Group II and the four-ρ remainders get no c.c.

B Three-ρ terms from reordering the four-ρ terms

B.1 Two Wilson lines, first term

In the notes this term reads $-\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{\log(\vee/\Lambda)}{2\pi^2}\frac{1}{k^+}\int(\dots)$, i.e. its prefactor is $-i\mathcal{N}$. It is the complete remainder of reordering the four-ρ terms, so no c.c. is added.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm A1} = &-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_1(x',x,y)\,\Big\{\\ &+ f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^d(x')\,\rho^a(x)\,\rho^c(y)\\ &- f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,\rho^d(x')\,\rho^a(x)\,\rho^c(y)\Big\}\end{aligned}\tag{B.1}$$

with $K_1(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+f^{abc}\,U^{de}(w')\,U^{eb}(w)\,\rho^d(x')\,\rho^a(x)\,\rho^c(y)$.

$$\begin{aligned}[A,B]:\ &[\rho^d(x'),\rho^a(x)] = if^{dah}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ &\tfrac14\{[A,B],C\} = \tfrac14\,i\,K_1(x',x',y)\,f^{abc}\,U^{de}(w')\,U^{eb}(w)\,f^{dah}\,\{\rho^h(x'),\rho^c(y)\}\\ &\qquad = \tfrac14\,i\,K_1(x',x',y)\,f^{abc}\,f^{dae}\,U^{dh}(w')\,U^{hb}(w)\,\{\rho^e(x'),\rho^c(y)\}\end{aligned}\tag{B.2}$$

$$\begin{aligned}[A,C]:\ &[\rho^d(x'),\rho^c(y)] = if^{dch}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\tfrac14\{[A,C],B\} = \tfrac14\,i\,K_1(x',x,x')\,f^{abc}\,U^{de}(w')\,U^{eb}(w)\,f^{dch}\,\{\rho^h(x'),\rho^a(x)\}\\ &\qquad = \tfrac14\,i\,K_1(x',x,x')\,f^{abc}\,f^{dce}\,U^{dh}(w')\,U^{hb}(w)\,\{\rho^e(x'),\rho^a(x)\}\end{aligned}\tag{B.3}$$

$$\begin{aligned}[B,C]:\ &[\rho^a(x),\rho^c(y)] = if^{ach}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\ &\tfrac14\{[B,C],A\} = \tfrac14\,i\,K_1(x',x,x)\,f^{abc}\,U^{de}(w')\,U^{eb}(w)\,f^{ach}\,\{\rho^h(x),\rho^d(x')\}\\ &\qquad = -\tfrac14\,iN_c\,K_1(x',x,x)\,U^{ab}(w')\,U^{bc}(w)\,\{\rho^c(x),\rho^a(x')\}\quad\text{(F)}\end{aligned}\tag{B.4}$$

Sub-term 2: $-f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,\rho^d(x')\,\rho^a(x)\,\rho^c(y)$.

$$\begin{aligned}[A,B]:\ &[\rho^d(x'),\rho^a(x)] = if^{dah}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ &\tfrac14\{[A,B],C\} = -\tfrac14\,i\,K_1(x',x',y)\,f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,f^{dah}\,\{\rho^h(x'),\rho^c(y)\}\\ &\qquad = -\tfrac14\,i\,K_1(x',x',y)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{hd}(x')\,\{\rho^e(x'),\rho^c(y)\}\end{aligned}\tag{B.5}$$

$$\begin{aligned}[A,C]:\ &[\rho^d(x'),\rho^c(y)] = if^{dch}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\tfrac14\{[A,C],B\} = -\tfrac14\,i\,K_1(x',x,x')\,f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,f^{dch}\,\{\rho^h(x'),\rho^a(x)\}\\ &\qquad = -\tfrac14\,i\,K_1(x',x,x')\,f^{abc}\,f^{dce}\,U^{hb}(w)\,U^{hd}(x')\,\{\rho^e(x'),\rho^a(x)\}\end{aligned}\tag{B.6}$$

$$\begin{aligned}[B,C]:\ &[\rho^a(x),\rho^c(y)] = if^{ach}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\ &\tfrac14\{[B,C],A\} = -\tfrac14\,i\,K_1(x',x,x)\,f^{abc}\,U^{eb}(w)\,U^{ed}(x')\,f^{ach}\,\{\rho^h(x),\rho^d(x')\}\\ &\qquad = \tfrac14\,iN_c\,K_1(x',x,x)\,U^{ab}(w)\,U^{ac}(x')\,\{\rho^b(x),\rho^c(x')\}\quad\text{(F)}\end{aligned}\tag{B.7}$$

Two-ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm A1} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{&\\ &+\tfrac14\,K_1(x',x',y)\,f^{abc}\,f^{dae}\,U^{dh}(w')\,U^{hb}(w)\,\{\rho^e(x'),\rho^c(y)\}\\ &+\tfrac14\,K_1(x',x,x')\,f^{abc}\,f^{dce}\,U^{dh}(w')\,U^{hb}(w)\,\{\rho^e(x'),\rho^a(x)\}\\ &-\tfrac14\,N_c\,K_1(x',x,x)\,U^{ab}(w')\,U^{bc}(w)\,\{\rho^c(x),\rho^a(x')\}\\ &-\tfrac14\,K_1(x',x',y)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{hd}(x')\,\{\rho^e(x'),\rho^c(y)\}\\ &-\tfrac14\,K_1(x',x,x')\,f^{abc}\,f^{dce}\,U^{hb}(w)\,U^{hd}(x')\,\{\rho^e(x'),\rho^a(x)\}\end{aligned}$$

$$+\tfrac{1}{4}\,N_c\,K_1(x',x,x)\,U^{ab}(w)\,U^{ac}(x')\,\{\rho^b(x),\rho^c(x')\}\Big\} \tag{B.8}$$

The kernel at the coincident points:

$$\begin{aligned}K_1(x',x',y)&=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\ K_1(x',x,x')&=\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\\ K_1(x',x,x)&=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2}\end{aligned}$$

B.2 Two Wilson lines, second term

Same prefactor $-i\mathcal{N}$ and no c.c.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\rm A2} =\ &-i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_2(x',x,y)\,\Big\{\\ &+\ f^{abc}\,U^{eb}(w')\,U^{ed}(y)\,\rho^c(x')\,\rho^a(x)\,\rho^d(y)\\ &-\ f^{abc}\,U^{be}(w')\,U^{ed}(w)\,\rho^c(x')\,\rho^a(x)\,\rho^d(y)\Big\}\end{aligned} \tag{B.9}$$

with $K_2(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,f^{abc}\,U^{eb}(w')\,U^{ed}(y)\,\rho^c(x')\,\rho^a(x)\,\rho^d(y)$.

$$\begin{aligned}&[A,B]:\ [\rho^c(x'),\rho^a(x)]=if^{cah}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ &\quad\tfrac{1}{4}\{[A,B],C\}=\tfrac{1}{4}\,i\,K_2(x',x',y)\,f^{abc}\,U^{eb}(w')\,U^{ed}(y)\,f^{cah}\,\{\rho^h(x'),\rho^d(y)\}\\ &\qquad=\tfrac{1}{4}\,iN_c\,K_2(x',x',y)\,U^{ab}(w')\,U^{ac}(y)\,\{\rho^b(x'),\rho^c(y)\}\quad\text{(F)}\end{aligned} \tag{B.10}$$

$$\begin{aligned}&[A,C]:\ [\rho^c(x'),\rho^d(y)]=if^{cdh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{4}\{[A,C],B\}=\tfrac{1}{4}\,i\,K_2(x',x,x')\,f^{abc}\,U^{eb}(w')\,U^{ed}(x')\,f^{cdh}\,\{\rho^h(x'),\rho^a(x)\}\\ &\qquad=\tfrac{1}{4}\,i\,K_2(x',x,x')\,f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{hd}(x')\,\{\rho^e(x'),\rho^a(x)\}\end{aligned} \tag{B.11}$$

$$\begin{aligned}&[B,C]:\ [\rho^a(x),\rho^d(y)]=if^{adh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\ &\quad\tfrac{1}{4}\{[B,C],A\}=\tfrac{1}{4}\,i\,K_2(x',x,x)\,f^{abc}\,U^{eb}(w')\,U^{ed}(x)\,f^{adh}\,\{\rho^h(x),\rho^c(x')\}\\ &\qquad=\tfrac{1}{4}\,i\,K_2(x',x,x)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\{\rho^e(x),\rho^c(x')\}\end{aligned} \tag{B.12}$$

Sub-term 2: $-\,f^{abc}\,U^{be}(w')\,U^{ed}(w)\,\rho^c(x')\,\rho^a(x)\,\rho^d(y)$.

$$\begin{aligned}&[A,B]:\ [\rho^c(x'),\rho^a(x)]=if^{cah}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ &\quad\tfrac{1}{4}\{[A,B],C\}=-\tfrac{1}{4}\,i\,K_2(x',x',y)\,f^{abc}\,U^{be}(w')\,U^{ed}(w)\,f^{cah}\,\{\rho^h(x'),\rho^d(y)\}\\ &\qquad=-\tfrac{1}{4}\,iN_c\,K_2(x',x',y)\,U^{ab}(w')\,U^{bc}(w)\,\{\rho^a(x'),\rho^c(y)\}\quad\text{(F)}\end{aligned} \tag{B.13}$$

$$\begin{aligned}&[A,C]:\ [\rho^c(x'),\rho^d(y)]=if^{cdh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{4}\{[A,C],B\}=-\tfrac{1}{4}\,i\,K_2(x',x,x')\,f^{abc}\,U^{be}(w')\,U^{ed}(w)\,f^{cdh}\,\{\rho^h(x'),\rho^a(x)\}\\ &\qquad=-\tfrac{1}{4}\,i\,K_2(x',x,x')\,f^{abc}\,f^{cde}\,U^{bh}(w')\,U^{hd}(w)\,\{\rho^e(x'),\rho^a(x)\}\end{aligned} \tag{B.14}$$

$$\begin{aligned}&[B,C]:\ [\rho^a(x),\rho^d(y)]=if^{adh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\ &\quad\tfrac{1}{4}\{[B,C],A\}=-\tfrac{1}{4}\,i\,K_2(x',x,x)\,f^{abc}\,U^{be}(w')\,U^{ed}(w)\,f^{adh}\,\{\rho^h(x),\rho^c(x')\}\\ &\qquad=-\tfrac{1}{4}\,i\,K_2(x',x,x)\,f^{abc}\,f^{ade}\,U^{bh}(w')\,U^{hd}(w)\,\{\rho^e(x),\rho^c(x')\}\end{aligned} \tag{B.15}$$

Two-ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm A2}=\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{\\ +\tfrac{1}{4}\,N_c\,K_2(x',x',y)\,U^{ab}(w')\,U^{ac}(y)\,\{\rho^b(x'),\rho^c(y)\}\end{aligned}$$

$$
\begin{aligned}
&+\tfrac{1}{4}\,K_2(x',x,x')\,f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{hd}(x')\,\big\{\rho^e(x'),\rho^a(x)\big\}\\
&+\tfrac{1}{4}\,K_2(x',x,x)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{hd}(x)\,\big\{\rho^e(x),\rho^c(x')\big\}\\
&-\tfrac{1}{4}\,N_c\,K_2(x',x',y)\,U^{ab}(w')\,U^{bc}(w)\,\big\{\rho^a(x'),\rho^c(y)\big\}\\
&-\tfrac{1}{4}\,K_2(x',x,x')\,f^{abc}\,f^{cde}\,U^{bh}(w')\,U^{hd}(w)\,\big\{\rho^e(x'),\rho^a(x)\big\}\\
&-\tfrac{1}{4}\,K_2(x',x,x)\,f^{abc}\,f^{ade}\,U^{bh}(w')\,U^{hd}(w)\,\big\{\rho^e(x),\rho^c(x')\big\}\Big\}
\end{aligned}
\tag{B.16}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K_2(x',x',y)&=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\frac{1}{(x'-z)^2}\\
K_2(x',x,x')&=\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\\
K_2(x',x,x)&=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)
\end{aligned}
$$

B.3 Three Wilson lines, kernel K_1

In the notes the three-Wilson-line remainder has prefactor $+\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{\log(\vee/\Lambda)}{2\pi^2}\frac{1}{k^+}=+i\mathcal{N}$. Its first and fourth terms share the kernel K_1 and are treated here; no c.c. is added.

The three-ρ term.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\rm B1}&=i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_1(x',x,y)\,\Big\{\\
&\quad-\;f^{abc}\,U^{db}(w')\,U^{ec}(y)\,U^{af}(x)\,\rho^d(x')\,\rho^f(x)\,\rho^e(y)\\
&\quad+\;f^{abc}\,U^{ec}(y)\,U^{bd}(x')\,U^{af}(x)\,\rho^d(x')\,\rho^f(x)\,\rho^e(y)\Big\}
\end{aligned}
\tag{B.17}
$$

with $K_1(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $-\,f^{abc}\,U^{db}(w')\,U^{ec}(y)\,U^{af}(x)\,\rho^d(x')\,\rho^f(x)\,\rho^e(y)$.

$$
\begin{aligned}
&[A,B]:\;[\rho^d(x'),\rho^f(x)]=if^{dfh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad\tfrac{1}{4}\{[A,B],C\}=-\tfrac{1}{4}\,i\;K_1(x',x',y)\,f^{abc}\,U^{db}(w')\,U^{ec}(y)\,U^{af}(x')\,f^{dfh}\,\big\{\rho^h(x'),\rho^e(y)\big\}\\
&\qquad\qquad=-\tfrac{1}{4}\,i\;K_1(x',x',y)\,f^{abc}\,f^{deh}\,U^{db}(w')\,U^{lc}(y)\,U^{ae}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}
\end{aligned}
\tag{B.18}
$$

$$
\begin{aligned}
&[A,C]:\;[\rho^d(x'),\rho^e(y)]=if^{deh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
&\quad\tfrac{1}{4}\{[A,C],B\}=-\tfrac{1}{4}\,i\;K_1(x',x,x')\,f^{abc}\,U^{db}(w')\,U^{ec}(x')\,U^{af}(x)\,f^{deh}\,\big\{\rho^h(x'),\rho^f(x)\big\}\\
&\qquad\qquad=-\tfrac{1}{4}\,i\;K_1(x',x,x')\,f^{abc}\,f^{deh}\,U^{db}(w')\,U^{ec}(x')\,U^{al}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}
\end{aligned}
\tag{B.19}
$$

$$
\begin{aligned}
&[B,C]:\;[\rho^f(x),\rho^e(y)]=if^{feh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad\tfrac{1}{4}\{[B,C],A\}=-\tfrac{1}{4}\,i\;K_1(x',x,x)\,f^{abc}\,U^{db}(w')\,U^{ec}(x)\,U^{af}(x)\,f^{feh}\,\big\{\rho^h(x),\rho^d(x')\big\}\\
&\qquad\qquad=\tfrac{1}{4}\,iN_c\;K_1(x',x,x)\,U^{ab}(w')\,U^{bc}(x)\,\big\{\rho^c(x),\rho^a(x')\big\}\quad(\mathrm{U2}^\star),\;(\mathrm{F})
\end{aligned}
\tag{B.20}
$$

Sub-term 2: $+\,f^{abc}\,U^{ec}(y)\,U^{bd}(x')\,U^{af}(x)\,\rho^d(x')\,\rho^f(x)\,\rho^e(y)$.

$$
\begin{aligned}
&[A,B]:\;[\rho^d(x'),\rho^f(x)]=if^{dfh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad\tfrac{1}{4}\{[A,B],C\}=\tfrac{1}{4}\,i\;K_1(x',x',y)\,f^{abc}\,U^{ec}(y)\,U^{bd}(x')\,U^{af}(x')\,f^{dfh}\,\big\{\rho^h(x'),\rho^e(y)\big\}\\
&\qquad\qquad=-\tfrac{1}{4}\,iN_c\;K_1(x',x',y)\,U^{ab}(y)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(y)\big\}\quad(\mathrm{U2}),\;(\mathrm{F})
\end{aligned}
\tag{B.21}
$$

$$
\begin{aligned}
&[A,C]:\;[\rho^d(x'),\rho^e(y)]=if^{deh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
&\quad\tfrac{1}{4}\{[A,C],B\}=\tfrac{1}{4}\,i\;K_1(x',x,x')\,f^{abc}\,U^{ec}(x')\,U^{bd}(x')\,U^{af}(x)\,f^{deh}\,\big\{\rho^h(x'),\rho^f(x)\big\}\\
&\qquad\qquad=\tfrac{1}{4}\,iN_c\;K_1(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}\quad(\mathrm{U2}^\star),\;(\mathrm{F})
\end{aligned}
\tag{B.22}
$$

$$
\begin{aligned}
&[B,C]:\;[\rho^f(x),\rho^e(y)]=if^{feh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad\tfrac{1}{4}\{[B,C],A\}=\tfrac{1}{4}\,i\;K_1(x',x,x)\,f^{abc}\,U^{ec}(x)\,U^{bd}(x')\,U^{af}(x)\,f^{feh}\,\big\{\rho^h(x),\rho^d(x')\big\}\\
&\qquad\qquad=-\tfrac{1}{4}\,iN_c\;K_1(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\quad(\mathrm{U2}^\star),\;(\mathrm{F})
\end{aligned}
\tag{B.23}
$$

Two-ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\mathrm{B1}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{ &\\ &+\tfrac{1}{4}\,K_1(x',x',y)\,f^{abc}\,f^{deh}\,U^{db}(w')\,U^{lc}(y)\,U^{ae}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}\\ &+\tfrac{1}{4}\,K_1(x',x,x')\,f^{abc}\,f^{deh}\,U^{db}(w')\,U^{ec}(x')\,U^{al}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}\\ &-\tfrac{1}{4}\,N_c\,K_1(x',x,x)\,U^{ab}(w')\,U^{bc}(x)\,\big\{\rho^c(x),\rho^a(x')\big\}\\ &+\tfrac{1}{4}\,N_c\,K_1(x',x',y)\,U^{ab}(y)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(y)\big\}\\ &-\tfrac{1}{4}\,N_c\,K_1(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}\\ &+\tfrac{1}{4}\,N_c\,K_1(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\Big\}\end{aligned} \tag{B.24}$$

The kernel at the coincident points:

$$\begin{aligned}K_1(x',x',y) &= \mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\\ K_1(x',x,x') &= \mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\\ K_1(x',x,x) &= \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2}\end{aligned}$$

B.4 Three Wilson lines, kernel K_3

The second and third terms of the three-Wilson-line remainder share the kernel K_3; prefactor $+i\mathcal{N}$, no c.c.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\mathrm{B2}} = i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_3(x',x,y)\Big\{ &\\ &+\ f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(y)\,\rho^e(x')\,\rho^d(x)\,\rho^f(y)\\ &-\ f^{abc}\,U^{ec}(x')\,U^{bd}(x)\,U^{af}(y)\,\rho^e(x')\,\rho^d(x)\,\rho^f(y)\Big\}\end{aligned} \tag{B.25}$$

with $K_3(x',x,y)=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–2 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(y)\,\rho^e(x')\,\rho^d(x)\,\rho^f(y)$.

$$\begin{aligned}&[A,B]:\ [\rho^e(x'),\rho^d(x)] = if^{edh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ &\quad\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\,i\,K_3(x',x',y)\,f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(y)\,f^{edh}\,\big\{\rho^h(x'),\rho^f(y)\big\}\\ &\qquad\qquad = \tfrac{1}{4}\,i\,K_3(x',x',y)\,f^{abc}\,f^{deh}\,U^{dc}(x')\,U^{eb}(w)\,U^{al}(y)\,\big\{\rho^h(x'),\rho^l(y)\big\}\end{aligned} \tag{B.26}$$

$$\begin{aligned}&[A,C]:\ [\rho^e(x'),\rho^f(y)] = if^{efh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\,i\,K_3(x',x,x')\,f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(x')\,f^{efh}\,\big\{\rho^h(x'),\rho^d(x)\big\}\\ &\qquad\qquad = \tfrac{1}{4}\,iN_c\,K_3(x',x,x')\,U^{ab}(w)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(x)\big\}\quad (\mathrm{U2}^\star),\ (\mathrm{F})\end{aligned} \tag{B.27}$$

$$\begin{aligned}&[B,C]:\ [\rho^d(x),\rho^f(y)] = if^{dfh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x\\ &\quad\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\,i\,K_3(x',x,x)\,f^{abc}\,U^{ec}(x')\,U^{db}(w)\,U^{af}(x)\,f^{dfh}\,\big\{\rho^h(x),\rho^e(x')\big\}\\ &\qquad\qquad = \tfrac{1}{4}\,i\,K_3(x',x,x)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{db}(w)\,U^{ae}(x)\,\big\{\rho^h(x),\rho^l(x')\big\}\end{aligned} \tag{B.28}$$

Sub-term 2: $-\,f^{abc}\,U^{ec}(x')\,U^{bd}(x)\,U^{af}(y)\,\rho^e(x')\,\rho^d(x)\,\rho^f(y)$.

$$\begin{aligned}&[A,B]:\ [\rho^e(x'),\rho^d(x)] = if^{edh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ &\quad\tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\,i\,K_3(x',x',y)\,f^{abc}\,U^{ec}(x')\,U^{bd}(x')\,U^{af}(y)\,f^{edh}\,\big\{\rho^h(x'),\rho^f(y)\big\}\\ &\qquad\qquad = \tfrac{1}{4}\,iN_c\,K_3(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\quad (\mathrm{U2}^\star),\ (\mathrm{F})\end{aligned} \tag{B.29}$$

$$\begin{aligned}&[A,C]:\ [\rho^e(x'),\rho^f(y)] = if^{efh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\,i\,K_3(x',x,x')\,f^{abc}\,U^{ec}(x')\,U^{bd}(x)\,U^{af}(x')\,f^{efh}\,\big\{\rho^h(x'),\rho^d(x)\big\}\end{aligned}$$

$$= -\tfrac{1}{4}\, iN_c\; K_3(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(x)\big\} \quad \text{(U2}^\star\text{), (F)} \tag{B.30}$$

$$[B,C]:\ [\rho^d(x),\rho^f(y)] = if^{dfh}\rho^h(x)\,\delta^{(2)}(x-y), \quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K_3(x',x,x)\, f^{abc}\, U^{ec}(x')\, U^{bd}(x)\, U^{af}(x)\, f^{dfh}\, \big\{\rho^h(x),\rho^e(x')\big\}$$
$$= \tfrac{1}{4}\, iN_c\; K_3(x',x,x)\, U^{ab}(x')\, U^{bc}(x)\, \big\{\rho^c(x),\rho^a(x')\big\} \quad \text{(U2), (F)} \tag{B.31}$$

Two-ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm B2} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{$$
$$-\tfrac{1}{4}\,K_3(x',x',y)\,f^{abc}\,f^{deh}\,U^{dc}(x')\,U^{eb}(w)\,U^{al}(y)\,\big\{\rho^h(x'),\rho^l(y)\big\}$$
$$-\tfrac{1}{4}\,N_c\,K_3(x',x,x')\,U^{ab}(w)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(x)\big\}$$
$$-\tfrac{1}{4}\,K_3(x',x,x)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{db}(w)\,U^{ae}(x)\,\big\{\rho^h(x),\rho^l(x')\big\}$$
$$-\tfrac{1}{4}\,N_c\,K_3(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}$$
$$+\tfrac{1}{4}\,N_c\,K_3(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}$$
$$-\tfrac{1}{4}\,N_c\,K_3(x',x,x)\,U^{ab}(x')\,U^{bc}(x)\,\big\{\rho^c(x),\rho^a(x')\big\}\Big\} \tag{B.32}$$

The kernel at the coincident points:

$$K_3(x',x',y) = \mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$$
$$K_3(x',x,x') = \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x'-z)^2}$$
$$K_3(x',x,x) = \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$$

C Group I (with c.c.)

C.1 Group I, Row II (two-charge part of $\mathbb{A}^{(3)}$)

Input. In the notes,

$$\frac{d^3N_2}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4f^{abc}}{64\pi^6}\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2\mathbf{p}\,d^2\mathbf{k}\int_{x,y} e^{i\mathbf{k}\cdot(w-y)}e^{i\mathbf{p}\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,B^i(\mathbf{k},\mathbf{p})\,\log\frac{\vee}{\Lambda}$$
$$\times\Big[U^{ab'}(x')\rho^{b'}(x') - U^{ab'}(w')\rho^{b'}(x')\Big]\Big[U^{ab}(w)\{\rho^b(x),\rho^c(y)\} - U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\}\Big] + \text{c.c.},$$

$$B^i(\mathbf{k},\mathbf{p}) \equiv -\frac{\mathbf{k}^2\mathbf{p}^i - (\mathbf{k}\cdot\mathbf{p})\mathbf{k}^i}{(\mathbf{k}-\mathbf{p})^2} - 2\mathbf{k}^i - \frac{\mathbf{p}^2\mathbf{k}^i + \mathbf{k}^2\mathbf{p}^i}{(\mathbf{k}-\mathbf{p})^2}.$$

Two remarks before reordering.

1. *Colour.* As written, the indices a and b each appear three times in the first term of the second bracket: in f^{abc}, $U^{ab'}(x')$, $U^{ab}(w)$ and $\rho^b(x)$. This term is $U^{ab}(w)\mathbb{A}^{(3)b}$ with $\mathbb{A}^{(3)b}\propto f^{bcd}\{\rho^c(x),\rho^d(y)\}$, so we read it as $U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\}$. The second term, $f^{abc}U^{bd}(x)U^{ce}(y)\{\rho^d(x),\rho^e(y)\} = \bar{\mathbb{A}}^{(3)a}$, is unchanged.
2. *Kernel.* The Fourier transform is done in App. 4. With $\int d^2\mathbf{p}\,d^2\mathbf{k}\,e^{i\mathbf{k}\cdot(w-y)}e^{i\mathbf{p}\cdot(y-x)}B^i/(\mathbf{p}^2\mathbf{k}^2) = 2\pi i\,\mathfrak{F}^i(w;x,y)$, and since both colour structures are odd under $x\leftrightarrow y$, only the part of $\mathfrak{F}^i$ odd under $x\leftrightarrow y$ contributes. It is
$$\mathfrak{F}^i(w;x,y) = \mathcal{K}^i(x-w)\int_{z'}\big[K(w,y,z') - K(x,y,z')\big] - \mathcal{K}^i(y-w)\int_{z'}\big[K(w,x,z') - K(x,y,z')\big],$$
with $K(a,b,z') = \mathcal{K}(a-z')\cdot\mathcal{K}(b-z')$; equivalently $\mathfrak{F}^i(w;x,y) = -\pi\big[\mathcal{K}^i(x-w)\log\frac{(w-y)^2}{(x-y)^2} - \mathcal{K}^i(y-w)\log\frac{(w-x)^2}{(x-y)^2}\big]$. The prefactor becomes $-\frac{g^4}{64\pi^6}\cdot 2\pi i = -\frac{i}{2}\cdot\frac{g^4}{16\pi^5}$, i.e. $-\frac{i}{2}\mathcal{N}$.

The three-ρ term.

$$\left.\frac{d^3N}{d^3k}\right|_{\rm GI.II} = -\tfrac{1}{2}i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,G_{\rm I.II}(x',x,y)\Big\{$$
$$+\,U^{ab'}(x')\,f^{hbc}\,U^{ah}(w)\,\rho^{b'}(x')\,\big\{\rho^b(x),\rho^c(y)\big\}$$
$$-\,U^{ab'}(x')\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\big\{\rho^d(x),\rho^e(y)\big\}$$
$$-\,U^{ab'}(w')\,f^{hbc}\,U^{ah}(w)\,\rho^{b'}(x')\,\big\{\rho^b(x),\rho^c(y)\big\}$$
$$+\,U^{ab'}(w')\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\big\{\rho^d(x),\rho^e(y)\big\}\Big\} + \text{c.c.} \tag{C.1}$$

with $G_{\rm I.II}(x',x,y) = \mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,y)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+U^{ab'}(x')\,f^{hbc}\,U^{ah}(w)\,\rho^{b'}(x')\,\{\rho^b(x),\rho^c(y)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^b(x)] &= if^{b'bl}\rho^l(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,U^{ab'}(x')\,f^{hbc}\,U^{ah}(w)\,f^{b'bl}\,\{\rho^l(x'),\rho^c(y)\}\\
&= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\{\rho^e(x'),\rho^c(y)\} \qquad \text{(C.2)}\\
[A,C]:\ [\rho^{b'}(x'),\rho^c(y)] &= if^{b'cl}\rho^l(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,U^{ab'}(x')\,f^{hbc}\,U^{ah}(w)\,f^{b'cl}\,\{\rho^l(x'),\rho^b(x)\}\\
&= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\,\{\rho^e(x'),\rho^b(x)\} \qquad \text{(C.3)}
\end{aligned}
$$

Sub-term 2: $-U^{ab'}(x')\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\{\rho^d(x),\rho^e(y)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^d(x)] &= if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,U^{ab'}(x')\,f^{abc}\,U^{bd}(x')\,U^{ce}(y)\,f^{b'dh}\,\{\rho^h(x'),\rho^e(y)\}\\
&= -\tfrac{1}{2}\,iN_c\,G_{\mathrm{I.II}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}\quad \text{(U2), (F)} \qquad \text{(C.4)}\\
[A,C]:\ [\rho^{b'}(x'),\rho^e(y)] &= if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,U^{ab'}(x')\,f^{abc}\,U^{bd}(x)\,U^{ce}(x')\,f^{b'eh}\,\{\rho^h(x'),\rho^d(x)\}\\
&= \tfrac{1}{2}\,iN_c\,G_{\mathrm{I.II}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(x)\}\quad \text{(U2), (F)} \qquad \text{(C.5)}
\end{aligned}
$$

Sub-term 3: $-U^{ab'}(w')\,f^{hbc}\,U^{ah}(w)\,\rho^{b'}(x')\,\{\rho^b(x),\rho^c(y)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^b(x)] &= if^{b'bl}\rho^l(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,U^{ab'}(w')\,f^{hbc}\,U^{ah}(w)\,f^{b'bl}\,\{\rho^l(x'),\rho^c(y)\}\\
&= -\tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,f^{abc}\,f^{dbe}\,U^{hd}(w')\,U^{ha}(w)\,\{\rho^e(x'),\rho^c(y)\} \qquad \text{(C.6)}\\
[A,C]:\ [\rho^{b'}(x'),\rho^c(y)] &= if^{b'cl}\rho^l(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,U^{ab'}(w')\,f^{hbc}\,U^{ah}(w)\,f^{b'cl}\,\{\rho^l(x'),\rho^b(x)\}\\
&= -\tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,f^{abc}\,f^{dce}\,U^{hd}(w')\,U^{ha}(w)\,\{\rho^e(x'),\rho^b(x)\} \qquad \text{(C.7)}
\end{aligned}
$$

Sub-term 4: $+U^{ab'}(w')\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\rho^{b'}(x')\,\{\rho^d(x),\rho^e(y)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^d(x)] &= if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,U^{ab'}(w')\,f^{abc}\,U^{bd}(x')\,U^{ce}(y)\,f^{b'dh}\,\{\rho^h(x'),\rho^e(y)\}\\
&= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x')\,U^{cl}(y)\,\{\rho^h(x'),\rho^l(y)\} \qquad \text{(C.8)}\\
[A,C]:\ [\rho^{b'}(x'),\rho^e(y)] &= if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,U^{ab'}(w')\,f^{abc}\,U^{bd}(x)\,U^{ce}(x')\,f^{b'eh}\,\{\rho^h(x'),\rho^d(x)\}\\
&= \tfrac{1}{2}\,i\,G_{\mathrm{I.II}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(x')\,\{\rho^h(x'),\rho^l(x)\} \qquad \text{(C.9)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-\frac{1}{2}i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\mathrm{GI.II}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{&\\
&+\tfrac{1}{4}\,G_{\mathrm{I.II}}(x',x',y)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\{\rho^e(x'),\rho^c(y)\}\\
&+\tfrac{1}{4}\,G_{\mathrm{I.II}}(x',x,x')\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\,\{\rho^e(x'),\rho^b(x)\}\\
&-\tfrac{1}{4}\,N_c\,G_{\mathrm{I.II}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}\\
&+\tfrac{1}{4}\,N_c\,G_{\mathrm{I.II}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(x)\}\\
&-\tfrac{1}{4}\,G_{\mathrm{I.II}}(x',x',y)\,f^{abc}\,f^{dbe}\,U^{hd}(w')\,U^{ha}(w)\,\{\rho^e(x'),\rho^c(y)\}\\
&-\tfrac{1}{4}\,G_{\mathrm{I.II}}(x',x,x')\,f^{abc}\,f^{dce}\,U^{hd}(w')\,U^{ha}(w)\,\{\rho^e(x'),\rho^b(x)\}
\end{aligned}
$$

$$+\tfrac{1}{4}\,G_{\rm I.II}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x')\,U^{cl}(y)\,\{\rho^h(x'),\rho^l(y)\}$$
$$+\tfrac{1}{4}\,G_{\rm I.II}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(x')\,\{\rho^h(x'),\rho^l(x)\}\Big\}+(w'\leftrightarrow w) \tag{C.10}$$

The kernel at the coincident points:

$$G_{\rm I.II}(x',x',y)=\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x',y)$$
$$G_{\rm I.II}(x',x,x')=\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,x')$$

C.2 Group I, Row III ($\bar{\mathbb{B}}_3^\dagger\mathbb{A}$)

Input. Part 1 of the notes has three terms with the same colour structure. The first two (red) are equal and opposite and cancel. The third has prefactor $\frac{ig^4f^{abd}}{16\pi^5}=i\mathcal{N}f^{abd}$ and kernel $K^{(1)}=\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)$. Part 2 has five terms, again with the same colour structure and prefactor $\pm\frac{ig^4f^{abd}}{8\pi^5}=\pm2i\mathcal{N}f^{abd}$; its terms 1 and 5 are equal, and so are terms 2 and 4. Together,

$$\text{Row III}=i\mathcal{N}\,f^{abd}\int K_{\rm I.III}(x',x,y)\,[\dots][\dots]+\text{c.c.},\qquad K_{\rm I.III}=K^{(1)}+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x),$$
$$\Phi(x)\equiv\frac{1}{(z-w)^2}\left[\frac{2\,(z-w)\cdot(x-z)\,(x-w)^2-2\,(z-w)\cdot(x-w)\,(x-z)^2}{\big[(x-w)^2-(x-z)^2\big]^2}-1\right].$$

The three-ρ term.

$$\left.\frac{d^3N}{d^3k}\right|_{\rm GI.III}=i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\rm I.III}(x',x,y)\,\Big\{$$
$$+\;f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$$
$$-\;f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$$
$$-\;f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$$
$$+\;f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)\Big\}+\text{c.c.} \tag{C.11}$$

with $K_{\rm I.III}(x',x,y)=\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$.

$$[A,B]:\;[\rho^{b'}(x'),\rho^e(x)]=if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$
$$\tfrac{1}{4}\{[A,B],C\}=\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x',y)\,f^{abd}\,U^{ab'}(x')\,U^{de}(x')\,U^{bc}(z)\,f^{b'eh}\,\{\rho^h(x'),\rho^c(y)\}$$
$$=-\tfrac{1}{4}\,iN_c\;K_{\rm I.III}(x',x',y)\,U^{ab}(z)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}\quad\text{(U2), (F)} \tag{C.12}$$
$$[A,C]:\;[\rho^{b'}(x'),\rho^c(y)]=if^{b'ch}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'$$
$$\tfrac{1}{4}\{[A,C],B\}=\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x,x')\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,f^{b'ch}\,\{\rho^h(x'),\rho^e(x)\}$$
$$=\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(z)\,\{\rho^h(x'),\rho^l(x)\} \tag{C.13}$$
$$[B,C]:\;[\rho^e(x),\rho^c(y)]=if^{ech}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x$$
$$\tfrac{1}{4}\{[B,C],A\}=\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x,x)\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,f^{ech}\,\{\rho^h(x),\rho^{b'}(x')\}$$
$$=\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(x)\,U^{be}(z)\,\{\rho^h(x),\rho^l(x')\} \tag{C.14}$$

Sub-term 2: $-\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^e(x)\,\rho^c(y)$.

$$[A,B]:\;[\rho^{b'}(x'),\rho^e(x)]=if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$
$$\tfrac{1}{4}\{[A,B],C\}=-\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x',y)\,f^{abd}\,U^{ab'}(x')\,U^{de}(x')\,U^{bc}(y)\,f^{b'eh}\,\{\rho^h(x'),\rho^c(y)\}$$
$$=\tfrac{1}{4}\,iN_c\;K_{\rm I.III}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}\quad\text{(U2), (F)} \tag{C.15}$$
$$[A,C]:\;[\rho^{b'}(x'),\rho^c(y)]=if^{b'ch}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'$$
$$\tfrac{1}{4}\{[A,C],B\}=-\tfrac{1}{4}\,i\;K_{\rm I.III}(x',x,x')\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(x')\,f^{b'ch}\,\{\rho^h(x'),\rho^e(x)\}$$

$$= -\tfrac{1}{4}\, iN_c\; K_{\mathrm{I.III}}(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(x)\big\} \quad \text{(U2), (F)} \tag{C.16}$$

$$[B,C]:\ [\rho^e(x),\rho^c(y)] = if^{ech}\rho^h(x)\,\delta^{(2)}(x-y), \quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(x)\, f^{ech}\, \big\{\rho^h(x),\rho^{b'}(x')\big\}$$

$$= \tfrac{1}{4}\, iN_c\; K_{\mathrm{I.III}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \big\{\rho^c(x),\rho^b(x')\big\} \quad \text{(U2), (F)} \tag{C.17}$$

Sub-term 3: $-\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(z)\, \rho^{b'}(x')\, \rho^e(x)\, \rho^c(y).$

$$[A,B]:\ [\rho^{b'}(x'),\rho^e(x)] = if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-x), \quad x\to x'$$

$$\tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x',y)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x')\, U^{bc}(z)\, f^{b'eh}\, \big\{\rho^h(x'),\rho^c(y)\big\}$$

$$= -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(z)\, \big\{\rho^h(x'),\rho^l(y)\big\} \tag{C.18}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^c(y)] = if^{b'ch}\rho^h(x')\,\delta^{(2)}(x'-y), \quad y\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x')\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(z)\, f^{b'ch}\, \big\{\rho^h(x'),\rho^e(x)\big\}$$

$$= -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(z)\, \big\{\rho^h(x'),\rho^l(x)\big\} \tag{C.19}$$

$$[B,C]:\ [\rho^e(x),\rho^c(y)] = if^{ech}\rho^h(x)\,\delta^{(2)}(x-y), \quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(z)\, f^{ech}\, \big\{\rho^h(x),\rho^{b'}(x')\big\}$$

$$= -\tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{cd}(x)\, U^{be}(z)\, \big\{\rho^h(x),\rho^l(x')\big\} \tag{C.20}$$

Sub-term 4: $+\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(y)\, \rho^{b'}(x')\, \rho^e(x)\, \rho^c(y).$

$$[A,B]:\ [\rho^{b'}(x'),\rho^e(x)] = if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-x), \quad x\to x'$$

$$\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x',y)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x')\, U^{bc}(y)\, f^{b'eh}\, \big\{\rho^h(x'),\rho^c(y)\big\}$$

$$= \tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(y)\, \big\{\rho^h(x'),\rho^l(y)\big\} \tag{C.21}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^c(y)] = if^{b'ch}\rho^h(x')\,\delta^{(2)}(x'-y), \quad y\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x')\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(x')\, f^{b'ch}\, \big\{\rho^h(x'),\rho^e(x)\big\}$$

$$= \tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(x')\, \big\{\rho^h(x'),\rho^l(x)\big\} \tag{C.22}$$

$$[B,C]:\ [\rho^e(x),\rho^c(y)] = if^{ech}\rho^h(x)\,\delta^{(2)}(x-y), \quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(x)\, f^{ech}\, \big\{\rho^h(x),\rho^{b'}(x')\big\}$$

$$= -\tfrac{1}{4}\, iN_c\; K_{\mathrm{I.III}}(x',x,x)\, U^{ab}(w')\, U^{ac}(x)\, \big\{\rho^c(x),\rho^b(x')\big\} \quad \text{(U2), (F)} \tag{C.23}$$

Two-ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
\frac{d^3N^{(2\rho)}}{d^3k}\bigg|_{\mathrm{GI.III}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{ &\\
&+\tfrac{1}{4}\,N_c\,K_{\mathrm{I.III}}(x',x',y)\,U^{ab}(z)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\\
&-\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(z)\,\big\{\rho^h(x'),\rho^l(x)\big\}\\
&-\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(x)\,U^{be}(z)\,\big\{\rho^h(x),\rho^l(x')\big\}\\
&-\tfrac{1}{4}\,N_c\,K_{\mathrm{I.III}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\\
&+\tfrac{1}{4}\,N_c\,K_{\mathrm{I.III}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}\\
&-\tfrac{1}{4}\,N_c\,K_{\mathrm{I.III}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\\
&+\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\
&+\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(x)\,U^{be}(z)\,\big\{\rho^h(x'),\rho^l(x)\big\}\\
&+\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(x)\,U^{be}(z)\,\big\{\rho^h(x),\rho^l(x')\big\}\\
&-\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\
&-\tfrac{1}{4}\,K_{\mathrm{I.III}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(x)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(x)\big\}\\
&+\tfrac{1}{4}\,N_c\,K_{\mathrm{I.III}}(x',x,x)\,U^{ab}(w')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\Big\} + \big(w'\leftrightarrow w\big)
\end{aligned} \tag{C.24}$$

The kernel at the coincident points:

$$K_{\mathrm{I.III}}(x',x',y) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x')\Big]$$

$$K_{\mathrm{I.III}}(x',x,x') = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\Phi(x)\Big]$$
$$K_{\mathrm{I.III}}(x',x,x) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-z)\,\Phi(x)\Big]$$

C.3 Group I, Row IV ($\bar{\mathbb{A}}^\dagger\mathbb{B}_2$)

Input. Prefactor $\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = 2i\mathcal{N}$. The kernel is the sum of the two kernels of the notes, $K_{\mathfrak{B}} \equiv \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \frac{1}{2}\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\mathrm{GI.IV}} = 2i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{ &\\ +\ U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{c'}(x) &\\ -\ U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{d}(x) &\\ -\ U^{ab'}(w')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{c'}(x) &\\ +\ U^{ab'}(w')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{d}(x)\Big\} + \text{c.c.}\end{aligned} \tag{C.25}$$

with $K_{\mathfrak{B}}(x',x,y) = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{c'}(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x'\\ \tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(x')\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{c'}(x)\}\\ &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{eb}(z)\,U^{hl}(x')\,\{\rho^{l}(x'),\rho^{c}(x)\}\quad \text{(U2)}\end{aligned} \tag{C.26}$$

$$\begin{aligned}[A,C]:\ [\rho^{b'}(x'),\rho^{c'}(x)] &= if^{b'c'h}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x'\\ \tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,f^{b'c'h}\,\{\rho^{h}(x'),\rho^{e}(y)\}\\ &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(y)\,\{\rho^{e}(x'),\rho^{m}(y)\}\end{aligned} \tag{C.27}$$

$$\begin{aligned}[B,C]:\ [\rho^{e}(y),\rho^{c'}(x)] &= if^{ec'h}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y\\ \tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,f^{ec'h}\,\{\rho^{h}(y),\rho^{b'}(x')\}\\ &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(y)\,\{\rho^{e}(y),\rho^{l}(x')\}\end{aligned} \tag{C.28}$$

Sub-term 2: $-\,U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{d}(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x'\\ \tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,f^{abc}\,U^{be}(x')\,U^{cd}(x)\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{d}(x)\}\\ &= -\tfrac{1}{4}\,iN_c\,K_{\mathfrak{B}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\}\quad \text{(U2), (F)}\end{aligned} \tag{C.29}$$

$$\begin{aligned}[A,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] &= if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x'\\ \tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x')\,f^{b'dh}\,\{\rho^{h}(x'),\rho^{e}(y)\}\\ &= \tfrac{1}{4}\,iN_c\,K_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\quad \text{(U2), (F)}\end{aligned} \tag{C.30}$$

$$\begin{aligned}[B,C]:\ [\rho^{e}(y),\rho^{d}(x)] &= if^{edh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y\\ \tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(y)\,f^{edh}\,\{\rho^{h}(y),\rho^{b'}(x')\}\\ &= -\tfrac{1}{4}\,iN_c\,K_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\{\rho^{c}(y),\rho^{b}(x')\}\quad \text{(U2), (F)}\end{aligned} \tag{C.31}$$

Sub-term 3: $-\,U^{ab'}(w')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^{e}(y)\,\rho^{c'}(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x'\\ \tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',x,x')\,U^{ab'}(w')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(x')\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{c'}(x)\}\end{aligned}$$

$$= -\tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ld}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{me}(x')\, \big\{\rho^h(x'),\rho^c(x)\big\} \tag{C.32}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^{c'}(x)] = if^{b'c'h}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x',y)\, U^{ab'}(w')\, f^{cdc'}\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, f^{b'c'h}\, \big\{\rho^h(x'),\rho^e(y)\big\}$$

$$= -\tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'),\rho^m(y)\big\} \tag{C.33}$$

$$[B,C]:\ [\rho^{e}(y),\rho^{c'}(x)] = if^{ec'h}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y$$

$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',y,y)\, U^{ab'}(w')\, f^{cdc'}\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, f^{ec'h}\, \big\{\rho^h(y),\rho^{b'}(x')\big\}$$

$$= -\tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y),\rho^l(x')\big\} \tag{C.34}$$

Sub-term 4: $+U^{ab'}(w')\, f^{abc}\, U^{be}(y)\, U^{cd}(x)\, \rho^{b'}(x')\, \rho^e(y)\, \rho^d(x)$.

$$[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)] = if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'$$

$$\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x,x')\, U^{ab'}(w')\, f^{abc}\, U^{be}(x')\, U^{cd}(x)\, f^{b'eh}\, \big\{\rho^h(x'),\rho^d(x)\big\}$$

$$= \tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x')\, U^{cl}(x)\, \big\{\rho^h(x'),\rho^l(x)\big\} \tag{C.35}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] = if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x',y)\, U^{ab'}(w')\, f^{abc}\, U^{be}(y)\, U^{cd}(x')\, f^{b'dh}\, \big\{\rho^h(x'),\rho^e(y)\big\}$$

$$= \tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(y)\, U^{ce}(x')\, \big\{\rho^h(x'),\rho^l(y)\big\} \tag{C.36}$$

$$[B,C]:\ [\rho^{e}(y),\rho^{d}(x)] = if^{edh}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y$$

$$\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K_{\mathfrak{B}}(x',y,y)\, U^{ab'}(w')\, f^{abc}\, U^{be}(y)\, U^{cd}(y)\, f^{edh}\, \big\{\rho^h(y),\rho^{b'}(x')\big\}$$

$$= \tfrac{1}{4}\, iN_c\; K_{\mathfrak{B}}(x',y,y)\, U^{ab}(w')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\}\quad \text{(U2), (F)} \tag{C.37}$$

Two-ρ terms of this source. Multiplying by the prefactor $2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GI.IV}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{ &\\
&-\tfrac{1}{2}\, K_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{da}(w)\, U^{eb}(z)\, U^{hl}(x')\, \big\{\rho^l(x'),\rho^c(x)\big\}\\
&-\tfrac{1}{2}\, K_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'),\rho^m(y)\big\}\\
&-\tfrac{1}{2}\, K_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y),\rho^l(x')\big\}\\
&+\tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(x)\big\}\\
&-\tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(y)\big\}\\
&+\tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',y,y)\, U^{ab}(x')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\}\\
&+\tfrac{1}{2}\, K_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ld}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{me}(x')\, \big\{\rho^h(x'),\rho^c(x)\big\}\\
&+\tfrac{1}{2}\, K_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'),\rho^m(y)\big\}\\
&+\tfrac{1}{2}\, K_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y),\rho^l(x')\big\}\\
&-\tfrac{1}{2}\, K_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x')\, U^{cl}(x)\, \big\{\rho^h(x'),\rho^l(x)\big\}\\
&-\tfrac{1}{2}\, K_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(y)\, U^{ce}(x')\, \big\{\rho^h(x'),\rho^l(y)\big\}\\
&-\tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',y,y)\, U^{ab}(w')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\}\Big\} + \big(w'\leftrightarrow w\big)
\end{aligned} \tag{C.38}$$

The kernel at the coincident points:

$$K_{\mathfrak{B}}(x',x,x') = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\right]$$

$$K_{\mathfrak{B}}(x',x',y) = \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\right]$$

$$K_{\mathfrak{B}}(x',y,y) = \mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\right]$$

C.4 Group I, Row V, first term (measured gluon at z)

Input. Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4}{32\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = -\frac{i}{2}\mathcal{N}$. After the w integration with the δ-function of $\mathbb{C}_1$ the measured gluon in the amplitude sits at z: the phase is $e^{-ik\cdot(w'-z)}$. The charges sit at x', y and y'.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GI.V1}} = & -\tfrac{1}{2}i\mathcal{N}\int e^{-ik\cdot(w'-z)}\,K_{\text{V1}}(x',y,y')\,\Big\{\\ & +\,U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^{e}(y),\rho^{b}(y')\big\}\\ & -\,U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^{d}(y),\rho^{e}(y')\big\}\\ & -\,U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^{e}(y),\rho^{b}(y')\big\}\\ & +\,U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^{d}(y),\rho^{e}(y')\big\}\Big\}+\text{c.c.}\end{aligned}\tag{C.39}$$

with $K_{\text{V1}}(x',y,y')=\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x', y, y'. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^{e}(y),\rho^{b}(y')\big\}$.

$$\begin{aligned}&[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)]=if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{2}\{[A,B],C\}=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'eh}\,\big\{\rho^{h}(x'),\rho^{b}(y')\big\}\\ &\qquad=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(x)\,U^{ce}(z)\,\big\{\rho^{h}(x'),\rho^{l}(y')\big\}\end{aligned}\tag{C.40}$$

$$\begin{aligned}&[A,C]:\ [\rho^{b'}(x'),\rho^{b}(y')]=if^{b'bh}\rho^{h}(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\ &\quad\tfrac{1}{2}\{[A,C],B\}=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'bh}\,\big\{\rho^{h}(x'),\rho^{e}(y)\big\}\\ &\qquad=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\}\end{aligned}\tag{C.41}$$

Sub-term 2: $-U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^{d}(y),\rho^{e}(y')\big\}$.

$$\begin{aligned}&[A,B]:\ [\rho^{b'}(x'),\rho^{d}(y)]=if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{2}\{[A,B],C\}=-\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,U^{ab'}(x')\,f^{abc}\,U^{cd}(x')\,U^{be}(y')\,f^{b'dh}\,\big\{\rho^{h}(x'),\rho^{e}(y')\big\}\\ &\qquad=\tfrac{1}{2}\,iN_c\,K_{\text{V1}}(x',x',y')\,U^{ab}(y')\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(y')\big\}\quad\text{(U2), (F)}\end{aligned}\tag{C.42}$$

$$\begin{aligned}&[A,C]:\ [\rho^{b'}(x'),\rho^{e}(y')]=if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\ &\quad\tfrac{1}{2}\{[A,C],B\}=-\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(x')\,f^{b'eh}\,\big\{\rho^{h}(x'),\rho^{d}(y)\big\}\\ &\qquad=-\tfrac{1}{2}\,iN_c\,K_{\text{V1}}(x',y,x')\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(y)\big\}\quad\text{(U2), (F)}\end{aligned}\tag{C.43}$$

Sub-term 3: $-U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^{e}(y),\rho^{b}(y')\big\}$.

$$\begin{aligned}&[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)]=if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{2}\{[A,B],C\}=-\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'eh}\,\big\{\rho^{h}(x'),\rho^{b}(y')\big\}\\ &\qquad=-\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(z)\,\big\{\rho^{h}(x'),\rho^{l}(y')\big\}\end{aligned}\tag{C.44}$$

$$\begin{aligned}&[A,C]:\ [\rho^{b'}(x'),\rho^{b}(y')]=if^{b'bh}\rho^{h}(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\ &\quad\tfrac{1}{2}\{[A,C],B\}=-\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'bh}\,\big\{\rho^{h}(x'),\rho^{e}(y)\big\}\\ &\qquad=-\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\}\end{aligned}\tag{C.45}$$

Sub-term 4: $+U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^{d}(y),\rho^{e}(y')\big\}$.

$$\begin{aligned}&[A,B]:\ [\rho^{b'}(x'),\rho^{d}(y)]=if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ &\quad\tfrac{1}{2}\{[A,B],C\}=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,U^{ab'}(w')\,f^{abc}\,U^{cd}(x')\,U^{be}(y')\,f^{b'dh}\,\big\{\rho^{h}(x'),\rho^{e}(y')\big\}\\ &\qquad=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y')\,\big\{\rho^{h}(x'),\rho^{l}(y')\big\}\end{aligned}\tag{C.46}$$

$$\begin{aligned}&[A,C]:\ [\rho^{b'}(x'),\rho^{e}(y')]=if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\ &\quad\tfrac{1}{2}\{[A,C],B\}=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(x')\,f^{b'eh}\,\big\{\rho^{h}(x'),\rho^{d}(y)\big\}\\ &\qquad=\tfrac{1}{2}\,i\,K_{\text{V1}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\}\end{aligned}\tag{C.47}$$

Two-ρ terms of this source. Multiplying by the prefactor $-\frac{1}{2}i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GI.V1}} = \mathcal{N}\int e^{-ik\cdot(w'-z)}\Big\{ &\\ &+\tfrac{1}{4}\,K_{\text{V1}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(x)\,U^{ce}(z)\,\big\{\rho^h(x'),\rho^l(y')\big\}\\ &+\tfrac{1}{4}\,K_{\text{V1}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\ &+\tfrac{1}{4}\,N_c\,K_{\text{V1}}(x',x',y')\,U^{ab}(y')\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y')\big\}\\ &-\tfrac{1}{4}\,N_c\,K_{\text{V1}}(x',y,x')\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\\ &-\tfrac{1}{4}\,K_{\text{V1}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(z)\,\big\{\rho^h(x'),\rho^l(y')\big\}\\ &-\tfrac{1}{4}\,K_{\text{V1}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\ &+\tfrac{1}{4}\,K_{\text{V1}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y')\,\big\{\rho^h(x'),\rho^l(y')\big\}\\ &+\tfrac{1}{4}\,K_{\text{V1}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}\Big\}+(w'\leftrightarrow z)\end{aligned}\tag{C.48}$$

The kernel at the coincident points:

$$\begin{aligned}K_{\text{V1}}(x',x',y') &= \mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x)\\ K_{\text{V1}}(x',y,x') &= \mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-x)\end{aligned}$$

C.5 Group I, Row V, second term (measured gluon at x)

Input. Prefactor $-\frac{i}{2}\mathcal{N}$; the measured gluon in the amplitude sits at x (phase $e^{-ik\cdot(w'-x)}$). Same colour structure as the first term.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GI.V2}} = &-\tfrac{1}{2}i\,\mathcal{N}\int e^{-ik\cdot(w'-x)}\,K_{\text{V2}}(x',y,y')\Big\{\\ &+U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}\\ &-U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}\\ &-U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}\\ &+U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}\Big\}+\text{c.c.}\end{aligned}\tag{C.49}$$

with $K_{\text{V2}}(x',y,y')=\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x', y, y'. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\big\{\rho^e(y),\rho^b(y')\big\}$.

$$\begin{aligned}[A,B]:\ [\rho^{b'}(x'),\rho^e(y)] &= if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ \tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\,K_{\text{V2}}(x',x',y')\,U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'eh}\,\big\{\rho^h(x'),\rho^b(y')\big\}\\ &= \tfrac{1}{2}\,i\,K_{\text{V2}}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(x)\,U^{ce}(z)\,\big\{\rho^h(x'),\rho^l(y')\big\}\end{aligned}\tag{C.50}$$

$$\begin{aligned}[A,C]:\ [\rho^{b'}(x'),\rho^b(y')] &= if^{b'bh}\rho^h(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\ \tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\,K_{\text{V2}}(x',y,x')\,U^{ab'}(x')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'bh}\,\big\{\rho^h(x'),\rho^e(y)\big\}\\ &= \tfrac{1}{2}\,i\,K_{\text{V2}}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\end{aligned}\tag{C.51}$$

Sub-term 2: $-U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\big\{\rho^d(y),\rho^e(y')\big\}$.

$$\begin{aligned}[A,B]:\ [\rho^{b'}(x'),\rho^d(y)] &= if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ \tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\,K_{\text{V2}}(x',x',y')\,U^{ab'}(x')\,f^{abc}\,U^{cd}(x')\,U^{be}(y')\,f^{b'dh}\,\big\{\rho^h(x'),\rho^e(y')\big\}\\ &= \tfrac{1}{2}\,iN_c\,K_{\text{V2}}(x',x',y')\,U^{ab}(y')\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y')\big\}\quad\text{(U2), (F)}\end{aligned}\tag{C.52}$$

$$\begin{aligned}[A,C]:\ [\rho^{b'}(x'),\rho^e(y')] &= if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\ \tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\,K_{\text{V2}}(x',y,x')\,U^{ab'}(x')\,f^{abc}\,U^{cd}(y)\,U^{be}(x')\,f^{b'eh}\,\big\{\rho^h(x'),\rho^d(y)\big\}\\ &= -\tfrac{1}{2}\,iN_c\,K_{\text{V2}}(x',y,x')\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\quad\text{(U2), (F)}\end{aligned}\tag{C.53}$$

Sub-term 3: $-U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\rho^{b'}(x')\,\{\rho^{e}(y),\rho^{b}(y')\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^{e}(y)] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\;K_{\rm V2}(x',x',y')\,U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{b}(y')\}\\
&= -\tfrac{1}{2}\,i\;K_{\rm V2}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(z)\,\{\rho^{h}(x'),\rho^{l}(y')\} \qquad \text{(C.54)}\\
[A,C]:\ [\rho^{b'}(x'),\rho^{b}(y')] &= if^{b'bh}\rho^{h}(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\;K_{\rm V2}(x',y,x')\,U^{ab'}(w')\,f^{adc}\,U^{db}(x)\,U^{ce}(z)\,f^{b'bh}\,\{\rho^{h}(x'),\rho^{e}(y)\}\\
&= -\tfrac{1}{2}\,i\;K_{\rm V2}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x)\,U^{cl}(z)\,\{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(C.55)}
\end{aligned}
$$

Sub-term 4: $+U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\rho^{b'}(x')\,\{\rho^{d}(y),\rho^{e}(y')\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^{d}(y)] &= if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\;K_{\rm V2}(x',x',y')\,U^{ab'}(w')\,f^{abc}\,U^{cd}(x')\,U^{be}(y')\,f^{b'dh}\,\{\rho^{h}(x'),\rho^{e}(y')\}\\
&= \tfrac{1}{2}\,i\;K_{\rm V2}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y')\,\{\rho^{h}(x'),\rho^{l}(y')\} \qquad \text{(C.56)}\\
[A,C]:\ [\rho^{b'}(x'),\rho^{e}(y')] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-y'),\quad y'\to x'\\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\;K_{\rm V2}(x',y,x')\,U^{ab'}(w')\,f^{abc}\,U^{cd}(y)\,U^{be}(x')\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\}\\
&= \tfrac{1}{2}\,i\;K_{\rm V2}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(y)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(C.57)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-\frac{1}{2}i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm GI.V2} = \mathcal{N}\int e^{-ik\cdot(w'-x)}\,\Big\{&\\
&+\tfrac{1}{4}\,K_{\rm V2}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(x)\,U^{ce}(z)\,\{\rho^{h}(x'),\rho^{l}(y')\}\\
&+\tfrac{1}{4}\,K_{\rm V2}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{be}(x)\,U^{cl}(z)\,\{\rho^{h}(x'),\rho^{l}(y)\}\\
&+\tfrac{1}{4}\,N_c\,K_{\rm V2}(x',x',y')\,U^{ab}(y')\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y')\}\\
&-\tfrac{1}{4}\,N_c\,K_{\rm V2}(x',y,x')\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\\
&-\tfrac{1}{4}\,K_{\rm V2}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(z)\,\{\rho^{h}(x'),\rho^{l}(y')\}\\
&-\tfrac{1}{4}\,K_{\rm V2}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x)\,U^{cl}(z)\,\{\rho^{h}(x'),\rho^{l}(y)\}\\
&+\tfrac{1}{4}\,K_{\rm V2}(x',x',y')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y')\,\{\rho^{h}(x'),\rho^{l}(y')\}\\
&+\tfrac{1}{4}\,K_{\rm V2}(x',y,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(y)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\}\Big\}+\big(w'\leftrightarrow x\big) \qquad \text{(C.58)}
\end{aligned}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K_{\rm V2}(x',x',y') &= \mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-z)\\
K_{\rm V2}(x',y,x') &= \mathcal{K}(x'-w')\cdot\mathcal{K}(x'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)
\end{aligned}
$$

D Group II (no c.c.)

D.1 Group II, Row I, first term

Input. The prefactor is $-\frac{1}{(2\pi)^3}\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=-i\mathcal{N}$ and the kernel is $K_{\mathfrak{B}}$, as in Row IV of Group I. The charges appear as $\{\rho(x'),\rho(y)\}\rho(x)$. Group II has no c.c.; the second term of the row is its conjugate.

The three-ρ term.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\rm GII.I1} = -\,i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{&\\
&+\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\{\rho^{c'}(x'),\rho^{b}(y)\}\,\rho^{d}(x)\\
&-\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\{\rho^{c'}(x'),\rho^{b}(y)\}\,\rho^{e}(x)\\
&-\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\,\rho^{d}(x)\\
&+\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\,\rho^{e}(x)\Big\} \qquad \text{(D.1)}
\end{aligned}
$$

with $K_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\frac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.7) with A, B, C the charges at x, x', y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\{\rho^{c'}(x'),\rho^{b}(y)\}\,\rho^{d}(x)$.

$$
\begin{aligned}
[A,B]:\ & [\rho^{d}(x),\rho^{c'}(x')] = i f^{dc'h}\rho^{h}(x)\,\delta^{(2)}(x-x'), \quad x'\to x\\
&-\tfrac{1}{2}\{[A,B],C\} = -\tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{dc'h}\,\{\rho^{h}(x),\rho^{b}(y)\}\\
&\qquad = -\tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\{\rho^{e}(x),\rho^{b}(y)\} \qquad (D.2)\\
[A,C]:\ & [\rho^{d}(x),\rho^{b}(y)] = i f^{dbh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x\\
&-\tfrac{1}{2}\{[A,C],B\} = -\tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x',x,x)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{dbh}\,\{\rho^{h}(x),\rho^{c'}(x')\}\\
&\qquad = \tfrac{1}{2}\,iN_c\;K_{\mathfrak{B}}(x',x,x)\,U^{ab}(w')\,U^{ac}(w)\,\{\rho^{c}(x),\rho^{b}(x')\} \quad \text{(F)} \qquad (D.3)
\end{aligned}
$$

Sub-term 2: $-\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\{\rho^{c'}(x'),\rho^{b}(y)\}\,\rho^{e}(x)$.

$$
\begin{aligned}
[A,B]:\ & [\rho^{e}(x),\rho^{c'}(x')] = i f^{ec'h}\rho^{h}(x)\,\delta^{(2)}(x-x'), \quad x'\to x\\
&-\tfrac{1}{2}\{[A,B],C\} = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ec'h}\,\{\rho^{h}(x),\rho^{b}(y)\}\\
&\qquad = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x)\,\{\rho^{h}(x),\rho^{l}(y)\} \qquad (D.4)\\
[A,C]:\ & [\rho^{e}(x),\rho^{b}(y)] = i f^{ebh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x\\
&-\tfrac{1}{2}\{[A,C],B\} = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x',x,x)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ebh}\,\{\rho^{h}(x),\rho^{c'}(x')\}\\
&\qquad = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\{\rho^{h}(x),\rho^{l}(x')\} \qquad (D.5)
\end{aligned}
$$

Sub-term 3: $-\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\,\rho^{d}(x)$.

$$
\begin{aligned}
[A,B]:\ & [\rho^{d}(x),\rho^{d'}(x')] = i f^{dd'h}\rho^{h}(x)\,\delta^{(2)}(x-x'), \quad x'\to x\\
&-\tfrac{1}{2}\{[A,B],C\} = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,U^{bc'}(z)\,U^{ad'}(x)\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,f^{dd'h}\,\{\rho^{h}(x),\rho^{e'}(y)\}\\
&\qquad = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{ld}(x)\,U^{hm}(y)\,U^{la}(w)\,\{\rho^{e}(x),\rho^{m}(y)\} \qquad (D.6)\\
[A,C]:\ & [\rho^{d}(x),\rho^{e'}(y)] = i f^{de'h}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x\\
&-\tfrac{1}{2}\{[A,C],B\} = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x',x,x)\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(x)\,f^{cbd}\,U^{ac}(w)\,f^{de'h}\,\{\rho^{h}(x),\rho^{d'}(x')\}\\
&\qquad = \tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{lm}(x')\,U^{hd}(x)\,U^{la}(w)\,\{\rho^{e}(x),\rho^{m}(x')\} \qquad (D.7)
\end{aligned}
$$

Sub-term 4: $+\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\,\rho^{e}(x)$.

$$
\begin{aligned}
[A,B]:\ & [\rho^{e}(x),\rho^{d'}(x')] = i f^{ed'h}\rho^{h}(x)\,\delta^{(2)}(x-x'), \quad x'\to x\\
&-\tfrac{1}{2}\{[A,B],C\} = -\tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x,x,y)\,U^{bc'}(z)\,U^{ad'}(x)\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ed'h}\,\{\rho^{h}(x),\rho^{e'}(y)\}\\
&\qquad = -\tfrac{1}{2}\,iN_c\;K_{\mathfrak{B}}(x,x,y)\,U^{ab}(y)\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(y)\} \quad \text{(U1), (U2), (F)} \qquad (D.8)\\
[A,C]:\ & [\rho^{e}(x),\rho^{e'}(y)] = i f^{ee'h}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x\\
&-\tfrac{1}{2}\{[A,C],B\} = -\tfrac{1}{2}\,i\;K_{\mathfrak{B}}(x',x,x)\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(x)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ee'h}\,\{\rho^{h}(x),\rho^{d'}(x')\}\\
&\qquad = \tfrac{1}{2}\,iN_c\;K_{\mathfrak{B}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\} \quad \text{(U1), (U2), (F)} \qquad (D.9)
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GII.I1}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{
&-\tfrac{1}{2}\,K_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\{\rho^{e}(x),\rho^{b}(y)\}\\
&+\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',x,x)\,U^{ab}(w')\,U^{ac}(w)\,\{\rho^{c}(x),\rho^{b}(x')\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x)\,\{\rho^{h}(x),\rho^{l}(y)\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\{\rho^{h}(x),\rho^{l}(x')\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{ld}(x)\,U^{hm}(y)\,U^{la}(w)\,\{\rho^{e}(x),\rho^{m}(y)\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{lm}(x')\,U^{hd}(x)\,U^{la}(w)\,\{\rho^{e}(x),\rho^{m}(x')\}
\end{aligned}
$$

$$
\begin{aligned}
&- \tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x,x,y)\, U^{ab}(y)\, U^{ac}(x)\, \big\{\rho^c(x), \rho^b(y)\big\} \\
&+ \tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \big\{\rho^c(x), \rho^b(x')\big\} \Big\}
\end{aligned} \tag{D.10}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K_{\mathfrak{B}}(x,x,y) &= \mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right] \\
K_{\mathfrak{B}}(x',x,x) &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(x-z) + \tfrac{1}{2}\,\frac{1}{(x-z)^2}\right]
\end{aligned}
$$

D.2 Group II, Row I, second term

Input. Prefactor $+i\mathcal{N}$ and kernel

$$
\bar{K}_{\mathfrak{B}} = K_{\mathfrak{B}}\big|_{w\leftrightarrow w'} = \mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\left[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right].
$$

The charges appear as $\rho(x)\{\rho(x'),\rho(y)\}$.

The three-ρ term.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GII.I2}} = i\,\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\bar{K}_{\mathfrak{B}}(x',x,y)\,\Big\{ \\
&+ f^{c'bd'}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x)\,\big\{\rho^c(x'),\rho^b(y)\big\} \\
&- f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{d'}(x)\,\big\{\rho^d(x'),\rho^e(y)\big\} \\
&- f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,\rho^{e'}(x)\,\big\{\rho^c(x'),\rho^b(y)\big\} \\
&+ f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{e'}(x)\,\big\{\rho^d(x'),\rho^e(y)\big\}\Big\}
\end{aligned} \tag{D.11}
$$

with $\bar{K}_{\mathfrak{B}}(x',x,y) = \mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\left[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x, x', y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'bd'}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x)\,\big\{\rho^c(x'),\rho^b(y)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{d'}(x),\rho^c(x')] &= i f^{d'ch}\rho^h(x)\,\delta^{(2)}(x-x'), \quad x'\to x \\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{c'bd'}\,U^{ac'}(w')\,U^{ac}(w)\,f^{d'ch}\,\big\{\rho^h(x),\rho^b(y)\big\} \\
&= \tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\big\{\rho^e(x),\rho^b(y)\big\}
\end{aligned} \tag{D.12}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{d'}(x),\rho^b(y)] &= i f^{d'bh}\rho^h(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{c'bd'}\,U^{ac'}(w')\,U^{ac}(w)\,f^{d'bh}\,\big\{\rho^h(x),\rho^c(x')\big\} \\
&= -\tfrac{1}{2}\,i N_c\,\bar{K}_{\mathfrak{B}}(x',x,x)\,U^{ab}(w')\,U^{ac}(w)\,\big\{\rho^b(x),\rho^c(x')\big\} \quad \text{(F)}
\end{aligned} \tag{D.13}
$$

Sub-term 2: $- f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{d'}(x)\,\big\{\rho^d(x'),\rho^e(y)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{d'}(x),\rho^d(x')] &= i f^{d'dh}\rho^h(x)\,\delta^{(2)}(x-x'), \quad x'\to x \\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,f^{d'dh}\,\big\{\rho^h(x),\rho^e(y)\big\} \\
&= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hd}(x)\,U^{lm}(y)\,\big\{\rho^e(x),\rho^m(y)\big\}
\end{aligned} \tag{D.14}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{d'}(x),\rho^e(y)] &= i f^{d'eh}\rho^h(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{d'eh}\,\big\{\rho^h(x),\rho^d(x')\big\} \\
&= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hm}(x')\,U^{ld}(x)\,\big\{\rho^e(x),\rho^m(x')\big\}
\end{aligned} \tag{D.15}
$$

Sub-term 3: $- f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,\rho^{e'}(x)\,\big\{\rho^c(x'),\rho^b(y)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x),\rho^c(x')] &= i f^{e'ch}\rho^h(x)\,\delta^{(2)}(x-x'), \quad x'\to x \\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,f^{e'ch}\,\big\{\rho^h(x),\rho^b(y)\big\} \\
&= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{cd}(x)\,U^{ae}(w)\,\big\{\rho^h(x),\rho^l(y)\big\}
\end{aligned} \tag{D.16}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{e'}(x),\rho^{b}(y)] &= i f^{e'bh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x\\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,f^{e'bh}\,\{\rho^{h}(x),\rho^{c}(x')\}\\
&= -\tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{cd}(x)\,U^{al}(w)\,\{\rho^{h}(x),\rho^{l}(x')\} \qquad \text{(D.17)}
\end{aligned}
$$

Sub-term 4: $+\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{e'}(x)\,\{\rho^{d}(x'),\rho^{e}(y)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x),\rho^{d}(x')] &= i f^{e'dh}\rho^{h}(x)\,\delta^{(2)}(x-x'), \quad x'\to x\\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,f^{e'dh}\,\{\rho^{h}(x),\rho^{e}(y)\}\\
&= \tfrac{1}{2}\,iN_c\,\bar{K}_{\mathfrak{B}}(x,x,y)\,U^{ab}(y)\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(y)\} \quad \text{(U1), (U2), (F)} \qquad \text{(D.18)}\\
[A,C]:\ [\rho^{e'}(x),\rho^{e}(y)] &= i f^{e'eh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x\\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'eh}\,\{\rho^{h}(x),\rho^{d}(x')\}\\
&= -\tfrac{1}{2}\,iN_c\,\bar{K}_{\mathfrak{B}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\} \quad \text{(U1), (U2), (F)} \qquad \text{(D.19)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GII.I2}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{ &\\
&-\tfrac{1}{2}\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\{\rho^{e}(x),\rho^{b}(y)\}\\
&+\tfrac{1}{2}\,N_c\,\bar{K}_{\mathfrak{B}}(x',x,x)\,U^{ab}(w')\,U^{ac}(w)\,\{\rho^{b}(x),\rho^{c}(x')\}\\
&+\tfrac{1}{2}\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hd}(x)\,U^{lm}(y)\,\{\rho^{e}(x),\rho^{m}(y)\}\\
&+\tfrac{1}{2}\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hm}(x')\,U^{ld}(x)\,\{\rho^{e}(x),\rho^{m}(x')\}\\
&+\tfrac{1}{2}\,\bar{K}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{cd}(x)\,U^{ae}(w)\,\{\rho^{h}(x),\rho^{l}(y)\}\\
&+\tfrac{1}{2}\,\bar{K}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{cd}(x)\,U^{al}(w)\,\{\rho^{h}(x),\rho^{l}(x')\}\\
&-\tfrac{1}{2}\,N_c\,\bar{K}_{\mathfrak{B}}(x,x,y)\,U^{ab}(y)\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(y)\}\\
&+\tfrac{1}{2}\,N_c\,\bar{K}_{\mathfrak{B}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\}\Big\} \qquad \text{(D.20)}
\end{aligned}
$$

The kernel at the coincident points:

$$
\begin{aligned}
\bar{K}_{\mathfrak{B}}(x,x,y) &= \mathcal{K}(x-w')\cdot\mathcal{K}(x-w)\left[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]\\
\bar{K}_{\mathfrak{B}}(x',x,x) &= \mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\left[\mathcal{K}(z-w')\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\right]
\end{aligned}
$$

E Group III (with c.c.)

E.1 Group III, Row I

Input. Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = -2i\mathcal{N}$, kernel $K_{\mathfrak{B}}$. Order of the charges: $\rho(x')\rho(y)\rho(x)$.

The three-ρ term.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.I}} = &-2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\,\Big\{\\
&+\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{d}(x)\\
&-\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{e}(x)\\
&-\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{d}(x)\\
&+\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{e}(x)\Big\}+\text{c.c.} \qquad \text{(E.1)}
\end{aligned}
$$

with $K_{\mathfrak{B}}(x',x,y) = \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{d}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^{d'}(y)] &= i f^{b'd'h}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,U^{c'd'}(x')\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,f^{b'd'h}\left\{\rho^{h}(x'),\rho^{d}(x)\right\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{da}(w)\,U^{hl}(x')\left\{\rho^{l}(x'),\rho^{c}(x)\right\} \quad \text{(U2)} \qquad \text{(E.2)} \\
[A,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] &= i f^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,f^{b'dh}\left\{\rho^{h}(x'),\rho^{d'}(y)\right\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(y)\,U^{lb}(z)\,U^{ha}(w)\left\{\rho^{e}(x'),\rho^{m}(y)\right\} \qquad \text{(E.3)} \\
[B,C]:\ [\rho^{d'}(y),\rho^{d}(x)] &= i f^{d'dh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,f^{d'dh}\left\{\rho^{h}(y),\rho^{b'}(x')\right\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{md}(y)\,U^{mb}(z)\,U^{ha}(w)\left\{\rho^{e}(y),\rho^{l}(x')\right\} \qquad \text{(E.4)}
\end{aligned}
$$

Sub-term 2: $-\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{e}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^{d'}(y)] &= i f^{b'd'h}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,U^{c'd'}(x')\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{b'd'h}\left\{\rho^{h}(x'),\rho^{e}(x)\right\} \\
&= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\left\{\rho^{c}(x'),\rho^{b}(x)\right\} \quad \text{(U1}^{\star}\text{), (U2), (F)} \qquad \text{(E.5)} \\
[A,C]:\ [\rho^{b'}(x'),\rho^{e}(x)] &= i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{b'eh}\left\{\rho^{h}(x'),\rho^{d'}(y)\right\} \\
&= \tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\left\{\rho^{c}(x'),\rho^{b}(y)\right\} \quad \text{(U1}^{\star}\text{), (U2), (F)} \qquad \text{(E.6)} \\
[B,C]:\ [\rho^{d'}(y),\rho^{e}(x)] &= i f^{d'eh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{d'eh}\left\{\rho^{h}(y),\rho^{b'}(x')\right\} \\
&= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\left\{\rho^{c}(y),\rho^{b}(x')\right\} \quad \text{(U1}^{\star}\text{), (U2), (F)} \qquad \text{(E.7)}
\end{aligned}
$$

Sub-term 3: $-\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{d}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{c'}(x'),\rho^{e'}(y)] &= i f^{c'e'h}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,U^{bd'}(z)\,U^{d'e'}(x')\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{c'e'h}\left\{\rho^{h}(x'),\rho^{d}(x)\right\} \\
&= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x')\,U^{md}(w')\,U^{ma}(w)\left\{\rho^{h}(x'),\rho^{c}(x)\right\} \qquad \text{(E.8)} \\
[A,C]:\ [\rho^{c'}(x'),\rho^{d}(x)] &= i f^{c'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{c'dh}\left\{\rho^{h}(x'),\rho^{e'}(y)\right\} \\
&= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(y)\,U^{md}(w')\,U^{ma}(w)\left\{\rho^{e}(x'),\rho^{l}(y)\right\} \qquad \text{(E.9)} \\
[B,C]:\ [\rho^{e'}(y),\rho^{d}(x)] &= i f^{e'dh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{e'dh}\left\{\rho^{h}(y),\rho^{c'}(x')\right\} \\
&= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(y)\,U^{lm}(w')\,U^{la}(w)\left\{\rho^{e}(y),\rho^{m}(x')\right\} \qquad \text{(E.10)}
\end{aligned}
$$

Sub-term 4: $+\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{e}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{c'}(x'),\rho^{e'}(y)] &= i f^{c'e'h}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,U^{bd'}(z)\,U^{d'e'}(x')\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{c'e'h}\left\{\rho^{h}(x'),\rho^{e}(x)\right\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{be}(x')\,U^{ad}(w')\,U^{cl}(x)\left\{\rho^{h}(x'),\rho^{l}(x)\right\} \quad \text{(U1)} \qquad \text{(E.11)} \\
[A,C]:\ [\rho^{c'}(x'),\rho^{e}(x)] &= i f^{c'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{c'eh}\left\{\rho^{h}(x'),\rho^{e'}(y)\right\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{bl}(y)\,U^{ad}(w')\,U^{ce}(x')\left\{\rho^{h}(x'),\rho^{l}(y)\right\} \quad \text{(U1)} \qquad \text{(E.12)} \\
[B,C]:\ [\rho^{e'}(y),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{e'eh}\left\{\rho^{h}(y),\rho^{c'}(x')\right\} \\
&= \tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',y,y)\,U^{ab}(w')\,U^{ac}(y)\left\{\rho^{c}(y),\rho^{b}(x')\right\} \quad \text{(U1), (U2), (F)} \qquad \text{(E.13)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.I}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{ & \\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{da}(w)\,U^{hl}(x')\,\big\{\rho^{l}(x'),\rho^{c}(x)\big\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(y)\,U^{lb}(z)\,U^{ha}(w)\,\big\{\rho^{e}(x'),\rho^{m}(y)\big\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{md}(y)\,U^{mb}(z)\,U^{ha}(w)\,\big\{\rho^{e}(y),\rho^{l}(x')\big\}\\
&-\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(x)\big\}\\
&+\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(y)\big\}\\
&-\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\big\{\rho^{c}(y),\rho^{b}(x')\big\}\\
&-\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x')\,U^{md}(w')\,U^{ma}(w)\,\big\{\rho^{h}(x'),\rho^{c}(x)\big\}\\
&-\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(y)\,U^{md}(w')\,U^{ma}(w)\,\big\{\rho^{e}(x'),\rho^{l}(y)\big\}\\
&-\tfrac{1}{2}\,K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(y)\,U^{lm}(w')\,U^{la}(w)\,\big\{\rho^{e}(y),\rho^{m}(x')\big\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{be}(x')\,U^{ad}(w')\,U^{cl}(x)\,\big\{\rho^{h}(x'),\rho^{l}(x)\big\}\\
&+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{bl}(y)\,U^{ad}(w')\,U^{ce}(x')\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\}\\
&+\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',y,y)\,U^{ab}(w')\,U^{ac}(y)\,\big\{\rho^{c}(y),\rho^{b}(x')\big\}\Big\}+\big(w'\leftrightarrow w\big)
\end{aligned}\tag{E.14}$$

The kernel at the coincident points:

$$\begin{aligned}
K_{\mathfrak{B}}(x',x,x') &= \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\right]\\
K_{\mathfrak{B}}(x',x',y) &= \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\right]\\
K_{\mathfrak{B}}(x',y,y) &= \mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\right]
\end{aligned}$$

E.2 Group III, Row II

Input. Prefactor $-2i\mathcal{N}$, kernel $K_{\mathfrak{B}}$. Order of the charges: $\rho(y)\rho(x')\rho(x)$.

The three-ρ term.

$$\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.II}} = & -2i\mathcal{N}\int e^{-ik\cdot(w'-w)}\,K_{\mathfrak{B}}(x',x,y)\Big\{\\
&+\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{d}(x)\\
&-\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\\
&-\,U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,\rho^{b}(y)\,\rho^{b'}(x')\,\rho^{d}(x)\\
&+\,U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\Big\}+\text{c.c.}
\end{aligned}\tag{E.15}$$

with $K_{\mathfrak{B}}(x',x,y)=\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at y, x', x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{d}(x)$.

$$\begin{aligned}
[A,B]:\ [\rho^{d'}(y),\rho^{b'}(x')] &= i f^{d'b'h}\rho^{h}(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(y,x,y)\,U^{c'd'}(y)\,U^{ab'}(y)\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,f^{d'b'h}\,\big\{\rho^{h}(y),\rho^{d}(x)\big\}\\
&= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w)\,U^{hl}(y)\,\big\{\rho^{l}(y),\rho^{c}(x)\big\}\quad\text{(U2)}
\end{aligned}\tag{E.16}$$

$$\begin{aligned}
[A,C]:\ [\rho^{d'}(y),\rho^{d}(x)] &= i f^{d'dh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',y,y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,f^{d'dh}\,\big\{\rho^{h}(y),\rho^{b'}(x')\big\}\\
&= \tfrac{1}{4}\,i\,K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hd}(y)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\big\{\rho^{e}(y),\rho^{m}(x')\big\}
\end{aligned}\tag{E.17}$$

$$[B,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] = i f^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$

$$\begin{aligned}\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,f^{b'dh}\left\{\rho^h(x'),\rho^{d'}(y)\right\}\\ &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hl}(y)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\left\{\rho^e(x'),\rho^l(y)\right\}\end{aligned}\tag{E.18}$$

Sub-term 2: $-\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^e(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{d'}(y),\rho^{b'}(x')] &= if^{d'b'h}\rho^h(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\ \tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(y,x,y)\,U^{c'd'}(y)\,U^{ab'}(y)\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{d'b'h}\left\{\rho^h(y),\rho^e(x)\right\}\\ &= \tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(y,x,y)\,U^{ab}(x)\,U^{ac}(y)\left\{\rho^c(y),\rho^b(x)\right\}\quad\text{(U1), (U2), (F)}\end{aligned}\tag{E.19}$$

$$\begin{aligned}[A,C]:\ [\rho^{d'}(y),\rho^{e}(x)] &= if^{d'eh}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y\\ \tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{d'eh}\left\{\rho^h(y),\rho^{b'}(x')\right\}\\ &= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\left\{\rho^c(y),\rho^b(x')\right\}\quad\text{(U1), (U2), (F)}\end{aligned}\tag{E.20}$$

$$\begin{aligned}[B,C]:\ [\rho^{b'}(x'),\rho^{e}(x)] &= if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ \tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{b'eh}\left\{\rho^h(x'),\rho^{d'}(y)\right\}\\ &= \tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\left\{\rho^c(x'),\rho^b(y)\right\}\quad\text{(U1), (U2), (F)}\end{aligned}\tag{E.21}$$

Sub-term 3: $-\,U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,\rho^b(y)\,\rho^{b'}(x')\,\rho^d(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{b}(y),\rho^{b'}(x')] &= if^{bb'h}\rho^h(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\ \tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(y,x,y)\,U^{ab'}(y)\,f^{cbd}\,U^{ac}(w)\,f^{bb'h}\left\{\rho^h(y),\rho^d(x)\right\}\\ &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{bde}\,U^{hd}(y)\,U^{ha}(w)\left\{\rho^e(y),\rho^c(x)\right\}\end{aligned}\tag{E.22}$$

$$\begin{aligned}[A,C]:\ [\rho^{b}(y),\rho^{d}(x)] &= if^{bdh}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y\\ \tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,f^{bdh}\left\{\rho^h(y),\rho^{b'}(x')\right\}\\ &= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(w)\left\{\rho^c(y),\rho^b(x')\right\}\quad\text{(F)}\end{aligned}\tag{E.23}$$

$$\begin{aligned}[B,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] &= if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ \tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,f^{b'dh}\left\{\rho^h(x'),\rho^b(y)\right\}\\ &= -\tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\left\{\rho^e(x'),\rho^b(y)\right\}\end{aligned}\tag{E.24}$$

Sub-term 4: $+\,U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^b(y)\,\rho^{b'}(x')\,\rho^e(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{b}(y),\rho^{b'}(x')] &= if^{bb'h}\rho^h(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\ \tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(y,x,y)\,U^{ab'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{bb'h}\left\{\rho^h(y),\rho^e(x)\right\}\\ &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{deh}\,U^{ae}(y)\,U^{db}(z)\,U^{cl}(x)\left\{\rho^h(y),\rho^l(x)\right\}\end{aligned}\tag{E.25}$$

$$\begin{aligned}[A,C]:\ [\rho^{b}(y),\rho^{e}(x)] &= if^{beh}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y\\ \tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{beh}\left\{\rho^h(y),\rho^{b'}(x')\right\}\\ &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{db}(z)\,U^{ce}(y)\left\{\rho^h(y),\rho^l(x')\right\}\end{aligned}\tag{E.26}$$

$$\begin{aligned}[B,C]:\ [\rho^{b'}(x'),\rho^{e}(x)] &= if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ \tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{b'eh}\left\{\rho^h(x'),\rho^b(y)\right\}\\ &= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{B}}(x',x',y)\,U^{ab}(z)\,U^{bc}(x')\left\{\rho^c(x'),\rho^a(y)\right\}\quad\text{(U2), (F)}\end{aligned}\tag{E.27}$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.II}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{&\\ &+\tfrac{1}{2}\,K_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w)\,U^{hl}(y)\left\{\rho^l(y),\rho^c(x)\right\}\\ &+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hd}(y)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\left\{\rho^e(y),\rho^m(x')\right\}\\ &+\tfrac{1}{2}\,K_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hl}(y)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\left\{\rho^e(x'),\rho^l(y)\right\}\\ &+\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(y,x,y)\,U^{ab}(x)\,U^{ac}(y)\left\{\rho^c(y),\rho^b(x)\right\}\\ &-\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\left\{\rho^c(y),\rho^b(x')\right\}\\ &+\tfrac{1}{2}\,N_c\,K_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\left\{\rho^c(x'),\rho^b(y)\right\}\end{aligned}$$

$$
\begin{aligned}
&- \tfrac{1}{2}\, K_{\mathfrak{B}}(y,x,y)\, f^{abc}\, f^{bde}\, U^{hd}(y)\, U^{ha}(w)\, \big\{\rho^{e}(y), \rho^{c}(x)\big\} \\
&- \tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',y,y)\, U^{ab}(x')\, U^{ac}(w)\, \big\{\rho^{c}(y), \rho^{b}(x')\big\} \\
&- \tfrac{1}{2}\, K_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w)\, \big\{\rho^{e}(x'), \rho^{b}(y)\big\} \\
&+ \tfrac{1}{2}\, K_{\mathfrak{B}}(y,x,y)\, f^{abc}\, f^{deh}\, U^{ae}(y)\, U^{db}(z)\, U^{cl}(x)\, \big\{\rho^{h}(y), \rho^{l}(x)\big\} \\
&+ \tfrac{1}{2}\, K_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{db}(z)\, U^{ce}(y)\, \big\{\rho^{h}(y), \rho^{l}(x')\big\} \\
&- \tfrac{1}{2}\, N_c\, K_{\mathfrak{B}}(x',x',y)\, U^{ab}(z)\, U^{bc}(x')\, \big\{\rho^{c}(x'), \rho^{a}(y)\big\}\Big\} + \big(w' \leftrightarrow w\big)
\end{aligned}
\tag{E.28}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K_{\mathfrak{B}}(y,x,y) &= \mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\right] \\
K_{\mathfrak{B}}(x',y,y) &= \mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\frac{1}{(y-z)^2}\right] \\
K_{\mathfrak{B}}(x',x',y) &= \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\left[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\right]
\end{aligned}
$$

E.3 Group III, Row III, first part

Input. Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4 f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda} = -i\mathcal{N} f^{c'd'a}$. After the w' integration the measured gluon of the conjugate amplitude sits at y': the phase is $e^{-ik\cdot(y'-w)}$, and the kernel is $K_{\mathfrak{C}} \equiv \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$.

The three-ρ term.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.III}} = &- i\mathcal{N}\int e^{-ik\cdot(y'-w)}\, K_{\mathfrak{C}}(x',x,y)\,\Big\{ \\
&+ f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{ac}(w)\, \rho^{e'}(x')\, \big\{\rho^{c}(y), \rho^{b}(x)\big\} \\
&- f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\, \rho^{e'}(x')\, \big\{\rho^{d}(y), \rho^{e}(x)\big\} \\
&- f^{c'd'a}\, U^{bd'}(z)\, U^{c'e'}(x')\, U^{ac}(w)\, \rho^{e'}(x')\, \big\{\rho^{c}(y), \rho^{b}(x)\big\} \\
&+ f^{c'd'a}\, U^{bd'}(z)\, U^{c'e'}(x')\, U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\, \rho^{e'}(x')\, \big\{\rho^{d}(y), \rho^{e}(x)\big\}\Big\} + \text{c.c.}
\end{aligned}
\tag{E.29}
$$

with $K_{\mathfrak{C}}(x',x,y) = \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{ac}(w)\, \rho^{e'}(x')\, \big\{\rho^{c}(y), \rho^{b}(x)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'), \rho^{c}(y)] &= i f^{e'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y \to x' \\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{ac}(w)\, f^{e'ch}\, \big\{\rho^{h}(x'), \rho^{b}(x)\big\} \\
&= \tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{bl}(z)\, U^{ce}(w)\, \big\{\rho^{h}(x'), \rho^{l}(x)\big\}
\end{aligned}
\tag{E.30}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{e'}(x'), \rho^{b}(x)] &= i f^{e'bh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x \to x' \\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{ac}(w)\, f^{e'bh}\, \big\{\rho^{h}(x'), \rho^{c}(y)\big\} \\
&= \tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{be}(z)\, U^{cl}(w)\, \big\{\rho^{h}(x'), \rho^{l}(y)\big\}
\end{aligned}
\tag{E.31}
$$

Sub-term 2: $- f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\, \rho^{e'}(x')\, \big\{\rho^{d}(y), \rho^{e}(x)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'), \rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y \to x' \\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{bc}(z)\, U^{ad}(x')\, U^{ce}(x)\, f^{e'dh}\, \big\{\rho^{h}(x'), \rho^{e}(x)\big\} \\
&= -\tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{ce}(x')\, U^{bl}(x)\, \big\{\rho^{h}(x'), \rho^{l}(x)\big\} \quad (\text{U1}^\star)
\end{aligned}
\tag{E.32}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{e'}(x'), \rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x \to x' \\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{c'd'a}\, U^{e'c'}(y')\, U^{d'b}(z)\, U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x')\, f^{e'eh}\, \big\{\rho^{h}(x'), \rho^{d}(y)\big\} \\
&= -\tfrac{1}{2}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{cl}(y)\, U^{be}(x')\, \big\{\rho^{h}(x'), \rho^{l}(y)\big\} \quad (\text{U1}^\star)
\end{aligned}
\tag{E.33}
$$

Sub-term 3: $-\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,\rho^{e'}(x')\,\{\rho^c(y),\rho^b(x)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^c(y)] &= if^{e'ch}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac12\{[A,B],C\} &= -\tfrac12\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,f^{e'ch}\,\{\rho^h(x'),\rho^b(x)\}\\
&= -\tfrac12\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w)\,\{\rho^h(x'),\rho^l(x)\} \qquad \text{(E.34)}\\
[A,C]:\ [\rho^{e'}(x'),\rho^b(x)] &= if^{e'bh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac12\{[A,C],B\} &= -\tfrac12\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,f^{e'bh}\,\{\rho^h(x'),\rho^c(y)\}\\
&= -\tfrac12\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(w)\,\{\rho^h(x'),\rho^l(y)\} \qquad \text{(E.35)}
\end{aligned}
$$

Sub-term 4: $+\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\{\rho^d(y),\rho^e(x)\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^d(y)] &= if^{e'dh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac12\{[A,B],C\} &= \tfrac12\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\{\rho^h(x'),\rho^e(x)\}\\
&= -\tfrac12\,iN_c\;K_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(x)\}\quad \text{(U1), (U2), (F)} \qquad \text{(E.36)}\\
[A,C]:\ [\rho^{e'}(x'),\rho^e(x)] &= if^{e'eh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac12\{[A,C],B\} &= \tfrac12\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\{\rho^h(x'),\rho^d(y)\}\\
&= \tfrac12\,iN_c\;K_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}\quad \text{(U1), (U2), (F)} \qquad \text{(E.37)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.III}} = \mathcal{N}\int e^{-ik\cdot(y'-w)}\,\Big\{&\\
&+\tfrac12\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{bl}(z)\,U^{ce}(w)\,\{\rho^h(x'),\rho^l(x)\}\\
&+\tfrac12\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{be}(z)\,U^{cl}(w)\,\{\rho^h(x'),\rho^l(y)\}\\
&-\tfrac12\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{ce}(x')\,U^{bl}(x)\,\{\rho^h(x'),\rho^l(x)\}\\
&-\tfrac12\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{cl}(y)\,U^{be}(x')\,\{\rho^h(x'),\rho^l(y)\}\\
&-\tfrac12\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w)\,\{\rho^h(x'),\rho^l(x)\}\\
&-\tfrac12\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(w)\,\{\rho^h(x'),\rho^l(y)\}\\
&-\tfrac12\,N_c\,K_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(x)\}\\
&+\tfrac12\,N_c\,K_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}\Big\}+\big(y'\leftrightarrow w\big) \qquad \text{(E.38)}
\end{aligned}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K_{\mathfrak{C}}(x',x,x') &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)\\
K_{\mathfrak{C}}(x',x',y) &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)
\end{aligned}
$$

E.4 Group III, Row V

Input. Prefactor $-\frac{1}{(2\pi)^3}\frac{ig^4f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\frac{\vee}{\Lambda}=-2i\mathcal{N}f^{c'ca}$, phase $e^{-ik\cdot(y'-w)}$, kernel $K_{\mathfrak{C}}$. Order of the charges: $\rho(x')\rho(x)\rho(y)$.

The three-ρ term.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.V}} = -\,2i\,\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K_{\mathfrak{C}}(x',x,y)\,\Big\{&\\
&+\;f^{c'ca}\,U^{e'c'}(y')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\\
&-\;f^{c'ca}\,U^{e'c'}(y')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\\
&-\;f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\\
&+\;f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^e(x)\,\rho^d(y)\Big\}+\text{c.c.} \qquad \text{(E.39)}
\end{aligned}
$$

with $K_{\mathfrak{C}}(x',x,y)=\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'ca}\, U^{e'c'}(y')\, U^{ce}(x)\, U^{ad}(y)\, \rho^{e'}(x')\, \rho^{e}(x)\, \rho^{d}(y).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\, U^{e'c'}(y')\, U^{ce}(x')\, U^{ad}(y)\, f^{e'eh}\, \{\rho^{h}(x'),\rho^{d}(y)\} \\
&= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{be}(x')\, U^{cl}(y)\, \{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(E.40)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\, U^{e'c'}(y')\, U^{ce}(x)\, U^{ad}(x')\, f^{e'dh}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{bl}(x)\, U^{ce}(x')\, \{\rho^{h}(x'),\rho^{l}(x)\} \qquad \text{(E.41)} \\
[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] &= i f^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\, U^{e'c'}(y')\, U^{ce}(x)\, U^{ad}(x)\, f^{edh}\, \{\rho^{h}(x),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\, i N_c\, K_{\mathfrak{C}}(x',x,x)\, U^{ab}(y')\, U^{bc}(x)\, \{\rho^{c}(x),\rho^{a}(x')\} \quad \text{(U2), (F)} \qquad \text{(E.42)}
\end{aligned}
$$

Sub-term 2: $- f^{c'ca}\, U^{e'c'}(y')\, U^{ad}(w)\, U^{ce}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\, \rho^{d}(y).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\, U^{e'c'}(y')\, U^{ad}(w)\, U^{ce}(x')\, f^{e'eh}\, \{\rho^{h}(x'),\rho^{d}(y)\} \\
&= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{cl}(w)\, U^{be}(x')\, \{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(E.43)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\, U^{e'c'}(y')\, U^{ad}(w)\, U^{ce}(x)\, f^{e'dh}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
&= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{da}(y')\, U^{ce}(w)\, U^{bl}(x)\, \{\rho^{h}(x'),\rho^{l}(x)\} \qquad \text{(E.44)} \\
[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] &= i f^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\, U^{e'c'}(y')\, U^{ad}(w)\, U^{ce}(x)\, f^{edh}\, \{\rho^{h}(x),\rho^{e'}(x')\} \\
&= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{la}(y')\, U^{ce}(w)\, U^{bd}(x)\, \{\rho^{h}(x),\rho^{l}(x')\} \qquad \text{(E.45)}
\end{aligned}
$$

Sub-term 3: $- f^{c'ca}\, U^{c'e'}(x')\, U^{ce}(x)\, U^{ad}(y)\, \rho^{e'}(x')\, \rho^{e}(x)\, \rho^{d}(y).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\, U^{c'e'}(x')\, U^{ce}(x')\, U^{ad}(y)\, f^{e'eh}\, \{\rho^{h}(x'),\rho^{d}(y)\} \\
&= -\tfrac{1}{4}\, i N_c\, K_{\mathfrak{C}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U2), (F)} \qquad \text{(E.46)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\, U^{c'e'}(x')\, U^{ce}(x)\, U^{ad}(x')\, f^{e'dh}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\, i N_c\, K_{\mathfrak{C}}(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \{\rho^{c}(x'),\rho^{b}(x)\} \quad \text{(U2), (F)} \qquad \text{(E.47)} \\
[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] &= i f^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\, U^{c'e'}(x')\, U^{ce}(x)\, U^{ad}(x)\, f^{edh}\, \{\rho^{h}(x),\rho^{e'}(x')\} \\
&= -\tfrac{1}{4}\, i N_c\, K_{\mathfrak{C}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \{\rho^{c}(x),\rho^{b}(x')\} \quad \text{(U2), (F)} \qquad \text{(E.48)}
\end{aligned}
$$

Sub-term 4: $+ f^{c'ca}\, U^{c'e'}(x')\, U^{ad}(w)\, U^{ce}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\, \rho^{d}(y).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\, U^{c'e'}(x')\, U^{ad}(w)\, U^{ce}(x')\, f^{e'eh}\, \{\rho^{h}(x'),\rho^{d}(y)\} \\
&= \tfrac{1}{4}\, i N_c\, K_{\mathfrak{C}}(x',x',y)\, U^{ab}(w)\, U^{ac}(x')\, \{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U2), (F)} \qquad \text{(E.49)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\, U^{c'e'}(x')\, U^{ad}(w)\, U^{ce}(x)\, f^{e'dh}\, \{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{ce}(w)\, U^{bl}(x)\, \{\rho^{h}(x'),\rho^{l}(x)\} \qquad \text{(E.50)} \\
[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] &= i f^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\, U^{c'e'}(x')\, U^{ad}(w)\, U^{ce}(x)\, f^{edh}\, \{\rho^{h}(x),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\, i\, K_{\mathfrak{C}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{ce}(w)\, U^{bd}(x)\, \{\rho^{h}(x),\rho^{l}(x')\} \qquad \text{(E.51)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned}\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.V}} = \mathcal{N}\int e^{-ik\cdot(y'-w)}\Big\{ &\\ &+\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{be}(x')\,U^{cl}(y)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\ &+\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{bl}(x)\,U^{ce}(x')\,\big\{\rho^h(x'),\rho^l(x)\big\}\\ &+\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x,x)\,U^{ab}(y')\,U^{bc}(x)\,\big\{\rho^c(x),\rho^a(x')\big\}\\ &-\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{cl}(w)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}\\ &-\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{ce}(w)\,U^{bl}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}\\ &-\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{la}(y')\,U^{ce}(w)\,U^{bd}(x)\,\big\{\rho^h(x),\rho^l(x')\big\}\\ &-\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\\ &+\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}\\ &-\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\\ &+\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x',y)\,U^{ab}(w)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\\ &+\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w)\,U^{bl}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}\\ &+\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(w)\,U^{bd}(x)\,\big\{\rho^h(x),\rho^l(x')\big\}\Big\} + \big(y'\leftrightarrow w\big)\end{aligned}\tag{E.52}$$

The kernel at the coincident points:

$$\begin{aligned}K_{\mathfrak{C}}(x',x',y) &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\\ K_{\mathfrak{C}}(x',x,x') &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)\\ K_{\mathfrak{C}}(x',x,x) &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x-w)\end{aligned}$$

E.5 Group III, Row VI

Input. Prefactor $-2i\mathcal{N}f^{c'd'a}$, phase $e^{-ik\cdot(y'-w)}$, kernel $K_{\mathfrak{C}}$. Order of the charges: $\rho(x')\rho(y)\rho(x)$.

The three-ρ term.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{\text{GIII.VI}} = &-2i\,\mathcal{N}\int e^{-ik\cdot(y'-w)}\,K_{\mathfrak{C}}(x',x,y)\Big\{\\ &+\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^e(x)\\ &-\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^b(x)\\ &-\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^e(x)\\ &+\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^b(x)\Big\}+\text{c.c.}\end{aligned}\tag{E.53}$$

with $K_{\mathfrak{C}}(x',x,y)=\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^d(y)\,\rho^e(x)$.

$$\begin{aligned}[A,B]:\ [\rho^{e'}(x'),\rho^d(y)] &= if^{e'dh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\ \tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\big\{\rho^h(x'),\rho^e(x)\big\}\\ &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{ce}(x')\,U^{bl}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}\quad(\text{U1}^\star)\end{aligned}\tag{E.54}$$

$$\begin{aligned}[A,C]:\ [\rho^{e'}(x'),\rho^e(x)] &= if^{e'eh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\ \tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\big\{\rho^h(x'),\rho^d(y)\big\}\\ &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}\quad(\text{U1}^\star)\end{aligned}\tag{E.55}$$

$$\begin{aligned}[B,C]:\ [\rho^d(y),\rho^e(x)] &= if^{deh}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y\\ \tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(y)\,f^{deh}\,\big\{\rho^h(y),\rho^{e'}(x')\big\}\\ &= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{C}}(x',y,y)\,U^{ab}(y')\,U^{bc}(y)\,\big\{\rho^c(y),\rho^a(x')\big\}\quad(\text{U1}^\star),\ (\text{U2}),\ (\text{F})\end{aligned}\tag{E.56}$$

Sub-term 2: $-\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= if^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(x')\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{b}(x)\} \\
&= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{bl}(z)\,U^{ce}(x')\,\{\rho^{h}(x'),\rho^{l}(x)\} \qquad \text{(E.57)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{b}(x)] &= if^{e'bh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,f^{e'bh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{be}(z)\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(E.58)} \\
[B,C]:\ [\rho^{d}(y),\rho^{b}(x)] &= if^{dbh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,U^{ad}(y)\,f^{dbh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',y,y)\,f^{abc}\,f^{deh}\,U^{la}(y')\,U^{be}(z)\,U^{cd}(y)\,\{\rho^{h}(y),\rho^{l}(x')\} \qquad \text{(E.59)}
\end{aligned}
$$

Sub-term 3: $-\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= if^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\,iN_c\;K_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\} \quad \text{(U1), (U2), (F)} \qquad \text{(E.60)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= if^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U1), (U2), (F)} \qquad \text{(E.61)} \\
[B,C]:\ [\rho^{d}(y),\rho^{e}(x)] &= if^{deh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(y)\,f^{deh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\,iN_c\;K_{\mathfrak{C}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\{\rho^{c}(y),\rho^{b}(x')\} \quad \text{(U1), (U2), (F)} \qquad \text{(E.62)}
\end{aligned}
$$

Sub-term 4: $+\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= if^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(x')\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{b}(x)\} \\
&= -\tfrac{1}{4}\,iN_c\;K_{\mathfrak{C}}(x',x,x')\,U^{ab}(z)\,U^{bc}(x')\,\{\rho^{c}(x'),\rho^{a}(x)\} \quad \text{(U2), (F)} \qquad \text{(E.63)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{b}(x)] &= if^{e'bh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,f^{e'bh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(E.64)} \\
[B,C]:\ [\rho^{d}(y),\rho^{b}(x)] &= if^{dbh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,f^{dbh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\,i\;K_{\mathfrak{C}}(x',y,y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(y)\,\{\rho^{h}(y),\rho^{l}(x')\} \qquad \text{(E.65)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.VI}} = \mathcal{N}\int e^{-ik\cdot(y'-w)}\,\Big\{ & \\
&+\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{ce}(x')\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\} \\
&+\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{cl}(y)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\} \\
&-\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',y,y)\,U^{ab}(y')\,U^{bc}(y)\,\{\rho^{c}(y),\rho^{a}(x')\} \\
&-\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{bl}(z)\,U^{ce}(x')\,\{\rho^{h}(x'),\rho^{l}(x)\} \\
&-\tfrac{1}{2}\,K_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{be}(z)\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\} \\
&-\tfrac{1}{2}\,K_{\mathfrak{C}}(x',y,y)\,f^{abc}\,f^{deh}\,U^{la}(y')\,U^{be}(z)\,U^{cd}(y)\,\{\rho^{h}(y),\rho^{l}(x')\} \\
&+\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\} \\
&-\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\} \\
&+\tfrac{1}{2}\,N_c\,K_{\mathfrak{C}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\{\rho^{c}(y),\rho^{b}(x')\}
\end{aligned}
$$

$$
\begin{aligned}
&- \tfrac{1}{2}\, N_c\, K_{\mathfrak{C}}(x',x,x')\, U^{ab}(z)\, U^{bc}(x')\, \big\{\rho^c(x'),\rho^a(x)\big\} \\
&+ \tfrac{1}{2}\, K_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{eb}(z)\, U^{ad}(x')\, U^{cl}(y)\, \big\{\rho^h(x'),\rho^l(y)\big\} \\
&+ \tfrac{1}{2}\, K_{\mathfrak{C}}(x',y,y)\, f^{abc}\, f^{deh}\, U^{eb}(z)\, U^{al}(x')\, U^{cd}(y)\, \big\{\rho^h(y),\rho^l(x')\big\}\Big\} + \big(y' \leftrightarrow w\big)
\end{aligned}
\tag{E.66}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K_{\mathfrak{C}}(x',x,x') &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w) \\
K_{\mathfrak{C}}(x',x',y) &= \mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w) \\
K_{\mathfrak{C}}(x',y,y) &= \mathcal{K}(y'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)
\end{aligned}
$$

F Summary: all two-ρ terms

Below are all two-ρ terms generated by the reordering, with the kernels written out. Every coefficient is real. For Groups I and III the complex conjugate is included as the exchange of the two measured-gluon positions in the phase, shown at the end of each line. The four-ρ remainders and Group II have no c.c. (Sec. 2). The symmetric three-ρ remainders are not listed (Sec. A).

Three-ρ terms from reordering the four-ρ terms

$$
\begin{aligned}
\frac{d^3N^{(2\rho)}}{d^3k}\bigg|_{\mathrm{A1}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{ &\\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) \\
&\quad\times f^{abc}\, f^{dae}\, U^{dh}(w')\, U^{hb}(w)\, \big\{\rho^e(x'),\rho^c(y)\big\} \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \\
&\quad\times f^{abc}\, f^{dce}\, U^{dh}(w')\, U^{hb}(w)\, \big\{\rho^e(x'),\rho^a(x)\big\} \\
&- \tfrac{1}{4}\, N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} \\
&\quad\times U^{ab}(w')\, U^{bc}(w)\, \big\{\rho^c(x),\rho^a(x')\big\} \\
&- \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) \\
&\quad\times f^{abc}\, f^{dae}\, U^{hb}(w)\, U^{hd}(x')\, \big\{\rho^e(x'),\rho^c(y)\big\} \\
&- \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \\
&\quad\times f^{abc}\, f^{dce}\, U^{hb}(w)\, U^{hd}(x')\, \big\{\rho^e(x'),\rho^a(x)\big\} \\
&+ \tfrac{1}{4}\, N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} \\
&\quad\times U^{ab}(w)\, U^{ac}(x')\, \big\{\rho^b(x),\rho^c(x')\big\}\Big\}
\end{aligned}
\tag{F.1}
$$

$$
\begin{aligned}
\frac{d^3N^{(2\rho)}}{d^3k}\bigg|_{\mathrm{A2}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{ &\\
&+ \tfrac{1}{4}\, N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\frac{1}{(x'-z)^2} \\
&\quad\times U^{ab}(w')\, U^{ac}(y)\, \big\{\rho^b(x'),\rho^c(y)\big\} \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \\
&\quad\times f^{abc}\, f^{cde}\, U^{hb}(w')\, U^{hd}(x')\, \big\{\rho^e(x'),\rho^a(x)\big\} \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \\
&\quad\times f^{abc}\, f^{ade}\, U^{hb}(w')\, U^{hd}(x)\, \big\{\rho^e(x),\rho^c(x')\big\} \\
&- \tfrac{1}{4}\, N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\frac{1}{(x'-z)^2} \\
&\quad\times U^{ab}(w')\, U^{bc}(w)\, \big\{\rho^a(x'),\rho^c(y)\big\} \\
&- \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \\
&\quad\times f^{abc}\, f^{cde}\, U^{bh}(w')\, U^{hd}(w)\, \big\{\rho^e(x'),\rho^a(x)\big\} \\
&- \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) \\
&\quad\times f^{abc}\, f^{ade}\, U^{bh}(w')\, U^{hd}(w)\, \big\{\rho^e(x),\rho^c(x')\big\}\Big\}
\end{aligned}
\tag{F.2}
$$

$$\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm B1} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{$$
$$+\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$$
$$\times f^{abc}\,f^{deh}\,U^{db}(w')\,U^{lc}(y)\,U^{ae}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}$$
$$+\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$$
$$\times f^{abc}\,f^{deh}\,U^{db}(w')\,U^{ec}(x')\,U^{al}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}$$
$$-\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2}$$
$$\times U^{ab}(w')\,U^{bc}(x)\,\big\{\rho^c(x),\rho^a(x')\big\}$$
$$+\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)$$
$$\times U^{ab}(y)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(y)\big\}$$
$$-\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$$
$$\times U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}$$
$$+\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2}$$
$$\times U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\Big\} \tag{F.3}$$

$$\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm B2} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{$$
$$-\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$$
$$\times f^{abc}\,f^{deh}\,U^{dc}(x')\,U^{eb}(w)\,U^{al}(y)\,\big\{\rho^h(x'),\rho^l(y)\big\}$$
$$-\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x'-z)^2}$$
$$\times U^{ab}(w)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(x)\big\}$$
$$-\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$$
$$\times f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{db}(w)\,U^{ae}(x)\,\big\{\rho^h(x),\rho^l(x')\big\}$$
$$-\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)$$
$$\times U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}$$
$$+\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x'-z)^2}$$
$$\times U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}$$
$$-\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)$$
$$\times U^{ab}(x')\,U^{bc}(x)\,\big\{\rho^c(x),\rho^a(x')\big\}\Big\} \tag{F.4}$$

Group I (with c.c.)

$$\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\rm GI.II} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{$$
$$+\tfrac{1}{4}\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x',y)$$
$$\times f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\big\{\rho^e(x'),\rho^c(y)\big\}$$
$$+\tfrac{1}{4}\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,x')$$
$$\times f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\,\big\{\rho^e(x'),\rho^b(x)\big\}$$
$$-\tfrac{1}{4}\,N_c\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x',y)$$
$$\times U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}$$
$$+\tfrac{1}{4}\,N_c\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,x')$$
$$\times U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}$$
$$-\tfrac{1}{4}\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x',y)$$
$$\times f^{abc}\,f^{dbe}\,U^{hd}(w')\,U^{ha}(w)\,\big\{\rho^e(x'),\rho^c(y)\big\}$$
$$-\tfrac{1}{4}\,\mathcal{K}^i(x'-w')\,\mathfrak{F}^i(w;x,x')$$

$$
\begin{aligned}
&\quad \times f^{abc}\, f^{dce}\, U^{hd}(w')\, U^{ha}(w)\, \big\{\rho^e(x'), \rho^b(x)\big\} \\
&+ \tfrac{1}{4}\, \mathcal{K}^i(x'-w')\, \mathfrak{F}^i(w; x', y) \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x')\, U^{cl}(y)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&+ \tfrac{1}{4}\, \mathcal{K}^i(x'-w')\, \mathfrak{F}^i(w; x, x') \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(x)\, U^{ce}(x')\, \big\{\rho^h(x'), \rho^l(x)\big\}\Big\} + \big(w' \leftrightarrow w\big)
\end{aligned}
\tag{F.5}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GI.III}} &= \mathcal{N} \int e^{-ik\cdot(w'-w)} \Big\{ \\
&+ \tfrac{1}{4}\, N_c\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x')\Big] \\
&\quad \times U^{ab}(z)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&- \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\Phi(x)\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{cl}(x)\, U^{be}(z)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&- \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-z)\,\Phi(x)\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{al}(x')\, U^{cd}(x)\, U^{be}(z)\, \big\{\rho^h(x), \rho^l(x')\big\} \\
&- \tfrac{1}{4}\, N_c\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x')\Big] \\
&\quad \times U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&+ \tfrac{1}{4}\, N_c\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\Phi(x)\Big] \\
&\quad \times U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(x)\big\} \\
&- \tfrac{1}{4}\, N_c\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-z)\,\Phi(x)\Big] \\
&\quad \times U^{ab}(x')\, U^{ac}(x)\, \big\{\rho^c(x), \rho^b(x')\big\} \\
&+ \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x')\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(z)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&+ \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\Phi(x)\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(z)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&+ \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-z)\,\Phi(x)\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{al}(w')\, U^{cd}(x)\, U^{be}(z)\, \big\{\rho^h(x), \rho^l(x')\big\} \\
&- \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\Phi(x')\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(y)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&- \tfrac{1}{4}\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\Phi(x)\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(x')\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&+ \tfrac{1}{4}\, N_c\, \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\frac{1}{(x-z)^2} + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-z)\,\Phi(x)\Big] \\
&\quad \times U^{ab}(w')\, U^{ac}(x)\, \big\{\rho^c(x), \rho^b(x')\big\}\Big\} + \big(w' \leftrightarrow w\big)
\end{aligned}
\tag{F.6}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GI.IV}} &= \mathcal{N} \int e^{-ik\cdot(w'-w)} \Big\{ \\
&- \tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big] \\
&\quad \times f^{abc}\, f^{deh}\, U^{da}(w)\, U^{eb}(z)\, U^{hl}(x')\, \big\{\rho^l(x'), \rho^c(x)\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big] \\
&\quad \times f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'), \rho^m(y)\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]
\end{aligned}
$$

$$
\begin{aligned}
&\times f^{abc}\, f^{dce}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y), \rho^l(x')\big\} \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big] \\
&\times U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(x)\big\} \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big] \\
&\times U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big] \\
&\times U^{ab}(x')\, U^{ac}(y)\, \big\{\rho^c(y), \rho^b(x')\big\} \\
&+ \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big] \\
&\times f^{abc}\, f^{deh}\, U^{ld}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{me}(x')\, \big\{\rho^h(x'), \rho^c(x)\big\} \\
&+ \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big] \\
&\times f^{abc}\, f^{dce}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'), \rho^m(y)\big\} \\
&+ \tfrac{1}{2}\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big] \\
&\times f^{abc}\, f^{dce}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y), \rho^l(x')\big\} \\
&- \tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z) + \tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big] \\
&\times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x')\, U^{cl}(x)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&- \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big] \\
&\times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(y)\, U^{ce}(x')\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\, \Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z) + \tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big] \\
&\times U^{ab}(w')\, U^{ac}(y)\, \big\{\rho^c(y), \rho^b(x')\big\}\Big\} + \big(w' \leftrightarrow w\big)
\end{aligned}
\tag{F.7}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GI.V1}} = \mathcal{N}\int e^{-ik\cdot(w'-z)}\,\Big\{ \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{bl}(x)\, U^{ce}(z)\, \big\{\rho^h(x'), \rho^l(y')\big\} \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-x) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{be}(x)\, U^{cl}(z)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&+ \tfrac{1}{4}\, N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x) \\
&\times U^{ab}(y')\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y')\big\} \\
&- \tfrac{1}{4}\, N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-x) \\
&\times U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&- \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(x)\, U^{ce}(z)\, \big\{\rho^h(x'), \rho^l(y')\big\} \\
&- \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-x) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x)\, U^{cl}(z)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y'-x) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(y')\, \big\{\rho^h(x'), \rho^l(y')\big\} \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-x) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(y)\, U^{be}(x')\, \big\{\rho^h(x'), \rho^l(y)\big\}\Big\} + \big(w' \leftrightarrow z\big)
\end{aligned}
\tag{F.8}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GI.V2}} = \mathcal{N}\int e^{-ik\cdot(w'-x)}\,\Big\{ \\
&+ \tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-z) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{bl}(x)\, U^{ce}(z)\, \big\{\rho^h(x'), \rho^l(y')\big\}
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\\
&\quad\times f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\
&+\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-z)\\
&\quad\times U^{ab}(y')\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y')\big\}\\
&-\tfrac{1}{4}\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\\
&\quad\times U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}\\
&-\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-z)\\
&\quad\times f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{bl}(x)\,U^{ce}(z)\,\big\{\rho^h(x'),\rho^l(y')\big\}\\
&-\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\\
&\quad\times f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{be}(x)\,U^{cl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\
&+\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(x'-z)\\
&\quad\times f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y')\,\big\{\rho^h(x'),\rho^l(y')\big\}\\
&+\tfrac{1}{4}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-x)\,\mathcal{K}(z-x)\cdot\mathcal{K}(y-z)\\
&\quad\times f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}\Big\}+\big(w'\leftrightarrow x\big)
\end{aligned}
\tag{F.9}
$$

Group II (no c.c.)

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GII.I1}}=\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{&\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\big\{\rho^e(x),\rho^b(y)\big\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times U^{ab}(w')\,U^{ac}(w)\,\big\{\rho^c(x),\rho^b(x')\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x)\,\big\{\rho^h(x),\rho^l(y)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\big\{\rho^h(x),\rho^l(x')\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{ld}(x)\,U^{hm}(y)\,U^{la}(w)\,\big\{\rho^e(x),\rho^m(y)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{lm}(x')\,U^{hd}(x)\,U^{la}(w)\,\big\{\rho^e(x),\rho^m(x')\big\}\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times U^{ab}(y)\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(y)\big\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\Big\}
\end{aligned}
\tag{F.10}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GII.I2}}=\mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{&\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x-w)\,\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\big\{\rho^e(x),\rho^b(y)\big\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\,\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times U^{ab}(w')\,U^{ac}(w)\,\big\{\rho^b(x),\rho^c(x')\big\}
\end{aligned}
$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hd}(x)\,U^{lm}(y)\,\big\{\rho^{e}(x),\rho^{m}(y)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hm}(x')\,U^{ld}(x)\,\big\{\rho^{e}(x),\rho^{m}(x')\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{cd}(x)\,U^{ae}(w)\,\big\{\rho^{h}(x),\rho^{l}(y)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{cd}(x)\,U^{al}(w)\,\big\{\rho^{h}(x),\rho^{l}(x')\big\}\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w')\cdot\mathcal{K}(x-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times U^{ab}(y)\,U^{ac}(x)\,\big\{\rho^{c}(x),\rho^{b}(y)\big\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w')\cdot\mathcal{K}(x'-w)\Big[\mathcal{K}(z-w')\cdot\mathcal{K}(x-z)+\tfrac{1}{2}\,\frac{1}{(x-z)^2}\Big]\\
&\quad\times U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^{c}(x),\rho^{b}(x')\big\}\Big\}
\end{aligned}
\tag{F.11}
$$

Group III (with c.c.)

$$
\begin{aligned}
\frac{d^3N^{(2\rho)}}{d^3k}\Big|_{\mathrm{GIII.I}} = \mathcal{N}\int e^{-ik\cdot(w'-w)}\,\Big\{&\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{da}(w)\,U^{hl}(x')\,\big\{\rho^{l}(x'),\rho^{c}(x)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(y)\,U^{lb}(z)\,U^{ha}(w)\,\big\{\rho^{e}(x'),\rho^{m}(y)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]\\
&\quad\times f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{md}(y)\,U^{mb}(z)\,U^{ha}(w)\,\big\{\rho^{e}(y),\rho^{l}(x')\big\}\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big]\\
&\quad\times U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(x)\big\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(y)\big\}\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]\\
&\quad\times U^{ab}(x')\,U^{ac}(y)\,\big\{\rho^{c}(y),\rho^{b}(x')\big\}\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x')\,U^{md}(w')\,U^{ma}(w)\,\big\{\rho^{h}(x'),\rho^{c}(x)\big\}\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(y)\,U^{md}(w')\,U^{ma}(w)\,\big\{\rho^{e}(x'),\rho^{l}(y)\big\}\\
&-\tfrac{1}{2}\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]\\
&\quad\times f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(y)\,U^{lm}(w')\,U^{la}(w)\,\big\{\rho^{e}(y),\rho^{m}(x')\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(x'-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{be}(x')\,U^{ad}(w')\,U^{cl}(x)\,\big\{\rho^{h}(x'),\rho^{l}(x)\big\}\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]\\
&\quad\times f^{abc}\,f^{deh}\,U^{bl}(y)\,U^{ad}(w')\,U^{ce}(x')\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\}
\end{aligned}
$$

$$+\tfrac{1}{2}\,N_c\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]$$
$$\times\,U^{ab}(w')\,U^{ac}(y)\,\big\{\rho^c(y),\rho^b(x')\big\}\Big\}+\big(w'\leftrightarrow w\big)\tag{F.12}$$

$$\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.II}}=\mathcal{N}\int e^{-ik\cdot(w'-w)}\Big\{$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w)\,U^{hl}(y)\,\big\{\rho^l(y),\rho^c(x)\big\}$$
$$+\tfrac{1}{2}\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]$$
$$\times\,f^{abc}\,f^{dce}\,U^{hd}(y)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\big\{\rho^e(y),\rho^m(x')\big\}$$
$$+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,f^{abc}\,f^{dce}\,U^{hl}(y)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\big\{\rho^e(x'),\rho^l(y)\big\}$$
$$+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,U^{ab}(x)\,U^{ac}(y)\,\big\{\rho^c(y),\rho^b(x)\big\}$$
$$-\tfrac{1}{2}\,N_c\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]$$
$$\times\,U^{ab}(x')\,U^{ac}(y)\,\big\{\rho^c(y),\rho^b(x')\big\}$$
$$+\tfrac{1}{2}\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(y)\big\}$$
$$-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,f^{abc}\,f^{bde}\,U^{hd}(y)\,U^{ha}(w)\,\big\{\rho^e(y),\rho^c(x)\big\}$$
$$-\tfrac{1}{2}\,N_c\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]$$
$$\times\,U^{ab}(x')\,U^{ac}(w)\,\big\{\rho^c(y),\rho^b(x')\big\}$$
$$-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\,\big\{\rho^e(x'),\rho^b(y)\big\}$$
$$+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,f^{abc}\,f^{deh}\,U^{ae}(y)\,U^{db}(z)\,U^{cl}(x)\,\big\{\rho^h(y),\rho^l(x)\big\}$$
$$+\tfrac{1}{2}\,\mathcal{K}(y-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\frac{1}{(y-z)^2}\Big]$$
$$\times\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{db}(z)\,U^{ce}(y)\,\big\{\rho^h(y),\rho^l(x')\big\}$$
$$-\tfrac{1}{2}\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\Big[\mathcal{K}(z-w)\cdot\mathcal{K}(y-z)+\tfrac{1}{2}\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]$$
$$\times\,U^{ab}(z)\,U^{bc}(x')\,\big\{\rho^c(x'),\rho^a(y)\big\}\Big\}+\big(w'\leftrightarrow w\big)\tag{F.13}$$

$$\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.III}}=\mathcal{N}\int e^{-ik\cdot(y'-w)}\Big\{$$
$$+\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)$$
$$\times\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{bl}(z)\,U^{ce}(w)\,\big\{\rho^h(x'),\rho^l(x)\big\}$$
$$+\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$$
$$\times\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{be}(z)\,U^{cl}(w)\,\big\{\rho^h(x'),\rho^l(y)\big\}$$
$$-\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)$$
$$\times\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{ce}(x')\,U^{bl}(x)\,\big\{\rho^h(x'),\rho^l(x)\big\}$$
$$-\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)$$
$$\times\,f^{abc}\,f^{deh}\,U^{da}(y')\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(y)\big\}$$
$$-\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)$$

$$
\begin{aligned}
&\times f^{abc}\, f^{deh}\, U^{lb}(z)\, U^{ad}(x')\, U^{ce}(w)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times f^{abc}\, f^{deh}\, U^{eb}(z)\, U^{ad}(x')\, U^{cl}(w)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(x)\big\} \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\}\Big\} + \big(y' \leftrightarrow w\big)
\end{aligned}
\tag{F.14}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.V}} = \mathcal{N}\int e^{-ik\cdot(y'-w)}\,\Big\{ \\
&+ \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{be}(x')\, U^{cl}(y)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&+ \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{bl}(x)\, U^{ce}(x')\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x-w) \\
&\times U^{ab}(y')\, U^{bc}(x)\, \big\{\rho^c(x), \rho^a(x')\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{cl}(w)\, U^{be}(x')\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{ce}(w)\, U^{bl}(x)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x-w) \\
&\times f^{abc}\, f^{deh}\, U^{la}(y')\, U^{ce}(w)\, U^{bd}(x)\, \big\{\rho^h(x), \rho^l(x')\big\} \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(x)\big\} \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x-w) \\
&\times U^{ab}(x')\, U^{ac}(x)\, \big\{\rho^c(x), \rho^b(x')\big\} \\
&+ \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times U^{ab}(w)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&+ \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{ce}(w)\, U^{bl}(x)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&+ \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x-w) \\
&\times f^{abc}\, f^{deh}\, U^{al}(x')\, U^{ce}(w)\, U^{bd}(x)\, \big\{\rho^h(x), \rho^l(x')\big\}\Big\} + \big(y' \leftrightarrow w\big)
\end{aligned}
\tag{F.15}
$$

$$
\begin{aligned}
\left.\frac{d^3N^{(2\rho)}}{d^3k}\right|_{\text{GIII.VI}} = \mathcal{N}\int e^{-ik\cdot(y'-w)}\,\Big\{ \\
&+ \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{ce}(x')\, U^{bl}(x)\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&+ \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{cl}(y)\, U^{be}(x')\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&- \tfrac{1}{2}\, N_c\, \mathcal{K}(y'-z)\cdot \mathcal{K}(y-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times U^{ab}(y')\, U^{bc}(y)\, \big\{\rho^c(y), \rho^a(x')\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(x'-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{bl}(z)\, U^{ce}(x')\, \big\{\rho^h(x'), \rho^l(x)\big\} \\
&- \tfrac{1}{2}\, \mathcal{K}(y'-z)\cdot \mathcal{K}(x'-z)\, \mathcal{K}(x'-y')\cdot \mathcal{K}(y-w) \\
&\times f^{abc}\, f^{deh}\, U^{da}(y')\, U^{be}(z)\, U^{cl}(y)\, \big\{\rho^h(x'), \rho^l(y)\big\}
\end{aligned}
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\\
&\quad\times f^{abc}\,f^{deh}\,U^{la}(y')\,U^{be}(z)\,U^{cd}(y)\,\{\rho^{h}(y),\rho^{l}(x')\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)\\
&\quad\times U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\}\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\\
&\quad\times U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(y'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\\
&\quad\times U^{ab}(x')\,U^{ac}(y)\,\{\rho^{c}(y),\rho^{b}(x')\}\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(x'-w)\\
&\quad\times U^{ab}(z)\,U^{bc}(x')\,\{\rho^{c}(x'),\rho^{a}(x)\}\\
&+\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(x'-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\\
&\quad\times f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\}\\
&+\tfrac{1}{2}\,\mathcal{K}(y'-z)\cdot\mathcal{K}(y-z)\,\mathcal{K}(x'-y')\cdot\mathcal{K}(y-w)\\
&\quad\times f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(y)\,\{\rho^{h}(y),\rho^{l}(x')\}\Big\}+\big(y'\leftrightarrow w\big)
\end{aligned}
\tag{F.16}
$$

G Region $p^+ \gg k^+$: reordering the three-ρ terms

The three-ρ terms proportional to $\log(\vee/k^+)$ are $B = P - A$ for the rows of Sec. 24 with three charges. The reordering is linear, so the two-ρ terms of B are those of P minus those of A:

- The two-ρ terms of A (the $\log(\vee/\Lambda)$ three-ρ terms) are worked out step by step in Apps. B–E (with the corrections of Part II: $\mathcal{D}$, and the α part of Row II). They enter here with a minus sign.
- The two-ρ terms of P are worked out below in the same way, for every P source. There the coefficient of $\log\vee$ is written as $P|_{\text{source}}$, in units of $\mathcal{N}_P$, like B. The colour structures are those of the corresponding A source, only the kernel changes (for Group III Rows III, V, VI also $U(y')\to U(z)$).

As written, Group I Rows II and V have $P=0$, and their $B=-A$; with F4, Row II has $P=-\alpha$ (Sec. 5). The fix P1 is included in the source "Group I, Row III (region P)" below: its kernel is $K^P_{\text{I.III}} = -K_b + 2K_h$, i.e. the corrected one. The kernels at the coincident points are listed after each source.

G.1 Group I, Row III ($\bar{\mathbb{B}}_3^\dagger\mathbb{A}$) (region P)

The three-ρ term.

$$
\begin{aligned}
P\Big|_{\text{GI.III.P}} = i\,\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K^P_{\text{I.III}}(x',x,y)\,\Big\{&\\
&+\ f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\
&-\ f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\
&-\ f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\\
&+\ f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)\Big\}+\text{c.c.}
\end{aligned}
\tag{G.1}
$$

with $K^P_{\text{I.III}}(x',x,y) = \Big[-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z) + 2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y)$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\,i\,K^P_{\text{I.III}}(x',x',y)\,f^{abd}\,U^{ab'}(x')\,U^{de}(x')\,U^{bc}(z)\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{c}(y)\}\\
&\qquad\qquad = -\tfrac{1}{4}\,iN_c\,K^P_{\text{I.III}}(x',x',y)\,U^{ab}(z)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\quad\text{(U2), (F)}\\
&[A,C]:\ [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
&\quad\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\,i\,K^P_{\text{I.III}}(x',x,x')\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(z)\,f^{b'ch}\,\{\rho^{h}(x'),\rho^{e}(x)\}
\end{aligned}
\tag{G.2}
$$

$$= \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{cl}(x)\, U^{be}(z)\, \big\{\rho^{h}(x'),\rho^{l}(x)\big\} \tag{G.3}$$

$$[B,C]:\ [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(z)\, f^{ech}\, \big\{\rho^{h}(x),\rho^{b'}(x')\big\}$$
$$= \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{cd}(x)\, U^{be}(z)\, \big\{\rho^{h}(x),\rho^{l}(x')\big\} \tag{G.4}$$

Sub-term 2: $-\,f^{abd}\,U^{ab'}(x')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y).$

$$[A,B]:\ [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$

$$\tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x',y)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x')\, U^{bc}(y)\, f^{b'eh}\, \big\{\rho^{h}(x'),\rho^{c}(y)\big\}$$
$$= \tfrac{1}{4}\, iN_c\; K^{P}_{\mathrm{I.III}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^{c}(x'),\rho^{b}(y)\big\}\quad \text{(U2), (F)} \tag{G.5}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(x')\, f^{b'ch}\, \big\{\rho^{h}(x'),\rho^{e}(x)\big\}$$
$$= -\tfrac{1}{4}\, iN_c\; K^{P}_{\mathrm{I.III}}(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^{c}(x'),\rho^{b}(x)\big\}\quad \text{(U2), (F)} \tag{G.6}$$

$$[B,C]:\ [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(x')\, U^{de}(x)\, U^{bc}(x)\, f^{ech}\, \big\{\rho^{h}(x),\rho^{b'}(x')\big\}$$
$$= \tfrac{1}{4}\, iN_c\; K^{P}_{\mathrm{I.III}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \big\{\rho^{c}(x),\rho^{b}(x')\big\}\quad \text{(U2), (F)} \tag{G.7}$$

Sub-term 3: $-\,f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(z)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y).$

$$[A,B]:\ [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$

$$\tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x',y)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x')\, U^{bc}(z)\, f^{b'eh}\, \big\{\rho^{h}(x'),\rho^{c}(y)\big\}$$
$$= -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(z)\, \big\{\rho^{h}(x'),\rho^{l}(y)\big\} \tag{G.8}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(z)\, f^{b'ch}\, \big\{\rho^{h}(x'),\rho^{e}(x)\big\}$$
$$= -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(z)\, \big\{\rho^{h}(x'),\rho^{l}(x)\big\} \tag{G.9}$$

$$[B,C]:\ [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(z)\, f^{ech}\, \big\{\rho^{h}(x),\rho^{b'}(x')\big\}$$
$$= -\tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{cd}(x)\, U^{be}(z)\, \big\{\rho^{h}(x),\rho^{l}(x')\big\} \tag{G.10}$$

Sub-term 4: $+\,f^{abd}\,U^{ab'}(w')\,U^{de}(x)\,U^{bc}(y)\,\rho^{b'}(x')\,\rho^{e}(x)\,\rho^{c}(y).$

$$[A,B]:\ [\rho^{b'}(x'),\rho^{e}(x)] = i f^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'$$

$$\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x',y)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x')\, U^{bc}(y)\, f^{b'eh}\, \big\{\rho^{h}(x'),\rho^{c}(y)\big\}$$
$$= \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{ce}(x')\, U^{bl}(y)\, \big\{\rho^{h}(x'),\rho^{l}(y)\big\} \tag{G.11}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^{c}(y)] = i f^{b'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'$$

$$\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(x')\, f^{b'ch}\, \big\{\rho^{h}(x'),\rho^{e}(x)\big\}$$
$$= \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{cl}(x)\, U^{be}(x')\, \big\{\rho^{h}(x'),\rho^{l}(x)\big\} \tag{G.12}$$

$$[B,C]:\ [\rho^{e}(x),\rho^{c}(y)] = i f^{ech}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x$$

$$\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abd}\, U^{ab'}(w')\, U^{de}(x)\, U^{bc}(x)\, f^{ech}\, \big\{\rho^{h}(x),\rho^{b'}(x')\big\}$$
$$= -\tfrac{1}{4}\, iN_c\; K^{P}_{\mathrm{I.III}}(x',x,x)\, U^{ab}(w')\, U^{ac}(x)\, \big\{\rho^{c}(x),\rho^{b}(x')\big\}\quad \text{(U2), (F)} \tag{G.13}$$

Two-ρ terms of this source. Multiplying by the prefactor $i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$P^{(2\rho)}\Big|_{\mathrm{GI.III.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,\Big\{$$
$$+\ \tfrac{1}{4}\, N_c\, K^{P}_{\mathrm{I.III}}(x',x',y)\, U^{ab}(z)\, U^{ac}(x')\, \big\{\rho^{c}(x'),\rho^{b}(y)\big\}$$
$$-\ \tfrac{1}{4}\, K^{P}_{\mathrm{I.III}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(x')\, U^{cl}(x)\, U^{be}(z)\, \big\{\rho^{h}(x'),\rho^{l}(x)\big\}$$
$$-\ \tfrac{1}{4}\, K^{P}_{\mathrm{I.III}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{cd}(x)\, U^{be}(z)\, \big\{\rho^{h}(x),\rho^{l}(x')\big\}$$
$$-\ \tfrac{1}{4}\, N_c\, K^{P}_{\mathrm{I.III}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^{c}(x'),\rho^{b}(y)\big\}$$

$$
\begin{aligned}
&+\tfrac{1}{4}\,N_c\,K^P_{\mathrm{I.III}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}\\
&-\tfrac{1}{4}\,N_c\,K^P_{\mathrm{I.III}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\\
&+\tfrac{1}{4}\,K^P_{\mathrm{I.III}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(z)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\
&+\tfrac{1}{4}\,K^P_{\mathrm{I.III}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(x)\,U^{be}(z)\,\big\{\rho^h(x'),\rho^l(x)\big\}\\
&+\tfrac{1}{4}\,K^P_{\mathrm{I.III}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(x)\,U^{be}(z)\,\big\{\rho^h(x),\rho^l(x')\big\}\\
&-\tfrac{1}{4}\,K^P_{\mathrm{I.III}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{ce}(x')\,U^{bl}(y)\,\big\{\rho^h(x'),\rho^l(y)\big\}\\
&-\tfrac{1}{4}\,K^P_{\mathrm{I.III}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{cl}(x)\,U^{be}(x')\,\big\{\rho^h(x'),\rho^l(x)\big\}\\
&+\tfrac{1}{4}\,N_c\,K^P_{\mathrm{I.III}}(x',x,x)\,U^{ab}(w')\,U^{ac}(x)\,\big\{\rho^c(x),\rho^b(x')\big\}\Big\}+\big(w'\leftrightarrow w\big)
\end{aligned}
\tag{G.14}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K^P_{\mathrm{I.III}}(x',x',y)&=\Big[-\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z)\Big]\\
K^P_{\mathrm{I.III}}(x',x,x')&=\Big[-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z)\Big]\\
K^P_{\mathrm{I.III}}(x',x,x)&=\frac{1}{(x-z)^2}\Big[-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\Big]
\end{aligned}
$$

G.2 Group I, Row IV ($\bar{\mathbb{A}}^\dagger\mathbb{B}_2$) (region P)

The three-ρ term.

$$
\begin{aligned}
P\Big|_{\mathrm{GI.IV.P}}&=2i\,\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K^P_{\mathfrak{B}}(x',x,y)\,\Big\{\\
&\quad+\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^{c'}(x)\\
&\quad-\,U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^d(x)\\
&\quad-\,U^{ab'}(w')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^{c'}(x)\\
&\quad+\,U^{ab'}(w')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^d(x)\Big\}+\text{c.c.}
\end{aligned}
\tag{G.15}
$$

with $K^P_{\mathfrak{B}}(x',x,y)=\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z)-\tfrac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^{c'}(x)$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^e(y)]&=if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{4}\{[A,B],C\}&=\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(x')\,f^{b'eh}\,\big\{\rho^h(x'),\rho^{c'}(x)\big\}\\
&=\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{eb}(z)\,U^{hl}(x')\,\big\{\rho^l(x'),\rho^c(x)\big\}\quad\text{(U2)}
\end{aligned}
\tag{G.16}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{b'}(x'),\rho^{c'}(x)]&=if^{b'c'h}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[A,C],B\}&=\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,f^{b'c'h}\,\big\{\rho^h(x'),\rho^e(y)\big\}\\
&=\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(y)\,\big\{\rho^e(x'),\rho^m(y)\big\}
\end{aligned}
\tag{G.17}
$$

$$
\begin{aligned}
[B,C]:\ [\rho^e(y),\rho^{c'}(x)]&=if^{ec'h}\rho^h(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[B,C],A\}&=\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,f^{ec'h}\,\big\{\rho^h(y),\rho^{b'}(x')\big\}\\
&=\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(y)\,\big\{\rho^e(y),\rho^l(x')\big\}
\end{aligned}
\tag{G.18}
$$

Sub-term 2: $-\,U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{b'}(x')\,\rho^e(y)\,\rho^d(x)$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^e(y)]&=if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{4}\{[A,B],C\}&=-\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,f^{abc}\,U^{be}(x')\,U^{cd}(x)\,f^{b'eh}\,\big\{\rho^h(x'),\rho^d(x)\big\}\\
&=-\tfrac{1}{4}\,iN_c\,K^P_{\mathfrak{B}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^c(x'),\rho^b(x)\big\}\quad\text{(U2), (F)}
\end{aligned}
\tag{G.19}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{b'}(x'),\rho^d(x)]&=if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[A,C],B\}&=-\tfrac{1}{4}\,i\,K^P_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{abc}\,U^{be}(y)\,U^{cd}(x')\,f^{b'dh}\,\big\{\rho^h(x'),\rho^e(y)\big\}
\end{aligned}
$$

$$= \tfrac{1}{4}\, iN_c\; K^P_{\mathfrak{B}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(y)\big\} \quad \text{(U2), (F)} \tag{G.20}$$

$$[B,C]:\ [\rho^e(y),\rho^d(x)] = if^{edh}\rho^h(y)\,\delta^{(2)}(y-x), \quad x\to y$$
$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',y,y)\, U^{ab'}(x')\, f^{abc}\, U^{be}(y)\, U^{cd}(y)\, f^{edh}\, \big\{\rho^h(y),\rho^{b'}(x')\big\}$$
$$= -\tfrac{1}{4}\, iN_c\; K^P_{\mathfrak{B}}(x',y,y)\, U^{ab}(x')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\} \quad \text{(U2), (F)} \tag{G.21}$$

Sub-term 3: $-\,U^{ab'}(w')\, f^{cdc'}\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, \rho^{b'}(x')\, \rho^e(y)\, \rho^{c'}(x)$.

$$[A,B]:\ [\rho^{b'}(x'),\rho^e(y)] = if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y), \quad y\to x'$$
$$\tfrac{1}{4}\{[A,B],C\} = -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x,x')\, U^{ab'}(w')\, f^{cdc'}\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(x')\, f^{b'eh}\, \big\{\rho^h(x'),\rho^{c'}(x)\big\}$$
$$= -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ld}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{me}(x')\, \big\{\rho^h(x'),\rho^c(x)\big\} \tag{G.22}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^{c'}(x)] = if^{b'c'h}\rho^h(x')\,\delta^{(2)}(x'-x), \quad x\to x'$$
$$\tfrac{1}{4}\{[A,C],B\} = -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x',y)\, U^{ab'}(w')\, f^{cdc'}\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, f^{b'c'h}\, \big\{\rho^h(x'),\rho^e(y)\big\}$$
$$= -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'),\rho^m(y)\big\} \tag{G.23}$$

$$[B,C]:\ [\rho^e(y),\rho^{c'}(x)] = if^{ec'h}\rho^h(y)\,\delta^{(2)}(y-x), \quad x\to y$$
$$\tfrac{1}{4}\{[B,C],A\} = -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',y,y)\, U^{ab'}(w')\, f^{cdc'}\, U^{ac}(w)\, U^{bd}(z)\, U^{be}(y)\, f^{ec'h}\, \big\{\rho^h(y),\rho^{b'}(x')\big\}$$
$$= -\tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y),\rho^l(x')\big\} \tag{G.24}$$

Sub-term 4: $+\,U^{ab'}(w')\, f^{abc}\, U^{be}(y)\, U^{cd}(x)\, \rho^{b'}(x')\, \rho^e(y)\, \rho^d(x)$.

$$[A,B]:\ [\rho^{b'}(x'),\rho^e(y)] = if^{b'eh}\rho^h(x')\,\delta^{(2)}(x'-y), \quad y\to x'$$
$$\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x,x')\, U^{ab'}(w')\, f^{abc}\, U^{be}(x')\, U^{cd}(x)\, f^{b'eh}\, \big\{\rho^h(x'),\rho^d(x)\big\}$$
$$= \tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x')\, U^{cl}(x)\, \big\{\rho^h(x'),\rho^l(x)\big\} \tag{G.25}$$

$$[A,C]:\ [\rho^{b'}(x'),\rho^d(x)] = if^{b'dh}\rho^h(x')\,\delta^{(2)}(x'-x), \quad x\to x'$$
$$\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x',y)\, U^{ab'}(w')\, f^{abc}\, U^{be}(y)\, U^{cd}(x')\, f^{b'dh}\, \big\{\rho^h(x'),\rho^e(y)\big\}$$
$$= \tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(y)\, U^{ce}(x')\, \big\{\rho^h(x'),\rho^l(y)\big\} \tag{G.26}$$

$$[B,C]:\ [\rho^e(y),\rho^d(x)] = if^{edh}\rho^h(y)\,\delta^{(2)}(y-x), \quad x\to y$$
$$\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\, i\; K^P_{\mathfrak{B}}(x',y,y)\, U^{ab'}(w')\, f^{abc}\, U^{be}(y)\, U^{cd}(y)\, f^{edh}\, \big\{\rho^h(y),\rho^{b'}(x')\big\}$$
$$= \tfrac{1}{4}\, iN_c\; K^P_{\mathfrak{B}}(x',y,y)\, U^{ab}(w')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\} \quad \text{(U2), (F)} \tag{G.27}$$

Two-ρ terms of this source. Multiplying by the prefactor $2i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$P^{(2\rho)}\Big|_{\text{GI.IV.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)} \Big\{$$
$$- \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{da}(w)\, U^{eb}(z)\, U^{hl}(x')\, \big\{\rho^l(x'),\rho^c(x)\big\}$$
$$- \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'),\rho^m(y)\big\}$$
$$- \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y),\rho^l(x')\big\}$$
$$+ \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',x,x')\, U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(x)\big\}$$
$$- \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'),\rho^b(y)\big\}$$
$$+ \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',y,y)\, U^{ab}(x')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\}$$
$$+ \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ld}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{me}(x')\, \big\{\rho^h(x'),\rho^c(x)\big\}$$
$$+ \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(y)\, \big\{\rho^e(x'),\rho^m(y)\big\}$$
$$+ \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(y)\, \big\{\rho^e(y),\rho^l(x')\big\}$$
$$- \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x,x')\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{be}(x')\, U^{cl}(x)\, \big\{\rho^h(x'),\rho^l(x)\big\}$$
$$- \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{deh}\, U^{ad}(w')\, U^{bl}(y)\, U^{ce}(x')\, \big\{\rho^h(x'),\rho^l(y)\big\}$$
$$- \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',y,y)\, U^{ab}(w')\, U^{ac}(y)\, \big\{\rho^c(y),\rho^b(x')\big\}\Big\} + \big(w'\leftrightarrow w\big) \tag{G.28}$$

The kernel at the coincident points:

$$K^P_{\mathfrak{B}}(x',x,x') = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big]$$

$$K^P_{\mathfrak{B}}(x',x',y) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]$$
$$K^P_{\mathfrak{B}}(x',y,y) = \frac{1}{(y-z)^2}\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\Big]$$

G.3 Group II, Row I, first term (region P)

The three-ρ term.

$$\begin{aligned} P\Big|_{\text{GII.I1.P}} = & -i\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K^P_{\mathfrak{B}}(x',x,y)\,\Big\{ \\ & + U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\big\{\rho^{c'}(x'),\rho^b(y)\big\}\,\rho^d(x) \\ & - U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\big\{\rho^{c'}(x'),\rho^b(y)\big\}\,\rho^e(x) \\ & - U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,\big\{\rho^{d'}(x'),\rho^{e'}(y)\big\}\,\rho^d(x) \\ & + U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\big\{\rho^{d'}(x'),\rho^{e'}(y)\big\}\,\rho^e(x)\Big\} \end{aligned} \tag{G.29}$$

with $K^P_{\mathfrak{B}}(x',x,y) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.7) with A, B, C the charges at x, x', y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\big\{\rho^{c'}(x'),\rho^b(y)\big\}\,\rho^d(x)$.

$$\begin{aligned} &[A,B]:\ [\rho^d(x),\rho^{c'}(x')] = if^{dc'h}\rho^h(x)\,\delta^{(2)}(x-x'),\quad x'\to x \\ &\quad -\tfrac{1}{2}\{[A,B],C\} = -\tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{dc'h}\,\big\{\rho^h(x),\rho^b(y)\big\} \\ &\qquad\qquad = -\tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\big\{\rho^e(x),\rho^b(y)\big\} \end{aligned} \tag{G.30}$$

$$\begin{aligned} &[A,C]:\ [\rho^d(x),\rho^b(y)] = if^{dbh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x \\ &\quad -\tfrac{1}{2}\{[A,C],B\} = -\tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x',x,x)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{dbh}\,\big\{\rho^h(x),\rho^{c'}(x')\big\} \\ &\qquad\qquad = \tfrac{1}{2}\,iN_c\,K^P_{\mathfrak{B}}(x',x,x)\,U^{ab}(w')\,U^{ac}(w)\,\big\{\rho^c(x),\rho^b(x')\big\}\quad \text{(F)} \end{aligned} \tag{G.31}$$

Sub-term 2: $-U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\big\{\rho^{c'}(x'),\rho^b(y)\big\}\,\rho^e(x)$.

$$\begin{aligned} &[A,B]:\ [\rho^e(x),\rho^{c'}(x')] = if^{ec'h}\rho^h(x)\,\delta^{(2)}(x-x'),\quad x'\to x \\ &\quad -\tfrac{1}{2}\{[A,B],C\} = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ec'h}\,\big\{\rho^h(x),\rho^b(y)\big\} \\ &\qquad\qquad = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x)\,\big\{\rho^h(x),\rho^l(y)\big\} \end{aligned} \tag{G.32}$$

$$\begin{aligned} &[A,C]:\ [\rho^e(x),\rho^b(y)] = if^{ebh}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x \\ &\quad -\tfrac{1}{2}\{[A,C],B\} = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x',x,x)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ebh}\,\big\{\rho^h(x),\rho^{c'}(x')\big\} \\ &\qquad\qquad = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\big\{\rho^h(x),\rho^l(x')\big\} \end{aligned} \tag{G.33}$$

Sub-term 3: $-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,\big\{\rho^{d'}(x'),\rho^{e'}(y)\big\}\,\rho^d(x)$.

$$\begin{aligned} &[A,B]:\ [\rho^d(x),\rho^{d'}(x')] = if^{dd'h}\rho^h(x)\,\delta^{(2)}(x-x'),\quad x'\to x \\ &\quad -\tfrac{1}{2}\{[A,B],C\} = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,U^{bc'}(z)\,U^{ad'}(x)\,U^{c'e'}(y)\,f^{cbd}\,U^{ac}(w)\,f^{dd'h}\,\big\{\rho^h(x),\rho^{e'}(y)\big\} \\ &\qquad\qquad = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{ld}(x)\,U^{hm}(y)\,U^{la}(w)\,\big\{\rho^e(x),\rho^m(y)\big\} \end{aligned} \tag{G.34}$$

$$\begin{aligned} &[A,C]:\ [\rho^d(x),\rho^{e'}(y)] = if^{de'h}\rho^h(x)\,\delta^{(2)}(x-y),\quad y\to x \\ &\quad -\tfrac{1}{2}\{[A,C],B\} = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x',x,x)\,U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(x)\,f^{cbd}\,U^{ac}(w)\,f^{de'h}\,\big\{\rho^h(x),\rho^{d'}(x')\big\} \\ &\qquad\qquad = \tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{lm}(x')\,U^{hd}(x)\,U^{la}(w)\,\big\{\rho^e(x),\rho^m(x')\big\} \end{aligned} \tag{G.35}$$

Sub-term 4: $+U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\big\{\rho^{d'}(x'),\rho^{e'}(y)\big\}\,\rho^e(x)$.

$$\begin{aligned} &[A,B]:\ [\rho^e(x),\rho^{d'}(x')] = if^{ed'h}\rho^h(x)\,\delta^{(2)}(x-x'),\quad x'\to x \\ &\quad -\tfrac{1}{2}\{[A,B],C\} = -\tfrac{1}{2}\,i\,K^P_{\mathfrak{B}}(x,x,y)\,U^{bc'}(z)\,U^{ad'}(x)\,U^{c'e'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{ed'h}\,\big\{\rho^h(x),\rho^{e'}(y)\big\} \end{aligned}$$

$$= -\tfrac{1}{2}\, iN_c\; K^P_{\mathfrak{B}}(x,x,y)\, U^{ab}(y)\, U^{ac}(x)\, \{\rho^c(x), \rho^b(y)\} \quad \text{(U1), (U2), (F)} \tag{G.36}$$

$$[A,C]:\ [\rho^e(x), \rho^{e'}(y)] = if^{ee'h}\rho^h(x)\,\delta^{(2)}(x-y), \quad y \to x$$
$$-\tfrac{1}{2}\{[A,C],B\} = -\tfrac{1}{2}\, i\; K^P_{\mathfrak{B}}(x',x,x)\, U^{bc'}(z)\, U^{ad'}(x')\, U^{c'e'}(x)\, f^{acd}\, U^{bc}(z)\, U^{de}(x)\, f^{ee'h}\, \{\rho^h(x), \rho^{d'}(x')\}$$
$$= \tfrac{1}{2}\, iN_c\; K^P_{\mathfrak{B}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \{\rho^c(x), \rho^b(x')\} \quad \text{(U1), (U2), (F)} \tag{G.37}$$

Two-ρ terms of this source. Multiplying by the prefactor $-i\,\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$\begin{aligned} P^{(2\rho)}\Big|_{\text{GII.I1.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)} \Big\{ & \\ & -\tfrac{1}{2}\, K^P_{\mathfrak{B}}(x,x,y)\, f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, \{\rho^e(x), \rho^b(y)\} \\ & +\tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',x,x)\, U^{ab}(w')\, U^{ac}(w)\, \{\rho^c(x), \rho^b(x')\} \\ & +\tfrac{1}{2}\, K^P_{\mathfrak{B}}(x,x,y)\, f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{lb}(z)\, U^{cd}(x)\, \{\rho^h(x), \rho^l(y)\} \\ & +\tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x,x)\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{eb}(z)\, U^{cd}(x)\, \{\rho^h(x), \rho^l(x')\} \\ & +\tfrac{1}{2}\, K^P_{\mathfrak{B}}(x,x,y)\, f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{ld}(x)\, U^{hm}(y)\, U^{la}(w)\, \{\rho^e(x), \rho^m(y)\} \\ & +\tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x,x)\, f^{abc}\, f^{cde}\, U^{bh}(z)\, U^{lm}(x')\, U^{hd}(x)\, U^{la}(w)\, \{\rho^e(x), \rho^m(x')\} \\ & -\tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x,x,y)\, U^{ab}(y)\, U^{ac}(x)\, \{\rho^c(x), \rho^b(y)\} \\ & +\tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',x,x)\, U^{ab}(x')\, U^{ac}(x)\, \{\rho^c(x), \rho^b(x')\} \Big\} \end{aligned} \tag{G.38}$$

The kernel at the coincident points:

$$K^P_{\mathfrak{B}}(x,x,y) = \Big[\mathcal{K}(x-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(x-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$$
$$K^P_{\mathfrak{B}}(x',x,x) = \frac{1}{(x-z)^2}\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\Big]$$

G.4 Group II, Row I, second term (region P)

The three-ρ term.

$$\begin{aligned} P\Big|_{\text{GII.I2.P}} = i\,\mathcal{N}_P \int e^{-ik\cdot(w'-w)}\, \bar{K}^P_{\mathfrak{B}}(x',x,y) \Big\{ & \\ & +\, f^{c'bd'}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x)\, \{\rho^c(x'), \rho^b(y)\} \\ & -\, f^{c'bd'}\, U^{ac'}(w')\, U^{bc}(z)\, U^{ad}(x')\, U^{ce}(y)\, \rho^{d'}(x)\, \{\rho^d(x'), \rho^e(y)\} \\ & -\, f^{ac'd'}\, U^{bc'}(z)\, U^{d'e'}(x)\, U^{ac}(w)\, \rho^{e'}(x)\, \{\rho^c(x'), \rho^b(y)\} \\ & +\, f^{ac'd'}\, U^{bc'}(z)\, U^{d'e'}(x)\, U^{bc}(z)\, U^{ad}(x')\, U^{ce}(y)\, \rho^{e'}(x)\, \{\rho^d(x'), \rho^e(y)\} \Big\} \end{aligned} \tag{G.39}$$

with $\bar{K}^P_{\mathfrak{B}}(x',x,y) = \Big[\mathcal{K}(x'-w)\cdot\mathcal{K}(z-w')\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z) - \tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x, x', y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\, f^{c'bd'}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x)\, \{\rho^c(x'), \rho^b(y)\}$.

$$[A,B]:\ [\rho^{d'}(x), \rho^c(x')] = if^{d'ch}\rho^h(x)\,\delta^{(2)}(x-x'), \quad x' \to x$$
$$\tfrac{1}{2}\{[A,B],C\} = \tfrac{1}{2}\, i\; \bar{K}^P_{\mathfrak{B}}(x,x,y)\, f^{c'bd'}\, U^{ac'}(w')\, U^{ac}(w)\, f^{d'ch}\, \{\rho^h(x), \rho^b(y)\}$$
$$= \tfrac{1}{2}\, i\; \bar{K}^P_{\mathfrak{B}}(x,x,y)\, f^{abc}\, f^{cde}\, U^{ha}(w')\, U^{hd}(w)\, \{\rho^e(x), \rho^b(y)\} \tag{G.40}$$
$$[A,C]:\ [\rho^{d'}(x), \rho^b(y)] = if^{d'bh}\rho^h(x)\,\delta^{(2)}(x-y), \quad y \to x$$
$$\tfrac{1}{2}\{[A,C],B\} = \tfrac{1}{2}\, i\; \bar{K}^P_{\mathfrak{B}}(x',x,x)\, f^{c'bd'}\, U^{ac'}(w')\, U^{ac}(w)\, f^{d'bh}\, \{\rho^h(x), \rho^c(x')\}$$
$$= -\tfrac{1}{2}\, iN_c\; \bar{K}^P_{\mathfrak{B}}(x',x,x)\, U^{ab}(w')\, U^{ac}(w)\, \{\rho^b(x), \rho^c(x')\} \quad \text{(F)} \tag{G.41}$$

Sub-term 2: $-\,f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{d'}(x)\,\{\rho^{d}(x'),\rho^{e}(y)\}$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{d'}(x),\rho^{d}(x')] = if^{d'dh}\rho^{h}(x)\,\delta^{(2)}(x-x'),\quad x'\to x\\
&\quad \tfrac{1}{2}\{[A,B],C\} = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,f^{d'dh}\,\{\rho^{h}(x),\rho^{e}(y)\}\\
&\qquad\qquad = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hd}(x)\,U^{lm}(y)\,\{\rho^{e}(x),\rho^{m}(y)\} \qquad \text{(G.42)}\\
&[A,C]:\ [\rho^{d'}(x),\rho^{e}(y)] = if^{d'eh}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad \tfrac{1}{2}\{[A,C],B\} = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{c'bd'}\,U^{ac'}(w')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{d'eh}\,\{\rho^{h}(x),\rho^{d}(x')\}\\
&\qquad\qquad = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hm}(x')\,U^{ld}(x)\,\{\rho^{e}(x),\rho^{m}(x')\} \qquad \text{(G.43)}
\end{aligned}
$$

Sub-term 3: $-\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,\rho^{e'}(x)\,\{\rho^{c}(x'),\rho^{b}(y)\}$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{e'}(x),\rho^{c}(x')] = if^{e'ch}\rho^{h}(x)\,\delta^{(2)}(x-x'),\quad x'\to x\\
&\quad \tfrac{1}{2}\{[A,B],C\} = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,f^{e'ch}\,\{\rho^{h}(x),\rho^{b}(y)\}\\
&\qquad\qquad = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{cd}(x)\,U^{ae}(w)\,\{\rho^{h}(x),\rho^{l}(y)\} \qquad \text{(G.44)}\\
&[A,C]:\ [\rho^{e'}(x),\rho^{b}(y)] = if^{e'bh}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad \tfrac{1}{2}\{[A,C],B\} = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{ac}(w)\,f^{e'bh}\,\{\rho^{h}(x),\rho^{c}(x')\}\\
&\qquad\qquad = -\tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{cd}(x)\,U^{al}(w)\,\{\rho^{h}(x),\rho^{l}(x')\} \qquad \text{(G.45)}
\end{aligned}
$$

Sub-term 4: $+\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\rho^{e'}(x)\,\{\rho^{d}(x'),\rho^{e}(y)\}$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{e'}(x),\rho^{d}(x')] = if^{e'dh}\rho^{h}(x)\,\delta^{(2)}(x-x'),\quad x'\to x\\
&\quad \tfrac{1}{2}\{[A,B],C\} = \tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x)\,U^{ce}(y)\,f^{e'dh}\,\{\rho^{h}(x),\rho^{e}(y)\}\\
&\qquad\qquad = \tfrac{1}{2}\,iN_c\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,U^{ab}(y)\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(y)\}\quad \text{(U1), (U2), (F)} \qquad \text{(G.46)}\\
&[A,C]:\ [\rho^{e'}(x),\rho^{e}(y)] = if^{e'eh}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad \tfrac{1}{2}\{[A,C],B\} = \tfrac{1}{2}\,i\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'eh}\,\{\rho^{h}(x),\rho^{d}(x')\}\\
&\qquad\qquad = -\tfrac{1}{2}\,iN_c\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\}\quad \text{(U1), (U2), (F)} \qquad \text{(G.47)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $i\,\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
P^{(2\rho)}\Big|_{\text{GII.I2.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,\Big\{
&-\tfrac{1}{2}\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{hd}(w)\,\{\rho^{e}(x),\rho^{b}(y)\}\\
&+\tfrac{1}{2}\,N_c\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,U^{ab}(w')\,U^{ac}(w)\,\{\rho^{b}(x),\rho^{c}(x')\}\\
&+\tfrac{1}{2}\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hd}(x)\,U^{lm}(y)\,\{\rho^{e}(x),\rho^{m}(y)\}\\
&+\tfrac{1}{2}\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{cde}\,U^{ha}(w')\,U^{bl}(z)\,U^{hm}(x')\,U^{ld}(x)\,\{\rho^{e}(x),\rho^{m}(x')\}\\
&+\tfrac{1}{2}\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{cd}(x)\,U^{ae}(w)\,\{\rho^{h}(x),\rho^{l}(y)\}\\
&+\tfrac{1}{2}\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{cd}(x)\,U^{al}(w)\,\{\rho^{h}(x),\rho^{l}(x')\}\\
&-\tfrac{1}{2}\,N_c\,\bar{K}^{P}_{\mathfrak{B}}(x,x,y)\,U^{ab}(y)\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(y)\}\\
&+\tfrac{1}{2}\,N_c\,\bar{K}^{P}_{\mathfrak{B}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\}\Big\}
\end{aligned}
\qquad \text{(G.48)}
$$

The kernel at the coincident points:

$$
\begin{aligned}
\bar{K}^{P}_{\mathfrak{B}}(x,x,y) &= \Big[\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]\\
\bar{K}^{P}_{\mathfrak{B}}(x',x,x) &= \frac{1}{(x-z)^2}\Big[\mathcal{K}(x'-w)\cdot\mathcal{K}(z-w') - \frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x-w')\Big]
\end{aligned}
$$

G.5 Group III, Row I (region P)

The three-ρ term.

$$
P\Big|_{\text{GIII.I.P}} = -\,2i\,\mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,K^{P}_{\mathfrak{B}}(x',x,y)\,\Big\{
$$

$$
\begin{aligned}
&+\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{d}(x)\\
&-\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{e}(x)\\
&-\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{d}(x)\\
&+\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{e}(x)\Big\}+\text{c.c.}
\end{aligned}
\tag{G.49}
$$

with $K^P_{\mathfrak{B}}(x',x,y)=\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z)-\frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{d}(x)$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^{d'}(y)]&=if^{b'd'h}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{4}\{[A,B],C\}&=\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,U^{c'd'}(x')\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,f^{b'd'h}\,\{\rho^{h}(x'),\rho^{d}(x)\}\\
&=\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{da}(w)\,U^{hl}(x')\,\{\rho^{l}(x'),\rho^{c}(x)\}\quad\text{(U2)}
\end{aligned}
\tag{G.50}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{b'}(x'),\rho^{d}(x)]&=if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[A,C],B\}&=\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,f^{b'dh}\,\{\rho^{h}(x'),\rho^{d'}(y)\}\\
&=\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(y)\,U^{lb}(z)\,U^{ha}(w)\,\{\rho^{e}(x'),\rho^{m}(y)\}
\end{aligned}
\tag{G.51}
$$

$$
\begin{aligned}
[B,C]:\ [\rho^{d'}(y),\rho^{d}(x)]&=if^{d'dh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[B,C],A\}&=\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{cbd}\,U^{ac}(w)\,f^{d'dh}\,\{\rho^{h}(y),\rho^{b'}(x')\}\\
&=\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{md}(y)\,U^{mb}(z)\,U^{ha}(w)\,\{\rho^{e}(y),\rho^{l}(x')\}
\end{aligned}
\tag{G.52}
$$

Sub-term 2: $-\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b'}(x')\,\rho^{d'}(y)\,\rho^{e}(x)$.

$$
\begin{aligned}
[A,B]:\ [\rho^{b'}(x'),\rho^{d'}(y)]&=if^{b'd'h}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{4}\{[A,B],C\}&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,U^{ab'}(x')\,U^{c'd'}(x')\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{b'd'h}\,\{\rho^{h}(x'),\rho^{e}(x)\}\\
&=-\tfrac{1}{4}\,iN_c\;K^P_{\mathfrak{B}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\}\quad\text{(U1}^\star\text{)},\ \text{(U2)},\ \text{(F)}
\end{aligned}
\tag{G.53}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{b'}(x'),\rho^{e}(x)]&=if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[A,C],B\}&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{b'eh}\,\{\rho^{h}(x'),\rho^{d'}(y)\}\\
&=\tfrac{1}{4}\,iN_c\;K^P_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\quad\text{(U1}^\star\text{)},\ \text{(U2)},\ \text{(F)}
\end{aligned}
\tag{G.54}
$$

$$
\begin{aligned}
[B,C]:\ [\rho^{d'}(y),\rho^{e}(x)]&=if^{d'eh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[B,C],A\}&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{d'eh}\,\{\rho^{h}(y),\rho^{b'}(x')\}\\
&=-\tfrac{1}{4}\,iN_c\;K^P_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\{\rho^{c}(y),\rho^{b}(x')\}\quad\text{(U1}^\star\text{)},\ \text{(U2)},\ \text{(F)}
\end{aligned}
\tag{G.55}
$$

Sub-term 3: $-\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{d}(x)$.

$$
\begin{aligned}
[A,B]:\ [\rho^{c'}(x'),\rho^{e'}(y)]&=if^{c'e'h}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
\tfrac{1}{4}\{[A,B],C\}&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,U^{bd'}(z)\,U^{d'e'}(x')\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{c'e'h}\,\{\rho^{h}(x'),\rho^{d}(x)\}\\
&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x')\,U^{md}(w')\,U^{ma}(w)\,\{\rho^{h}(x'),\rho^{c}(x)\}
\end{aligned}
\tag{G.56}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{c'}(x'),\rho^{d}(x)]&=if^{c'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[A,C],B\}&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{c'dh}\,\{\rho^{h}(x'),\rho^{e'}(y)\}\\
&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(y)\,U^{md}(w')\,U^{ma}(w)\,\{\rho^{e}(x'),\rho^{l}(y)\}
\end{aligned}
\tag{G.57}
$$

$$
\begin{aligned}
[B,C]:\ [\rho^{e'}(y),\rho^{d}(x)]&=if^{e'dh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[B,C],A\}&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',y,y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{cbd}\,U^{ac}(w)\,f^{e'dh}\,\{\rho^{h}(y),\rho^{c'}(x')\}\\
&=-\tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(y)\,U^{lm}(w')\,U^{la}(w)\,\{\rho^{e}(y),\rho^{m}(x')\}
\end{aligned}
\tag{G.58}
$$

Sub-term 4: $+\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{c'}(x')\,\rho^{e'}(y)\,\rho^{e}(x)$.

$$
[A,B]:\ [\rho^{c'}(x'),\rho^{e'}(y)]=if^{c'e'h}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'
$$

$$\tfrac{1}{4}\{[A,B],C\} = \tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,U^{bd'}(z)\,U^{d'e'}(x')\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{c'e'h}\,\{\rho^h(x'),\rho^e(x)\}$$
$$= \tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{be}(x')\,U^{ad}(w')\,U^{cl}(x)\,\{\rho^h(x'),\rho^l(x)\} \quad \text{(U1)} \tag{G.59}$$

$$[A,C]:\ [\rho^{c'}(x'),\rho^e(x)] = if^{c'eh}\rho^h(x')\,\delta^{(2)}(x'-x), \quad x\to x'$$
$$\tfrac{1}{4}\{[A,C],B\} = \tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{c'eh}\,\{\rho^h(x'),\rho^{e'}(y)\}$$
$$= \tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{bl}(y)\,U^{ad}(w')\,U^{ce}(x')\,\{\rho^h(x'),\rho^l(y)\} \quad \text{(U1)} \tag{G.60}$$

$$[B,C]:\ [\rho^{e'}(y),\rho^e(x)] = if^{e'eh}\rho^h(y)\,\delta^{(2)}(y-x), \quad x\to y$$
$$\tfrac{1}{4}\{[B,C],A\} = \tfrac{1}{4}\,i\;K^P_{\mathfrak{B}}(x',y,y)\,U^{bd'}(z)\,U^{d'e'}(y)\,U^{ac'}(w')\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{e'eh}\,\{\rho^h(y),\rho^{c'}(x')\}$$
$$= \tfrac{1}{4}\,iN_c\;K^P_{\mathfrak{B}}(x',y,y)\,U^{ab}(w')\,U^{ac}(y)\,\{\rho^c(y),\rho^b(x')\} \quad \text{(U1), (U2), (F)} \tag{G.61}$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$P^{(2\rho)}\Big|_{\text{GIII.I.P}} = \mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,\Big\{$$
$$+\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{da}(w)\,U^{hl}(x')\,\{\rho^l(x'),\rho^c(x)\}$$
$$+\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(y)\,U^{lb}(z)\,U^{ha}(w)\,\{\rho^e(x'),\rho^m(y)\}$$
$$+\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hl}(x')\,U^{md}(y)\,U^{mb}(z)\,U^{ha}(w)\,\{\rho^e(y),\rho^l(x')\}$$
$$-\tfrac{1}{2}\,N_c\,K^P_{\mathfrak{B}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(x)\}$$
$$+\tfrac{1}{2}\,N_c\,K^P_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^c(x'),\rho^b(y)\}$$
$$-\tfrac{1}{2}\,N_c\,K^P_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\{\rho^c(y),\rho^b(x')\}$$
$$-\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x')\,U^{md}(w')\,U^{ma}(w)\,\{\rho^h(x'),\rho^c(x)\}$$
$$-\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(y)\,U^{md}(w')\,U^{ma}(w)\,\{\rho^e(x'),\rho^l(y)\}$$
$$-\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(y)\,U^{lm}(w')\,U^{la}(w)\,\{\rho^e(y),\rho^m(x')\}$$
$$+\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{be}(x')\,U^{ad}(w')\,U^{cl}(x)\,\{\rho^h(x'),\rho^l(x)\}$$
$$+\tfrac{1}{2}\,K^P_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{bl}(y)\,U^{ad}(w')\,U^{ce}(x')\,\{\rho^h(x'),\rho^l(y)\}$$
$$+\tfrac{1}{2}\,N_c\,K^P_{\mathfrak{B}}(x',y,y)\,U^{ab}(w')\,U^{ac}(y)\,\{\rho^c(y),\rho^b(x')\}\Big\} + \big(w'\leftrightarrow w\big) \tag{G.62}$$

The kernel at the coincident points:

$$K^P_{\mathfrak{B}}(x',x,x') = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\Big]$$
$$K^P_{\mathfrak{B}}(x',x',y) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]$$
$$K^P_{\mathfrak{B}}(x',y,y) = \frac{1}{(y-z)^2}\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\Big]$$

G.6 Group III, Row II (region P)

The three-ρ term.

$$P\Big|_{\text{GIII.II.P}} = -2i\mathcal{N}_P\int e^{-ik\cdot(w'-w)}\,K^P_{\mathfrak{B}}(x',x,y)\,\Big\{$$
$$+\;U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^d(x)$$
$$-\;U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^e(x)$$
$$-\;U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,\rho^b(y)\,\rho^{b'}(x')\,\rho^d(x)$$
$$+\;U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^b(y)\,\rho^{b'}(x')\,\rho^e(x)\Big\} + \text{c.c.} \tag{G.63}$$

with $K^P_{\mathfrak{B}}(x',x,y) = \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big]$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at y, x', x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{d}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{d'}(y),\rho^{b'}(x')] &= if^{d'b'h}\rho^{h}(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,U^{c'd'}(y)\,U^{ab'}(y)\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,f^{d'b'h}\left\{\rho^{h}(y),\rho^{d}(x)\right\}\\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w)\,U^{hl}(y)\left\{\rho^{l}(y),\rho^{c}(x)\right\}\quad\text{(U2)} \qquad \text{(G.64)}\\
[A,C]:\ [\rho^{d'}(y),\rho^{d}(x)] &= if^{d'dh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',y,y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,f^{d'dh}\left\{\rho^{h}(y),\rho^{b'}(x')\right\}\\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{dce}\,U^{hd}(y)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\left\{\rho^{e}(y),\rho^{m}(x')\right\} \qquad \text{(G.65)}\\
[B,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] &= if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',x',y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{cbd}\,U^{ac}(w)\,f^{b'dh}\left\{\rho^{h}(x'),\rho^{d'}(y)\right\}\\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hl}(y)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\left\{\rho^{e}(x'),\rho^{l}(y)\right\} \qquad \text{(G.66)}
\end{aligned}
$$

Sub-term 2: $-U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(y)\,\rho^{b'}(x')\,\rho^{e}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{d'}(y),\rho^{b'}(x')] &= if^{d'b'h}\rho^{h}(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,U^{c'd'}(y)\,U^{ab'}(y)\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{d'b'h}\left\{\rho^{h}(y),\rho^{e}(x)\right\}\\
&= \tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{B}}(y,x,y)\,U^{ab}(x)\,U^{ac}(y)\left\{\rho^{c}(y),\rho^{b}(x)\right\}\quad\text{(U1), (U2), (F)} \qquad \text{(G.67)}\\
[A,C]:\ [\rho^{d'}(y),\rho^{e}(x)] &= if^{d'eh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',y,y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{d'eh}\left\{\rho^{h}(y),\rho^{b'}(x')\right\}\\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\left\{\rho^{c}(y),\rho^{b}(x')\right\}\quad\text{(U1), (U2), (F)} \qquad \text{(G.68)}\\
[B,C]:\ [\rho^{b'}(x'),\rho^{e}(x)] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',x',y)\,U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{b'eh}\left\{\rho^{h}(x'),\rho^{d'}(y)\right\}\\
&= \tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{B}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\left\{\rho^{c}(x'),\rho^{b}(y)\right\}\quad\text{(U1), (U2), (F)} \qquad \text{(G.69)}
\end{aligned}
$$

Sub-term 3: $-U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,\rho^{b}(y)\,\rho^{b'}(x')\,\rho^{d}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{b}(y),\rho^{b'}(x')] &= if^{bb'h}\rho^{h}(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,U^{ab'}(y)\,f^{cbd}\,U^{ac}(w)\,f^{bb'h}\left\{\rho^{h}(y),\rho^{d}(x)\right\}\\
&= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{bde}\,U^{hd}(y)\,U^{ha}(w)\left\{\rho^{e}(y),\rho^{c}(x)\right\} \qquad \text{(G.70)}\\
[A,C]:\ [\rho^{b}(y),\rho^{d}(x)] &= if^{bdh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,f^{bdh}\left\{\rho^{h}(y),\rho^{b'}(x')\right\}\\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{B}}(x',y,y)\,U^{ab}(x')\,U^{ac}(w)\left\{\rho^{c}(y),\rho^{b}(x')\right\}\quad\text{(F)} \qquad \text{(G.71)}\\
[B,C]:\ [\rho^{b'}(x'),\rho^{d}(x)] &= if^{b'dh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{cbd}\,U^{ac}(w)\,f^{b'dh}\left\{\rho^{h}(x'),\rho^{b}(y)\right\}\\
&= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',x',y)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w)\left\{\rho^{e}(x'),\rho^{b}(y)\right\} \qquad \text{(G.72)}
\end{aligned}
$$

Sub-term 4: $+U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{b}(y)\,\rho^{b'}(x')\,\rho^{e}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{b}(y),\rho^{b'}(x')] &= if^{bb'h}\rho^{h}(y)\,\delta^{(2)}(y-x'),\quad x'\to y\\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,U^{ab'}(y)\,f^{acd}\,U^{bc}(z)\,U^{de}(x)\,f^{bb'h}\left\{\rho^{h}(y),\rho^{e}(x)\right\}\\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(y,x,y)\,f^{abc}\,f^{deh}\,U^{ae}(y)\,U^{db}(z)\,U^{cl}(x)\left\{\rho^{h}(y),\rho^{l}(x)\right\} \qquad \text{(G.73)}\\
[A,C]:\ [\rho^{b}(y),\rho^{e}(x)] &= if^{beh}\rho^{h}(y)\,\delta^{(2)}(y-x),\quad x\to y\\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',y,y)\,U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(y)\,f^{beh}\left\{\rho^{h}(y),\rho^{b'}(x')\right\}\\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',y,y)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{db}(z)\,U^{ce}(y)\left\{\rho^{h}(y),\rho^{l}(x')\right\} \qquad \text{(G.74)}\\
[B,C]:\ [\rho^{b'}(x'),\rho^{e}(x)] &= if^{b'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{B}}(x',x',y)\,U^{ab'}(x')\,f^{acd}\,U^{bc}(z)\,U^{de}(x')\,f^{b'eh}\left\{\rho^{h}(x'),\rho^{b}(y)\right\}\\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{B}}(x',x',y)\,U^{ab}(z)\,U^{bc}(x')\left\{\rho^{c}(x'),\rho^{a}(y)\right\}\quad\text{(U2), (F)} \qquad \text{(G.75)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\,\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
P^{(2\rho)}\Big|_{\text{GIII.II.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)} \Big\{ & \\
& + \tfrac{1}{2}\, K^P_{\mathfrak{B}}(y,x,y)\, f^{abc}\, f^{deh}\, U^{bd}(z)\, U^{ea}(w)\, U^{hl}(y)\, \big\{\rho^l(y), \rho^c(x)\big\} \\
& + \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{dce}\, U^{hd}(y)\, U^{lm}(x')\, U^{bh}(z)\, U^{la}(w)\, \big\{\rho^e(y), \rho^m(x')\big\} \\
& + \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hl}(y)\, U^{md}(x')\, U^{bh}(z)\, U^{ma}(w)\, \big\{\rho^e(x'), \rho^l(y)\big\} \\
& + \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(y,x,y)\, U^{ab}(x)\, U^{ac}(y)\, \big\{\rho^c(y), \rho^b(x)\big\} \\
& - \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',y,y)\, U^{ab}(x')\, U^{ac}(y)\, \big\{\rho^c(y), \rho^b(x')\big\} \\
& + \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',x',y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
& - \tfrac{1}{2}\, K^P_{\mathfrak{B}}(y,x,y)\, f^{abc}\, f^{bde}\, U^{hd}(y)\, U^{ha}(w)\, \big\{\rho^e(y), \rho^c(x)\big\} \\
& - \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',y,y)\, U^{ab}(x')\, U^{ac}(w)\, \big\{\rho^c(y), \rho^b(x')\big\} \\
& - \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',x',y)\, f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w)\, \big\{\rho^e(x'), \rho^b(y)\big\} \\
& + \tfrac{1}{2}\, K^P_{\mathfrak{B}}(y,x,y)\, f^{abc}\, f^{deh}\, U^{ae}(y)\, U^{db}(z)\, U^{cl}(x)\, \big\{\rho^h(y), \rho^l(x)\big\} \\
& + \tfrac{1}{2}\, K^P_{\mathfrak{B}}(x',y,y)\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{db}(z)\, U^{ce}(y)\, \big\{\rho^h(y), \rho^l(x')\big\} \\
& - \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{B}}(x',x',y)\, U^{ab}(z)\, U^{bc}(x')\, \big\{\rho^c(x'), \rho^a(y)\big\} \Big\} + \big(w' \leftrightarrow w\big)
\end{aligned}
\tag{G.76}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K^P_{\mathfrak{B}}(y,x,y) &= \Big[\mathcal{K}(y-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x-z) - \frac{1}{2}\,\mathcal{K}(y-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\Big] \\
K^P_{\mathfrak{B}}(x',y,y) &= \frac{1}{(y-z)^2}\Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\Big] \\
K^P_{\mathfrak{B}}(x',x',y) &= \Big[\mathcal{K}(x'-w')\cdot\mathcal{K}(z-w)\,\mathcal{K}(y-z)\cdot\mathcal{K}(x'-z) - \frac{1}{2}\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\Big]
\end{aligned}
$$

G.7 Group III, Row III, first part (region P)

The three-ρ term.

$$
\begin{aligned}
P\Big|_{\text{GIII.III.P}} = & -i\,\mathcal{N}_P \int e^{-ik\cdot(w'-w)}\, K^P_{\mathfrak{C}}(x',x,y)\, \Big\{ \\
& + f^{c'd'a}\, U^{e'c'}(z)\, U^{d'b}(z)\, U^{ac}(w)\, \rho^{e'}(x')\, \big\{\rho^c(y), \rho^b(x)\big\} \\
& - f^{c'd'a}\, U^{e'c'}(z)\, U^{d'b}(z)\, U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\, \rho^{e'}(x')\, \big\{\rho^d(y), \rho^e(x)\big\} \\
& - f^{c'd'a}\, U^{bd'}(z)\, U^{c'e'}(x')\, U^{ac}(w)\, \rho^{e'}(x')\, \big\{\rho^c(y), \rho^b(x)\big\} \\
& + f^{c'd'a}\, U^{bd'}(z)\, U^{c'e'}(x')\, U^{bc}(z)\, U^{ad}(y)\, U^{ce}(x)\, \rho^{e'}(x')\, \big\{\rho^d(y), \rho^e(x)\big\} \Big\} + \text{c.c.}
\end{aligned}
\tag{G.77}
$$

with $K^P_{\mathfrak{C}}(x',x,y) = \mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.6) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a}\, U^{e'c'}(z)\, U^{d'b}(z)\, U^{ac}(w)\, \rho^{e'}(x')\, \big\{\rho^c(y), \rho^b(x)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'), \rho^c(y)] &= i f^{e'ch}\rho^h(x')\,\delta^{(2)}(x'-y), \quad y \to x' \\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{c'd'a}\, U^{e'c'}(z)\, U^{d'b}(z)\, U^{ac}(w)\, f^{e'ch}\, \big\{\rho^h(x'), \rho^b(x)\big\} \\
&= \tfrac{1}{2}\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{ade}\, U^{hb}(w)\, U^{eh}(z)\, \big\{\rho^c(x'), \rho^d(x)\big\} \quad \text{(U2}^\star\text{)}
\end{aligned}
\tag{G.78}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{e'}(x'), \rho^b(x)] &= i f^{e'bh}\rho^h(x')\,\delta^{(2)}(x'-x), \quad x \to x' \\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{c'd'a}\, U^{e'c'}(z)\, U^{d'b}(z)\, U^{ac}(w)\, f^{e'bh}\, \big\{\rho^h(x'), \rho^c(y)\big\} \\
&= \tfrac{1}{2}\, i N_c\, K^P_{\mathfrak{C}}(x',x',y)\, U^{ab}(w)\, U^{ca}(z)\, \big\{\rho^c(x'), \rho^b(y)\big\} \quad \text{(U2}^\star\text{), (F)}
\end{aligned}
\tag{G.79}
$$

Sub-term 2: $-\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\big\{\rho^{d}(y),\rho^{e}(x)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\big\{\rho^{h}(x'),\rho^{e}(x)\big\} \\
&= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(x')\,U^{bl}(x)\,\big\{\rho^{h}(x'),\rho^{l}(x)\big\} \quad (\mathrm{U1}^{\star}) \qquad \text{(G.80)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\big\{\rho^{h}(x'),\rho^{d}(y)\big\} \\
&= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\} \quad (\mathrm{U1}^{\star}) \qquad \text{(G.81)}
\end{aligned}
$$

Sub-term 3: $-\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,\rho^{e'}(x')\,\big\{\rho^{c}(y),\rho^{b}(x)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{c}(y)] &= i f^{e'ch}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{2}\{[A,B],C\} &= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,f^{e'ch}\,\big\{\rho^{h}(x'),\rho^{b}(x)\big\} \\
&= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w)\,\big\{\rho^{h}(x'),\rho^{l}(x)\big\} \qquad \text{(G.82)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{b}(x)] &= i f^{e'bh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{2}\{[A,C],B\} &= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ac}(w)\,f^{e'bh}\,\big\{\rho^{h}(x'),\rho^{c}(y)\big\} \\
&= -\tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(w)\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\} \qquad \text{(G.83)}
\end{aligned}
$$

Sub-term 4: $+\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\big\{\rho^{d}(y),\rho^{e}(x)\big\}$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{2}\{[A,B],C\} &= \tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\big\{\rho^{h}(x'),\rho^{e}(x)\big\} \\
&= -\tfrac{1}{2}\,iN_c\;K^{P}_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(x)\big\} \quad (\mathrm{U1}),\ (\mathrm{U2}),\ (\mathrm{F}) \qquad \text{(G.84)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{2}\{[A,C],B\} &= \tfrac{1}{2}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\big\{\rho^{h}(x'),\rho^{d}(y)\big\} \\
&= \tfrac{1}{2}\,iN_c\;K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(y)\big\} \quad (\mathrm{U1}),\ (\mathrm{U2}),\ (\mathrm{F}) \qquad \text{(G.85)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-i\,\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
P^{(2\rho)}\Big|_{\text{GIII.III.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,\Big\{ & \\
& + \tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{eh}(z)\,\big\{\rho^{c}(x'),\rho^{d}(x)\big\} \\
& + \tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(w)\,U^{ca}(z)\,\big\{\rho^{c}(x'),\rho^{b}(y)\big\} \\
& - \tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(x')\,U^{bl}(x)\,\big\{\rho^{h}(x'),\rho^{l}(x)\big\} \\
& - \tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(y)\,U^{be}(x')\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\} \\
& - \tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w)\,\big\{\rho^{h}(x'),\rho^{l}(x)\big\} \\
& - \tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(w)\,\big\{\rho^{h}(x'),\rho^{l}(y)\big\} \\
& - \tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(x)\big\} \\
& + \tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\big\{\rho^{c}(x'),\rho^{b}(y)\big\}\Big\} + \big(w'\leftrightarrow w\big) \qquad \text{(G.86)}
\end{aligned}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K^{P}_{\mathfrak{C}}(x',x,x') &= \mathcal{K}(w'-z)\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z) \\
K^{P}_{\mathfrak{C}}(x',x',y) &= \mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,\frac{1}{(x'-z)^2}
\end{aligned}
$$

G.8 Group III, Row V (region P)

The three-ρ term.

$$
\begin{aligned}
P\Big|_{\text{GIII.V.P}} = & -2i\,\mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,K^{P}_{\mathfrak{C}}(x',x,y)\,\Big\{ \\
& + f^{c'ca}\,U^{e'c'}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)
\end{aligned}
$$

$$
\begin{aligned}
&- f^{c'ca}\,U^{e'c'}(z)\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\\
&- f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\\
&+ f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)\Big\} + \text{c.c.}
\end{aligned}
\tag{G.87}
$$

with $K^P_{\mathfrak{C}}(x',x,y) = \mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', x, y. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'ca}\,U^{e'c'}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] = if^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad \tfrac14\{[A,B],C\} = \tfrac14\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\,U^{e'c'}(z)\,U^{ce}(x')\,U^{ad}(y)\, f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\}\\
&\qquad\qquad = \tfrac14\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\,U^{da}(z)\,U^{be}(x')\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\}
\end{aligned}
\tag{G.88}
$$

$$
\begin{aligned}
&[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] = if^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
&\quad \tfrac14\{[A,C],B\} = \tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\,U^{e'c'}(z)\,U^{ce}(x)\,U^{ad}(x')\, f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\}\\
&\qquad\qquad = \tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\,U^{da}(z)\,U^{bl}(x)\,U^{ce}(x')\,\{\rho^{h}(x'),\rho^{l}(x)\}
\end{aligned}
\tag{G.89}
$$

$$
\begin{aligned}
&[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] = if^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad \tfrac14\{[B,C],A\} = \tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\,U^{e'c'}(z)\,U^{ce}(x)\,U^{ad}(x)\, f^{edh}\,\{\rho^{h}(x),\rho^{e'}(x')\}\\
&\qquad\qquad = \tfrac14\, iN_c\, K^P_{\mathfrak{C}}(x',x,x)\,U^{ab}(z)\,U^{bc}(x)\,\{\rho^{c}(x),\rho^{a}(x')\}\quad \text{(U2), (F)}
\end{aligned}
\tag{G.90}
$$

Sub-term 2: $- f^{c'ca}\,U^{e'c'}(z)\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] = if^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad \tfrac14\{[A,B],C\} = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\,U^{e'c'}(z)\,U^{ad}(w)\,U^{ce}(x')\, f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\}\\
&\qquad\qquad = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{abc}\, f^{deh}\,U^{da}(z)\,U^{cl}(w)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\}
\end{aligned}
\tag{G.91}
$$

$$
\begin{aligned}
&[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] = if^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
&\quad \tfrac14\{[A,C],B\} = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\,U^{e'c'}(z)\,U^{ad}(w)\,U^{ce}(x)\, f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\}\\
&\qquad\qquad = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{abc}\, f^{deh}\,U^{da}(z)\,U^{ce}(w)\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\}
\end{aligned}
\tag{G.92}
$$

$$
\begin{aligned}
&[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] = if^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad \tfrac14\{[B,C],A\} = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\,U^{e'c'}(z)\,U^{ad}(w)\,U^{ce}(x)\, f^{edh}\,\{\rho^{h}(x),\rho^{e'}(x')\}\\
&\qquad\qquad = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x)\, f^{abc}\, f^{deh}\,U^{la}(z)\,U^{ce}(w)\,U^{bd}(x)\,\{\rho^{h}(x),\rho^{l}(x')\}
\end{aligned}
\tag{G.93}
$$

Sub-term 3: $- f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] = if^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad \tfrac14\{[A,B],C\} = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x')\,U^{ad}(y)\, f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\}\\
&\qquad\qquad = -\tfrac14\, iN_c\, K^P_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\quad \text{(U2), (F)}
\end{aligned}
\tag{G.94}
$$

$$
\begin{aligned}
&[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] = if^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y),\quad y\to x'\\
&\quad \tfrac14\{[A,C],B\} = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x')\, f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(x')\, f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\}\\
&\qquad\qquad = \tfrac14\, iN_c\, K^P_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\}\quad \text{(U2), (F)}
\end{aligned}
\tag{G.95}
$$

$$
\begin{aligned}
&[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] = if^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y),\quad y\to x\\
&\quad \tfrac14\{[B,C],A\} = -\tfrac14\, i\, K^P_{\mathfrak{C}}(x',x,x)\, f^{c'ca}\,U^{c'e'}(x')\,U^{ce}(x)\,U^{ad}(x)\, f^{edh}\,\{\rho^{h}(x),\rho^{e'}(x')\}\\
&\qquad\qquad = -\tfrac14\, iN_c\, K^P_{\mathfrak{C}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\}\quad \text{(U2), (F)}
\end{aligned}
\tag{G.96}
$$

Sub-term 4: $+ f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\,\rho^{d}(y)$.

$$
\begin{aligned}
&[A,B]:\ [\rho^{e'}(x'),\rho^{e}(x)] = if^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x),\quad x\to x'\\
&\quad \tfrac14\{[A,B],C\} = \tfrac14\, i\, K^P_{\mathfrak{C}}(x',x',y)\, f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x')\, f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\}\\
&\qquad\qquad = \tfrac14\, iN_c\, K^P_{\mathfrak{C}}(x',x',y)\,U^{ab}(w)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\}\quad \text{(U2), (F)}
\end{aligned}
\tag{G.97}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w)\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\} \qquad \text{(G.98)} \\
[B,C]:\ [\rho^{e}(x),\rho^{d}(y)] &= i f^{edh}\rho^{h}(x)\,\delta^{(2)}(x-y), \quad y\to x \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x)\,f^{c'ca}\,U^{c'e'}(x')\,U^{ad}(w)\,U^{ce}(x)\,f^{edh}\,\{\rho^{h}(x),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(w)\,U^{bd}(x)\,\{\rho^{h}(x),\rho^{l}(x')\} \qquad \text{(G.99)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\,\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
P^{(2\rho)}\Big|_{\text{GIII.V.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,\Big\{ & \\
&+\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{be}(x')\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\} \\
&+\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{bl}(x)\,U^{ce}(x')\,\{\rho^{h}(x'),\rho^{l}(x)\} \\
&+\tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x,x)\,U^{ab}(z)\,U^{bc}(x)\,\{\rho^{c}(x),\rho^{a}(x')\} \\
&-\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(w)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\} \\
&-\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(w)\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\} \\
&-\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{la}(z)\,U^{ce}(w)\,U^{bd}(x)\,\{\rho^{h}(x),\rho^{l}(x')\} \\
&-\tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\} \\
&+\tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\} \\
&-\tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x,x)\,U^{ab}(x')\,U^{ac}(x)\,\{\rho^{c}(x),\rho^{b}(x')\} \\
&+\tfrac{1}{2}\,N_c\,K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(w)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\} \\
&+\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w)\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\} \\
&+\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(w)\,U^{bd}(x)\,\{\rho^{h}(x),\rho^{l}(x')\}\Big\} + (w'\leftrightarrow w) \qquad \text{(G.100)}
\end{aligned}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K^{P}_{\mathfrak{C}}(x',x',y) &= \mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,\frac{1}{(x'-z)^2} \\
K^{P}_{\mathfrak{C}}(x',x,x') &= \mathcal{K}(w'-z)\cdot\mathcal{K}(x'-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z) \\
K^{P}_{\mathfrak{C}}(x',x,x) &= \mathcal{K}(w'-z)\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z)
\end{aligned}
$$

G.9 Group III, Row VI (region P)

The three-ρ term.

$$
\begin{aligned}
P\Big|_{\text{GIII.VI.P}} = & -2i\mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,K^{P}_{\mathfrak{C}}(x',x,y)\,\Big\{ \\
&+ f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x) \\
&- f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x) \\
&- f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x) \\
&+ f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x)\Big\} + \text{c.c.} \qquad \text{(G.101)}
\end{aligned}
$$

with $K^{P}_{\mathfrak{C}}(x',x,y) = \mathcal{K}(w'-z)\cdot\mathcal{K}(y-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(x-z)$.

Reordering. In every sub-term (one product of the two brackets) the charges appear in the order shown, so we apply Eq. (A.4) with A, B, C the charges at x', y, x. The sub-terms are numbered 1–4 in the order above. For each sub-term we list the commutators, and each resulting two-ρ piece before and after the colour algebra.

Sub-term 1: $+ f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x)$.

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(x')\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\} \quad \text{(U1}^{\star}\text{)} \qquad \text{(G.102)}
\end{aligned}
$$

$$
\begin{aligned}
[A,C]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(y)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\} \quad \text{(U1}^{\star}\text{)} \qquad \text{(G.103)} \\
[B,C]:\ [\rho^{d}(y),\rho^{e}(x)] &= i f^{deh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(y)\,f^{deh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{C}}(x',y,y)\,U^{ab}(z)\,U^{bc}(y)\,\{\rho^{c}(y),\rho^{a}(x')\} \quad \text{(U1}^{\star}\text{), (U2), (F)} \qquad \text{(G.104)}
\end{aligned}
$$

Sub-term 2: $-\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{ad}(x')\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{b}(x)\} \\
&= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\{\rho^{c}(x'),\rho^{d}(x)\} \quad \text{(U2}^{\star}\text{)} \qquad \text{(G.105)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{b}(x)] &= i f^{e'bh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{ad}(y)\,f^{e'bh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ca}(z)\,\{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U2}^{\star}\text{), (F)} \qquad \text{(G.106)} \\
[B,C]:\ [\rho^{d}(y),\rho^{b}(x)] &= i f^{dbh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,U^{ad}(y)\,f^{dbh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',y,y)\,f^{abc}\,f^{dbe}\,U^{ha}(y)\,U^{eh}(z)\,\{\rho^{c}(y),\rho^{d}(x')\} \quad \text{(U2}^{\star}\text{)} \qquad \text{(G.107)}
\end{aligned}
$$

Sub-term 3: $-\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{e}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(x')\,U^{ce}(x)\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{e}(x)\} \\
&= \tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{C}}(x',x,x')\,U^{ab}(x)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(x)\} \quad \text{(U1), (U2), (F)} \qquad \text{(G.108)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{e}(x)] &= i f^{e'eh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x')\,f^{e'eh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{C}}(x',x',y)\,U^{ab}(y)\,U^{ac}(x')\,\{\rho^{c}(x'),\rho^{b}(y)\} \quad \text{(U1), (U2), (F)} \qquad \text{(G.109)} \\
[B,C]:\ [\rho^{d}(y),\rho^{e}(x)] &= i f^{deh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= -\tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{bc}(z)\,U^{ad}(y)\,U^{ce}(y)\,f^{deh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{C}}(x',y,y)\,U^{ab}(x')\,U^{ac}(y)\,\{\rho^{c}(y),\rho^{b}(x')\} \quad \text{(U1), (U2), (F)} \qquad \text{(G.110)}
\end{aligned}
$$

Sub-term 4: $+\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,\rho^{e'}(x')\,\rho^{d}(y)\,\rho^{b}(x).$

$$
\begin{aligned}
[A,B]:\ [\rho^{e'}(x'),\rho^{d}(y)] &= i f^{e'dh}\rho^{h}(x')\,\delta^{(2)}(x'-y), \quad y\to x' \\
\tfrac{1}{4}\{[A,B],C\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x,x')\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(x')\,f^{e'dh}\,\{\rho^{h}(x'),\rho^{b}(x)\} \\
&= -\tfrac{1}{4}\,iN_c\;K^{P}_{\mathfrak{C}}(x',x,x')\,U^{ab}(z)\,U^{bc}(x')\,\{\rho^{c}(x'),\rho^{a}(x)\} \quad \text{(U2), (F)} \qquad \text{(G.111)} \\
[A,C]:\ [\rho^{e'}(x'),\rho^{b}(x)] &= i f^{e'bh}\rho^{h}(x')\,\delta^{(2)}(x'-x), \quad x\to x' \\
\tfrac{1}{4}\{[A,C],B\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,f^{e'bh}\,\{\rho^{h}(x'),\rho^{d}(y)\} \\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x')\,U^{cl}(y)\,\{\rho^{h}(x'),\rho^{l}(y)\} \qquad \text{(G.112)} \\
[B,C]:\ [\rho^{d}(y),\rho^{b}(x)] &= i f^{dbh}\rho^{h}(y)\,\delta^{(2)}(y-x), \quad x\to y \\
\tfrac{1}{4}\{[B,C],A\} &= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',y,y)\,f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,U^{ad}(y)\,f^{dbh}\,\{\rho^{h}(y),\rho^{e'}(x')\} \\
&= \tfrac{1}{4}\,i\;K^{P}_{\mathfrak{C}}(x',y,y)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{al}(x')\,U^{cd}(y)\,\{\rho^{h}(y),\rho^{l}(x')\} \qquad \text{(G.113)}
\end{aligned}
$$

Two-ρ terms of this source. Multiplying by the prefactor $-2i\,\mathcal{N}_P$ and collecting equal terms (dummy colour indices renamed),

$$
\begin{aligned}
P^{(2\rho)}\Big|_{\text{GIII.VI.P}} = \mathcal{N}_P \int e^{-ik\cdot(w'-w)}\,\Big\{ & \\
&+\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x,x')\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{ce}(x')\,U^{bl}(x)\,\{\rho^{h}(x'),\rho^{l}(x)\} \\
&+\tfrac{1}{2}\,K^{P}_{\mathfrak{C}}(x',x',y)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{cl}(y)\,U^{be}(x')\,\{\rho^{h}(x'),\rho^{l}(y)\}
\end{aligned}
$$

$$
\begin{aligned}
&- \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{C}}(x', y, y)\, U^{ab}(z)\, U^{bc}(y)\, \big\{\rho^c(y), \rho^a(x')\big\} \\
&- \tfrac{1}{2}\, K^P_{\mathfrak{C}}(x', x, x')\, f^{abc}\, f^{ade}\, U^{hb}(x')\, U^{eh}(z)\, \big\{\rho^c(x'), \rho^d(x)\big\} \\
&- \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{C}}(x', x', y)\, U^{ab}(y)\, U^{ca}(z)\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&- \tfrac{1}{2}\, K^P_{\mathfrak{C}}(x', y, y)\, f^{abc}\, f^{dbe}\, U^{ha}(y)\, U^{eh}(z)\, \big\{\rho^c(y), \rho^d(x')\big\} \\
&+ \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{C}}(x', x, x')\, U^{ab}(x)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(x)\big\} \\
&- \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{C}}(x', x', y)\, U^{ab}(y)\, U^{ac}(x')\, \big\{\rho^c(x'), \rho^b(y)\big\} \\
&+ \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{C}}(x', y, y)\, U^{ab}(x')\, U^{ac}(y)\, \big\{\rho^c(y), \rho^b(x')\big\} \\
&- \tfrac{1}{2}\, N_c\, K^P_{\mathfrak{C}}(x', x, x')\, U^{ab}(z)\, U^{bc}(x')\, \big\{\rho^c(x'), \rho^a(x)\big\} \\
&+ \tfrac{1}{2}\, K^P_{\mathfrak{C}}(x', x', y)\, f^{abc}\, f^{deh}\, U^{eb}(z)\, U^{ad}(x')\, U^{cl}(y)\, \big\{\rho^h(x'), \rho^l(y)\big\} \\
&+ \tfrac{1}{2}\, K^P_{\mathfrak{C}}(x', y, y)\, f^{abc}\, f^{deh}\, U^{eb}(z)\, U^{al}(x')\, U^{cd}(y)\, \big\{\rho^h(y), \rho^l(x')\big\}\Big\} + \big(w' \leftrightarrow w\big)
\end{aligned}
\tag{G.114}
$$

The kernel at the coincident points:

$$
\begin{aligned}
K^P_{\mathfrak{C}}(x', x, x') &= \mathcal{K}(w' - z) \cdot \mathcal{K}(x' - w)\, \mathcal{K}(x' - z) \cdot \mathcal{K}(x - z) \\
K^P_{\mathfrak{C}}(x', x', y) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w)\, \frac{1}{(x' - z)^2} \\
K^P_{\mathfrak{C}}(x', y, y) &= \mathcal{K}(w' - z) \cdot \mathcal{K}(y - w)\, \mathcal{K}(x' - z) \cdot \mathcal{K}(y - z)
\end{aligned}
$$

H Relations between colour networks

For the kernel structures in which the simplified colour structures are linearly dependent, each dependent one equals the following combination of basis structures (exact; checked for SU(3) and SU(4)).

H.1 Part III: coefficient of $\log(\vee/\Lambda)$, two-ρ

For the kernel structures in which the simplified colour structures are linearly dependent, each dependent one equals the following combination of basis structures (exact; checked for SU(3) and SU(4)):

Structure 1, $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w')\; K(x, x, z)$**.**

$$
\begin{aligned}
&f^{abc}\, f^{cde}\, U^{ah}(z)\, U^{dl}(w)\, U^{lm}(w')\, U^{hb}(x)\, \rho^m(x')\, \rho^e(x) \\
&\quad = -1\, f^{abc}\, f^{cde}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^l(x')\, \rho^e(x) + 1\, f^{abc}\, f^{cde}\, U^{ah}(w)\, U^{bl}(x)\, U^{dl}(z)\, U^{mh}(w')\, \rho^m(x')\, \rho^e(x) \\
&f^{abc}\, f^{cde}\, U^{hd}(x)\, U^{hl}(w')\, U^{ma}(z)\, U^{mb}(x)\, \rho^l(x')\, \rho^e(x) \\
&\quad = -1\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{ce}(x)\, U^{bd}(z)\, \rho^l(x')\, \rho^h(x) + 1\, f^{abc}\, f^{cde}\, U^{ah}(z)\, U^{bl}(x)\, U^{dh}(x)\, U^{ml}(w')\, \rho^m(x')\, \rho^e(x) \\
&f^{abc}\, f^{cde}\, U^{hd}(x)\, U^{hl}(x')\, U^{ma}(z)\, U^{mb}(x)\, \rho^l(x')\, \rho^e(x) \\
&\quad = -1\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{ce}(x)\, U^{bd}(z)\, \rho^l(x')\, \rho^h(x) + 1\, f^{abc}\, f^{cde}\, U^{ah}(z)\, U^{bl}(x)\, U^{dh}(x)\, U^{ml}(x')\, \rho^m(x')\, \rho^e(x) \\
&f^{abc}\, f^{cde}\, U^{hb}(w)\, U^{hl}(x')\, U^{ma}(x)\, U^{md}(z)\, \rho^l(x')\, \rho^e(x) \\
&\quad = -1\, f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^l(x')\, \rho^e(x) - 1\, f^{abc}\, f^{cde}\, U^{hd}(w)\, U^{hl}(x')\, U^{ma}(z)\, U^{mb}(x)\, \rho^l(x')\, \rho^e(x)
\end{aligned}
$$

Structure 4, $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w')\; K(x', x', z)$**.**

$$
\begin{aligned}
&f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{hl}(w)\, U^{ma}(z)\, U^{mb}(x')\, \rho^l(x)\, \rho^e(x') \\
&\quad = 1\, f^{abc}\, f^{deh}\, U^{al}(w)\, U^{cd}(x')\, U^{be}(z)\, \rho^h(x')\, \rho^l(x) + 1\, f^{abc}\, f^{cde}\, U^{hb}(x')\, U^{hl}(w)\, U^{ma}(z)\, U^{md}(x')\, \rho^l(x)\, \rho^e(x') \\
&f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{hl}(x)\, U^{ma}(z)\, U^{mb}(x')\, \rho^l(x)\, \rho^e(x') \\
&\quad = 1\, f^{abc}\, f^{deh}\, U^{al}(x)\, U^{cd}(x')\, U^{be}(z)\, \rho^h(x')\, \rho^l(x) + 1\, f^{abc}\, f^{cde}\, U^{hb}(x')\, U^{hl}(x)\, U^{ma}(z)\, U^{md}(x')\, \rho^l(x)\, \rho^e(x')
\end{aligned}
$$

Structure 11, $\mathcal{K}(x - w) \cdot \mathcal{K}(x' - w')\; K(w, w, z)$**.**

$$
\begin{aligned}
&f^{abc}\, f^{deb}\, U^{hd}(z)\, U^{ha}(w)\, U^{lm}(w')\, U^{le}(w)\, \rho^m(x')\, \rho^c(x) \\
&\quad = -1\, f^{abc}\, f^{deh}\, U^{al}(w')\, U^{be}(z)\, U^{ch}(w)\, \rho^l(x')\, \rho^d(x) - 1\, f^{abc}\, f^{dea}\, U^{hd}(z)\, U^{he}(w)\, U^{lm}(w')\, U^{lb}(w)\, \rho^m(x')\, \rho^c(x) \\
&f^{abc}\, f^{deb}\, U^{hd}(z)\, U^{ha}(w)\, U^{lm}(x')\, U^{le}(w)\, \rho^m(x')\, \rho^c(x) \\
&\quad = -1\, f^{abc}\, f^{deh}\, U^{al}(x')\, U^{be}(z)\, U^{ch}(w)\, \rho^l(x')\, \rho^d(x) - 1\, f^{abc}\, f^{dea}\, U^{hd}(z)\, U^{he}(w)\, U^{lm}(x')\, U^{lb}(w)\, \rho^m(x')\, \rho^c(x)
\end{aligned}
$$

Structure 12, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$**.**

$$f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(w)\,\rho^{c}(x')\,\rho^{m}(x)$$
$$= -1\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x')\,\rho^{l}(x) - 1\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(w)\,\rho^{c}(x')\,\rho^{m}(x)$$
$$f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(x)\,\rho^{c}(x')\,\rho^{m}(x)$$
$$= -1\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^{d}(x')\,\rho^{l}(x) - 1\,f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(x)\,\rho^{c}(x')\,\rho^{m}(x)$$

H.2 Part IV: coefficient of $\log\vee$

Structure 1, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$**.**

$$f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)$$
$$= -f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x) + f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)$$
$$f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)$$
$$= -f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x) + f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)$$
$$f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)$$
$$= -f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x) + f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)$$
$$f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)$$
$$= -f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x) - f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)$$

Structure 3, $\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$**.**

$$f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{ha}(w)\,\rho^{d}(x)\,\rho^{e}(x')$$
$$= -f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^{a}(x)\,\rho^{e}(x') - f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{hd}(w)\,\rho^{a}(x)\,\rho^{e}(x')$$
$$f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(x')\,\rho^{b}(x)\,\rho^{e}(x')$$
$$= f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{hd}(x')\,\rho^{a}(x)\,\rho^{e}(x') + f^{abc}\,f^{cde}\,U^{hb}(w')\,U^{ha}(x')\,\rho^{d}(x)\,\rho^{e}(x')$$
$$f^{abc}\,f^{dce}\,U^{ha}(x')\,U^{hd}(w)\,\rho^{b}(x)\,\rho^{e}(x')$$
$$= f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hd}(x')\,\rho^{a}(x)\,\rho^{e}(x') + f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{ha}(w)\,\rho^{d}(x)\,\rho^{e}(x')$$

Structure 4, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$**.**

$$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')$$
$$= f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x) + f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')$$
$$f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')$$
$$= f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x) + f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')$$

I Checks of the reordering

The algebra was generated by a short program, which also checked it numerically.

1. *Reordering identity.* Eqs. (A.4)–(A.7), including the explicit one-ρ remainder R_1, hold to machine precision for random matrices A, B, C.

2. *Colour algebra.* Each of the 144 pieces was evaluated before and after the colour algebra, for $SU(3)$ with random classical charges and random symmetric adjoint Wilson lines (U of $V = W\,\mathrm{diag}(1,-1,-1)\,W^\dagger$, W random). The two agree to 2×10^{-15} for every piece. With general, non-symmetric Wilson lines, 14 pieces differ. These are exactly the pieces whose simplification uses (U1*) or (U2*), i.e. the convention $U^{bc} = U^{cb}$.

3. *The whole reordering.* For every sub-term we took the charges as explicit operators on three sites (the δ-function becomes a Kronecker delta), with random kernels and random symmetric Wilson lines. We then compared

$$\sum_{\text{singlets}} \langle s|\,O - S(O)\,|s\rangle \qquad \text{and} \qquad \sum_{\text{singlets}} \langle s|\,(\text{two-}\rho\text{ pieces})\,|s\rangle,$$

where O is the three-ρ operator of the sub-term, $S(O)$ its fully symmetrized form, and $|s\rangle$ runs over the colour singlets, in which every one-ρ term has zero expectation value. We used $SU(3)$ with one quark per site (one singlet) and $SU(2)$ with two spin-$\frac{1}{2}$ per site (five singlets), three random draws each. The largest difference was 4×10^{-14}, for values up to 25. This checks the commutators, the signs, the factors $\frac{1}{4}$ and $\frac{1}{2}$, the δ-functions and the colour algebra together.

J Scripts

The numbers and tables come from the scripts of the folders `three_rho`, `two_rho`, `bfkl` (see their README files) and of this folder:

- `ft/ft_checks.py`: Part I (partial fractions, the split of the bracket, the ξ integral of `main.tex`, the commutator identity with explicit charges);
- `allsets.py`: all terms by row, for the coefficients of $\log(\vee/\Lambda)$ (A) and of $\log\vee$ (P), as written and corrected;
- `check_all.py`: pointwise totals of A vs JIMWLK$\otimes$LO and of P vs BFKL$\otimes$LO (two-ρ; symmetrized three-ρ);
- `build_P.py` and `reduce_run.py`: the exact tables of Secs. 26, 27;
- `make_combined.py`: writes this file from `combined_static.tex`, `ft_part.tex` and the notes of the other folders.